

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/364226657>

ArcGIS 10x PRACTICAL MANUAL

Book · October 2022

CITATIONS

2

READS

12,819

1 author:



[Gezahegn Weldu](#)

The Space Science Institute

21 PUBLICATIONS 347 CITATIONS

SEE PROFILE



ArcGIS 10x

PRACTICAL MANUAL

Prepared by: Gezahegn Weldu W/Mariam (M.Sc.)

Content Editors: Kalid Hassen (B.Sc.)
Tadele Bedo (B.Sc.)

Haramaya University, 2015

Acknowledgments

I would like thank all professionals that contributed in many ways to the preparation of this manual. The shared technical knowledge, experiences, and perspectives have positive impact for the betterment of this particular manual. Hence, the preparation of this manual would not have been possible without technical and editorial reviewers who have forwarded their critical and helpful comment on early drafts of this manual: Mr.Kalid Hassen and Tadele Bedo (from, GISc Program, School of Geography and Environmental Science, Haramaya University). Thanks to all my friends for sharing my happiness when starting this manual and following with encouragement when it seemed too difficult to be completed. I would have probably give up without their support and example on what to do when you really want something.

TABLE OF CONTENTS

Acknowledgments.....	ii
INTRODUCTION	1
CHAPTER ONE	2
DATA MANAGEMENT – ArcCATALOG	2
1.1.Starting ArcCatalog	3
1.2.Components of ArcCatalog	3
1.3.Minimize/Expand directories and folder connection.....	5
1.3.1.Minimize directories	5
1.3.2.Folder Connection.....	5
1.4.Appearance of GIS data in ArcCatalog	7
CHAPTER TWO	9
WORKING WITH ARCMAP INTERFECE	9
2.1.Launching ArcMap and Saving Map Document	10
2.1.1.Launching ArcMap	10
2.1.2.Save the map document to a new location.....	13
2.2.Folder Connection and Add data to ArcMap.....	14
2.3.Customizing the Interface.....	15
2.4.Adding Tools and Toolbars	16
2.5.Opening ArcToolbox	17
2.6.Working with Catalog Window	17
2.7.Making Logical Arrangement in the TOC.....	19
2.7.1.Change a Layer’s Display Order.....	19
2.7.2.Turning on and off of layers	20
2.7.3.Removing Layers	21
2.7.4. Click the Catalog window button in the standard toolbar.....	22
2.7.5.Coloring and styling the layer.....	22
2.7.6.Labeling Layers	23
2.8. Navigate in a Map Document.....	24
2.8.1.Interactive Zoom in/out	25
2.8.2.The fixed zoom in/fixed zoom out.....	25
2.8.3. Panning	25
2.8.4. Full extent	26
2.8.5. Previous/Next.....	26
2.8.6. Identify features	26
2.8.7. Data view and layout view.....	27
2.8.8. The measuring Tool	27
2.8.9. Finding Features.....	28
2.8.10.Use the Select Features Tool.....	30
2.8.11.Clear Selected Features.....	30

CHAPTER THREE	31
DISPLAYING DATA	31
3.1. Getting information about features	31
3.1.2. Display a complete attribute table	31
3.1.3. Sorting records in an attribute table	32
3.2. Attribute Data and Spatial Querying	33
3.2.1. Selection by Attribute	33
3.2.2. Selection by location	35
CHAPTER FOUR	37
THEMATIC MAPPING USING SYMBOLOGY	37
4.1. Symbolizing Features	37
4.2. Symbolize Features with Different Colour	38
4.3. Display Features with Graduated Color	39
4.4. Display Features with Graduated Symbol	42
4.5. Using Query to Display Selected Categories	43
CHAPTER FIVE	45
SPATIAL ANALYSIS WITH VECTOR DATA	45
5.1. Creating Layers by Selection	45
5.2. Saving the Selected Layer	45
5.3. Clipping	46
5.4. Buffering	48
CHAPTER SIX	53
CREATING AND EDITING DATA	53
6.2. Spatial Referencing	54
6.3. Georeferencing	56
6.4. Data Entry	58
6.4.1. Creating Personal Geodatabase	58
6.4.2. Creating Feature Dataset	59
6.4.3. Creating Feature Classes	61
6.5. Digitizing in Arc Map	64
6.5.1. Load the Feature Data Set in ArcMap	64
6.5.2. Digitizing Polygon Features	65
6.5.3. Digitizing lines and snapping	69
6.5.4. Digitizing Point Features	72
6.5.5. Add Attribute Data	72
6.5.6. Calculating Areas and Lengths	73
CHAPTER EIGHT	84
WORKING WITH XY DATA	84
8.1. Adding XY data into the ArcMap	84
8.2. Exporting XY data feature class to shape file	88
8.3. Adding a field to a table	89
CHAPTER NINE	91
WORKING WITH RASTER DATA	91
9.1. Creating Slope, Aspect, Hillshade and Contour from DEM	92
9.1.1. Extract by Mask	93
9.1.2. Generating Slope	94

9.1.3. Generating Aspect.....	106
9.1.4. Generating Hillshade Data	108
9.1.5. Visualize Reclassified slope and the Hillshade.....	109
9.2. Display Landscape data in 3D	112
9.2.1. Create Contour and Triangulated Irregular Network.....	116
3.10. Spatial Interpolation.....	121
CHAPTER TEN.....	124
IMAGE CLASSIFICATION USING ArcGIS	124
10.1. Supervised classification	124
10.1.1. Adding the Image Classification toolbar	124
10.1.2. Adding a multiband image for the classification	125
10.2. Creating Training Samples	126
10.2.1. Managing training samples	126
10.2.2. Managing training samples	127
10.3. Executing the Maximum Likelihood Classification tool.....	128
CHAPTER ELEVEN.....	130
SPATIAL ANALYSIS AND MODELING	130
11.1. Tutorial scenario	130
11.2. Preparing for analysis	132
11.3. Starting New ArcMap.....	132
11.4. Saving your map document	133
11.5. Accessing Spatial Analyst and Data Exploration	134
11.1. Finding a site for a new Housing Sites	135
11.1.1. Creating a new toolbox	135
11.1.2. Creating a new model	137
11.1.3. Renaming the model	137
11.1.4. Specifying environment settings.....	139
11.2. Deriving datasets	140
11.3. Deriving Slope from Elevation	142
11.3.1. Deriving distance from Road	143
11.3.2. Deriving distance from River.....	144
11.3.3. Deriving Distance from Built-up (Developed area	145
11.4. Run Model to Derive Datasets.....	146
11.5. Reclassifying Datasets	147
11.5.1. Reclassifying slope	148
11.5.2. Reclassifying distance from Road	149
11.5.3. Reclassifying Distance from River	150
11.5.4. Reclassifying Distance to Built-up area.....	151
11.6. Executing Reclassify	152
11.7. Weighting and Combining Datasets	153
11.8. Executing Weighted Overlay.....	156
11.9. Selecting optimal sites	157
11.10. Selecting the Best site.....	159
11.10.1. Raster to Polygon.....	159
11.10.2. Select By Location.....	161
11.10.3. Select Layer By Attribute	162

11.10.4.Copy Features Tool.....	163
CHAPTER TWELVE.....	165
WATERSHED DELINEATION USING ARC HYDRO TOOLS	165
12.1.Getting Started	165
12.2.Dataset Setup	166
12.3.Adding Elevation data to ArcMap	167
12.4.The process of Watershed Delineation	168
12.4.1.Fill Sinks	169
12.4.2.Flow Direction	170
12.4.3.Flow Accumulation.....	171
12.4.4.Stream Definition.....	172
12.4.6.Catchment Grid Delineation	175
12.4.7.Catchment Polygon Processing.....	176
12.4.8.Drainage Line Processing	177
12.4.9.Adjoint Catchment Processing.....	178
12.4.10.Drainage Point Processing	179
12.5.Longest flow path for catchments.....	180
12.6.Watershed Processing.....	181
12.6..Batch watershed delineation	181
12.6.2.Delineating the watershed using batch point.	183

INTRODUCTION

Dear trainee, I am very pleased to be able to introduce a Geographic Information System (GIS). GIS has been formally defined by the Association for GIS as a powerful tool that integrates hardware, software, and data with functionality for capturing, storing, checking, integrating, manipulating, analyzing and displaying spatial data. Spatial data are data that are spatially referenced to the Earth. Hence, it integrates variety of data with a spatial component regardless of its source. It stores and manages the data, analyzes the data as a whole and display it in various formats, leverages visualization to make critical business and planning decisions, and saves tremendous time and money.

Moreover, today's GIS involves collaboration of various technologies and disciplines like geography, cartography, surveying, remote sensing, satellite imagery, photogrammetry, spatial statistics, mathematics, geometry, topology, computer science, information science, library science, web-technology, etc. As such this training manual aims to guide practitioners/beginners how to explore ArcGIS Desktop 10 version of a popular GIS software produced by ESRI (Environmental System Research Institute) for different analysis. This training manual is meant to train some fundamental and advanced GIS operations using ArcGIS. It is not meant to be a comprehensive training in GIS or ArcGIS. However, I hope the manual will let you start using GIS and excited about learning more. Moreover, it will guide you how to use the ArcMap module to create maps, query data, perform analysis, and most of the other basic and advanced GIS operations.

ArcMap is the interface window that you will mainly work with, during this introduction to GIS. Whereas, ArcCatalog allows you to browse and manage spatial data files. As well as ArcScene (for 3D visualization) works in a way similar to windows explorer but specialized for GIS data. ArcCatalog can be run as a standalone application or as ArcCatalog window in ArcMap. For further information about ArcMap, go to **Help-ArcGIS Desktop Help - Contents tab - ArcMap**. Thus, I urge interested students, staffs, researchers and all concerned to use this manual and familiarize themselves with application of GIS in particular to their field of specialization.

CHAPTER ONE

DATA MANAGEMENT – ARCCATALOG

The ArcCatalog application provides a catalog window that is used to organize and manage various types of geographic information for ArcGIS for Desktop you explore and manage your data. After connecting to your data, use the Catalog to explore its contents. When you find the data you want to use, you can add it to a map. Often, when you get data for a project, you can't use it right away; you may need to change its projection or format, modify its attributes, or link geographic features to attributes stored in another table. When the data is finally ready, you should document its contents and the changes you have made. These data management tasks can all be accomplished using tools that are available in the Catalog. Arc Catalog organizes these contents into a tree view that you can work with to organize your GIS datasets and ArcGIS documents, search and find information items, and to manage them. Arc Catalog presents this information in a tree view and allows you to select a GIS item, view its properties, and to access tools to operate on the selected items.

Dear trainee, at the end of this chapter the following are the major tasks you are expected to execute in ArcCatalog environment.

- ♣ Browse data
- ♣ Explore data
- ♣ View and Create Meta Data
- ♣ Search for data
- ♣ Manage data sources
- ♣ Use arc tool box
- ♣ To arc Map
- ♣ Add hyperlinks and photographs on the map

1.1. Starting ArcCatalog



Data management application. ArcCatalog helps users to browse, search, explore, view, and manage data (with metadata creation and editing capabilities) and maps. It also helps GIS database administrators maintain spatial and tabular GIS data for use by others in their organization. In this exercise you will explore Arc Catalog and learn how it is used within the wider ArcGIS environment. You will practice the most important functions only. Moreover, additional functions will be introduced later in the training. First you will be familiarized with the interface, then with specific functions such as browsing and exploring data, creating metadata, etc. Finally you will learn that Arc Catalog can be used as a gateway to Arc Map. The first step to explore in Arc Catalog is how to open the program.

Go the start menu on your computer and find the ArcGIS program tab. Open the ArcGIS folder and click on Arc Catalog. The following be used in the lessons. The following for moving through files may be used in the lessons.

To access Arc Catalog, **click Start → All Programs → ArcGIS → Arc Catalog**



1.2. Components of ArcCatalog

Before you start working with the data, first have look at the interface. We assume that this is the first time that you use the software. We will therefore briefly show the components of Arc Catalog's desktop. When **Arc Catalog** starts, you will see two panels in the **Arc Catalog** window. A list of folders and files appear on the left side of the **Arc Catalog** window and the window is known as catalog tree. The content of the currently selected folder are displayed on the right side. When Arc Catalog starts, the main menu and the standard Toolbar appear

by default. Toolbars are areas in the interface of ArcGIS with buttons that activate various features and functions. After opening the software, you will be greeted by screen like this one:

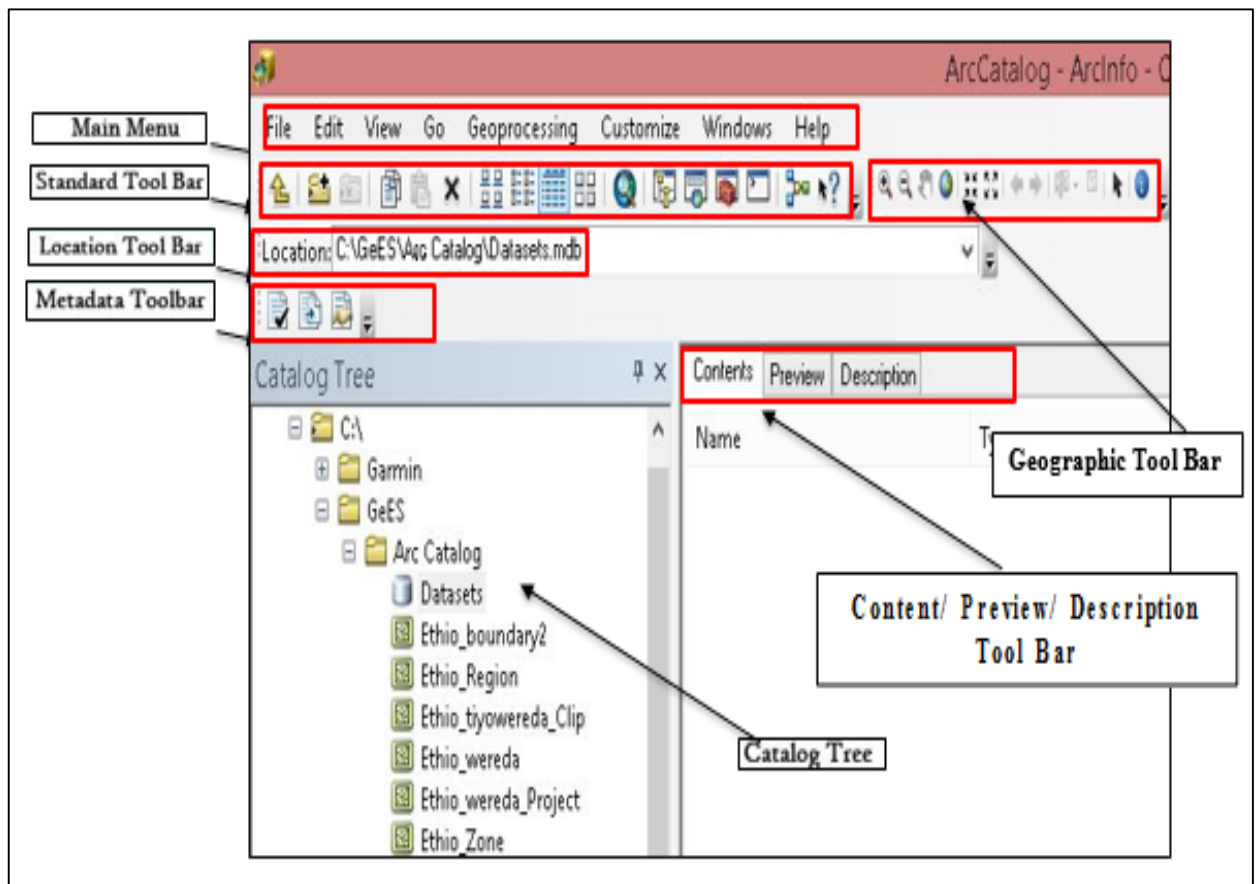


Figure 1: Components of Arc Catalogue

- **Menu:** default tool bar with pull-down menu that contains: File, Edit, View, Go, Tools, Windows and Help.
- **Standard:** allows connecting to any folders on hard drive as well as folders on the network drives, copying, pasting, deleting, viewing and searching files and folders.
- **Catalog Tree:** displays all connected files and folders, and their directory tree.
- **CPD Window:** allows examining name and type of the file, previewing content (geography and tabular data) and viewing metadata of the file.

1.3. Minimize/Expand directories and folder connection

1.3.1. Minimize directories

Clicking the plus signs on the left of the directories expands the directory (i.e., displays all the directories it contains). Selecting the name of the directory on the left hand side of **Arc Catalog** displays its contents on the right hand side.

- a) Click on the plus sign next to **C:**. Now, click on the folder “**GIS_GeES**” → “**Oromia_featurea**”.
- b) Click the **plus sign** next to the **GIS_GeES “Oromia_featurea”** folder to expand it.
- c) Click on the name “**Oromia_featurea**” to display the contents of the directory on the right hand side.
- d) Minimize the Data directory by clicking on the minus sign.

If you are a Windows user, you probably already know how to use the buttons in the upper right corner of the window. If not, practice using these buttons. Also practice resizing and moving application windows. Ask your instructor if this is unfamiliar to you.

- a) This button will minimize your window: 
- b) Click the middle button  to maximize the window:  If it is already maximized, the button will look like this:  Click it to make the window smaller and moveable.
- c) When a window is not maximized, you can drag on a corner of the window to resize it. Try it.
- d) This button will close your window: 
- e) Click and drag on the bar at the top of the window to move it around your screen.

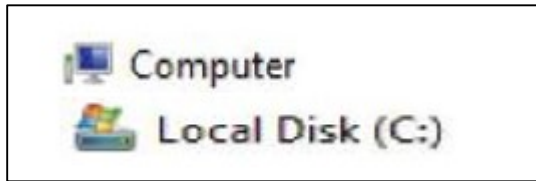
1.3.2. Folder Connection

When you select a folder connection in the Catalog tree, the Contents tab lists the items it contains. Unlike Windows Explorer, the Catalog doesn't list all files stored on disk; a folder might appear empty even though it contains several files. Folders containing geographic data sources have a different icon to make your data easier to find. Look in a folder connection and locating the training folder in your catalog.

Click a folder connection  in the Catalog tree. The items it contains appear in the Contents tab.



Expand the sign of  the C drive in the Catalog tree



Look for C drive and “C:\GIS_GeES” folder and click on Arc catalog

Now you should look the folder containing the training data listed in the catalog tree. To see also what is contained in each of the folders, expand the plus sign behind each of the see the different types of Arc GIS compatible Geographic datasets listed below each of the folders as well as displayed below the contents tab.

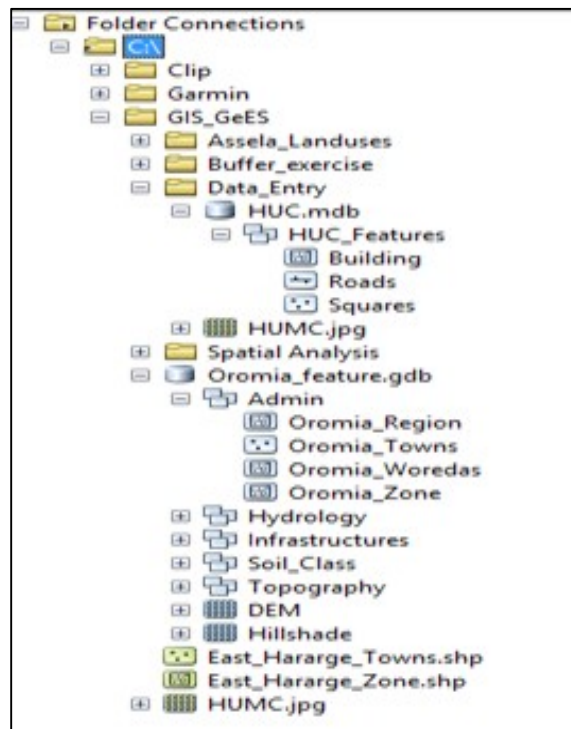


Figure 2: ArcCatalog Tree

1.4. Appearance of GIS data in ArcCatalog

In this part you are going to practice how to explore different types of GIS data i.e. (Point, line and polygon)  base in Arc Catalog. You are also expected to recognize data base compatible with ArcGIS software. ArcGIS lets you work with a number of data formats from many sources. Some of the Common GIS file Types are the followings.

- **Personal Geodatabase:**  Personal Geodatabase is a database that stores geographic information. The concept “Geodatabase” can be simply explained as a container (database) that stores data. It is datasets and feature classes.
- A Geodatabase under a folder is a single user Geodatabase. Connect to C:\GeES_GIS\Arc Catalog Folder and expand the plus sign.
- **Feature dataset:**  the feature classes that share the same geographic extent and coordinate system will be grouped together in a Feature Data Set. A feature dataset is a collection of feature classes, graphs, and relationship classes that share a common spatial reference. Connect to C:\GeES_GIS\ expand the plus sign behind the **Oromia_featuress.mdb**. The Hydrology, Infrastructure and Administration are three feature data set which contains feature classes.
- **Feature classes:**  are collections of geographic are collections of geographic features that share the same geometry type (point, line or polygon), attributes and spatial reference. Expand the plus sign behind one of the feature datasets (the hydrology, infrastructure or administration). The list of items in each of the feature datasets are feature classes.
- **Shape files:** ArcGIS Shape files format is a widely adopted standard and comprises three or more associated files. These are vector systems of storing geographic datasets in the Arc GIS software:

To see the appearance of shape files connect to C:\GeES_GIS\ArcCatalog and Click on the Arc Catalog folder. The green icons ordered inside the arc Catalog folder are shape files.

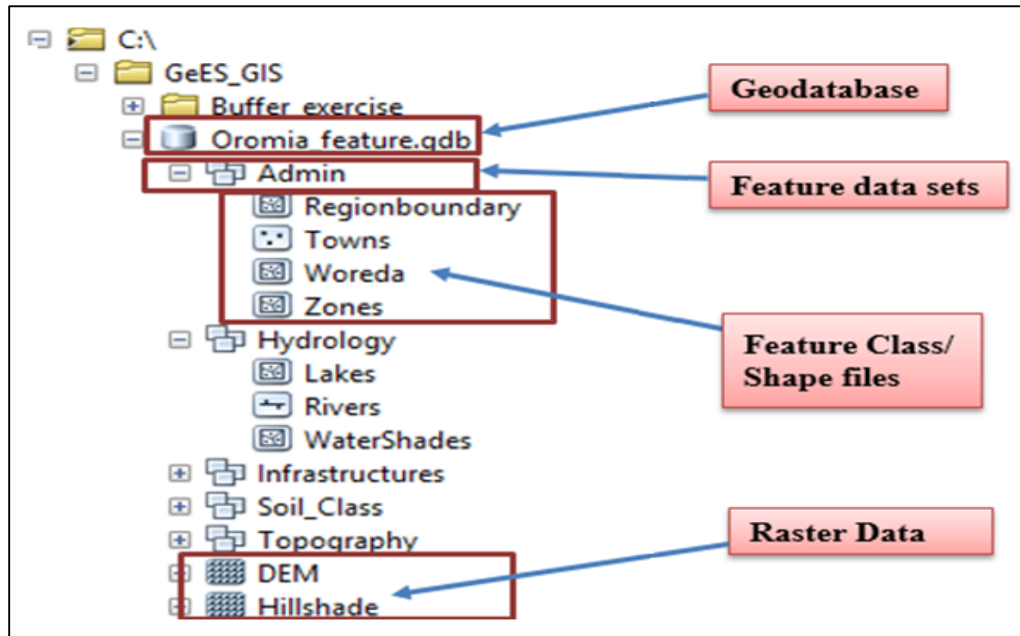


Figure 3: GIS data format appearance inn ArcCatalogue

CHAPTER TWO

WORKING WITH ARCMAP INTERFECE

ArcGIS has four separate applications, ArcMap, ArcCatalog, ArcToolbox, ArcScene and ArcGlobe. In the previous exercise you have seen the Arc Catalog and the functions in it which help to browse, explore and manage data. In this part we will focus on ArcMap interface. ArcGIS 10 is a desktop Geographic Information System from ESRI (Environmental System Research Institute).

Objectives

At the end of this task the trainees are expected to:

- ♣ Identify parts of ArcMap
- ♣ Describe basic functions of ArcMap
- ♣ Launch and save map documents
- ♣ Navigate in ArcMap using tools from Standard Tool bar
- ♣ Select features using Query Language

Why Arc Map?

ArcMap is the major mapping component in ARCGIS. It is a tool for creating, viewing / display data, querying, and editing, assign symbols composing, and publishing maps. Hence, Arc Map is where you create maps and access most of the ArcGIS functionality. Arc Map works with map documents. A map document is a collection of links to different spatial data layers and tables, along with instructions for how the layers will be displayed. Since ArcMap is used to perform most of the GIS operations, it requires more than one exercise to be understood. But you do not have to master all the functions and processes available for the first results. The purpose of this exercise is to guide your first steps through the basic functionalities of ArcMap, and to show you how you can create a map document, change the representation of a map and explore map data. Most of the remaining exercises will introduce additional features of ArcMap. But even then, the course will have covered only a portion of all functions available. To learn more about ArcMap, you can use the ***ArcMap online Help*** System. It is available at any time, and directly accessible via the ArcMap user interface. The Arc Map environment and functionalities will be practiced in this part of the training document. You will get the data for respective activities in the directories guided by the manual.

2.1. Launching ArcMap and Saving Map Document

2.1.1. Launching ArcMap

You can launch /Start ArcMap in several ways.

Steps

- a) From the windows taskbar Click the **Start button** → **All Programs** → **ArcGIS** → Click **Arc Map** or
- b) If you were on Arc Catalogue Click the Launch **Arc Map** button in **Arc Catalog**.

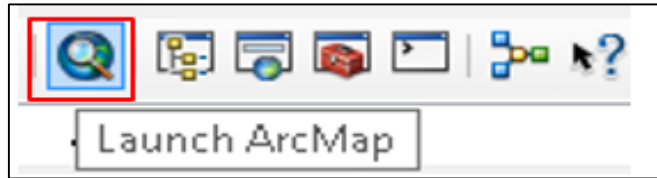


Figure 4: Launching ArcMap

- c) In the resulting ArcMap – getting started window, click ***open empty map*** or you can prevent this dialog from displaying again by putting a check mark in the “do not show this dialog again” box
- d) ArcMap will show a welcome screen. Click Ok and ArcMap Will display and empty map portrayed as follows.

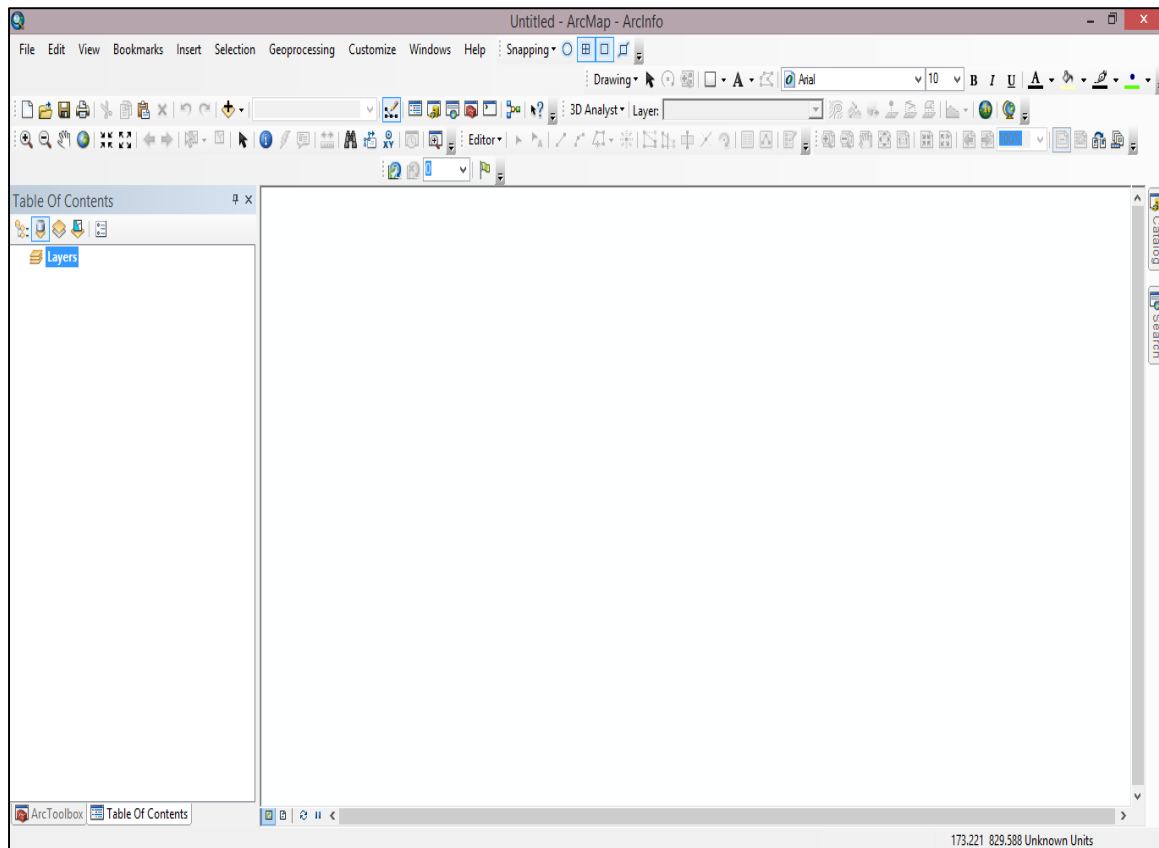


Figure 5: ArcMap Window

The following section will go through adding data to ArcMap, navigate through Arc map using Tools, and finally colouring and styling Layers. ArcMap is where you create maps and access most of the ArcGIS functionality. You can access some of the same functions through ArcCatalog and will need to decide for yourself which approach you find works best for you. It helps to display data, make selections, analyze data, manipulate data, create data, assign symbols and produce map.

There are many parts and functionality in ArcMap interface.

- a) Table of Content (TOC)
- b) ArcToolbox
- c) Data View
- d) Search window
- e) Arc Catalog

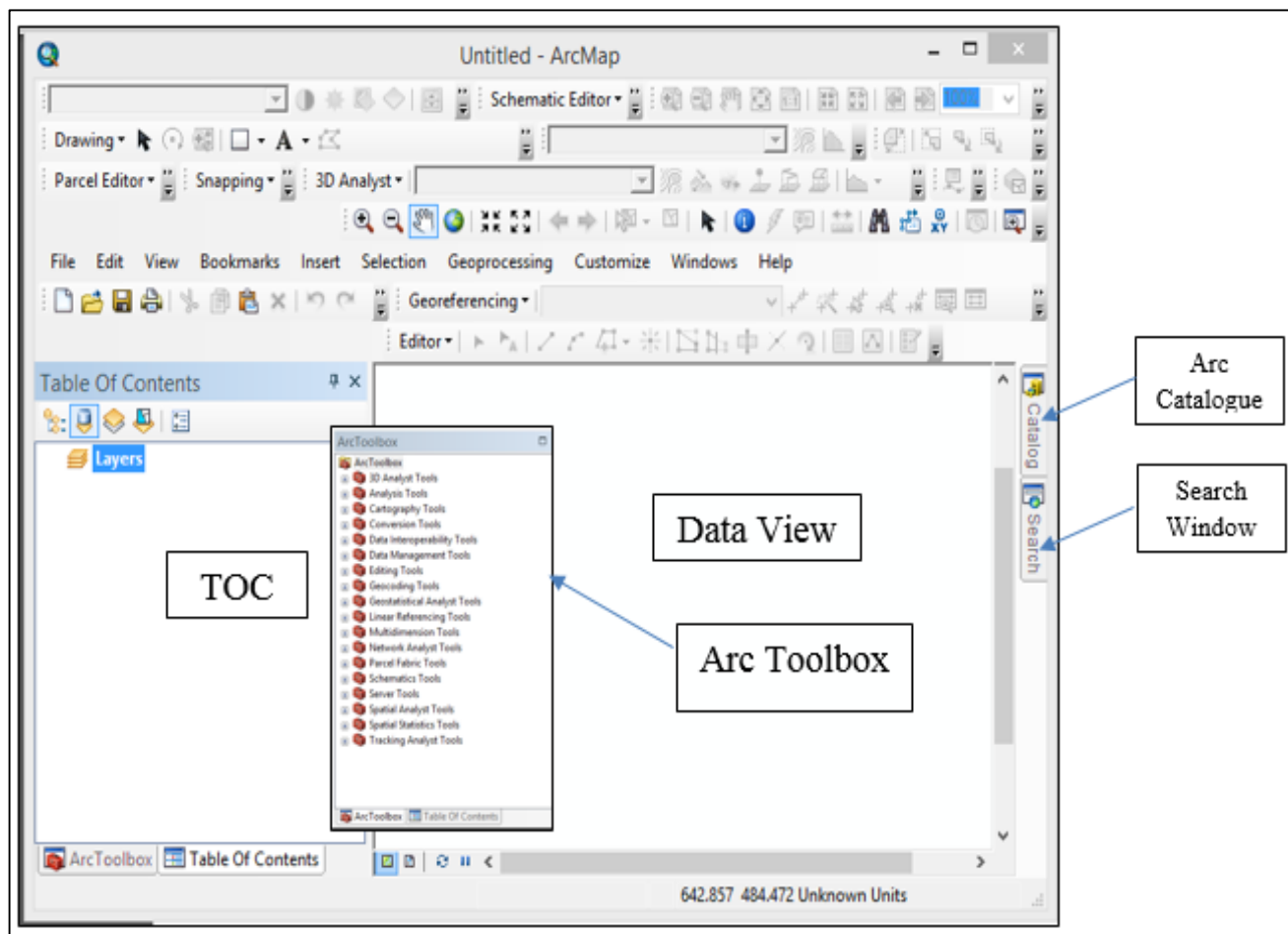


Figure 6: Components of ArcMap

- **Layout:** allows display, zoom and scale layouts.
- **Standard:** allows creating a new document, opening an existing document, saving and printing a document, editing functions, adding/editing map and data, opening ArcCatalog and ArcToolbox, opening Model Builder, and activating help.
- **Tools:** allows working with data view frame or work space viz. displaying, zooming, selecting, identifying, finding and measuring features. Several other toolbars can be made active by right-clicking on the Main Menu toolbar or by clicking on Main Menu/View/Toolbars...
- **TOC:** The left panel of the ArcMap and serves as a legend for the map (it also has other uses which we will discover later).The TOC also shows the order in which the layers are drawn.

- Table of Contents displays and manages (group, ungroup, activate, access layer properties, etc.) all the layers (and its source) and symbology of the layers. The layers can also be displayed by their source location.

What can you do in the Table of Contents?

You can do the following!

- a) Manage layer properties and characteristics
- b) Change the order of layer display

Turn on/off the visualization of a layer

- **Data Frame:** work space that allows displaying and working with maps. Data Frame displays maps as layers that are checked in the Table of Contents. Each displayed layer does not store the actual geographic data but it references data stored in shapefiles, Geodatabase, etc.
- **Status Bar:** displays coordinates/units of the cursor in the Data View or Layout View.
- **Layers:** A layer contains information about how to display the dataset, but not the dataset itself. Each data frame contains one or more layers. Any datasets you add to the map are represented as layers in the Table of Contents. Checking the box next to each layer will display the map elements. The order of the layers will display map elements in that order.

2.1.2. Save the map document to a new location

You can save all files in C:\GIS_GeES\ Create new folder and rename it as Exercise Data.

- a) In the menu bar click file, save As.
- b) Navigate to the C:\GIS_GeES folder and save the map as ***Introduction to ArcMap.mxd***.
- c) Click save.

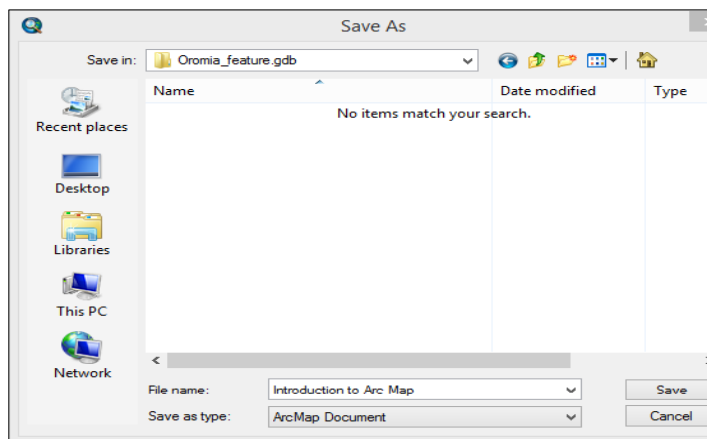


Figure 7: Saving ArcMap

2.2. Folder Connection and Add data to ArcMap

In order to add data to ArcMap interface:

- Click on the **Add data** icon in the standard tool bar 
- In the add data dialogue box Click Folder Connection  | From the dropdown arrow and select **C**
- Double Click **GIS_GeES** folder
- Double click on the **ArcMap_Catalog**
- Double Click on the **Oromia_featurea** Geodatabase
- Click on the **Administration** feature dataset | Select all data layers by clicking once on each features by holding down **Control** or **Shift** key
- Click on the add button.

You may use the following steps as an Alternative!

You can also add the above data by clicking on File from menu Bar → **Add data**. Result: In the Arc Map environment you get the major administrative boundaries of Oromia Regional State.

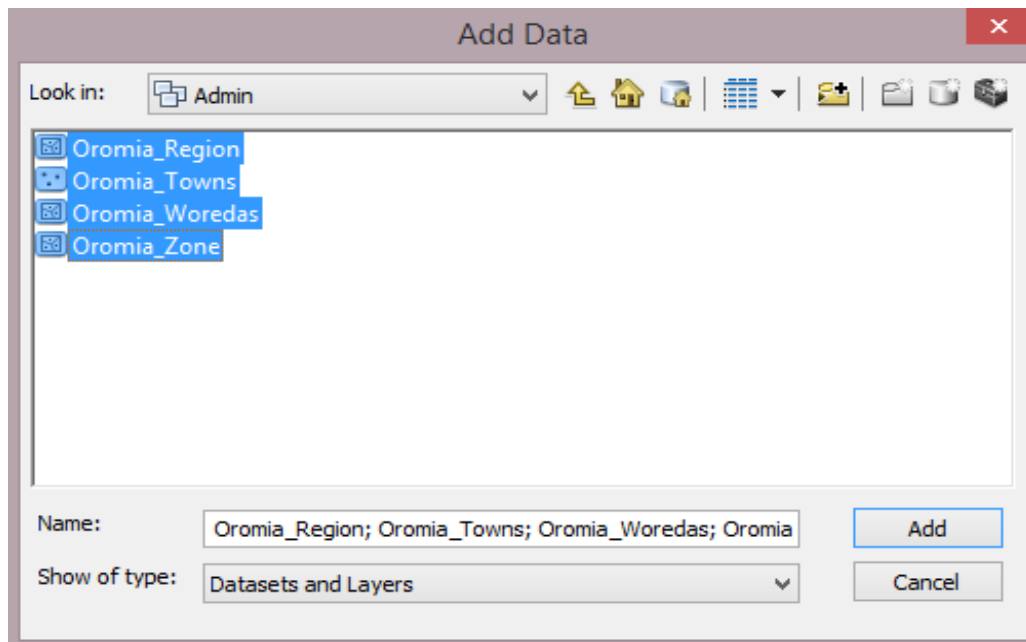


Figure 8: Add data to ArcMap

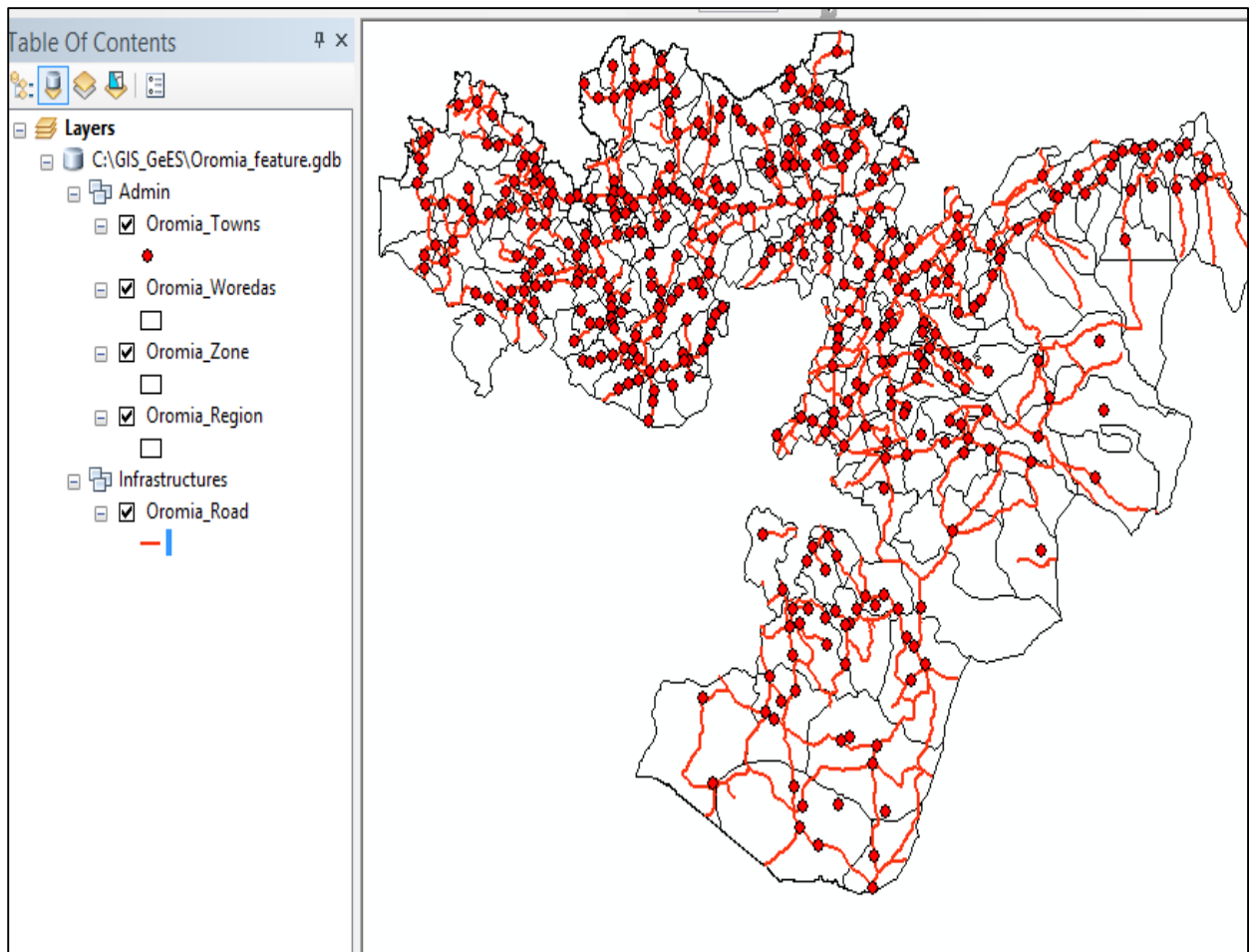


Figure 9: Adding spatial data in ArcMap

2.3. Customizing the Interface

ArcMap is made up of many different windows and (dock-able) toolbars that you can resize and move around, so don't be surprised if ArcMap looks slightly different each time you open it. The window that lists your map layers is the table of contents, the window that shows your map is the map display. You can close the table of contents by clicking on the "x" and resize it by holding your cursor over the right edge until your cursor changes to a two-headed arrow. Left-click and drag the edge to resize this window. To move one of your toolbars, left click on the end (top or left), hold down, and move to a new location.

2.4. Adding Tools and Toolbars

To Add or remove a toolbar

- a) Left-click anywhere outside of the table of contents, map display, anything with a check mark next to it will be displayed.
- b) You can choose any of the toolbars you want to add you can also add new buttons to your toolbars from the “customize” option in from the menu bars.

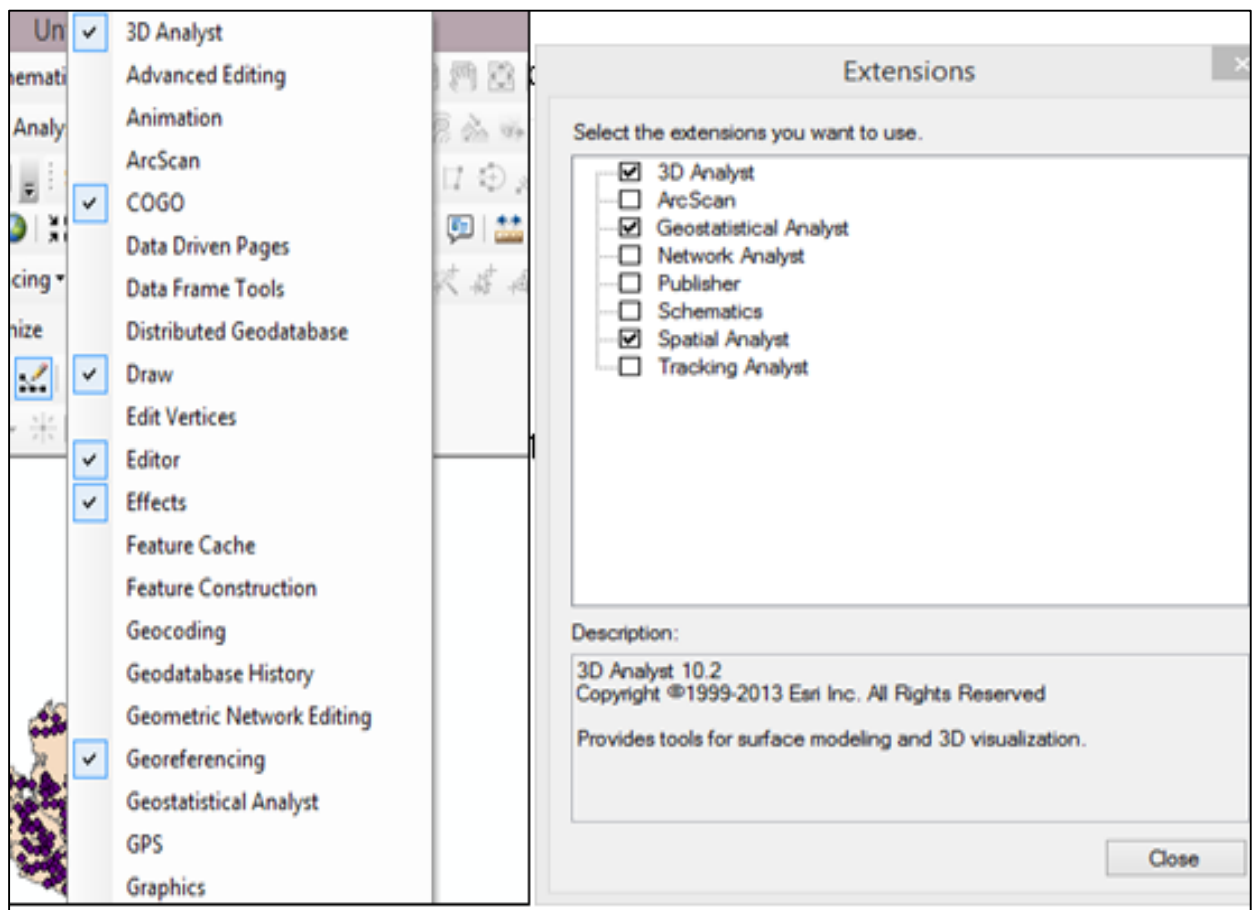


Figure 10: Customizing Tools in ArcMap

2.5. Opening ArcToolbox

ArcToolbox helps users to perform Geoprocessing operations such as data conversion, overlay processing, buffering, proximity analysis and map transformation. Each tool has a menu-driven interface with wizards or dialogs. Arc Toolbox contains the ArcGIS Spatial analyst extension makes nearly 170 Geoprocessing tools available to you, grouped into toolsets of related functionality for ease of access. In any ArcGIS for Desktop application, you can open the ArcToolbox window with the Show/Hide ArcToolbox Window button  found on the standard toolbar or by clicking Geoprocessing → ArcToolbox.

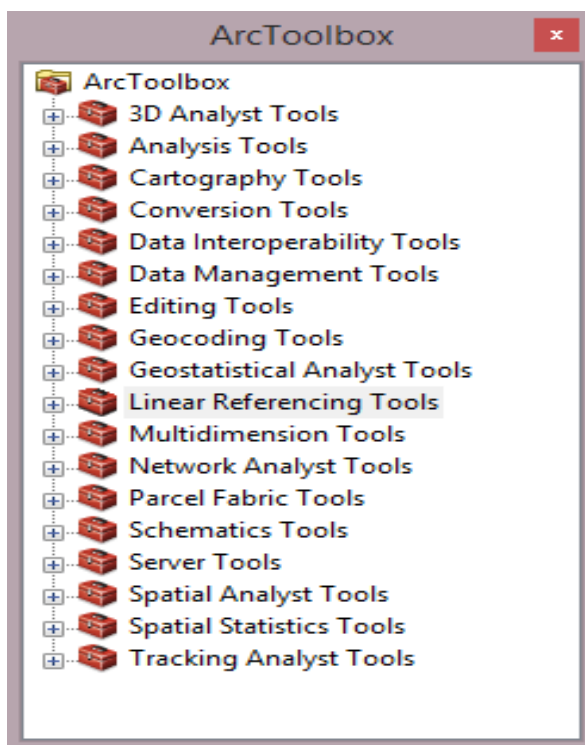


Figure 11: ArcToolBox

2.6. Working with Catalog Window

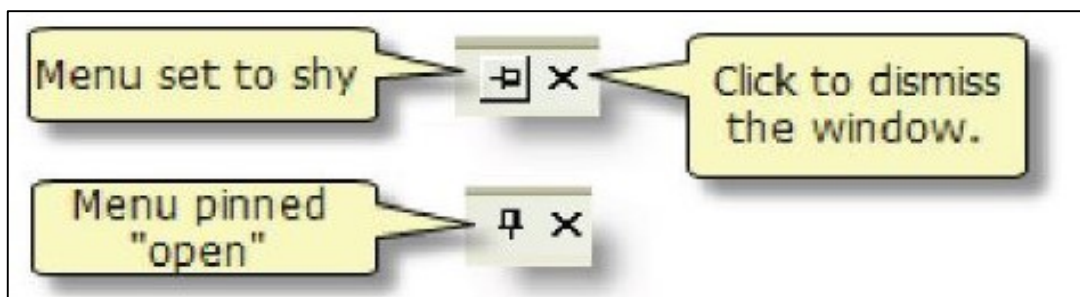
ArcCatalog is a standalone application incorporated in Arc GIS software. But it can be accessed easily in Arc Map rather than opening it as a standalone application as Catalog Window. The Catalog window provides an integrated and unified tree view of all the data files databases, and ArcGIS for Server documents available to you.

The main menu of the Catalog window has tools for navigating to common locations such as your map's Home folder and the default Geodatabase.

In ArcGIS, you can interactively dock and retract windows against the sides of the application window and slide them open as you need them. Retracting the Catalog window enables you to have more of your application window devoted to your map display.

To opening the Catalog window:

- a) Click the Catalog  button on the Standard toolbar or by clicking Windows Catalog on the main menu. Click the pushpin to hold the Catalog window open. You can close the Catalog window by clicking the X in the upper right corner.



- b) Adding Data in Arc Map from the Catalogue Window. You can add and look the data stored in the respective folders pointed in the catalog tree in the Arc Map display window.
- c) Click the Catalog window button in the standard toolbar. This will open the ArcCatalog window on the right side of the screen. ArcCatalog allows you to explore, maintain, and create GIS data files with its many ArcCatalog utility functions. From the ArcCatalog window you will drag and drop a map layer into the TOC as an alternative method of adding data.
- d) Click on the Oromia_Wereda shape file in the Catalogue window
- e) Drag the file and put it in the Table of Contents Window

f) You can add the rest of the data set in the same way and explore their attributes

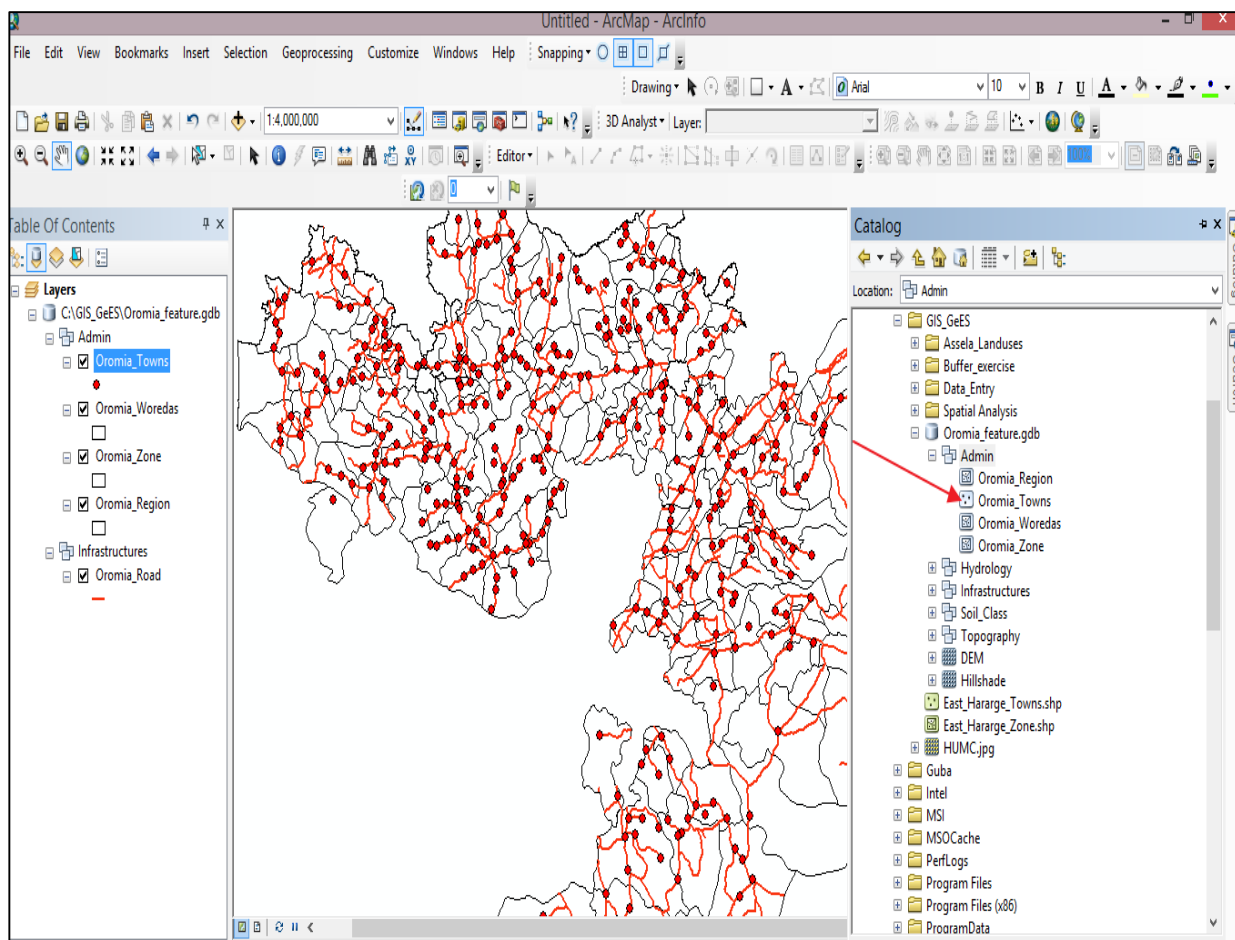


Figure 12: Adding Data from ArcCatalog

2.7. Making Logical Arrangement in the TOC

2.7.1. Change a Layer's Display Order

Changing the order of the map layers in the TOC will change the order in which the map layers are drawn in the map.

- Make sure the List by drawing order button is selected in the TOC.
- Check the **Zone layer** in the TOC to make it visible.
- Drag the **Zone layer** on the top of TOC

The arrangement of the layers in table of contents must be logical (the smallest features at the top and the largest at the top, so that the largest feature may not cover the smallest feature).

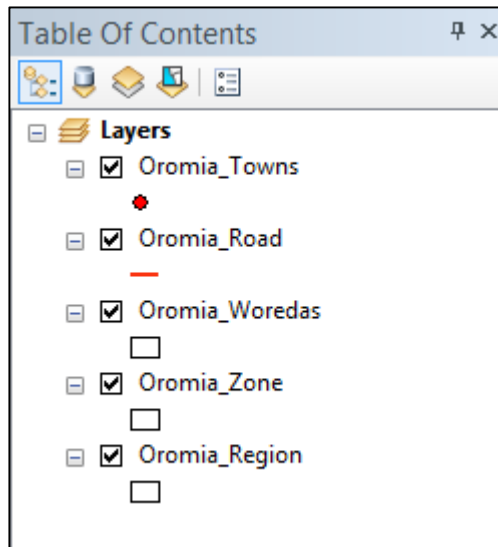


Figure 13: Arranging Layers in Logical order

Alternatively,

- a) Click on the Towns layer
- b) Drag them at the bottom of all others

Now, because towns are point features, they are covered by the largest Woreda and Zone layer, they are invisible in the display window, so you should have to return them to their original location as follows. Now you can look the town's layer visible in the display window. Try to practice rearranging the other layers in the same way.

2.7.2. Turning on and off of layers

With List by Visibility, you have visual cues to indicate the layer's visibility. Each type of layer has its own icon, and the symbol is either colored to indicate the layer is on or gray when it is not, so you can quickly look at the icon to determine whether the layers are visible.

- a) Click the small check box to the left of the Native American lands layer in the TOC to turn that layer on.
- b) If the TOC accidentally closes, click windows, Table of Contents to reopen it.

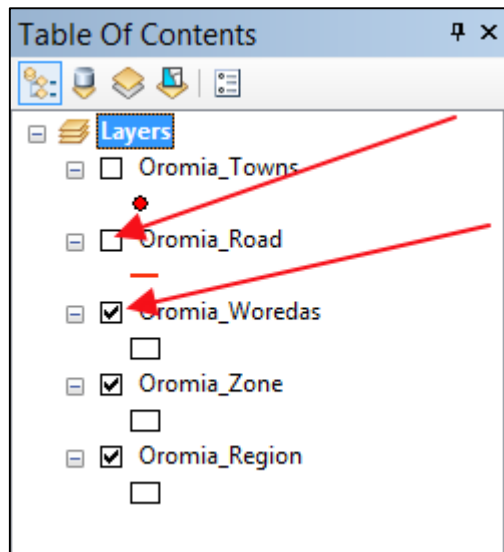


Figure 14: Turning off/On Layers in TOC

2.7.3. Removing Layers

If you don't want the layer to be found in Arc Map you can remove it by right click on the layer to be removed and click on the remove option. This removes the map layer from the map document but does not delete it from its storage location.

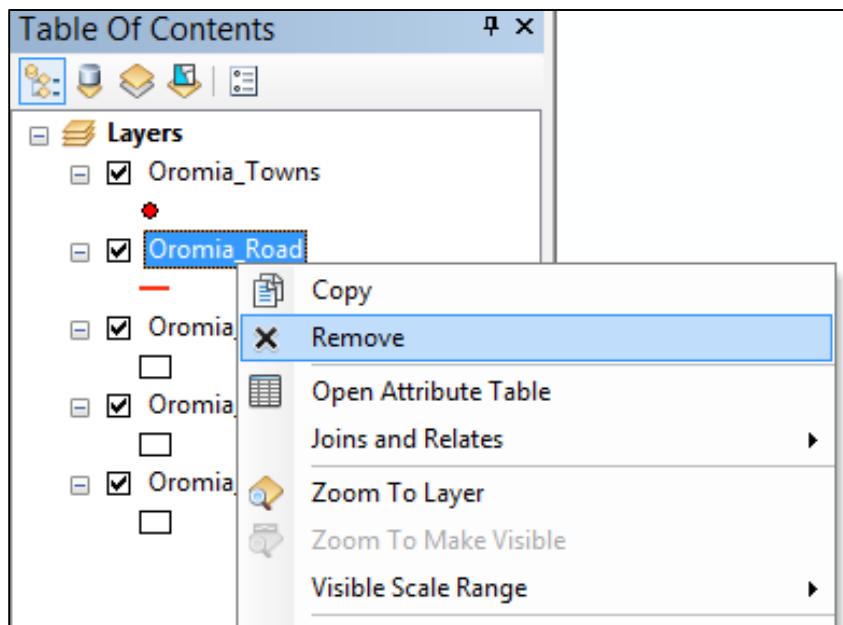


Figure 15: Removing Layers in TOC

2.7.4. Click the Catalog window button in the standard toolbar

This will open the ArcCatalog window on the right side of the screen. ArcCatalog allows you to explore, maintain, and create GIS data files with its many ArcCatalog utility functions. From the ArcCatalog window you will drag and drop a map layer into the TOC as an alternative method of adding data.

- a) In the Catalog window, click on folder connections and navigate to C:\GeES_GIS\Oromia_feature.gdb and click on “**Regionboundary.shp**” layer.
- b) Drag and drop the Regionboundary into the top of the TOC window.
- c) Use the Auto Hide  button in the catalog window to auto hide the window.

The Auto-Hide button lets you keep application windows (such as TOC, toolbox or the Search window) available for immediate use, but hides them in between uses so that you have more room for the map.

2.7.5. Coloring and styling the layer

The ability to change the color of a map layer is an important part of the map creation process. ArcMap automatically assigns a random colour and style to each layer. These are rarely suitable, but can easily be changed using the symbol colour selector.

To change the display colour and style of layers:

- a) Click the **Road** Layers legend symbol in the TOC. The legend symbol is the line below the layer name in the TOC.
- b) Click the Fill Color button in the Current symbol selection of the Symbol Selector window.
- c) Click the **Poinsettia Red** in the Color Palette (or any other color you like).
- d) Change the colour Width in to two.
- e) **Click OK.**

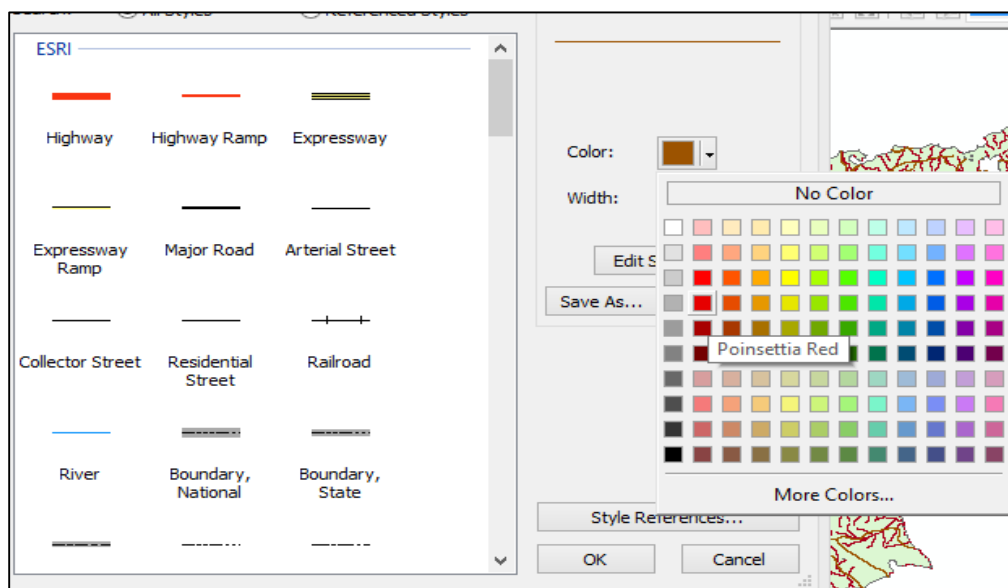


Figure16:Colouring feature

? Dear trainees now it is your turn: Use the Symbol selector and its Search function to Display the layers such as river, Lake Town, and Woreda with appropriate colour.

2.7.6. Labeling Layers

To display labels for layer.

- In the TOC right-click the layer you want to label and click label features (do this for Zone and Woreda)
- In the TOC right-click the Woreda and zones and click Zoom to layer and view the labels.

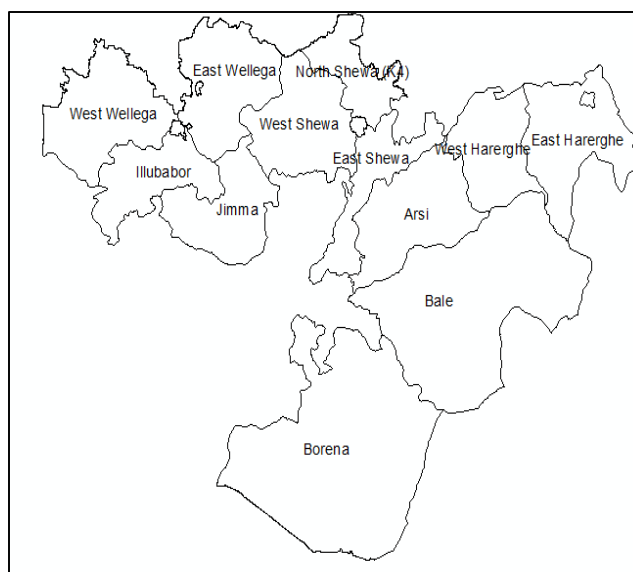


Figure 16: Labeling features in ArcMap interface

2.8. Navigate in a Map Document

Another essential function of GIS is to be able to view map data in different scales and to be able to move the extent of the map. The tools toolbar lets you zoom in, zoom out, pan, go to full extends of the map, go to last extend. The data cannot always be explored simply by looking at a map. If you need information about specific features, examine the attributes of a layer. You can explore the data sets in Arc Map using the Tools toolbar used for map navigation and query within the active data frame



What can you do in the Map Display?

You can do the following!

- a) Identify features
- b) View the attributes of the data in a layer
- c) View the properties of a layer
- d) View the properties of the data frame

Button	Name	Function
	Zoom In	Allows you to zoom in to a geographic window by clicking a point or dragging a box
	Zoom Out	Allows you to zoom out from a geographic
	Fixed Zoom In	Allows you to zoom in on the center of your data
	Fixed Zoom Out	Allows you to zoom out on the center of your data
	Pan	Allows you to pan the data frame

	Full Extent	Allows you Back Allows you to go back to the previous extent
	Forward	Allows you to o forward to the next extent
	Select Features	Allows you to select features by clicking or dragging a box
	Clear Selected Features	Deselects all of the currently selected features in the active data frame
	Identify	Identifies the geographic feature or place on which you click
	Find	Finds features in the map

2.8.1. Interactive Zoom in/out



There are several tools available for zooming in and out of your map. The continuous zoom tool gives you the greatest amount of control.

- Click on the zoom in/out button
- Then left click on your map and hold down the mouse button, and
- Move your mouse away from you to zoom in and toward you to zoom out.

2.8.2. The fixed zoom in/fixed zoom out



The fixed zoom in/fixed zoom out tools work like the zoom tools when you click them rather than draw a box. Each time you click, you will zoom in or out a fixed amount.

2.8.3. Panning



This helps to explore the map in zoomed level

- Click on the pan button
- Left-click on your map display and hold the mouse button down Shift your map, your extent (amount zoomed in or out) stays the same while your map moves.

Note: The pan tool works like the continuous zoom tool when you right click on it. Think of the pan tool as a sticky hand.

2.8.4. Full extent



The full extent button will zoom in or out so that all of your active (checked) can be viewed. You can also zoom in to a single layer by right clicking on the layer in the table of contents and choosing “zoom to layer.”

2.8.5. Previous/Next



The previous extent buttons allow you to return to the extent you had before using one of the zoom tools. The next extent button allows you to jump forward an extent (after you have used the previous extent button). The previous and next extent buttons work similar to the back and forward buttons in MS Internet explorer.

2.8.6. Identify features



This helps to identify about the feature. It tells all the feature

- a) Click the Identify button on the Tools toolbar.
- b) In the map display window, click on a feature (e.g. in one of the information about it.)
- c) After identifying features, close the Results window.

You can also specify from which specific layer you want following procedures.

- d) Using the identify button click on one of the layers in ArcMap
- e) Click on the dropdown arrow in front of identify dialog box
- f) Select Woreda layer as a target feature
- g) From now on, when you click in one of the layers on ArcMap using the identify button, the coming information is about our towards you to zoom out

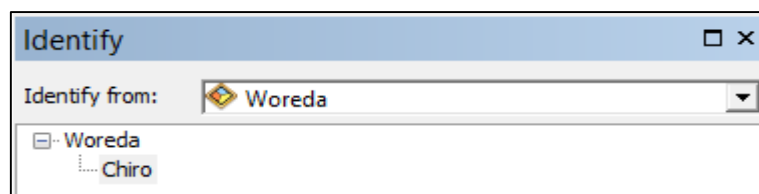


Figure 17: Using identify Tool

2.8.7. Data view and layout view

There are two ways to view a map: in Data view and in Layout view.

- a) Data view is used to display, explore, select and analyse data. You will use this view most of the time that you analyse or model your data in ArcMap.
- b) Layout view is used to prepare the final layout of a map for dissemination or printing. You can add all the items that complete your map (like title, legend, text, scale bars, north arrow etc.), color the background and move the items around to design a balanced layout.

***Note:** when you open ArcMap the default view is the Data view*

2.8.8. The measuring Tool

You can get a quick idea about the size of different features by using the measure tool.

Change measurement units

Maps use coordinate systems to display the earth surface into 2 Dimensions on a piece of paper or a monitor screen. Frequently the map units used by these coordinate systems are feet or meters. Regardless of the map coordinate system, you can set the measurement tool to reflect measures in any units (like miles or kilometers).

- a) Zoom to the extent.
- b) On the tools toolbar, click the Measure button 
- c) The Measure window opens. You will now set the measure units to Kilometers.
- d) In the Measure window, click the Units drop-down button.
- e) Click Distance and Kilometers.
- f) Click the Units drop-down button again.
- g) Select the measure tool, and then click at the beginning and end of the distance you want to measure. The total distance is shown in the status bar at the bottom of the window.
- h) On the **Oromia Region** shape, click at the extreme North and South tips of Oromia's border. (Double-click to finish measuring a feature.) What is the distance between two points? Are these units very meaningful?

***Notes:** If you don't see numbers changing at the bottom of your screen as you move around in the view, your status bar is probably turned off. Turn it on by going to the **VIEW** menu and putting a check mark next to **STATUS BAR**. Try measuring again.*

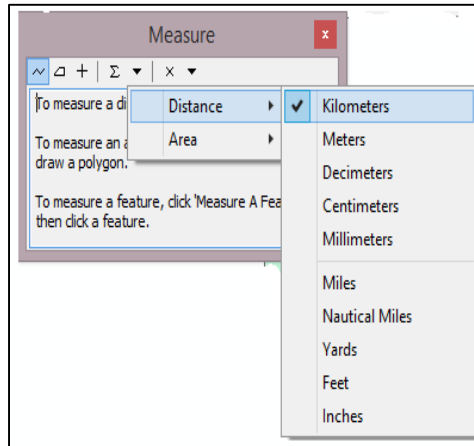


Figure 18: Working with measurement tool

2.8.9. Finding Features



- a) You can use the **FIND** tool to search for a particular attribute, like a town with a certain name.
- b) Click on the **FIND** Tool in the toolbar.
- c) In the **FIND DIALOGUE** that appears, type in or select the following values, then click **FIND**.
- d) Type **Ambo** in **FIND** box
- e) Select “**Oromia_town.shp**” from the **LAYERS** menu
- f) Search in **FIELD** “**TOWN_NAME**”.
- g) Move the dialogue box to a corner of the screen so that you can see your view window as well.
- h) Right click on the result and select **FLASH FEATURE**.
- i) Now right click on the result and select **IDENTIFY FEATURE**.
- j) Click on the **IDENTIFY RESULTS** window to bring it to the foreground.
- k) Right click on the result and choose **SELECT FEATURE**.
- l) Close the **FIND** window. Note that **Ambo** is still selected in the view window.

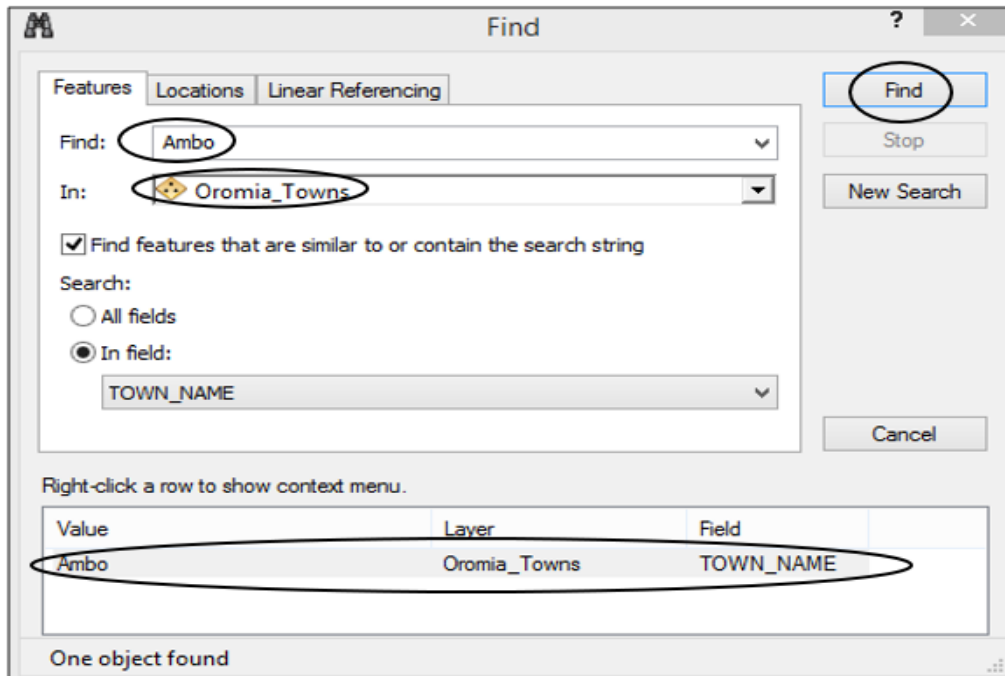


Figure 19: Find Tool

- m) Right click on *Ambo_town.shp* and select **Click Identify**.
- n) At the bottom of the table, it should say that (one object is found).
- o) Click on the **FIND** button, and see that **Ambo** is, in fact, selected in the table.
- p) Close the attribute table.

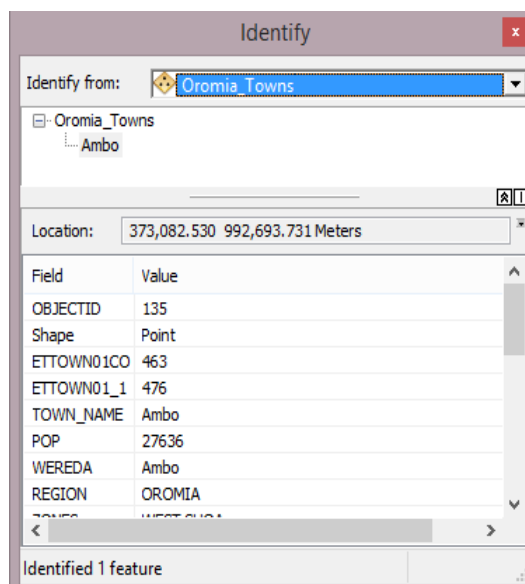


Figure 20: Open Attribute Table using Find Tool

2.8.10. Use the Select Features Tool

You can also select features interactively in the display area by using the Select Features tool. Remember, this will select features from whatever layer or layers you have set as “selectable.” Before you select features you will label all the Oromia towns Point data.

- a) Right click on the **Oromia_town.shp** and select Label Features to label all the towns with their corresponding names.
- b) Click on the **SELECT FEATURES TOOL** Click on some towns. Notice that the selected towns are highlighted, and that as you click on a new town, the old one is no longer selected.
- c) To select **multiple** towns, hold down the **SHIFT** key while you click on towns.
- d) You can also click and drag to draw a box with the tool and select all of the towns that are within or partially within that box.
- e) Select **Addis Ababa, Metu, and Harar**, using the **Select Feature Tool** and holding down the **Shift Key**. Do not forget to use the Zoom and Pan tools at the same time holding the shift key.

2.8.11. Clear Selected Features

- a) Select the **Selection** menu and click on **Clear Selected Features tool**. This will enable you to stop what you have selected using query builder or the **Select Feature Tool** and start a new selection.
- b) If you have features selected from **multiple** layers, this would clear all selected features from all layers -- which may not be the result you want. In that case, you would be better off clearing selected features by right-clicking on the layer name, and choosing **SELECTION - CLEAR SELECTED FEATURES**.

CHAPTER THREE

DISPLAYING DATA

Dear trainee under this task, you will explore more features of attribute tables, including how to relate attribute table to a geographic feature, sort data in tables, provide alias names for fields, add a new field, and calculate the value of selected field based on values in other fields.

3.1. Getting information about features

Individual features on a map have associated information held in tabular format and known as attribute table. To work with feature attributes is one of the most important capabilities of GIS and what makes it so different from a Database system or simple drawing Software like CAD. GIS brings) graphic features together with database tables. The Attribute table shows you all the data records stored for all the features in a feature class (or layer). It appears much like a spreadsheet (in MS Excel for example).

Each row in the table is linked to one feature in the layer. Selecting rows (called records) in the table will select a feature in the layer, deleting a complete row in the table will delete the connected feature in the layer. You will use tools to view database tables connected to the features on your map, query them and label the map with the values in the database tables.

3.1.1. Accessing attribute data using identify tool



Identify

- a) Browse to C:\GIS_GeES\Oromia
features\Administration\Zones
- b) Right-click on Zone layer and label the feature
- c) Click on identify button on the tools floating toolbar and Click on *East Hararghe* in map viewer.

3.1.2. Display a complete attribute table

The complete attribute table for a layer can be accessed by right clicking on the TOC entry for the layer and selecting open attribute table from the menu.

Do this for Oromia Zones?

Steps:

- a) Start a new ArcMap Session
- b) Browse to **C:\GIS_GeES\Administration**

- c) Select the **Oromia_Zones.shp** Shapefile
- d) Right click on **Oromia Zones** in the TOC
- e) Click on **Open Attribute Table** from the drop down list – this will display the attribute table of the **Oromia_Zones.shp** Shapefile.
- f) Close table when finished.

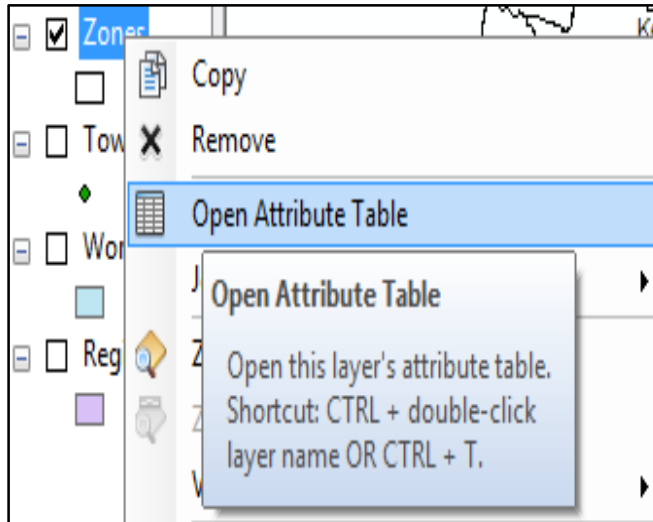


Figure 21: Working with Attribute Table

3.1.3. Sorting records in an attribute table

Steps

- a) Right click the Oromia Zone in the TOC.
- b) Click Open Attribute table from the drop down options.
- c) Right click on the heading of the Oromia **Zones** field (that is, on the word “**Zones**” at the top of the column).
- d) Select **SORT ASCENDING**. All of the Zone names are now in alphabetical order.

Zones					
OBJECTID *	Shape *	Zone	Perimetre Length	Shape_Area	
45	Polygon		796.435164	398234607.4	
48	Polygon		53.341392	2129192699.5	
49	Polygon		53.605205	103399371.1	
50	Polygon		199.76397	170906126.5	
51	Polygon		78.138846	4116929211	
34	Polygon	Amaro Special	534.11991	114198064.6	
12	Polygon	Arsi	5278.24971	23648329812	
17	Polygon	Asosa	460.963174	1888373205.6	
15	Polygon	Bale	906.691775	60163422426.1	
16	Polygon	Borena	907.690721	69440076594.1	
35	Polygon	Burji Special	937.325957	239430712.4	
44	Polygon	Dire Dawa	461.983314	22651724	
3	Polygon	East Gojam	1004.928043	103248402.7	
14	Polygon	East Harerge	246.060124	22728705307.8	
11	Polygon	East Shewa	257.300394	12756832972.2	
6	Polygon	East Wellega	1828.87771	21375325641.5	
32	Polygon	Gamo Gofa	85887.157215	139958463.6	0710
27	Polygon	Gedeo	213250.771895	431538403.9	0705
22	Polygon	Guraghe	257519.084973	590164921	0701
23	Polygon	Hadiya	24251.127406	19693925.0	0702
24	Polygon	Hadiya	34740.636996	22858925.5	0702
39	Polygon	Hundene	73299.62979	142824184.2	1301
7	Polygon	Illubabor	857394.486385	14888394125.6	0403

Figure 22: Sorting Records in Ascending/Descending orders

3.2. Attribute Data and Spatial Querying

GIS datasets often contain much more data than you need. There are several methods available to reduce or extract data from larger, more complex datasets. You can use the following options to select features.

3.2.1. Selection by Attribute

Step1.

- Open **ArcMap** with a new, empty map document
- Click the Add Data button, to add the **Oromia_Woreda.shp** Shapefile from **C:\GIS_GeES\Administration**
- Right click the Oromia Woreda in the TOC.
- Click Open Attribute table from the drop down options.
- Right click the header of the Name field.
- Click Sort Ascending

- g) Scroll down the attribute table and click the record selector (gray cells on the left side of the table) and highlight **Abay Chomen** Woreda
- h) In the table click the Clear Selected Features button  when finished.

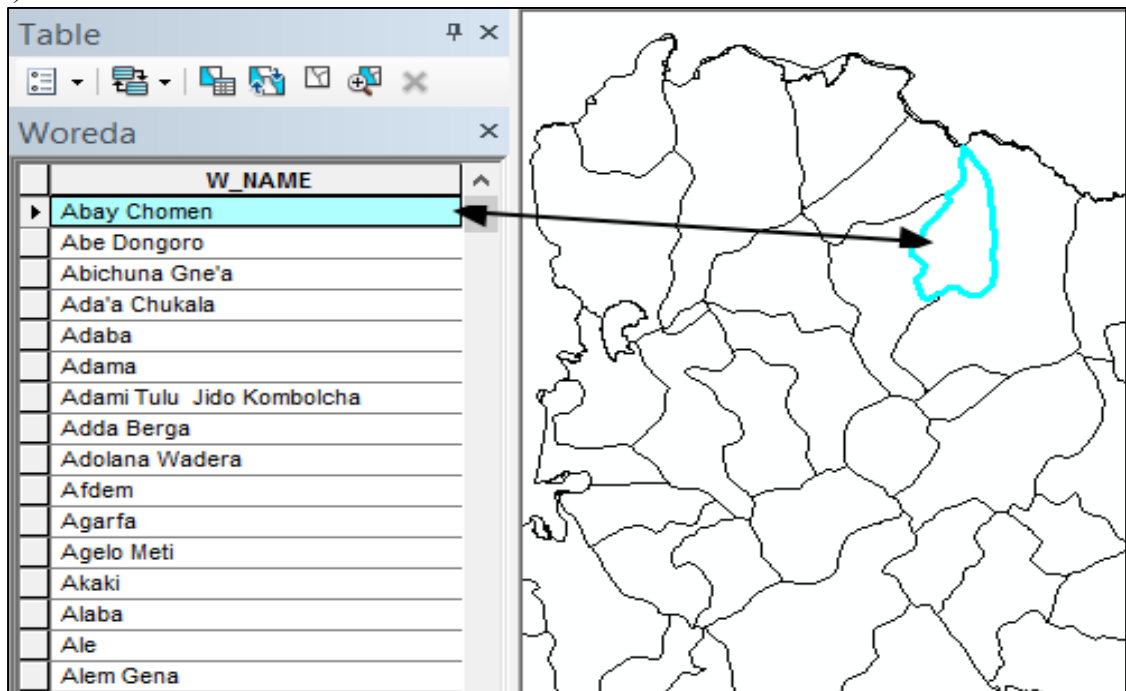


Figure 23: Querying features using Attribute Table

Step2:

An attribute table can be queried (i.e. each for records) using select by attribute menu.

- a) Click the Add Data button, to add the **Oromia_Roads.shp** map from **C:\GIS_GeES\Infrastructure**
- b) Highlight the **Oromia_Road** map in the TOC
- c) From the **Selection** menu select “**Select By Attribute**”
- d) In the Layer field select **Oromia_Road**
- e) From the Fields list double click “**Type**”
- f) Click on “**=**” sign on the tab
- g) Click on **Get Unique Values** tab
- h) Click on “**All-weather roads (gravel)**”
- i) When finished the dialogue box should be filled out as below. Click the verify tab and ensure that the syntax is correct.
- j) Click on **Apply** tab and **Close**

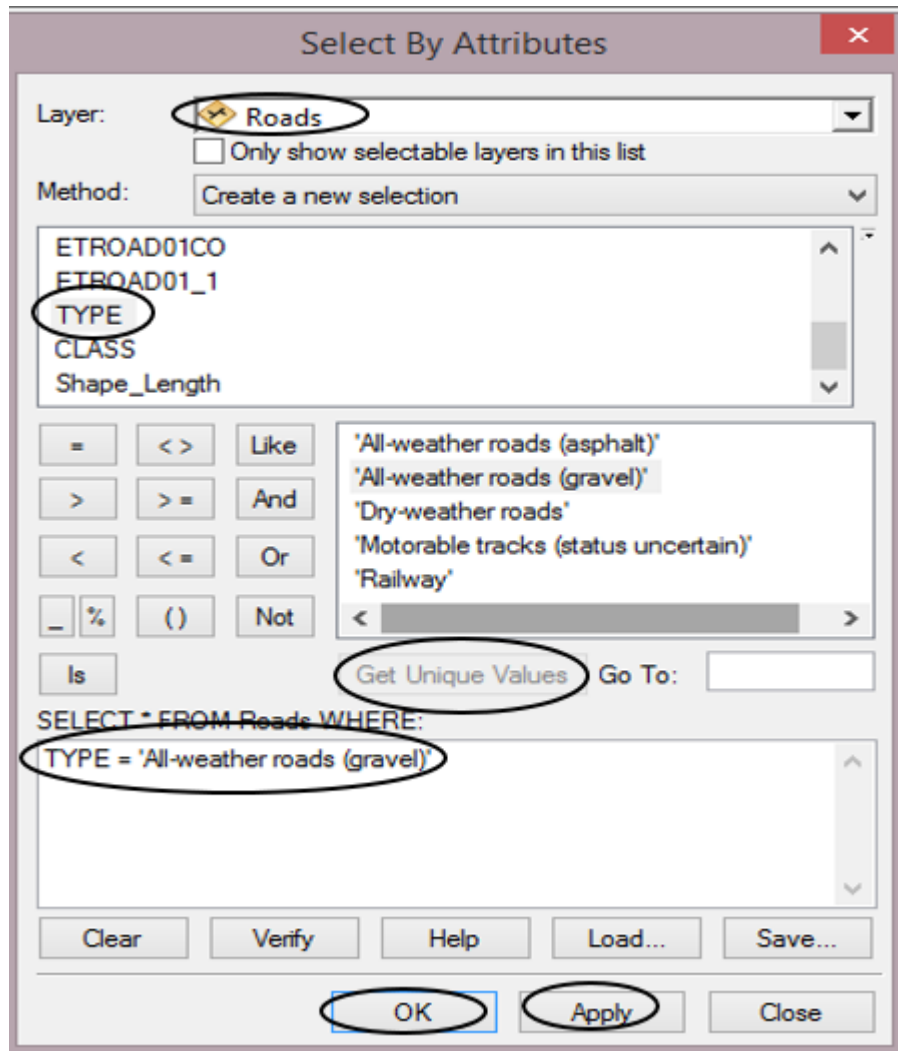


Figure 24: Selection by attribute Table

3.2.2. Selection by location

The select functions let you query features by attributes (Example, **Towns** that Intersect East Harge Zone).

- Add the **Oromia_Zones.shp** Shapefile from the folder **C:\GIS_GeES\Administration**
- From the **SELECTION** menu, choose **SELECT BY ATTRIBUTES**. While you open the **Select by Attributes** window in this way, and not through the attribute table, you need to specify which layer you want to query. You can also use a Query Wizard, but it is probably easier just to get comfortable with the operators and build your own queries.
- Make sure **Ethio_Zone** is the layer to be queried.
- Build the query, “**Zone_Name**” = “**East Harerghe**” and click **APPLY**.

- e) Close the **SELECT BY ATTRIBUTES** window.
- f) From the **SELECTION** menu, choose **SELECT BY LOCATION**.
- g) Put a check mark to **Oromia Town**, in target layer, select **Oromia_Zone** as the source layer and choose *Intersect (3d) the source layer feature* or *have centroids in the source layer feature* in spatial selection method.

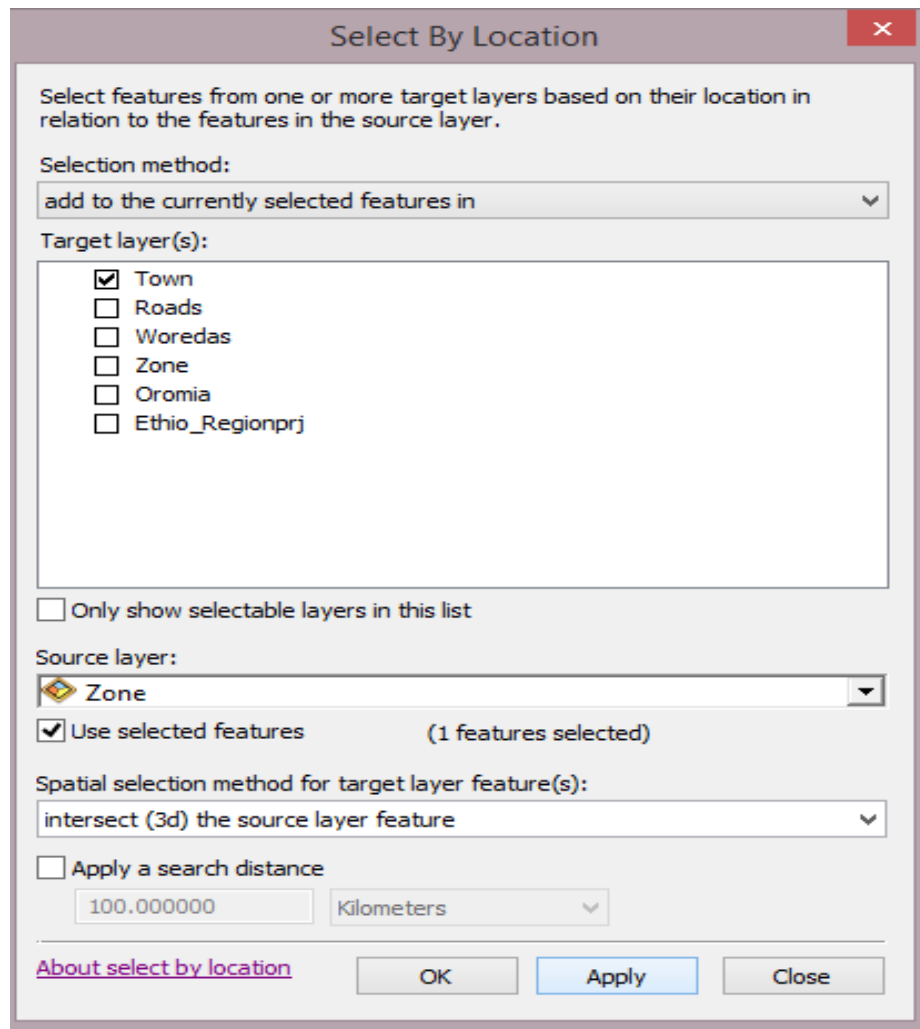


Figure 25: Feature Selection based on Location

CHAPTER FOUR

THEMATIC MAPPING USING SYMBOLOGY

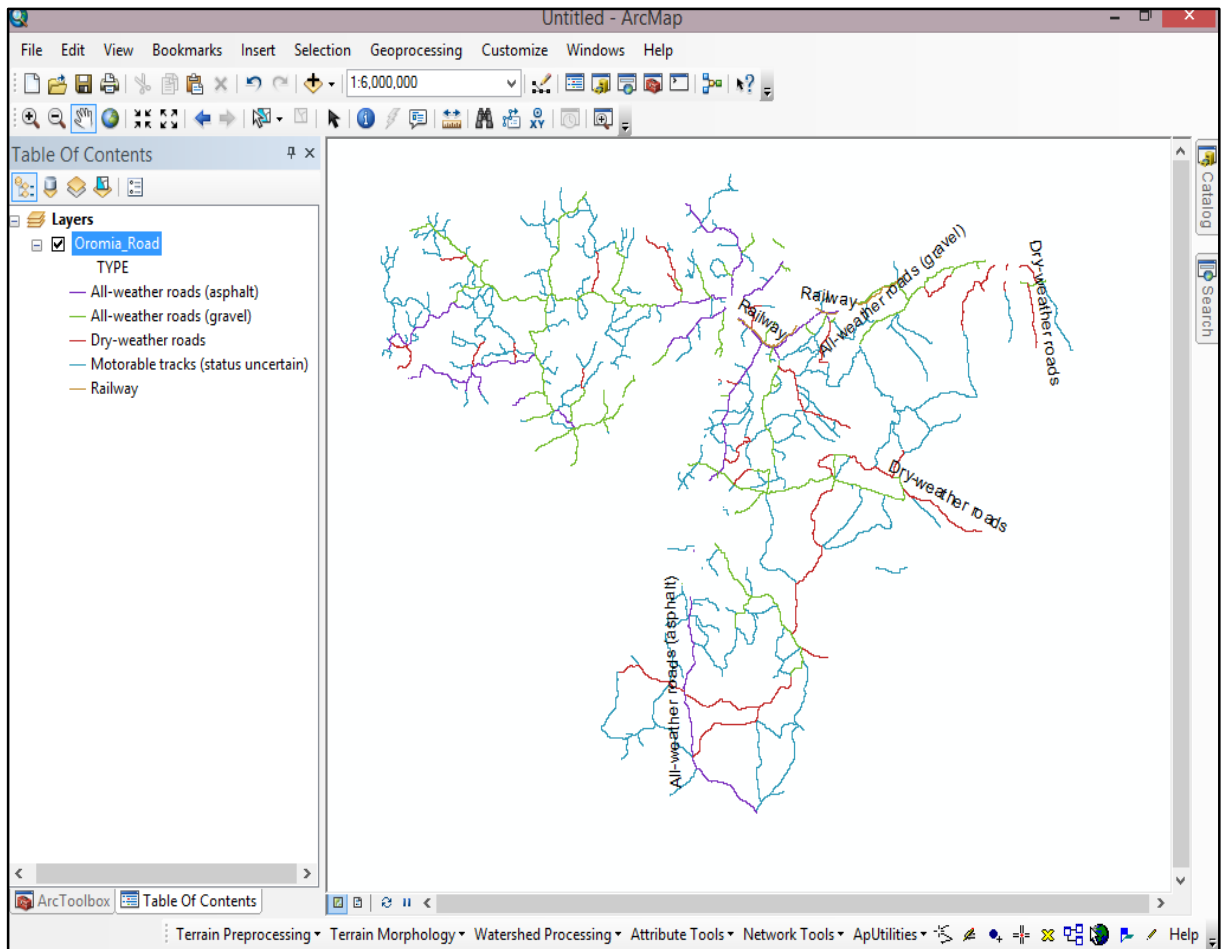
4.1. Symbolizing Features

- a. Start a new **ArcMap** session
- b. Add the **Oromia_Roads.shp** Shapefile from the folder
C:\Oromiafeatures\Infrastructure
- c. *To symbolize the type of road with a different color*
- d. Right click on **Oromia_Road**, and select **PROPERTIES**.
- e. Click on the **SYMBOLLOGY** tab.

NOTE: *There are a number of different ways to symbolize your data, and you can select the one you are interested in from the list at the left: **FEATURES**, **CATEGORIES**, **QUANTITIES**, **CHARTS**, or **MULTIPLE ATTRIBUTES**. There are also sub-options that appear when you click on one of these headings. **FEATURES-SINGLE SYMBOL** is the Symbolology style used by default.*

- a. Choose the first sub-option under **CATEGORIES**, which is **UNIQUE VALUES**.
- b. Choose the value field, “**Type.**”
- c. Click on the tab **ADD ALL VALUES**. Notice the number of the segments uniquely categorized into four types of road.
- d. Remove the check mark next to “all other values.”
- e. Double click on each of the road categories and select the corresponding symbols in the table below.
- d. Click on Apply. Notice the unique classification both in the Table of Contents and the map

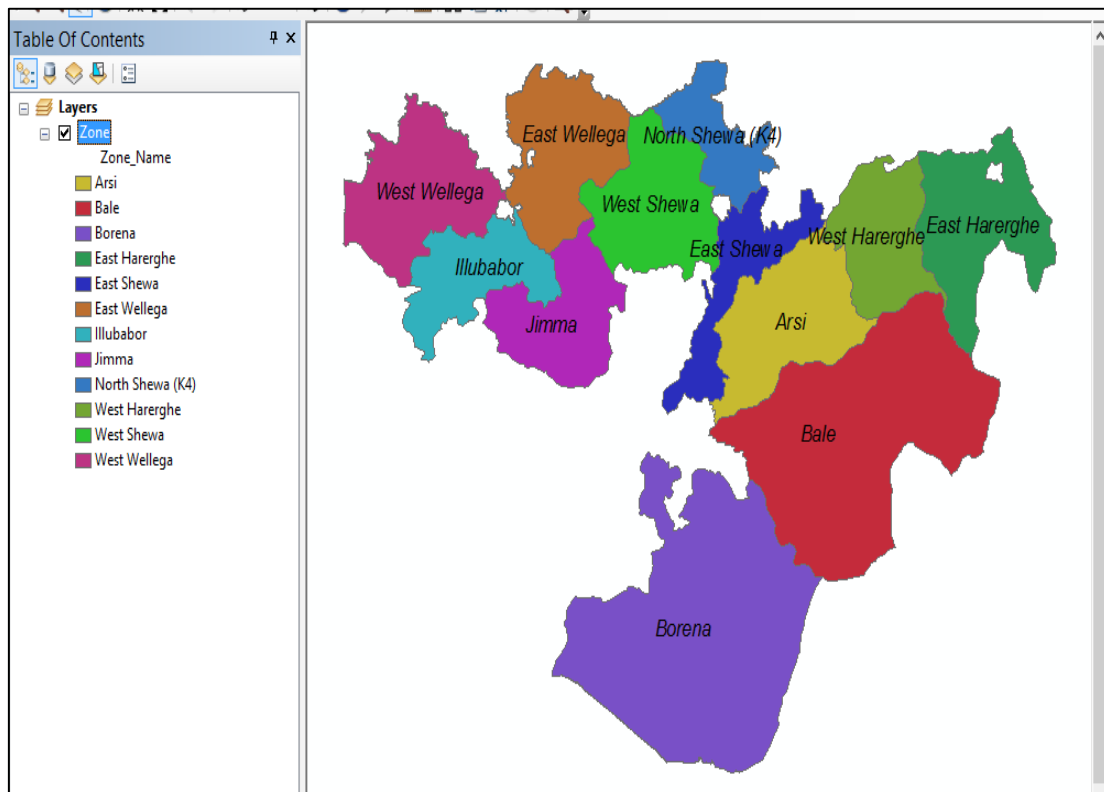
Road Category	Symbol
All-weather road (asphalt)	Highway
All-weather road (gravel)	Major road
Dry-weather roads	Dashed 2:1
Motorable tracks (status uncertain)	Arterial street
Railroad	Multi-track



4.2. Symbolize Features with Different Colour

- Right click on **Zone** in TOC, and select **PROPERTIES**.
- Click on the **SYMBOLOLOGY** tab. Choose the first sub-option under **CATEGORIES**, which is **UNIQUE VALUES**.
- Choose the value field, “**Zone Name.**”
- Click on the tab **ADD ALL VALUES**.
- Remove the check mark next to “all other values.”
- Click on Apply.

Notice the unique classification both in the Table of Contents and the map, label as well.

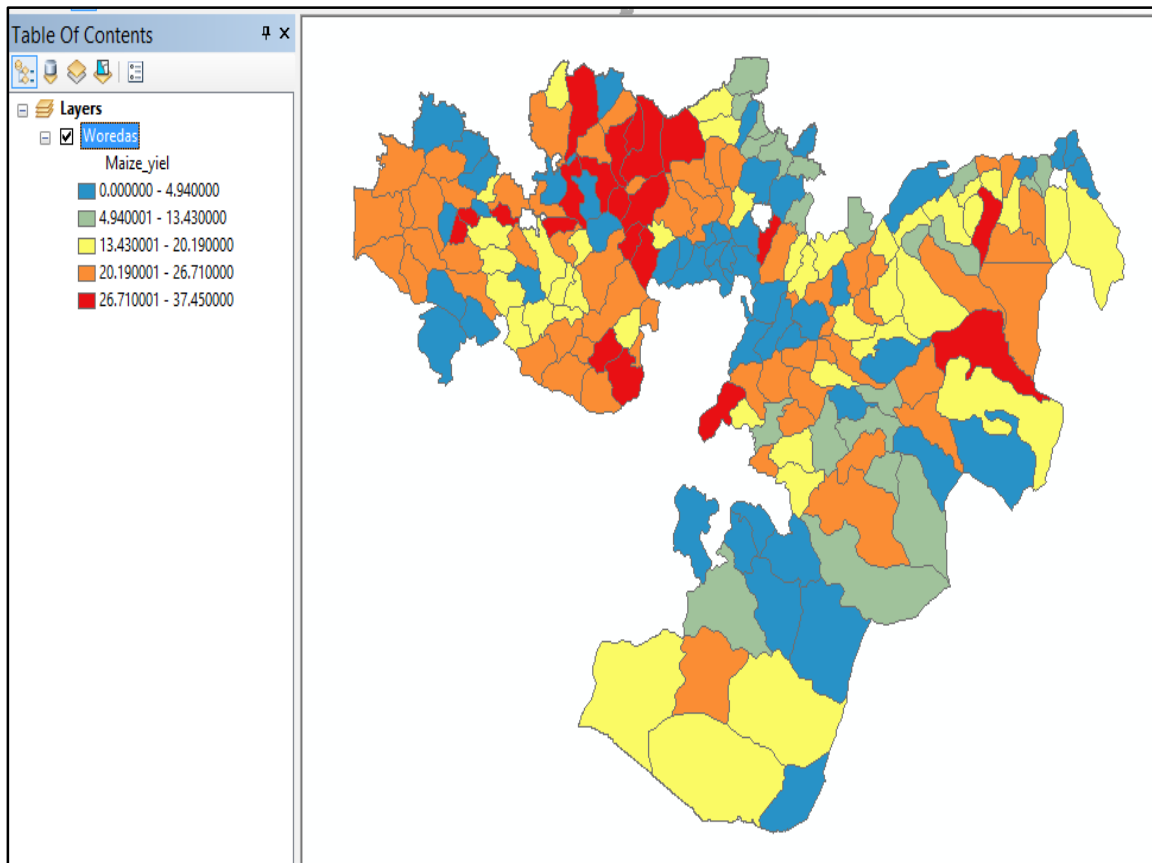


4.3. Display Features with Graduated Color

In this particular task, you will classify Oromia Weredas based on their population.

- Click the **ADD DATA** button.
Navigate to **C:\GIS_GeES\Administration** and add **Oromia _Wereda.shp**.
- Right click on **Oromia _Wereda** and select **PROPERTIES**.
- Click on the **SYMBOLOLOGY** tab.
- Select **QUANTITIES – GRADUATED COLOR** from the left hand side of the window.
- For the **VALUE FIELD**, select “**Maize yield**”.

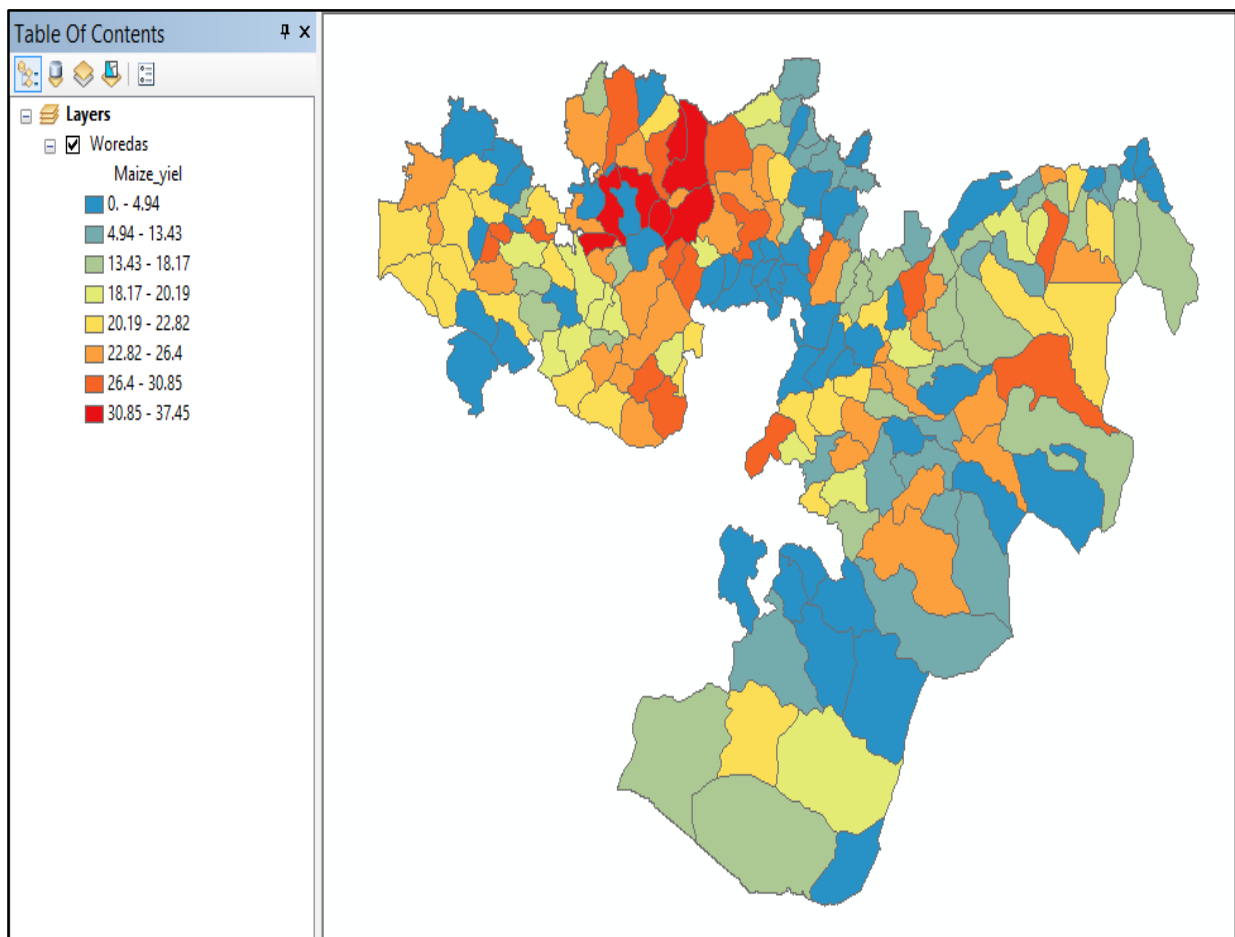
Click **APPLY** and move the window out of the way so you can view the result. Do you like the effect? It probably looks a little strange, by default, **ArcMap** assigns a light color to those Woredas with low maize yields, and a darker color to the higher yields.



You have a lot of control over how this map appears. **ArcMap** has defaulted to breaking the data down into five categories, and it chose where to put the breaks between these categories. You can change each of these parameters.

- Click the **CLASSIFY** button to alter the **Classification** method and change the class to 8.
- You can manually alter the breaks value by entering break values in the list at the right or as desired. You may want to do this so that the breaks are at round numbers, which

- will make the legend easier to interpret, set break values as :*(write only the maximum values of the class)*.
- From the **CLASSIFICATION METHOD** drop down select Define Intervals and set the break values: Click **OK**, and then **APPLY**. Move the Properties window so you can view the results.
 - Now you will change the number of classes to 8. Ordinarily, you can do this from the **Properties** window itself (in the box next to the *Classify* button), but now that you have manually defined classes, you must first change the classification method.
 - Click the **CLASSIFY** button.
 - Change the classification method back to **NATURAL BREAKS**.
 - Change the number of classes to 8, and click **OK** and **APPLY**.
 - View your results. Does the map look very different



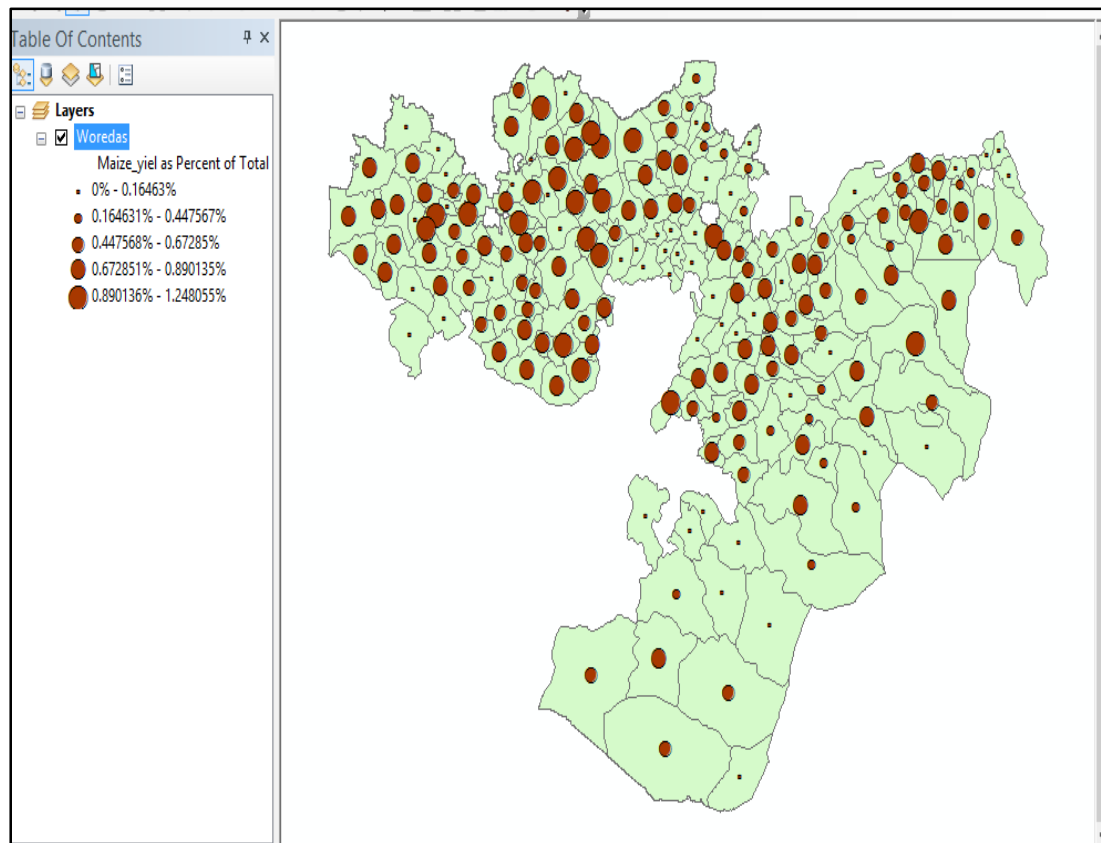
***NOTE:** Remember that as you increase the number of classes on your map, it may become more difficult to visually distinguish adjacent classes from each other. You need to find a happy medium between having enough classes to represent the variability in your data, and having symbols that are easy to differentiate.*

Tips of classification system!

- ♣ **Natural Breaks (Jenks):** This is the default in ArcGIS. It uses a formal (Jenks Optimization) to minimize the variance within classes and maximize the variance between classes. It usually works well, but be sure to round off the values manually.
- ♣ **Equal Interval:** This sounds like a good idea because it breaks your data into classes of equal size. The problem is, most data are not distributed evenly so this usually makes maps with lots of observations (map features) in only one or two classes.
- ♣ **Quantile:** This divides the observations (map features) into even groups of 4 (quartile) or 5 (quintile). By definition, it does a good job of showing variability..
- ♣ **Standard Deviation:** This shows how far values are from the mean. It works well when you want to show how extreme high and low values are, but you'll probably want to modify the legend to show actual values and not standard deviations which don't make sense to most audiences.

4.4. Display Features with Graduated Symbol

- b. Right click on **Oromia_Wereda** and select **PROPERTIES**.
- c. Click on the **SYMBOLOLOGY** tab.
- d. Select **QUANTITIES – GRADUATED COLOR** from the left hand side of the window. **For the VALUE FIELD, select “Maize yield” and Normalization field as “PERCENT OF TOTAL”.**
- e. **Apply | OK**



4.5. Using Query to Display Selected Categories

Suppose you wanted to display only the major three roads categories i.e. all weather road (Asphalt), All-weather road (gravel) and Railway.

- add Oromia roads
- Right click on **Ethio_Road** and select **Open Attribute Table**.
- Review the attribute fields in the table. You will notice that the road categories are coded with numbers in the 'Class' field as follows

Road Category	Class
Railroad	7
All-weather road (asphalt)	6
All-weather road (gravel)	5
Dry-weather roads	4

- Close the attribute table.
- Double-click on **Oromia_Road** to open the **PROPERTIES** window.

- f. Go to the **DEFINITION QUERY** tab and click on **QUERY BUILDER**.
- g. Double-click on Class, single-click on = and double-click on Get Unique Values tab, and finally click on to yield the SQL statement "Class" < 5. This statement restricts the **Oromia_Road** that appears in your map view to only those records where the road class value is less than 5 and those records that are dry weather and Motorable tracks will be selected.
- h. Click OK in the **QUERY BUILDER**, and **OK** again in the **DEFINITION QUERY**

CHAPTER FIVE

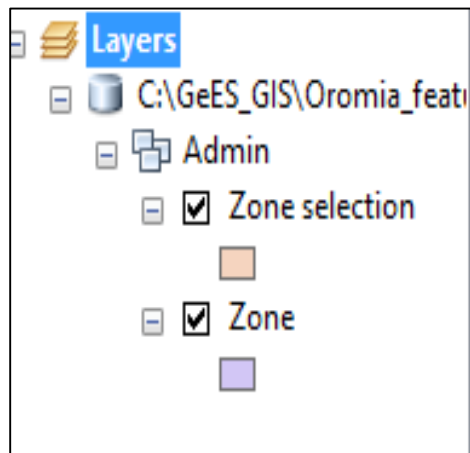
SPATIAL ANALYSIS WITH VECTOR DATA

Analysis is a key function of a GIS that separates it from a typical visualization software package. The ability to dig deeper into the data and locate things you cannot see during visualization is vital to understanding relationships in your data and can play an important role in decision making.

5.1. Creating Layers by Selection

In this exercise you will learn how to clip your area of interest from large data set using the clip analysis toolset

- a. Open a new empty map
- b. Add Oromia_Zone Shapefile from **C: / GIS_GeES/administration**
- c. Select “**East Hararge**” zone using one of the selection method you have practiced.
- d. Write Click on the “**Zones**”
- e. Go to Selection Create Layer from Selected Features
- f. You should see “**Zone**” selected layer added as a new layer in the table of contents

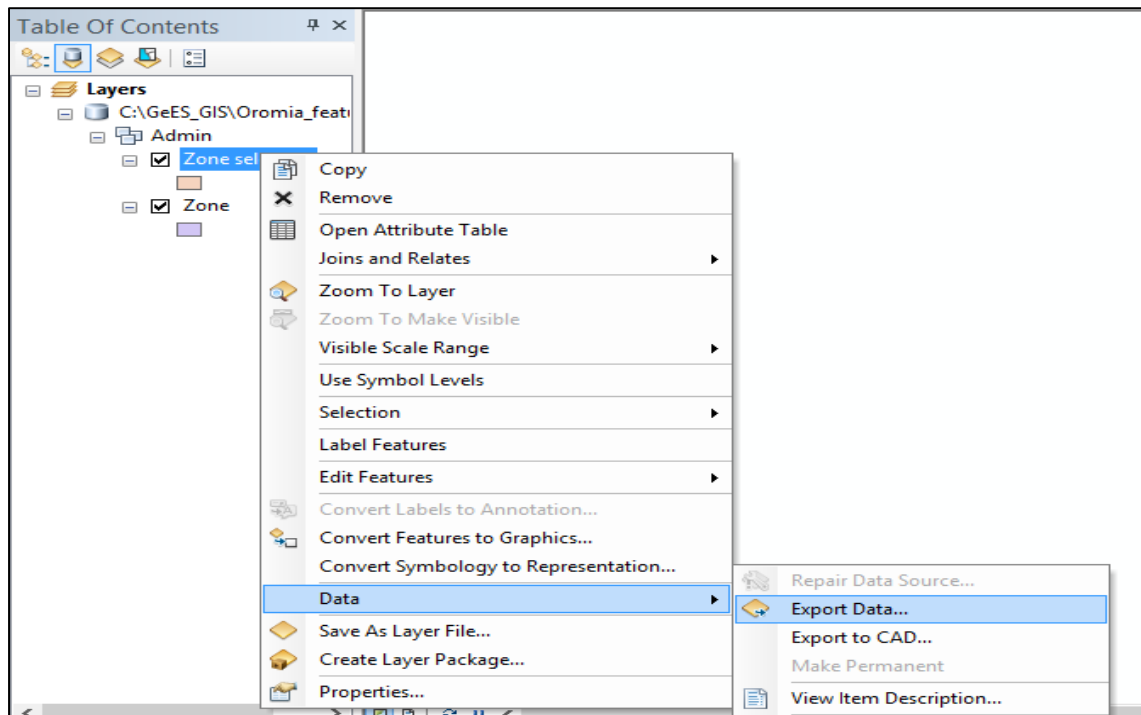


- g. Turn off all the layers other than “**East Harge_Zones**” and explore the new selected layer

5.2. Saving the Selected Layer

The “Zone Selection” layer in TOC is a temporary file, it will be lost when you close ArcMap. So you should save it permanently in your working directory.

- a. Right Click on “Zone Selection” layer in TOC
- b. Click on Export Data.

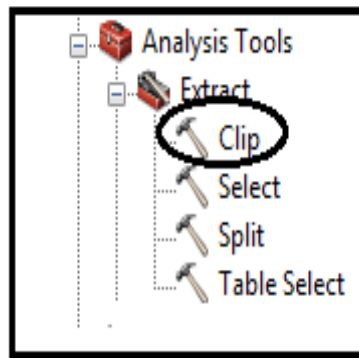


- c. In the export data dialog box, click on the Browse folder icon
- d. In the coming dialog box, navigate to **C: / GIS_GeES/administration** and write E-Hararge zone as a file name
- e. Click on Save
- f. Click ok in the Export Data dialog box
- g. Click Yes when it asks you to add the new layer in your ArcMap Environment
- h. Now you can see the new layer “E.Hararge Zone” displayed in ArcMap.

5.3. Clipping

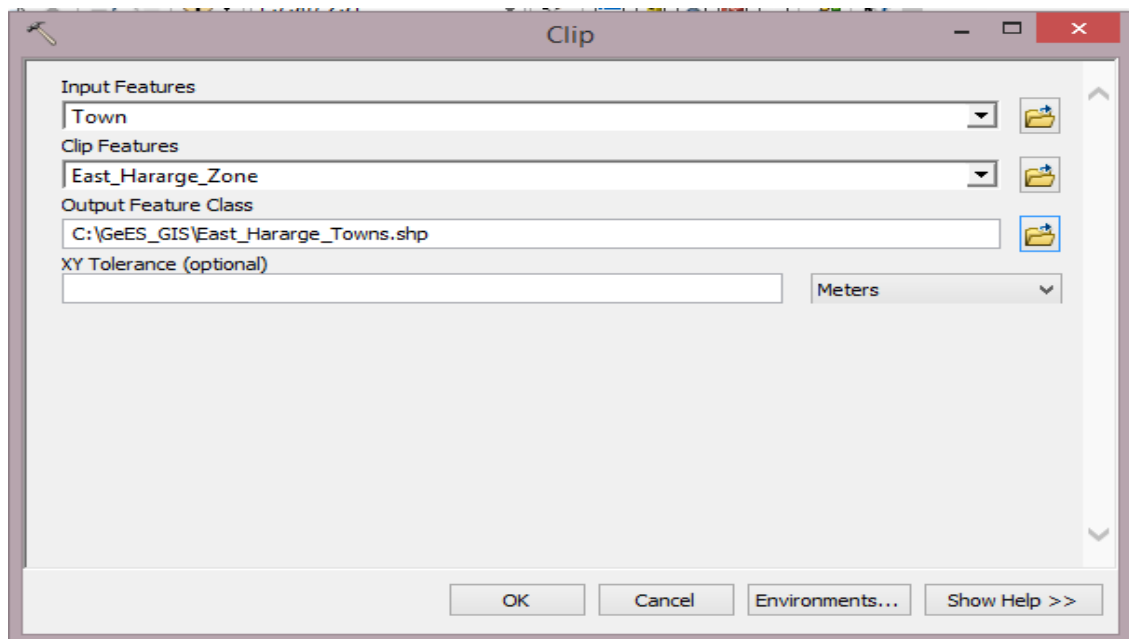
Clipping allows you to turn one Shapefile into a cookie cutter in order to cut out part of a larger Shapefile. In the following exercise you extract the towns found in East Hararge Zone from the Oromia Towns Shapefile.

- f. Add Oromia_town.shp Layer From **C: / GIS_GeES/administration**
- g. From the ArcToolbox Window, Expand the “Analysis” tools.
- h. Expand the Extract toolset on “Clip” tool element.



In the input Clip dialog box do the following arrangements.

- In the input Features, use the dropdown arrow and select **Oromia_town.shp**(this is the layer which contains the dataset to be clipped)
- In the Clip Features box, use the dropdown arrow and select “East Hararge”
- In the output feature class option, click on the browse folder and save the data on **C:\ GIS_GeES** using the file name “**East Hararge Zone Towns**”

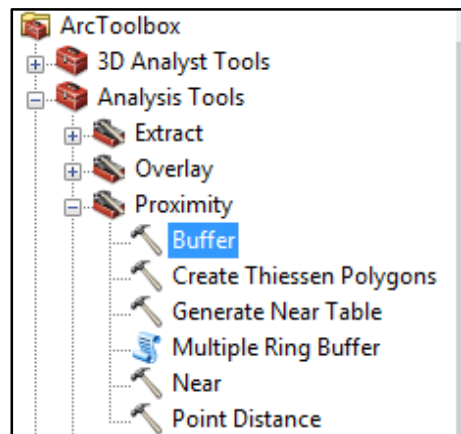


- Click OK when you finish
- Turn off all the layers except East Hararge Towns.
- Explore your clipped layers, which must look the following snapshot

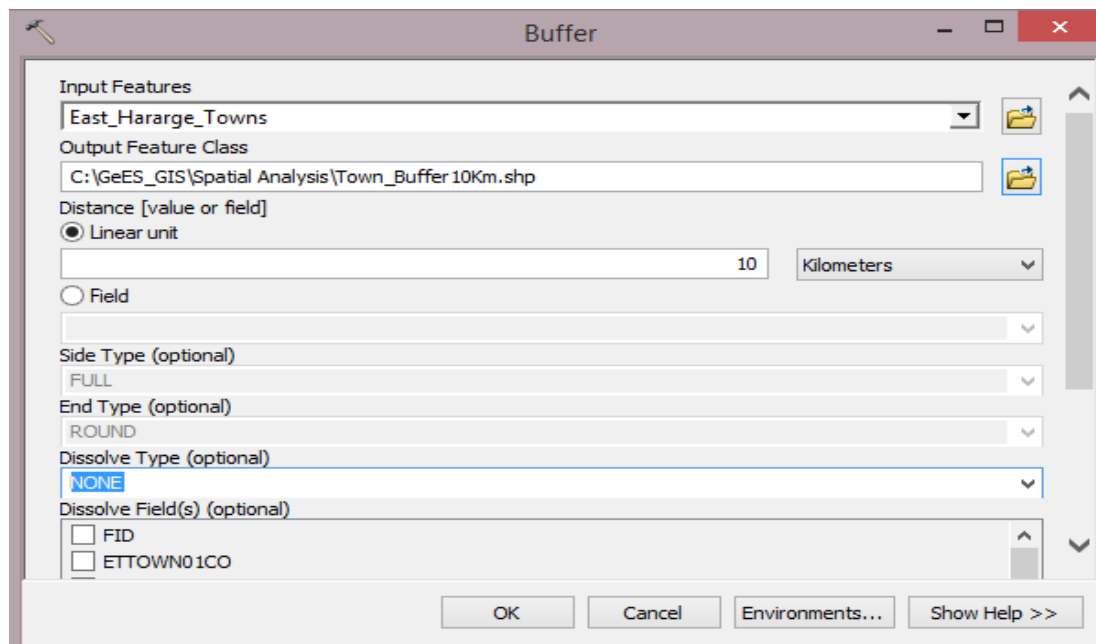
5.4. Buffering

Buffer is a zone around a map feature measured in units of distance or time. A buffer is useful for proximity analysis. In the coming exercise you will identify the areas of East Hararge zone found within 10 Kilometer surrounding the towns.

- Add Oromia roads from **GIS_GeES/Oromia_featurea/Infrastructures**
- From the Analysis tools of ArcToolbox expand the Proximity toolset,
- . Double Click on the Buffer tool element from the proximity toolset



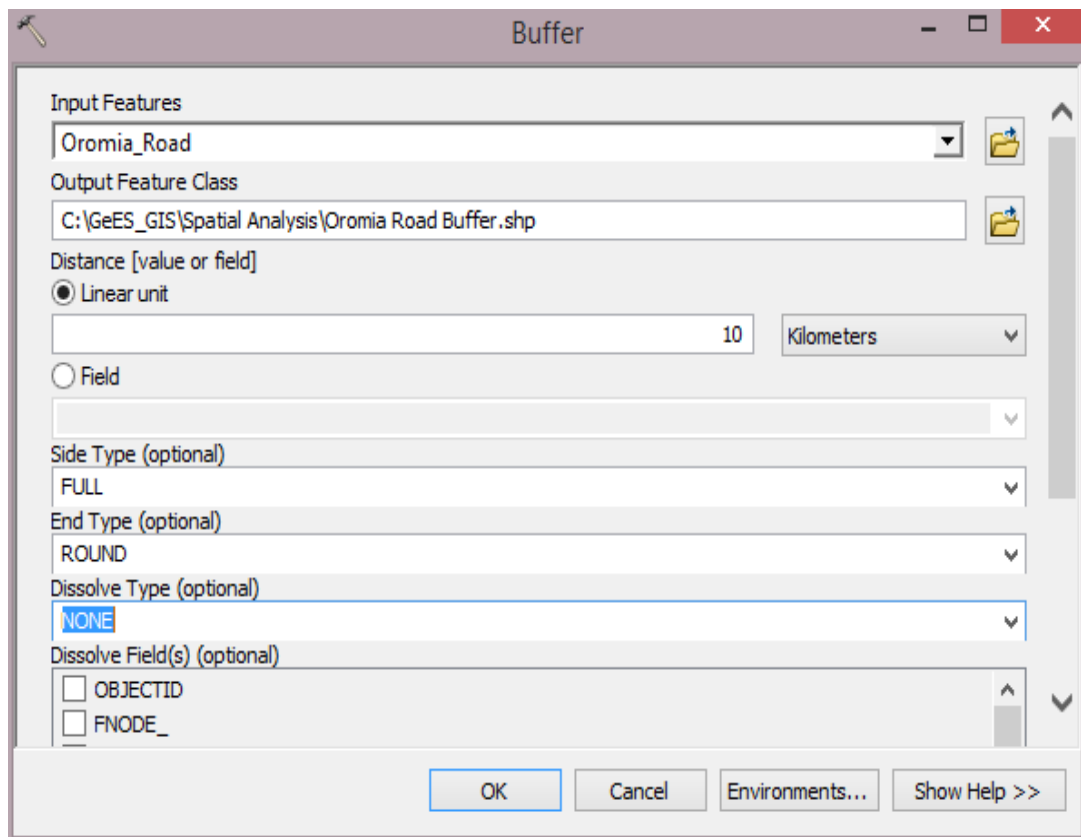
- Select East Hararge _ Zone_ Towns as input layers.
- In the Output feature class options save the output data on **C: / GIS_GeES/Spatial analysis** folder as “Town_Buffer10km

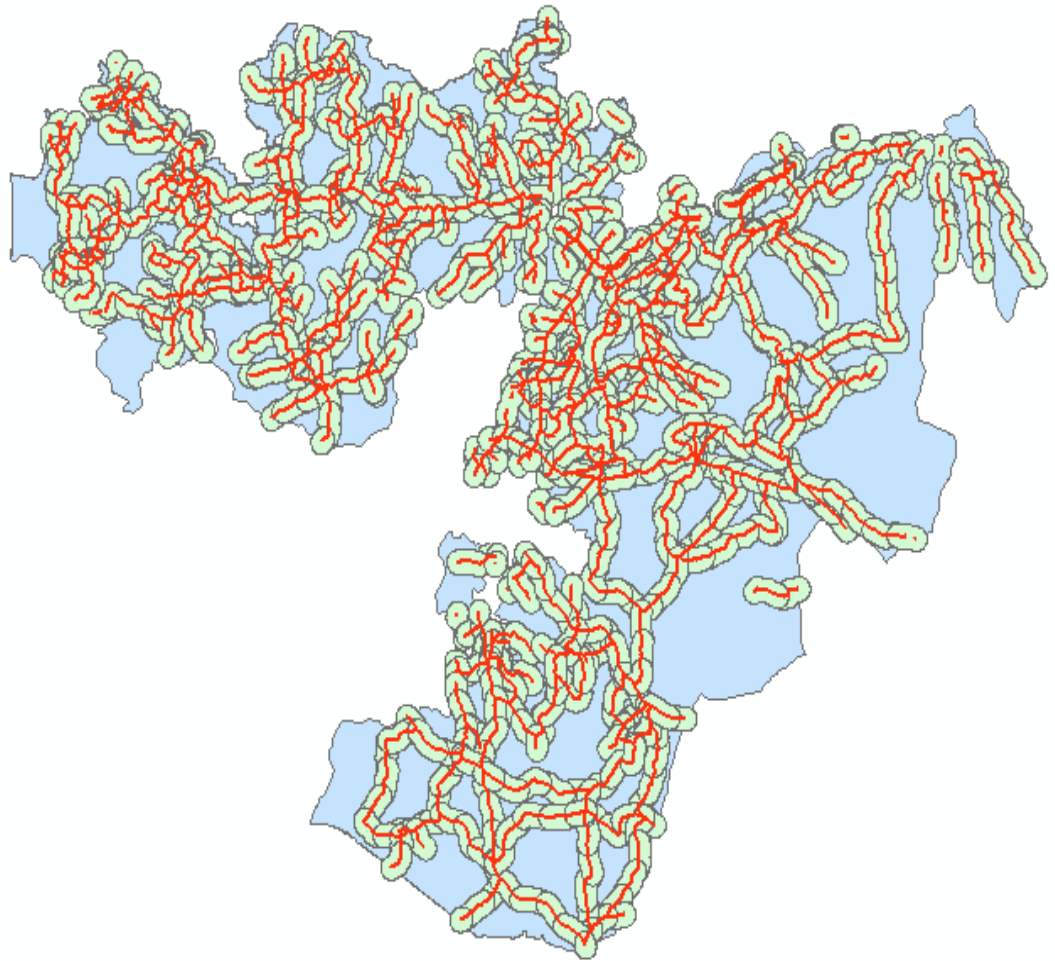


Question: Identify Oromia Region woreda's found within the 10 Km roads Buffer?

Hint! Follow the following Steps:

- a. From the Arc Toolbox window expand the Analysis Tools toolbox
- b. Expand the **Proximity toolset**
- c. From the proximity toolset **Double click the buffer**
- d. In the coming Buffer dialog box, under the input features option; select Oromia Road
- e. In the output feature class option; click on the browse folder icon and proceed to **C:\GIS_GeES/Spatial Analysis** sub folder.
- f. Set the distance field to **10km**.
- g. Click-Ok When you finish.

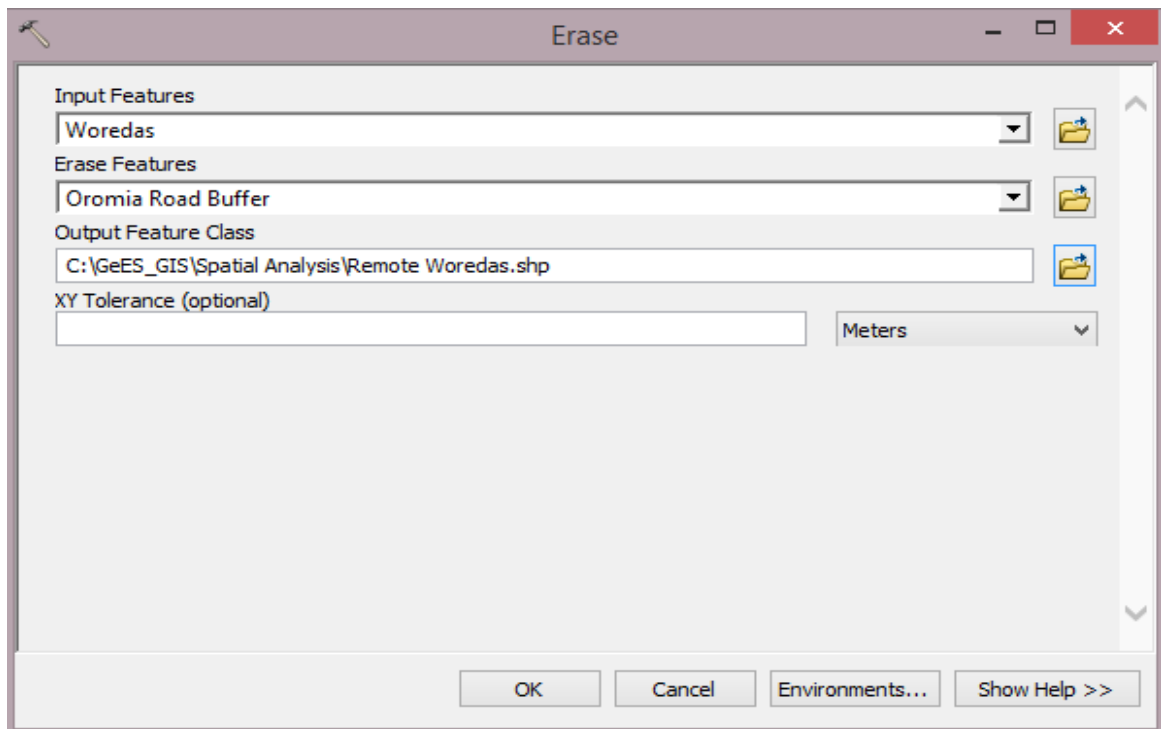




The outputs of your analysis will be displayed in ArcMap. In the map you see the 10KM buffer zone of the roads. From this map, you can analyze, how many Woreda's of the region are remote and close to roads. Based on the Buffer of roads you have in this exercise you are going to identify measure and analyze woreda's of Oromia region which are 10 km away from roads.

- a. From the ArcToolbox window expand the Analysis Tools toolbox
- b. Expand the Overlay toolset
- c. Double click on the Erase tool
- d. In the coming erase dialogue box
- e. In the Input features dialogue box; use the dropdown arrow and select "Woreda" Shapefile as an input data sets.

- f. In the Erase Features select Roads 10km buffer using the dropdown arrow Click on the browse folder icon and store the output feature class in C:\ GIS_GeES/ Spatial Analysis Give “Remote_ woreda’s” as a file name of your output feature.
- g. Click ok when finished.



Your output feature class has been added to ArcMap automatically. Turnoff the remaining layer and see the Remote Woreda’s layer only.

- ***Identifying the Percentage Coverage of Remote woreda’s in each Woreda’s of Oromia Regional State?***

By now you have been able to identify woreda’s of Oromia region which are remote from existing roads. By using this data you will identify the percentage coverage of remote woreda’s from the total areas covered by each of woreda’s of Oromia regional state. Follow the following procedures to do such things: Creating New Fields: now you will create a new column which will store the area Coverage of Remote areas in the region.

- a. Open the attribute table of Remote
- b. At the bottom of the dialogue box, click on Options

- c. From the options menu, click on “Add Field”
- d. In the Add field dialog box, write “Inaccessss woreda’s” as a name of the field
- e. In the Type option, use the drop down arrow and select “Double”
- f. Click on OK when you finish

The image shows the 'Add Field' dialog box with the following details:

- Name:** Inaccess_Woreda
- Type:** Double
- Field Properties:**

Precision	0
Scale	0
- Buttons:** OK, Cancel

• ***Calculating the area Coverage of Remote Areas?***

- a. Open the attribute table of the Remote areas
- b. Right click on the heading of “area” column you have crated
- c. Click on Calculate geometry option
- d. Click Yes to the incoming warning popup window
- e. From the calculate Geometry Dialog box, proceed to the Units option and select KM by using the dropdown arrow
- f. Click OK
- g. Analyze the result and close the attribute table

Result: the new area you have calculated contains the area coverage of each woreda’s of Oromia regional State which are remote from existing Roads.

CHAPTER SIX

CREATING AND EDITING DATA

Objectives

Dear trainees under this task you will be able to familiarize with:

- Spatial referencing and Georeferencing Imageries and other vector datasets
- Add simple spatial (points, lines or polygons) and attribute data to Geodatabase by using various techniques.
- Edit features in a Geodatabase by using various editing tools.
- Validate the topology within a Geodatabase

It is easy to create a geodatabase and add behavior to it, and no programming is required when you use the data management tools in ArcGIS Desktop. When querying and editing the geodatabase in ArcMap, which is the application for editing, analyzing, and creating maps from your data, you can easily take advantage of the data and behavior in your geodatabase without any customization. This tutorial lets you explore the capabilities of the geodatabase using an ArcEditor or ArcInfo licensed seat of ArcGIS Desktop.

Thus, Data entry is a task that complements all GIS work. Data has to be entered and stored in a proper digital format (in the computer) to apply any GIS analysis. Different methods are used for data input. Data can be recorded directly from the field in a digital format by using devices as Global Positioning Systems (GPS) and Satellite Imagery. Data also can be input from analogue format (hard copy) by mean of digitizing or scanning. Spatial data and its attributed data are stored in the computer in different ways depending on the GIS software in which it was created. However most commonly it is arranged in the form of tables. You are going to practice how to how to build spatial and attribute data base using ArcGIS spatial and

attribute data to the empty feature classes of the Geodatabase. In the following exercise you will practice data entry by digitizing though which you will produce impartial base maps of Haramaya University Campus include **three basic elements i.e. roads, building and squares of the campus.**

In the following exercise impartial high resolution satellite image of Haramaya University campus was downloaded from Google Earth and provided in **C: /GIS_GeES/ Data Entry Folder**. Toolbars which you are going to use are the “Editor Toolbar and the Topology toolbar”. You will get familiar with the toolbars in this part of the document. The major activities that you are going to perform are summarized as follows:

- Spatial Referencing
- Georeferencing
- Prepare the digitizing session
- Spatial Referencing
- Digitize line features
- Digitize polygon features
- Enter point features
- Add attribute data
- Edit features

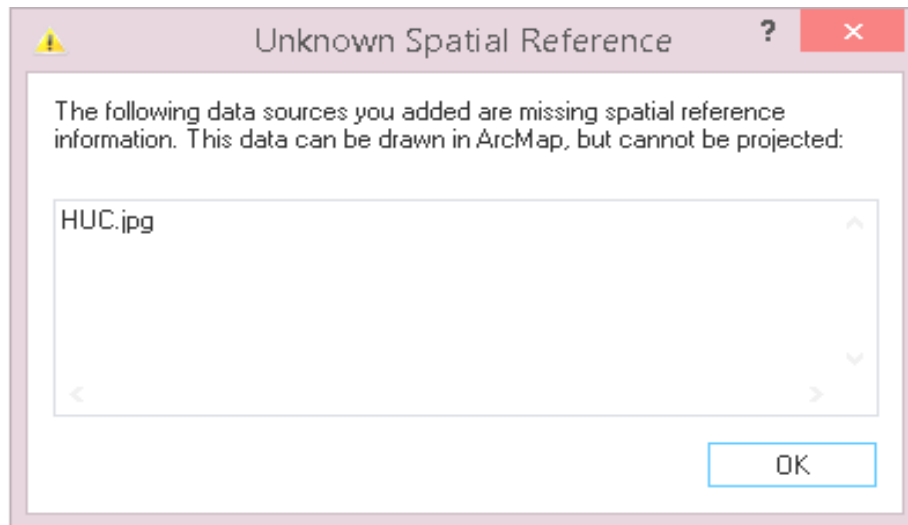
6.2. Spatial Referencing

Note: *The raster map is in pixel space with no spatial coordinate system information. ArcMap assumes that the data are without projection, and therefore it reads out geographic coordinates with map units expressed in degrees, minutes and seconds (see lower right corner). This is obviously wrong! We deal here with a scanned paper map, and any flat paper map is projected!*

Setting Ups

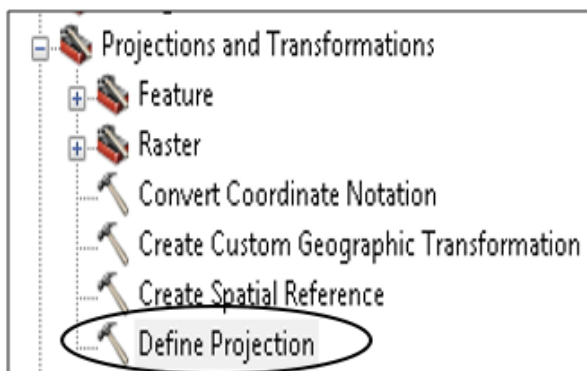
- A. Open a new empty map (Start >All Programs >ArcGIS >Arc map)
- B. Browse and to Haramaya University Campus Google Earth Image from **C:\GIS_GeES\Data_Entry\HUC.jpg**.

- C. When you added a data set which misses a spatial reference system, the following dialog data set you are adding do not have a coordinate system

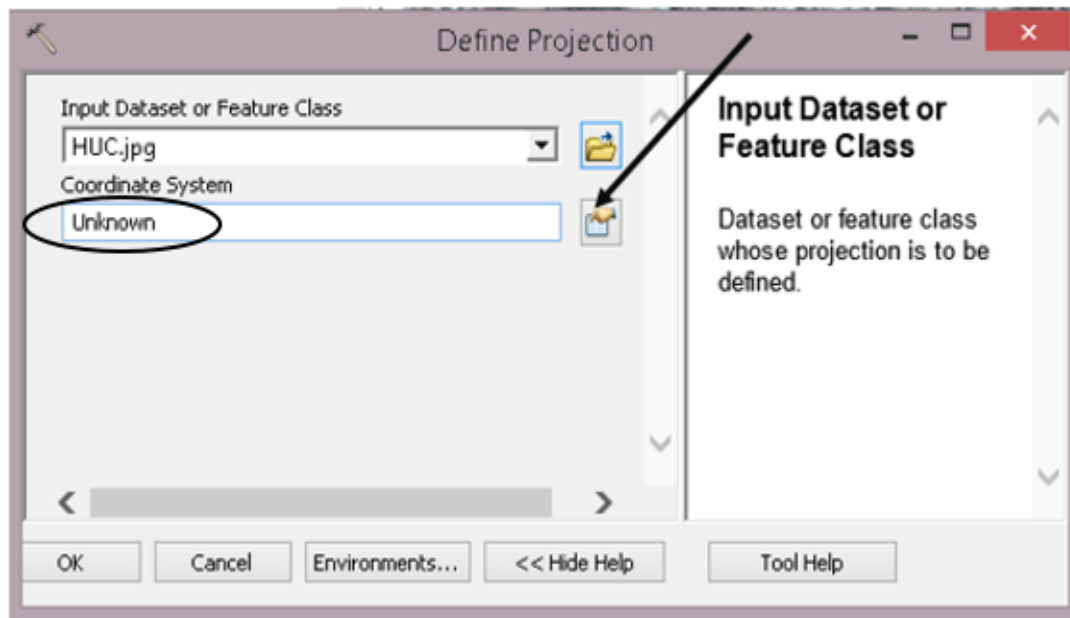


When the warning about the missing of coordinate system comes, the message is warning you as the data is missing a spatial reference system. Doing GIS analysis tasks with an unknown spatial reference system will create an error in the analysis result. To understand the effect of using a data set with an unknown spatial reference system.

- D. Click on the Zoom to extent button on the tools menu 
- E. Open ArcToolBox if it is not present in the ArcMap environment
- F. From the arc toolbox window, expand the Data management toolbox
- G. Expand the Projection and Transformation Tool
- H. Click on the Raster → Double click on the Define Projection toolset



- I. In the define projection dialogue box, use the drop down arrow to select the HUC image. Click on the select coordinate system arrow



- J. In the coming Spatial reference properties dialogue tab, click on the select tab.
- K. In the Browse for coordinate system dialog box, double click on projected Coordinate System and select WGS 1984 UTM Zone 37.

The above procedures that you have go through has made the HUC image to be projected in to WGS 1984, Zone 37 N coordinates system. From now onwards, when you add the Arc map, the missing of spatial reference system warning dialogue box will not come again.

6.3. Georeferencing

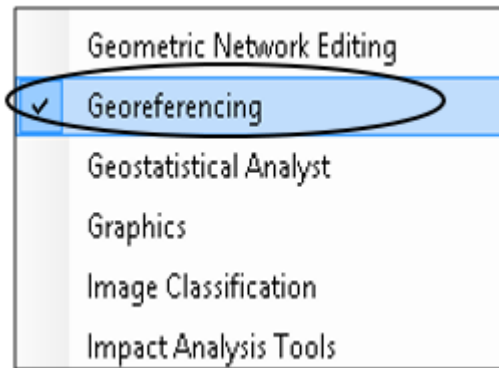
This part explains the procedure passing coordinates to a raster image (e.g. Google earth images, Aerial Photographs and a scanned map) by means of a number of control points, followed by the rectification of the raster image. When you georeference your raster dataset, you define its location using map coordinates and assign the coordinate system of the data frame. Georeferencing raster data allows it to be viewed, queried, and analyzed with other geographic data. After having the raster dataset, in order to georeference an image, we should have sample ground control points of the features which are depicted on the map. In other word we should have to know the absolute location of some representative features on the raster data set to be georeferenced. This is usually achieved by using ground GPS collection

if the image to be georeferenced is a satellite image or aerial photo absolute locations depicted with grids, which could be used as Ground Control Points (GCPs) in the Georeferencing process.

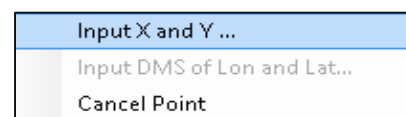
The table below shows the four corners coordinates of **HUC** image.

Easting (X)	Northing(Y)	Placement
174106.10	1043435.63	Left Top
174801.55	1043439.00	Right Top
174147.11	1043051.76	Left Bottom
174457.78	1043060.00	Right Bottom

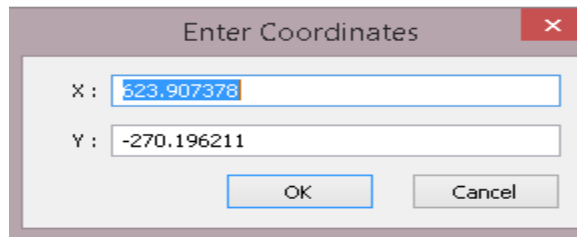
- a. Activate the Georeferencing tool bar (click on the top of the map document, around the main menu and thick mark the Georeferencing tool bar.



- b. Zoom in on the Top Left Corner of the image to the extent that you can see the X and Y numbers written at the edge of the image.
- c. Zoom in specific to the vertex represented by the X and Y points you have written (you have to zoom in to the extent that you can identify the center of the vertex)
- d. Click on the added control points icon from the Georeferencing tool box 
- e. Place the incoming cross at the center of the vertex you have zoomed in 
- f. Right click and press in put X and Y points



- g. In the incoming small dialog box, enter the X and Y points that you have saved



- h. When you finish, press ok, then the image may disappear from the data frame
- i. Click on the zoom to full extent button to repair the appearance of the image 
- j. First, take coordinates from the rest three corners of the map (Top right, top left bottom left corners of the map).
- k. See the attributes of your sample GCPs by clicking on the View link table in the View Link tool 
- l. Check the total RMS accuracy is less than 1.
- m. If it is the total accuracy is greater than 1 m you should have to redo the Georeferencing process.
- n. Click 'Georeferencing' tool → Rectify
- o. In the rectify dialog box, click on the browse folder icon In the select workspace dialog box, proceed to C:\GIS_GeES\Data_Entry
- p. From the list of folders under C:\GIS_GeES\Data_Entry and edit the output filename to be HUC_rectified.
- q. Using the drop down arrow in front of the format option, select TIFF
- r. Click Save when you finish

6.4. Data Entry

The aim of this exercise is to produce from high resolution satellite image. To do structure which will hold (contain) the necessary in Arc Catalogue, where you will create a Personal and a feature Class.

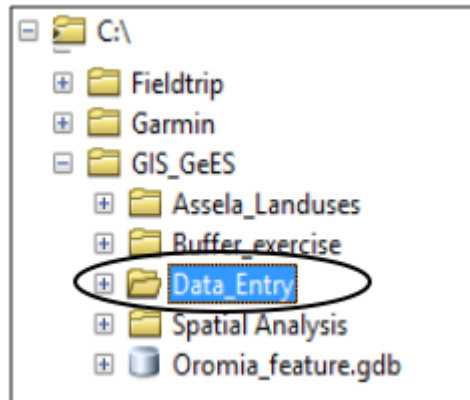
6.4.1. Creating Personal Geodatabase

Steps

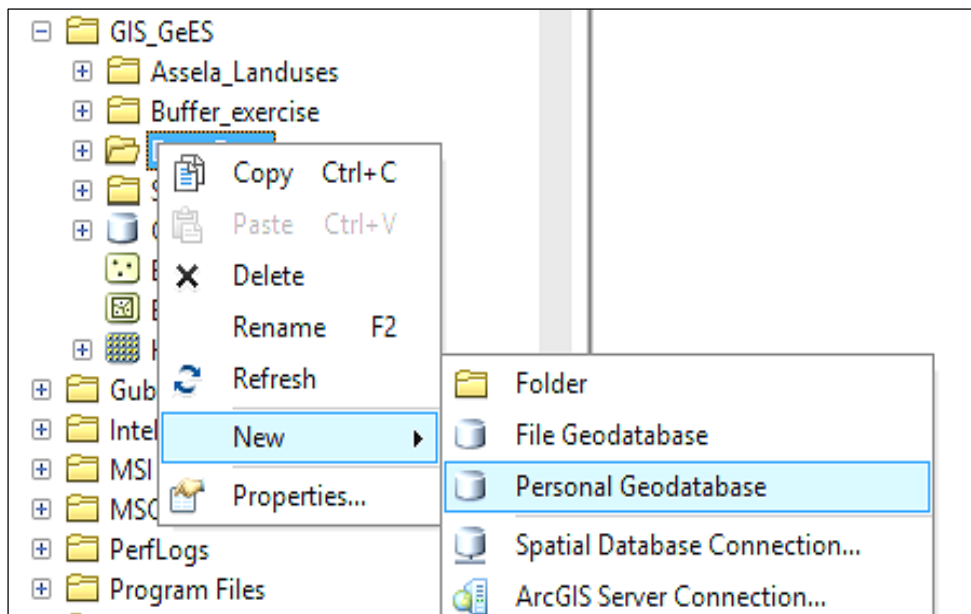
- a. Open new Arc Map and Expand the Catalog Window

b. Connect to **C: /GIS_GeES/ Data Entry Folder**.

b) In the Catalogue tree right click on the Data Entry Folder



e. Click on New and then Personal Geodatabase



e. Rename the New Personal geodatabase to “HUC”

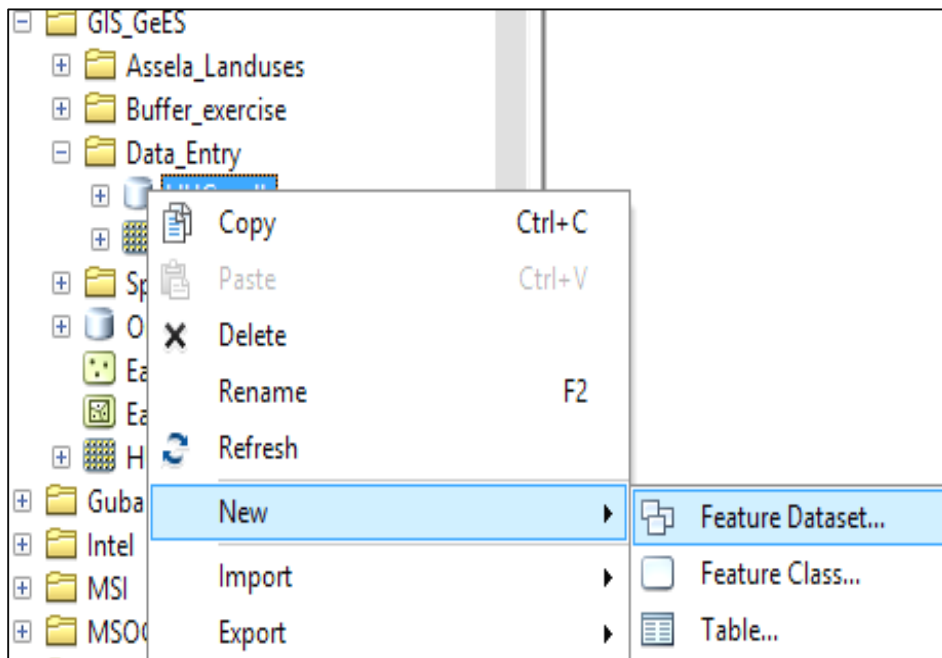
f. By now you can see the new HUC Geodatabase listed under Data Entry folder

6.4.2. Creating Feature Dataset

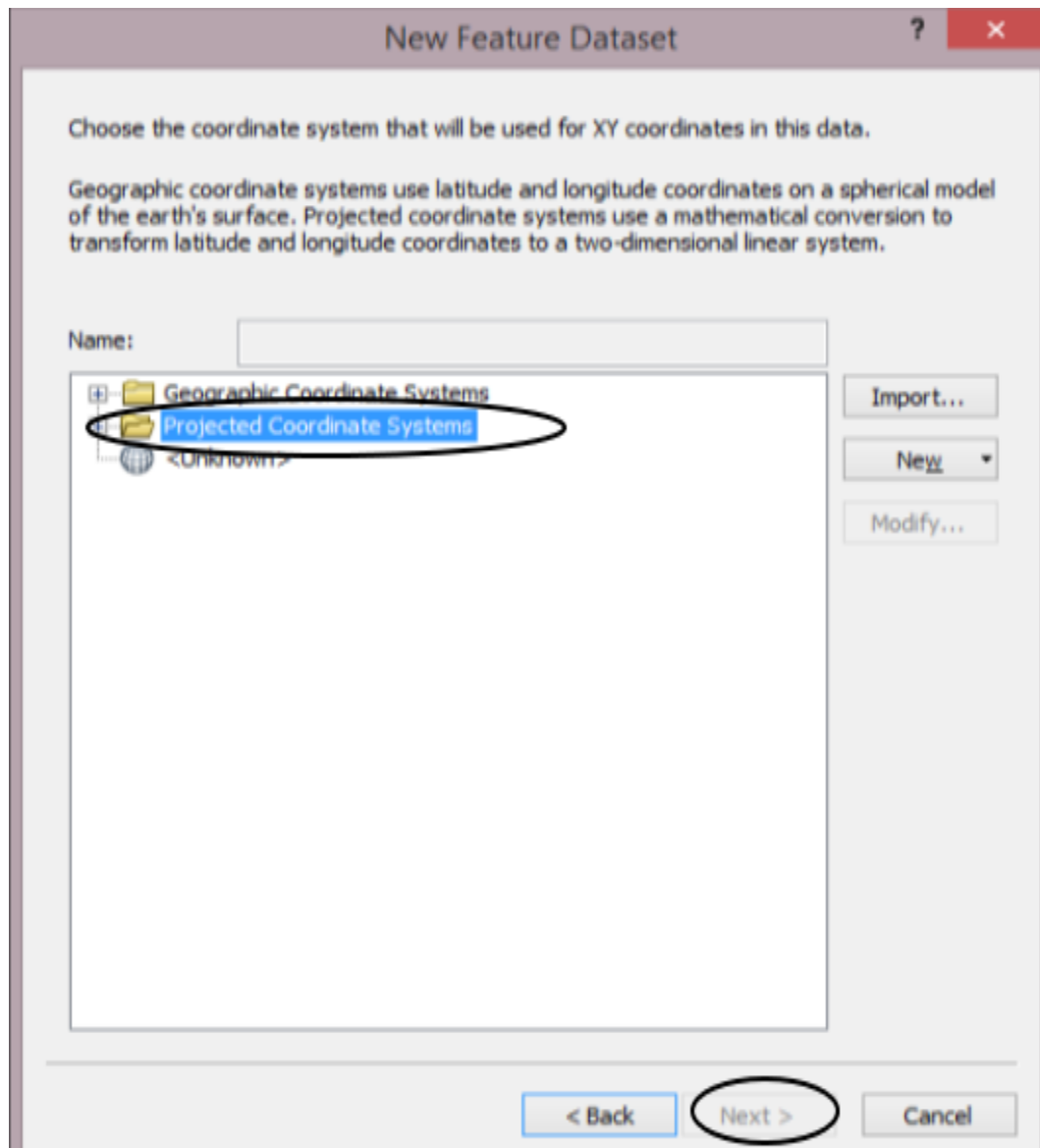
Steps

a. Right Click on the HUC geodatabase

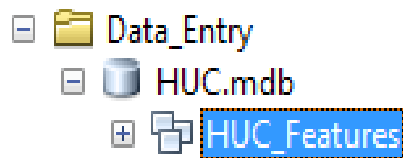
- b. Click on new and then Feature Dataset



- c. In the coming new feature data set dialogue box, write, HUC_ Features as a name of your new feature dataset.
- d. Click on next.
- e. Select from Projected coordinate systems-UTM and then click on WGS 1984. For Ethiopia, you click on the WGS 1984 UTM Zone 37N.



- f. Now you can see your new feature data set under the GPTCC personal geodatabase



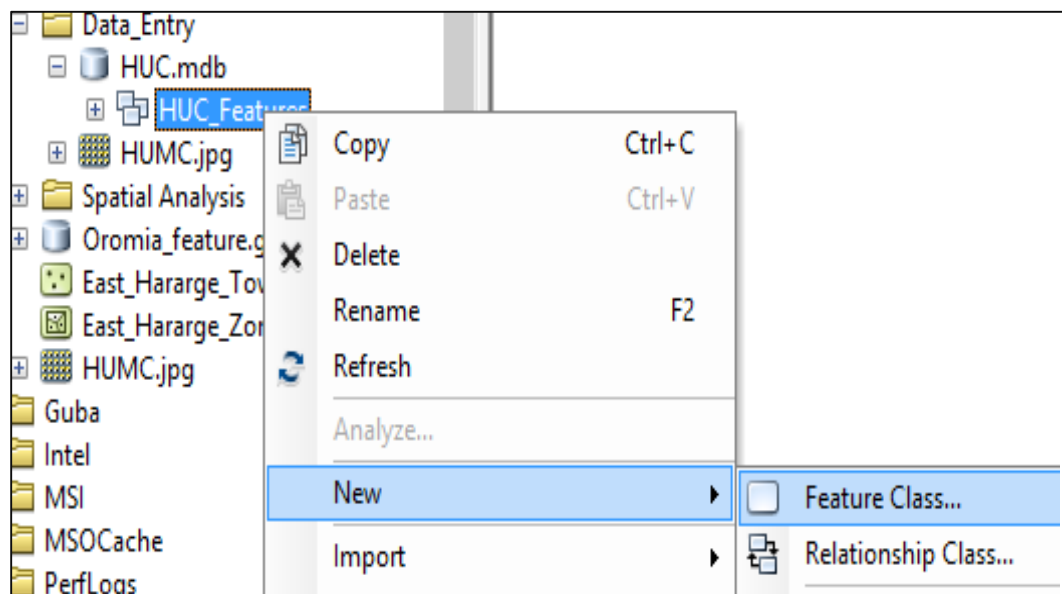
6.4.3 Creating Feature Classes

Until now you have built a feature data set which will hold all geometries to be created in the feature. Following this, you are going to create feature classes in which the land uses, the roads and rural villages will be stored.

6.4.3.1 . Creating Polygon Feature

Steps

- a. In the catalogue tree, Right click on the **HUC_ features** data set that you have created
- b. Click on New → click on Feature Class.
- c. On the New feature class dialogue box, write feature class on the space in front of it as “Blocks”.
- d. Select the type of feature to be stored in the feature class to be Polygon using the drop down arrow located in front of the type radio button and set spatial reference.



- e. Click Next
- f. In the incoming dialog box, you will add attribute fields to the new polygon features
- g. Below the Shape raw function supposed that you know the function of each building visible on the image and you will add this information with this attribute field).
- h. In the Shape dialog box, select Text, Double for area and Text for Types of
- i. Click on Finish, check whether the new feature class is add to ArcMap.

NEW FEATURE CLASS

Field Name	Data Type
OBJECTID	Object ID
SHAPE	Geometry
Area	Double
Type	Text

Click any field to see its properties.

Field Properties

Alias	
Allow NULL values	Yes
Default Value	
Length	50

Import...

To add a new field, type the name into an empty row in the Field Name column, click in the Data Type column to choose the data type, then edit the Field Properties.

< Back Finish Cancel

6.4.3.2. Create a point Feature Class

In the above exercise you have created a polygon feature class which will hold building with additional information such as building types and area. In the point features you will store squares found within Haramaya University Campus and visible on Satellite image.

- In the catalogue tree, Right click on the HUC_ features data set that you have created
- Click on New → click on Feature Class.
- On the New feature class dialogue box, write feature class on the space in front of it as “Square”.
- Select the type of feature to be stored in the feature class to be Point using the drop down arrow located in front of the type radio button and set spatial reference.
- Click on Finish, the new feature class is added to ArcMap.

6.4.3.3.Create Line Feature Class

There are also roads found in the compound and we will represent it as line feature.

To do that you must also create an empty road feature class in under your feature dataset. To create line features please follow the following steps.

- a. In the catalogue tree, Right click on the HUC_ features data set that you have created
- b. Click on New → click on Feature Class.
- c. On the New feature class dialogue box, write feature class on the space in front of it as “Roads”.
- d. Select the type of feature to be stored in the feature class to be Point using the drop down arrow located in front of the type radio button and set spatial reference.
- e. Click Next
- f. In the incoming dialog box, you will add attribute fields to the new polygon features
- g. Below the Shape raw function supposed that you know the function of each building visible on the image and you will add this information with this attribute field)
- h. In the Shape dialog box, select Text for road Types and Double for road lengths.

6.5. Digitizing in Arc Map

So far you have created the data structure which will hold the features classes to be created. The feature classes you have created are however empty, which does not contain any spatial data. In this exercise, you will use Google Earth image to create a new feature impartially representing Haramaya University main Campus. Once the feature is created, you will then add attribute values to the features. You are introduced to the Editor Toolbar, the Create Features window, and the Attributes window, which are the main elements of the ArcMap user interface when editing. To start this exercise, you first need to zoom the map to your area of interest.

6.5.1. Load the Feature Data Set in ArcMap

You can start ArcMap with a new map document and add the just created Feature Data Set to the map in order to see its attribute fields and coded value domains.

- Start ArcMap with a new empty map.

- Add the feature class you have created from **C:/GIS_GeES/Data Entry Folder** (In the HUC_Features Geodatabase) to the map: the feature classes will appear in the table of contents.
- Open the attribute table of all the feature classes to see which attribute fields you created. Then close the attribute tables.

6.5.2. Digitizing Polygon Features

By now you are going to digitize the buildings foot print found on the satellite image.

● Opening the Editor Tool Bar

- Right-Click on the upper top part of the page.
- You will get a menu of toolbars. Select Editor.
- The Editor Toolbar will appear on the window.

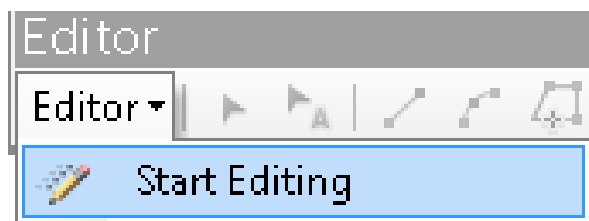


In digitizing buildings, you should have to store the buildings that each polygon will contain. To do so, do the following arrangements in the editor menu.

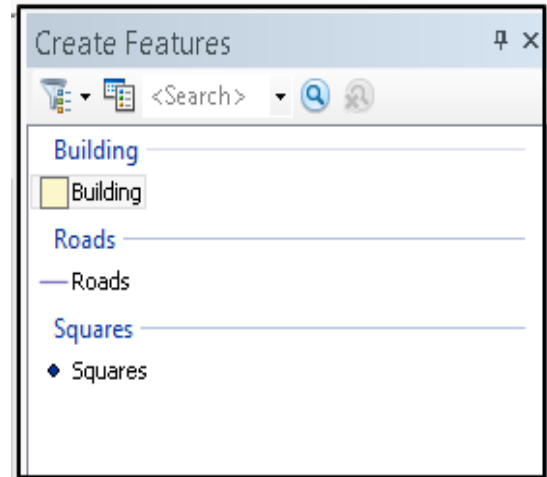
- Click on the Dropdown arrow of the Editor menu
- Click on Options
- Hit on the attribute tab
- Check on Display the Attribute Data before Storing the Features
- Check on For the following layers
- Turn on Land use layer and leave the remaining ones as they are.

● Start Editing

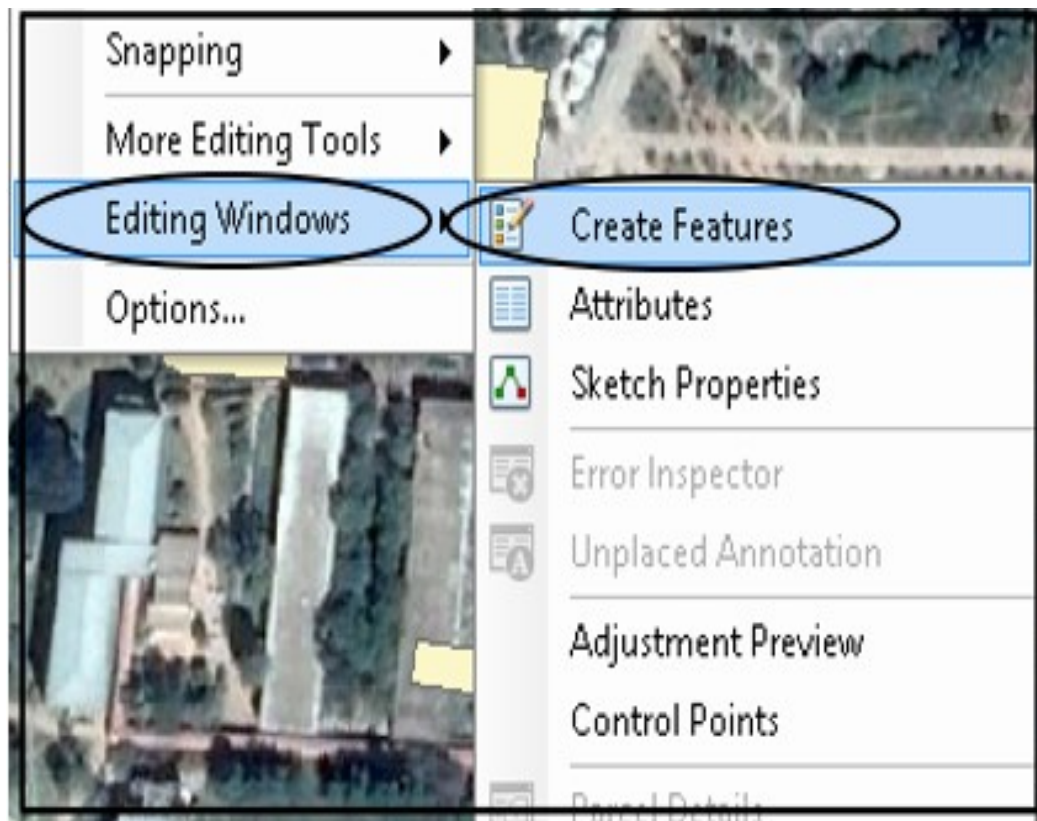
- Click on Start editing from the Dropdown arrow of the Editor menu



- b. Expand the Editor window at the left part of ArcMap, (if it's not presented, go to the editor menu.



- c. The editor window is not present in ArcMap, use the drop down arrow of the editor menu and go to the editing Windows menu.
- d. Click on the Create Features option in the editing Window option

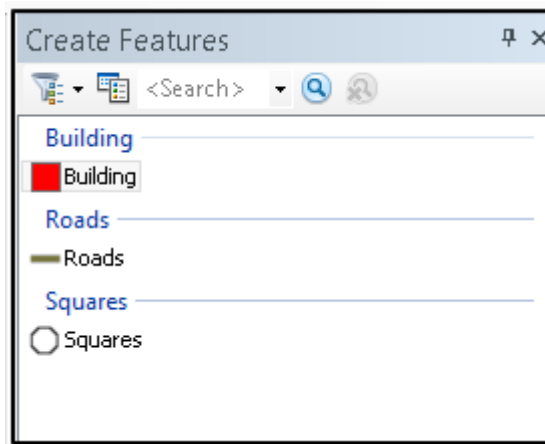


- e. Zoom in to the building you see on the screen and try to identify different buildings with their function before you proceed to actual digitizing.



Figure 58: The view of the satellite image of HU_Campuse

- f. Go to the editing window and Select Buildings on this features that the feature you will create will be stored)



- g. Start to digitize the surroundings of the buildings by clicking the boundaries of buildings.
- h. Double click when you finish digitizing the surroundings of the building.



Figure 58: The view of the satellite image for the area of the building polygons

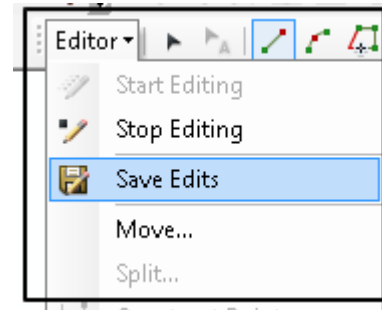
- i. In the coming Attribute Editor dialog box write the name of the building as the function of the land use type (it is asking you to add attribute information as you have already activated display the attribute data before storing the features option on the editor menu.

OBJECTID	16
Area	<Null>
Type	<Null>
SHAPE_Length	86.001779
SHAPE_Area	454.749764

OBJECTID
Object ID
Null values not allowed

OK

- j. Save your edits by clicking on the editor tool bar then click on save
- k. Try to digitize more polygon features as much as you can



6.5.3. Digitizing lines and snapping

In the first exercise, you digitized a polygon of Haramaya University main Campus building and a point data of the squares; in this one, you will trace over the image to create a new line representing a road. Because part of the road has already been created, you should use snapping to help ensure the new road feature connects to the existing roads. When snapping is turned on, your pointer will jump, or *snap to, edges, vertices, and other geometric elements* when it is near them.

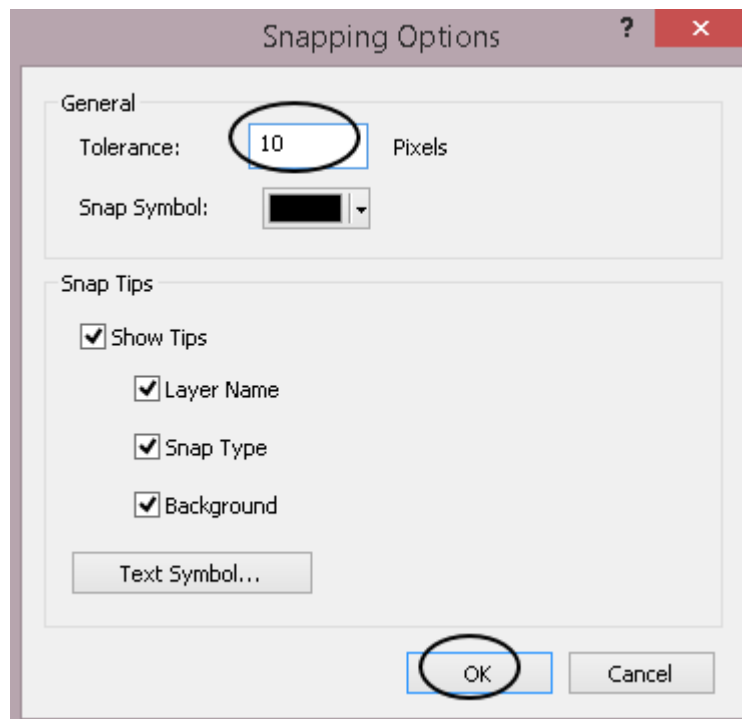
This enables you to position a feature easily in relation to the locations of other features. All the settings you need to work with snapping are located on the Snapping toolbar.

- a. Navigate to the **Digitizing roads** bookmark. The extent is just south of the point feature you created in the previous exercise.
- b. Add the **Snapping** toolbar to ArcMap. You can add a toolbar by clicking the Customize menu, pointing to Toolbars, then clicking the toolbar's name in the list. You can also add the Snapping toolbar by clicking the Editor menu, pointing to Snapping, then clicking Snapping Toolbar.
- c. On the **Snapping** toolbar, click the **Snapping** menu and confirm that **Use Snapping** is checked.
- d. If it is already checked, do not click it again, since that will turn off snapping. If **Use Snapping** is not checked, click it to enable snapping.
- e. Look on the **Snapping** toolbar and confirm that **End**, **Vertex**, and **Edge** snapping types are active. When enabled, the buttons are highlighted. If they are not enabled, click each button to enable those agents.
- f. Click the **Snapping** menu and click **Options**. From this dialog box, you can specify settings for Snapping in ArcMap.

- g. Ensure the snap tolerance is at least 10 pixels. The snapping tolerance is the distance within which the pointer or a feature is snapped to another location. If the element being snapped to—such as a vertex or edge—is within the distance you set, the pointer automatically snaps to the location. *For example, if a snapping tolerance of 8 map units is assigned: then whenever you digitize a vertex or point within this tolerance distance (e.g. 8 map units) from an existing vertex, the two vertices will be joined.*



Figure 59: The snapping window



To start the editing session

- a. Zoom in to the source image to the extent that you can clearly identify certain road
- b. In the Editing window, activate the road layer to be the layer being edited.



- c. Use the Edit Tool button to select one of the newly digitized lines and click the Attributes button. The Attribute window will open, in that window click on the value field of Road class and type Street then close the window.
- d. Continue to enter the other road class Type (street) for the roads you digitized
- e. Use the identify tool on the new lines and you will see that the road class attribute of these lines is street.

View ▾	
Lenght	<Null>
Type	<Null>

g. Save the new lines: **Editor** → **save edits**

6.5.4. Digitizing Point Features

Squares formed by the junction of three or more roads will be digitized as point features in the new data base being constructed. You will digitize these features as follows.

- a. Zoom in to square on the image, formed by the junction of one or more roads



- b. Click on square features on the edit
- c. Put the sketch tool on the center of each square
- d. Try to digitize more point features as much as you can.
- e. When you stop editing in the editor tool bar by saving your edits images, so you will store the roads on the image.

6.5.5. Add Attribute Data

You were adding attribute values to the buildings layer automatically when you finish digitizing the polygons. That is the first option available to add attribute data. In the following exercise, you will practice other ways of adding attribute data to the data base.

- a. Start Editing if you are not on the editing session
- b. Right Click on the Building layer and Open attribute table
- c. Select the first row of the layer

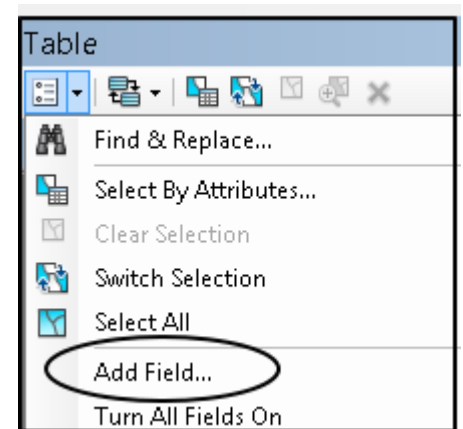
- Look at the selected polygon and determine what will be its function
- Type the function you have identified (in the Function field of the selected row)
- Using similar way, change or add the attributes of the other building polygons.
- In the same way, type the name of the squares in the attribute field of the square feature class layer.

Building						
	OBJECTID *	SHAPE_Length	SHAPE *	SHAPE_Area	Area	Type
	8	155.140637	Polygon	1309.506834	<Null>	Class Room
	9	312.145761	Polygon	5016.670224	<Null>	Resource Center
▶	10	120.162514	Polygon	664.482574	<Null>	Staff Launge
	13	98.346578	Polygon	532.035284	<Null>	<Null>
	14	168.009872	Polygon	1761.491418	<Null>	<Null>
	15	138.260223	Polygon	835.462166	<Null>	<Null>
	16	86.001779	Polygon	454.749764	<Null>	<Null>

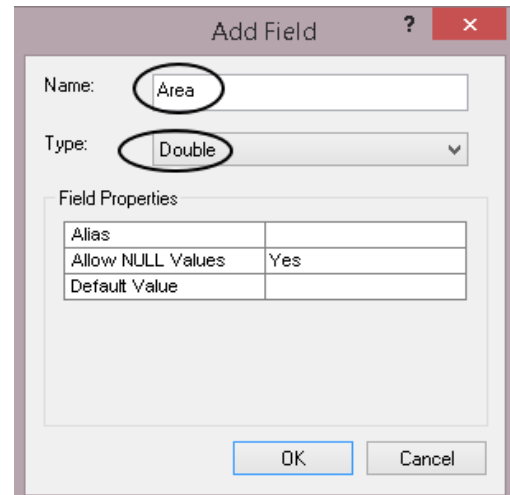
6.5.6. Calculating Areas and Lengths

To add a new attribute information, you should create a new attribute field which will contain the new attribute information to be added. You have added new attribute fields to the Buildings and Roads layer when you create the feature classes at the very beginning. If you want to add more attributes than you have created earlier, you can use the ArcMap's add attribute field option described below.

- If you are in the editing season close your edits by clicking on the stop editing tool on the editor tool bar
- Right click on the road layer in the table of contents
- Open attribute
- Click on the Table option s tab drop down arrow and click on added fields.

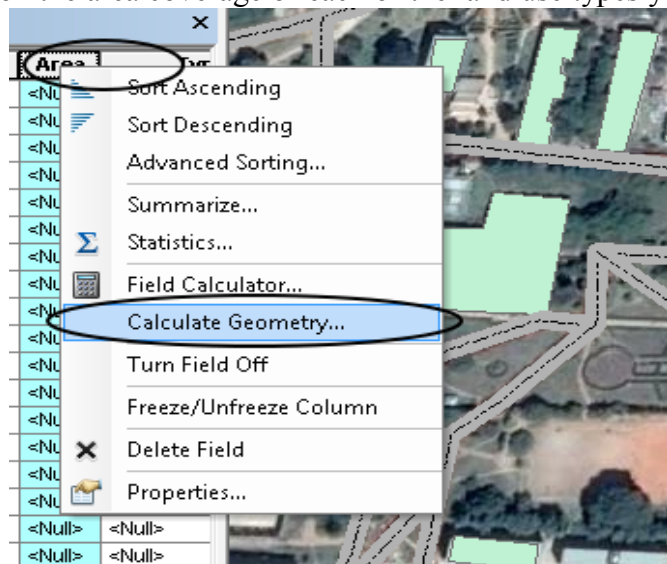


- f. Type “Area “in the name of the field layer
- g. Select “Double” as the field type
- h. Click Ok
- i. Using the similar procedure buildings Roads and rename the field of field to ‘Length’; make it to be in Double type.



Here you will calculate the area coverage of each the buildings we have digitized using the field Geometry Calculator option.

- a. Open the attribute tables of
- b. On the attribute table, Right click at the top of Area field
- c. Click on Calculate Geometry Option
- d. Click OK on the coming warning Box and select unit of Measurement “Square Meters (sq. m)”
- e. Click Ok
- f. On the attribute table you must look the area coverage of each of the land use types you have digitized.



Activity. Using similar procedure try to calculate road length?

TASK 7

DATA VISUALIZATION

Objectives

In the previous exercise, you worked in the Data View of ArcMap. The other view option in ArcMap is the Lay out View. The layout view is used when a GIS analysis is complete and you wish to display the layers as a finished map product. You create a layout when you want to actually create a map for printing or inclusion in another document. It is a view of your data, much like viewing the page layout when you are working in a word processing software. You should use the layout view when you are ready to create a map - do all the preliminary work and analysis in the data frame view (where you have been up to now in this tutorial). When you create a map, you should include the map itself (which is the same as your data frame), a title, a legend, a scale, and a north arrow.

In general the main difference between Data view and Layout view is:	
Data view	Layout view
• view and display data • symbolize data • selection –query • analysis • edit data	• Make printable map • Define Printer • Insert Map Element ✓ Legend ✓ Scale bar ✓ North Arrow ✓ Title etc.

In a GIS environment, maps can be used to:

- Input for GIS
- Communicate GIS results
- Support spatial analysis

7.1. Setting up a layout

Before you start a layout, it is important to have thought through what you want to do and how you want your map to look. What do you want to show? How large do you want your final map to be? Portrait or landscape? Do you need space for additional text or graphics?

- In ArcMap, choose **View - Layout View** from the main menu.
- The view changes to show your data frame on a page layout and a new toolbar appears – the *Layout* toolbar.

7.2. Elements of a map

In a “good” map, there are six elements that people agree are helpful, even if not always necessary:

- *Scale indicators*
- *Legend*
- *Title*
- *Authors*
- *Orientation (North Arrow)*
- *Date*
- **Data in maps:**
 - ♣ The data should take up a majority of the area; you should avoid excess white space.
 - ♣ Inclusion of unnecessary data should be avoided- most often **simple is better**.
 - ♣ Bright, flashy colors such as red should not be used for areas unless you specifically need to do so.
- **Titles:**

When you insert a title or other element, it is often very small and hard to see if it inserts into the data frame area. If you don't see your title at first, just type your title anyway, press Enter. It will automatically be selected, so that if you make its font bigger using the Draw tools, you can see it better and then move it.

- a. Should usually be in upper case.
- b. Should not be sentences, but should be simple and to the point.
- c. Should not be the focal point of your map.
- d. Should almost always be black or dark text.

- e. Should be placed in a location on the map so that they do not obstruct any other portion of the map.
- f. Sub-titles should be smaller than main titles.

➤ **Scale bars:**

The scale bar you select will be in the Display units of your data frame (e.g. feet). If you want a different unit, click on *Properties* in the *Scale Bar Selector* dialogbox, and select a different *division Unit*.

- **NEVER** have a scale bar that extends all the way across a page. Scale bars should not be the focal point of the map, they are for reference only.
- Scale bars should use appropriate measurement systems. Example: Km for Regional maps, meters for the **HARAMAYA UNIVERSITY** campus.
- Metric measurements are more suitable for scientific work, and are more acceptable for international distribution. Use intervals that make sense. Units of 2,5,10, 20 are common. For example, do not use 25.4.

➤ **Borders:**

- a. Maps need borders, they should usually be black.
- b. Borders are known as "Neatlines."
- c. Neatlines should be a large enough width to be clearly seen from a distance

➤ **North Arrows:**

- North Arrows should be unobtrusive.
- North Arrows are not always needed. For Example - If mapping Harar, an arrow is not needed, because we can usually expect people to know which way is North.

➤ **Legends:**

- a. There are lots of ways to improve your legend - we will work on a *Legend Tips* section of its own. Note that in the Legend Wizard you can control what items go into the legend, and the order in which they are listed. You can modify a legend by double-clicking on it to bring up the legend properties.
- b. Legends should be unobtrusive
- c. Legends should only show defined features that are not obvious. For example black points are usually associated with landmarks such as cities, particularly if labeled. You can assume that people understand this.

- d. Legends do not need a heading 'Legend' or 'Key', but a legend title may be appropriate. This is particularly true if the legend units or classes are not obvious.
- **Text on Map:**
 - Text should never cross other text or other features of the SAME color.
 - Labels for natural features such as streams, lakes etc. should be written in italics.
 - Text should be readable!
- **Source:** - It is appropriate to mention the projection and source of your data if known, and if at appropriate scale. This information should appear in SMALL lettering near the bottom of your map if possible.
- **Name on Map:**

Unless map is being published, names should be kept outside of the map. Your name can go underneath the Neatlines on the bottom right side. Since this is a class, include the date, time and lab section with your name.

- **White Space:**

Do not waste space. Try to find the balance between too much white space, and cluttered data.

Setting Up

- a. Open a new empty map
- b. Add the Oromia Features _feature data set
- c. Click the small paper icon at the bottom of the screen to switch to you can also do that by going to View → lay out view on the main menu).

The page setup you will do your maps has an important role in making your maps more visible. So try to choose the appropriate page setups either from portrait and Landscape. The choice will depends up on the shape of the data to map.

- a. Right click in the Arc map environment outside of the Layout view
- b. Proceed to Page and Print Setup; and do the following arrangements:
 - ✓ Click on the drop down arrow in front of Size and select A3 as size paper (the choice of paper size orientation will depend up on the scale of the map to be produced).
 - ✓ In the orientation option click on Landscape (since it best visualizes the map of Oromia Region)

- ✓ Click OK, when you finish.

When you change the paper size, the layout frame becomes reduced and placed at the center of the layout view, which means that you have more unused spaces which will potentially enlarge the visibility of the map elements.

To do so,

- a. Select the map frame by clicking on it
- b. By using the arrows of the frame, drag the edge of the layout view to be within the limited boundary
- c. Zoom in the Map layer to the extent that the boundary of the map best fits to the margins of the data frame.

Note: For detailed information about layouts, see *Help - ArcGIS Desktop Help - Contents tab - ArcMap – Laying Out and Printing Maps*.

7.3. Symbolizing the Layers

7.2.1. Symbolizing Polygon Layers:

Following this you will practice in to different colors

- a. Right click on Zone Layer
- b. Click on Properties →Symbology
- c. At the right of the Symbology dialog box, click on values
- d. Click on the Drop down arrow below the value field option and select Zone_ as the attribute table used to classify the data.
- e. Click on Add All values at the bottom of the dialog box (it will list out all the layers found within Tigray region and assign a color for each of them).
- f. You can change the color combination by clicking
- g. Click Apply to change the color options you have changed
- h. Click OK when you finish

7.2.3. Symbolizing Lines Layer

- a. Click on the line icon below the Road Layer
- b. Select roads Symbology from the set of listed symbols
- c. You can edit the width and color of the line symbol using the color and width option
- d. Click OK when you finish Symbolizing Point Layers:

In the same way you have symbolized Road layers try to change the Symbology of “Towns”
Labeling features:

7.2.4. Inserting Different Map Elements:

You can add different types of map elements to the layout map using the Insert menu from the main menu bar.

7.2.4.1. Add title

All maps have titles, and many have subtitles as well. A good title helps the reader understand what to look for in the map.

- a. Go to insert in the main menu
- b. Click on Title
- c. In the coming box writes the title as “Basic Feature of Tigray”
- d. Drag and put the title box on the top of the layout frame

7.2.4.2. North Arrow

- a. From the INSERT menu, select NORTH ARROW
- b. Choose a style you like, and click ok.
- c. Move the north arrow by dragging with the mouse when the pointer changes to a 4 way arrow.

7.2.4.3. Add Scale Bar

- a. From the INSERT menu, select scale bar
- b. Choose a style you like (Alternate Scale bar 1 will be good) and click OK
- c. Drag the scale bar and place it at the bottom of the layout frame.
- d. Right click on the scale bar.
- e. You can readjust lots of scale map properties using this the Scale bar properties.
- f. Put the division unit in Kilometers.
- g. When you finish editing scale bars, click Apply and OK
- h. Check your map is look like the figure below. Do not worry about the color, but look carefully about the content of your map and the figure.

7.2.4.4. Add a Legends

- a. From the INSERT menu, select legend
- b. Click NEXT, and work your way through the wizard.
- c. You can change any of these properties later, if you wish.

d. Drag and drop the Legend box at the bottom left corner of the map frame

Note: You can resize the legend by using the resize icons found in the corners of the legend box

7.2.4.5. Add a Grid to a Map

- a. Right Click on the Layout frame and click on Properties
- b. Click on Grid in the Properties dialog box
- c. Click on New Grid
- d. Select Graticules select by Maps to be the type of Grid to be created
- e. Click Next, next until you finish (you can readjust this things latter if you need)
- f. Click Apply and Ok in the Data Frame Properties.
- g. If you need to make any modification for the Grids, you can right click on the Layout data frame, proceeded to Grids
- h. Here in the grid properties dialog box, you can readjust so many things about the properties of the grid.
- i. Check your map is look like the figure below. Do not worry about the color, but look carefully about the content of your map and the figure.

7.2.4.6. Adding a data frame to show two or more maps on a layout

You can add a second (or more) data frames to your ArcMap session. Only one data frame will appear at a time in the data frame view (right-click on the data frame name and choose Activate to see the data frame you want). But in the layout, all the data frames will appear. This can be handy for showing two on one poster, or for putting in a small "locator" map as in the example below:

7.2.4.7. Insert Locator

You can do that by inserting a locator map in your data frame. Follow the following procedures to do that.

- a. Go to the main menu and click on
- b. Reduce the size of Oromia region Data frame and create a space for the new data frame at the right top corner of the map w
- c. Drag the new data frame to the space you have created
- d. In the table of contents tab Data.
- e. Add the Ethio_region_shapfile found in the directory.

You can insert the needed map. To do that, first you should have to select the new layer that you want to add map elements)

- f. You can draw an arrow showing the position of Oromia region in Ethiopia (to do that, find the Draw tool from ArcMap a draw)
- g. Using the line icon, draw a line from the position of Tigray region in Ethiopia to the outer boundary of the region in the main map.
- h. Check if your map looks like the map below the main content of the map. Now, you wish to show the location of Oromia Region in Ethiopia.

7.3. Saving and Exporting the Map

7.3.1. Saving the Map Document

If you need to modify the map elements you were inserting for the future, you can do that by saving the ArcMap document you have made a layout and reopen it latter. But, be aware that Saving ArcMap document saves only the map elements and the Symbology you have made.

Steps.

On the main Menu, click on Save as and save in the previous directory: /GIS-GeES/Oromia features.

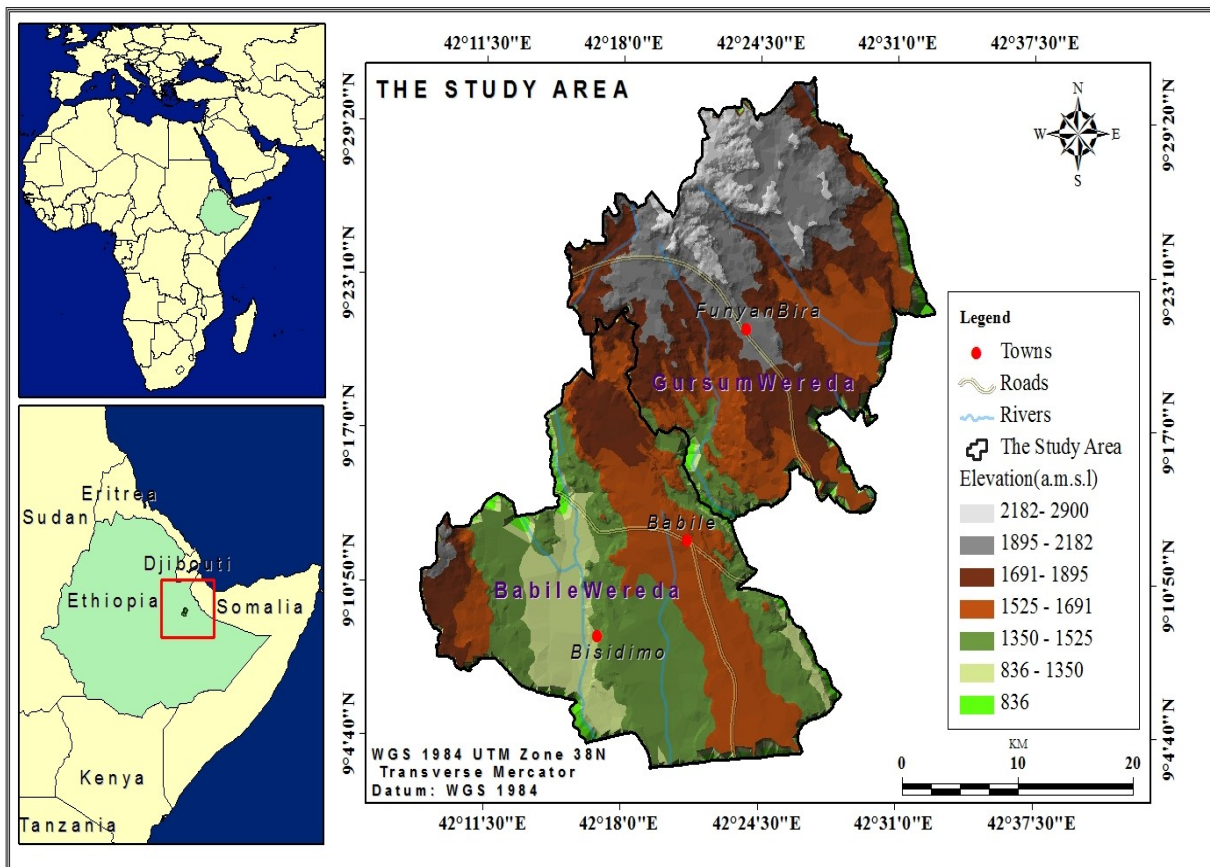
7.3.2. Exporting Map Document

If you need to insert the map in to a document or to print out it, or to transfer to another person, you should have to open other than GIS packages. This file formats may be PDF, image file formats like tiff or JPEG. So exporting the map helps you to make your maps open in any computer which might not have GIS software I transfer your data for others who cannot work with GIS software. Go through the following steps to export your maps.

- a. On the main menu click on file
- b. Chose the destination folder you want to save your maps (you can choose Desktop)
- c. Write Oromia Basic Map a file name
- d. Use the Save as File type drop down arrow and select Tiff to be the type of file format you will export your map.
- e. You can minimize or maximize resolution option
- f. Click on save when you finish

g. Open the map you have kept on the desk top.

 *That's the basics. Now practice what you have learned by creating several maps showing different aspects of the Oromia region.*



CHAPTER EIGHT

WORKING WITH XY DATA

8.1. Adding XY data into the ArcMap

The Global Positioning System (GPS) is a satellite-based navigation system made up of a network of 24 satellites placed into orbit by the U.S. Department of Defense. GPS was originally intended for military applications, but in the 1980s, the government made the system available for civilian use. GPS works in any weather conditions, anywhere in the world, 24 hours a day and, it is a fundamental GIS data collection tool. GPS satellites circle the earth twice a day in a very precise orbit and transmit signal information to earth. GPS receivers take this information and use triangulation to calculate the user's exact location. GPS consists of three segments: the space segment, the control segment, and the user segment.

The space segment: consists of the 24-satellite constellation described above. Each GPS satellite transmits a signal, which has a number of components: two sine waves (also known as carrier frequencies), two digital codes, and a navigation message. The codes and the navigation message are added to the carriers as binary bi phase modulations. The carriers and the codes are used mainly to determine the distance from the user's receiver to the GPS satellites. The navigation message contains, along with other information, the coordinates (the location) of the satellites as a function of time. The transmitted signals are controlled by highly accurate atomic clocks onboard the satellites.

The control segment: GPS system consists of a worldwide network of tracking stations, with a master control station (MCS) located in the United States at Colorado Springs, Colorado.

The user segment: includes all military and civilian users. With a GPS receiver connected to a GPS antenna, a user can receive the GPS signals, which can be used to determine his or her position anywhere in the world. GPS is currently available to all users worldwide at no direct charge.

In this portion of the tutorial, you will be learning how to import your GPS data files into ArcGIS. You can add data that are organized in columns to ArcMap to make contours, surfaces, or just display them as point maps.

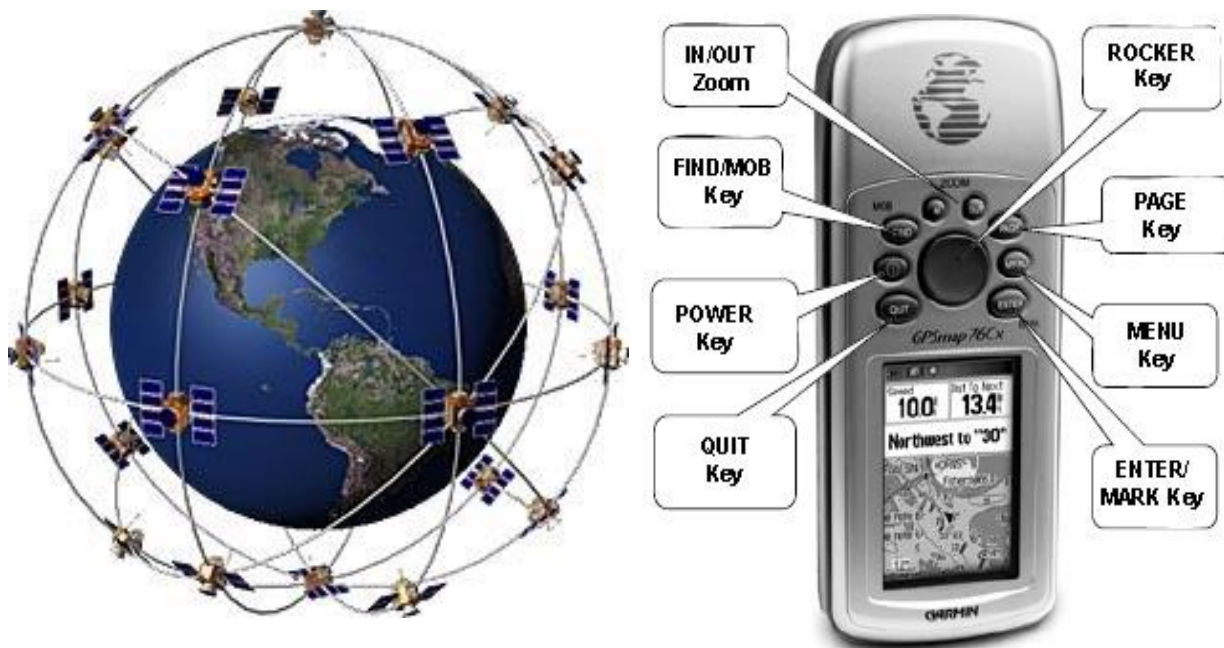


Figure GPS constellation.

This type of data may come from GPS readings from field survey, location of meteorological stations, sampling points, school locations, land-use class samples etc. The data should be organized in such a way that it has two columns that contain geographic locations (in lat/long or UTM). The table can be made with Excel and should be saved as .dbf or .txt format.

The file in *C/GIS_GeES Assela_Landuses* and open *GPSpts-2013.XLS* folder contain Coordinate points collected by GPS from Assela Town. These data will be added into Arc map to overlay with the town boundary. In some cases where it is not avoidable you might have to process GPS coordinates which are given to you in a spread-sheet (or table). You will now import an (Microsoft excel) spreadsheet into ArcMap and display their x/y coordinate columns in the map as point.

Browse to the directory that contains your table *C/GIS_GeES /Assela Landuse* folder and open *GPSpts-2013*. The table contains sampled Land-use Class in in Assela town.

- a. Close Microsoft Excel
- b. Open the new ArcMap
- c. Click add data button

d. Navigate to C:/GIS_GeES Assela_GPSpts2013.dbf and add the table.

When you add a standalone table to ArcMap the TOC will switch to the list by source view, since standalone tables do not show in the —list by drawing Order view.

e. Click display XY Data.

f. In the Display XY Data window choose “X” for the X field and “Y” for Y field, Altitude for and “Altitude for “Z” field.

g. Click Edit for the Coordinate System In the “XY” Coordinate System window click select

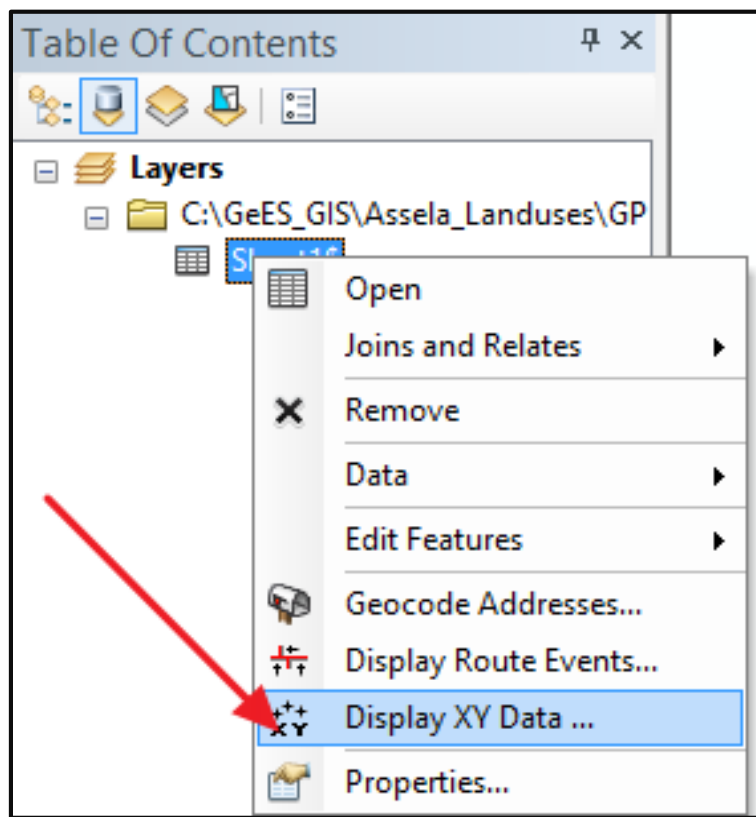
h. Navigate to and choose projected Coordinate Systems and set as WGS1984_UTM_ZONE_37N.

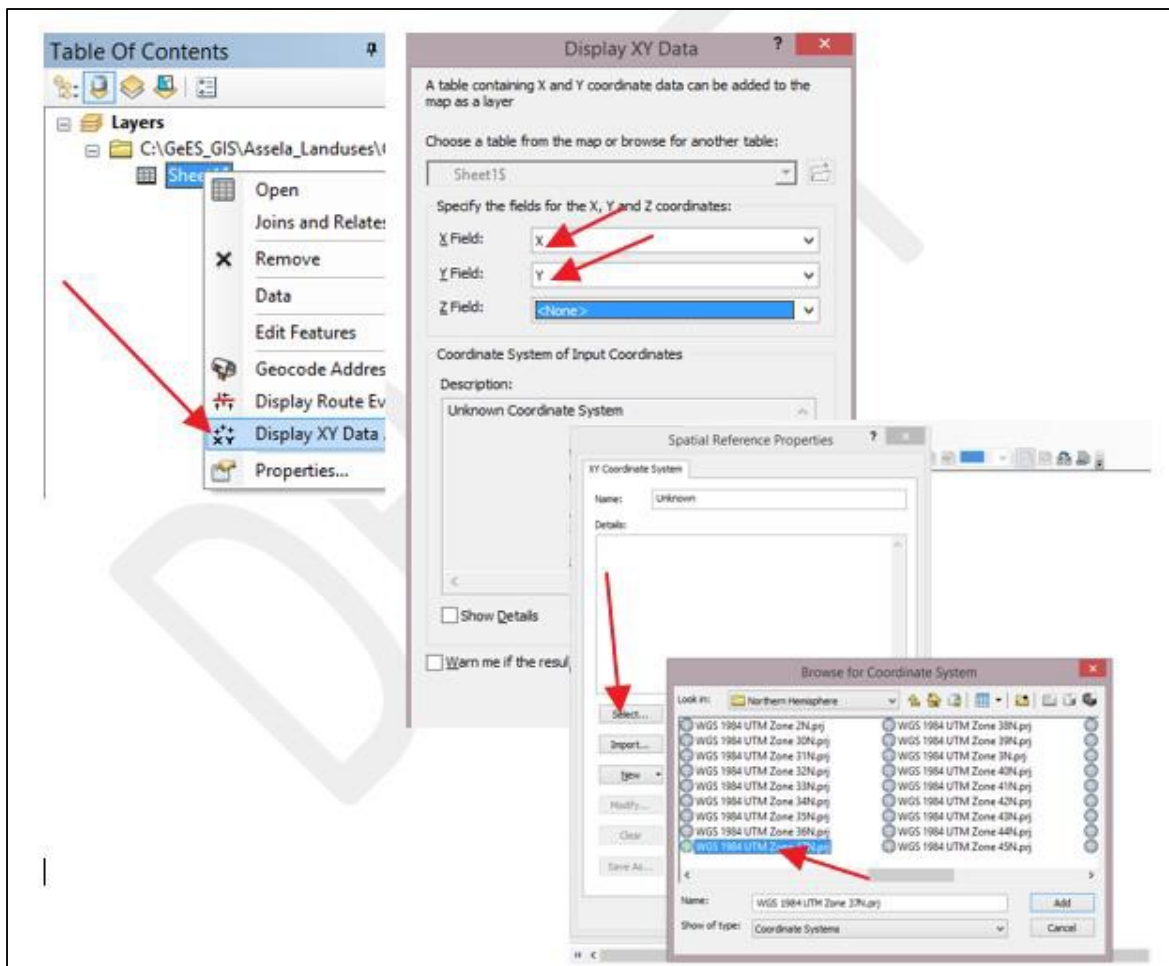
i. Click OK to close the Spatial Reference window.

j. Click Add to close the Browse for Coordinate System window.

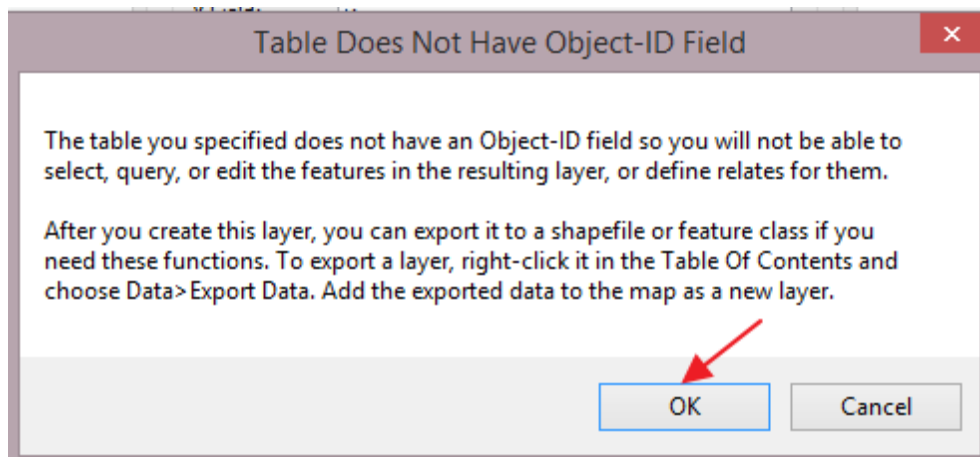
k. Click OK to close the Spatial Reference window.

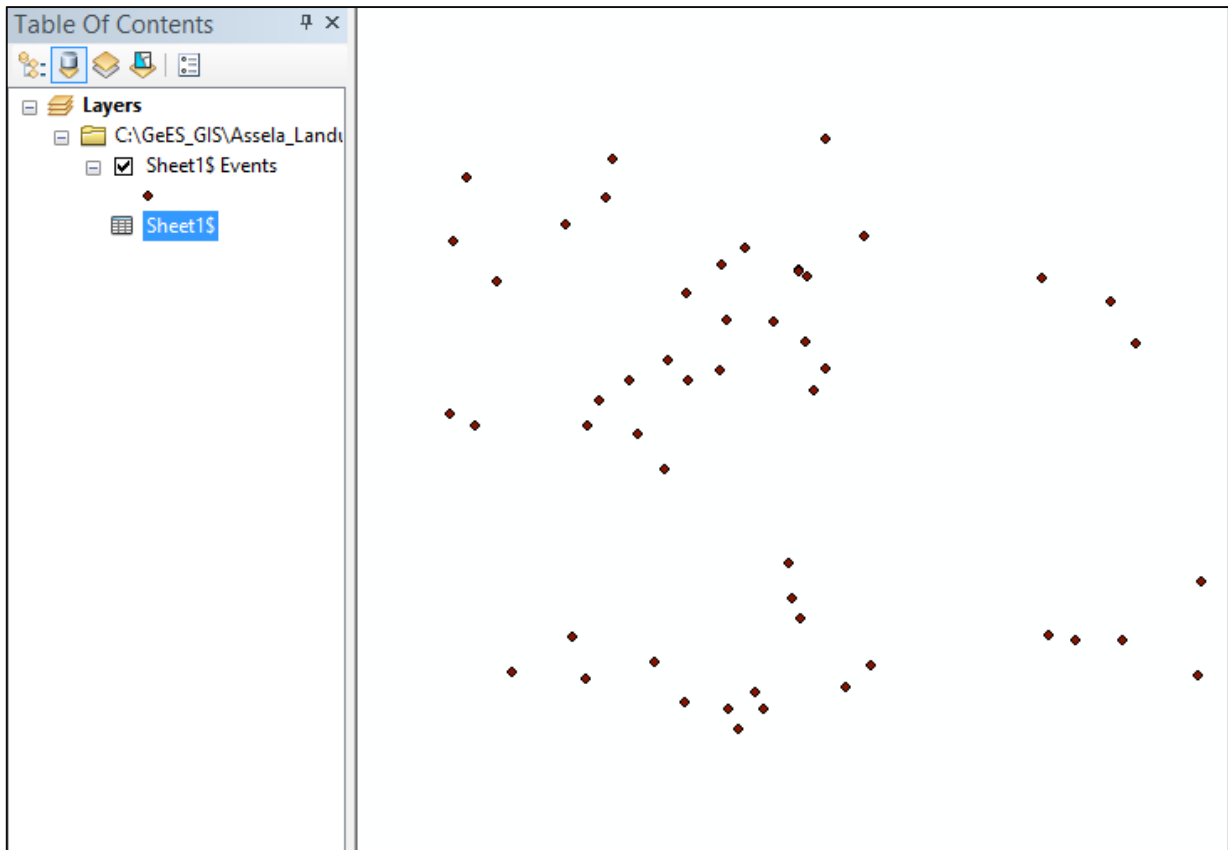
l. Click OK to close the XY Data window and run the pro





A message will show that the table does not have Object-ID field → Click OK



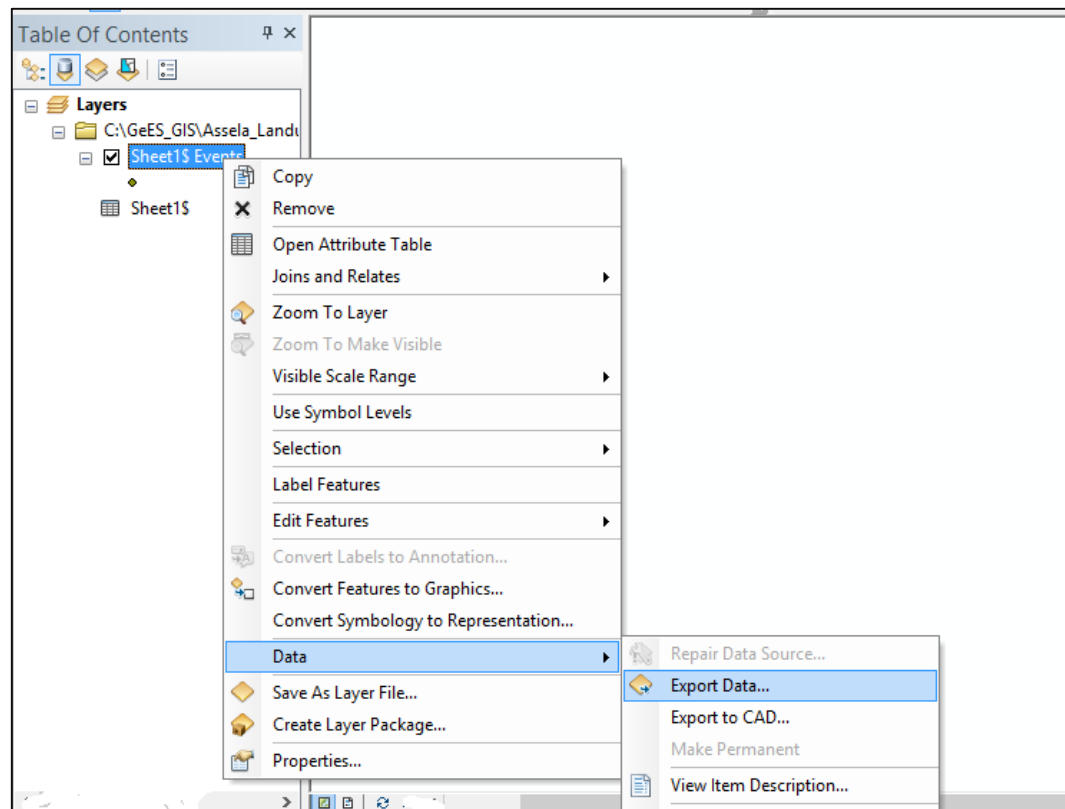


8.2. Exporting XY data feature class to shape file

After processing the map should now show the locations from the table as points. So far the new layer exists just as an event layer in the map document. In the next steps you will make this layer permanent by converting it to a feature class in your geo-database.

- a. Click the Sheet_event layer in the TOC.
- b. From the drop down list click DATA → Export Data.
- c. For the output feature class navigate to the C: /GIS_GeES/Assela-Landuse folder.

- d. Name the feature class AsselaLU_Shf.



- e. Click Save.
- f. Click OK. When asked if you want to add the exported data as a map layer click yes.
- g. The TOC in the list by source view should now show the Asselaxy_shapefiles.

8.3. Adding a field to a table

- Open the table (right-click on the layer rainfall ☐ **Open Attribute Table**)
- Click on Options from the lower part of the spreadsheet window
- Type = "Text" → **Ok**
- Click Add Field.
- Write the name of the new field as Landuse_type. The New field will be added as the last column.

Table

Asselaxy

	FID	Shape *	X	Y	Laduse_Ty	Altitude
	0	Point	513855	876	0	0
	1	Point	513925	876155	0	0
	2	Point	513715	876020	0	0
	3	Point	513985	876020	0	0
	4	Point	513795	875860	0	0
	5	Point	51381	875940	0	0
	6	Point	513125	876390	0	0
	7	Point	513360	876070	0	0
	8	Point	512590	876260	0	0

0 (0 out of 71 Selected)

Asselaxy

CHAPTER NINE

WORKING WITH RASTER DATA

Raster Data are Cell-based representation of map features. Each cell has a value. A group of cells with the same value represent a feature. Examples - satellite imagery, aerial photography and some come from software packages like ArcGIS and ERDAS IMAGINE.

Notice: The Spatial Analysis Extension and 3D Analyst must be turned on when working with raster data. Add the 3D and Spatial Analyst Extension from the Tools – Extensions menu. Check Spatial Analyst.

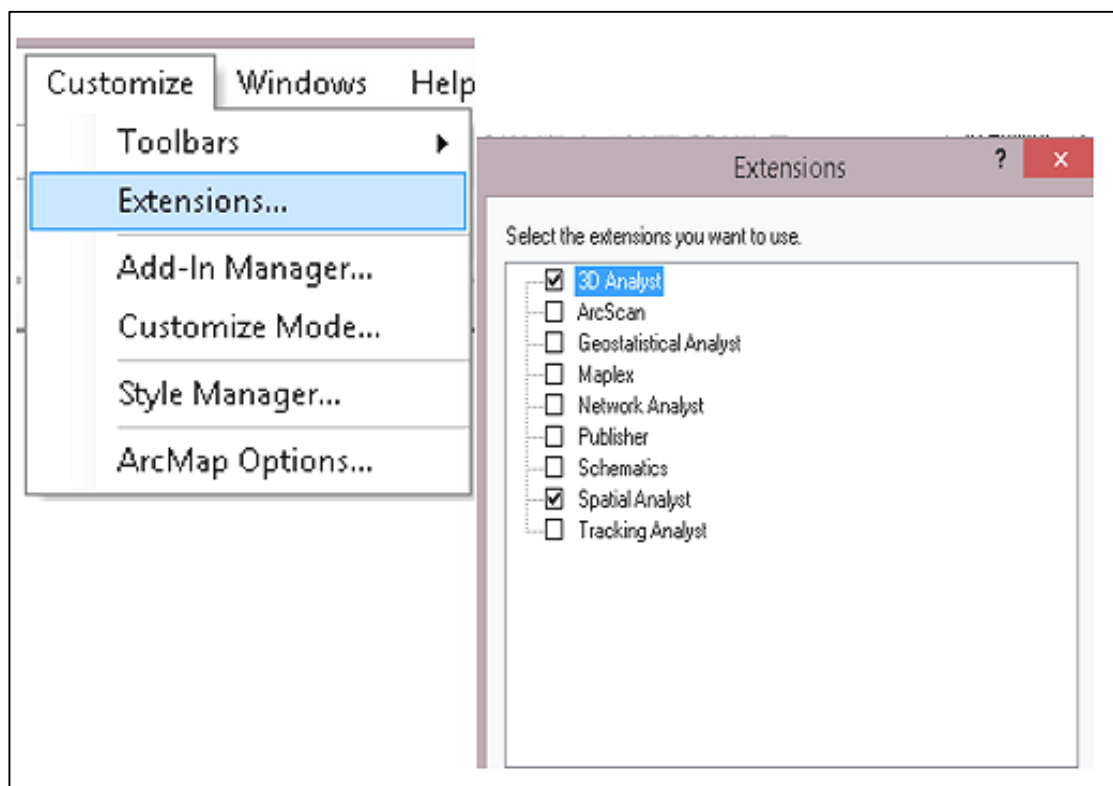
Dear trainee, under this chapter we will execute the following activities.

- ***Extract a raster data***
- ***3D Surface Analysis***
- ***Grid attributes and Histograms***
- ***Reclassify a grid***
- ***Calculate area using Raster Calculator***
- ***View various types of raster data sets***
- ***Map display options for raster data***
- ***Interpolation***

The digital Elevation Model or Digital terrain Model (DEM/DTM) is one of Raster data with availability of elevation continuously at each location in the study area. A DEM is a digital representation of ground surface topography or terrain represented as raster (a grid squares) used in GIS and basis for digitally produced relief maps. This exercise will demonstrate how to work with a digital elevation model for Agarfa Woreda. Wirth 30x 30meter resolution Freely Downloaded from ASTER Global DEM website <http://www.gdem.aster.ersdac.or.jp/search.jsp>. We will then derive slope, aspect, hillshade, contours and TIN from the Elevation model (DEM) and perform queries and analysis on raster data.

9.1. Creating Slope, Aspect, Hillshade and Contour from DEM

- Create a new map file in Arc Map
- Activate ArcGIS: Start → Program → ArcGIS → ArcMap
- Choose “**A new empty Map**” **when** the startup dialogue prompt Up.
- Click the “**Add Data**”  icon from the toolbar.
- Go to the path C:\GIS_GeES/ Raster folder and Add “Agarfa.shp, Ethio_DEM”.
- Make the “3D Analyst” Toolbar viewable in ArcMap (**Customize → Extensions → 3D Analyst→Make Active**).



A toolbar of **3D Analyst** will appear as shown below. Please check “Customize → Toolbars **3D Analyst** if the toolbar does not shown up.



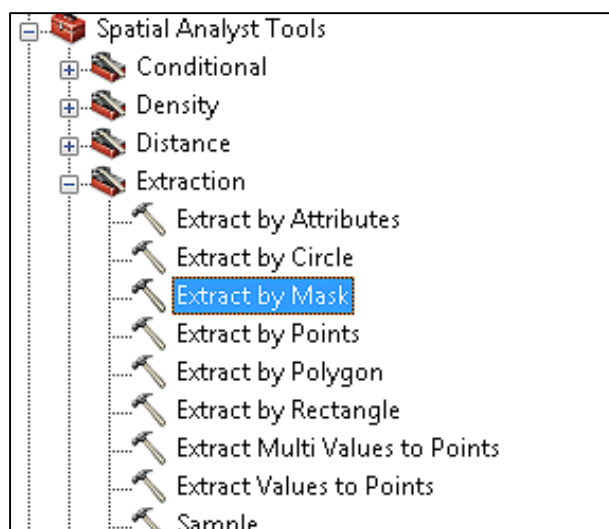
9.1.1. Extract by Mask

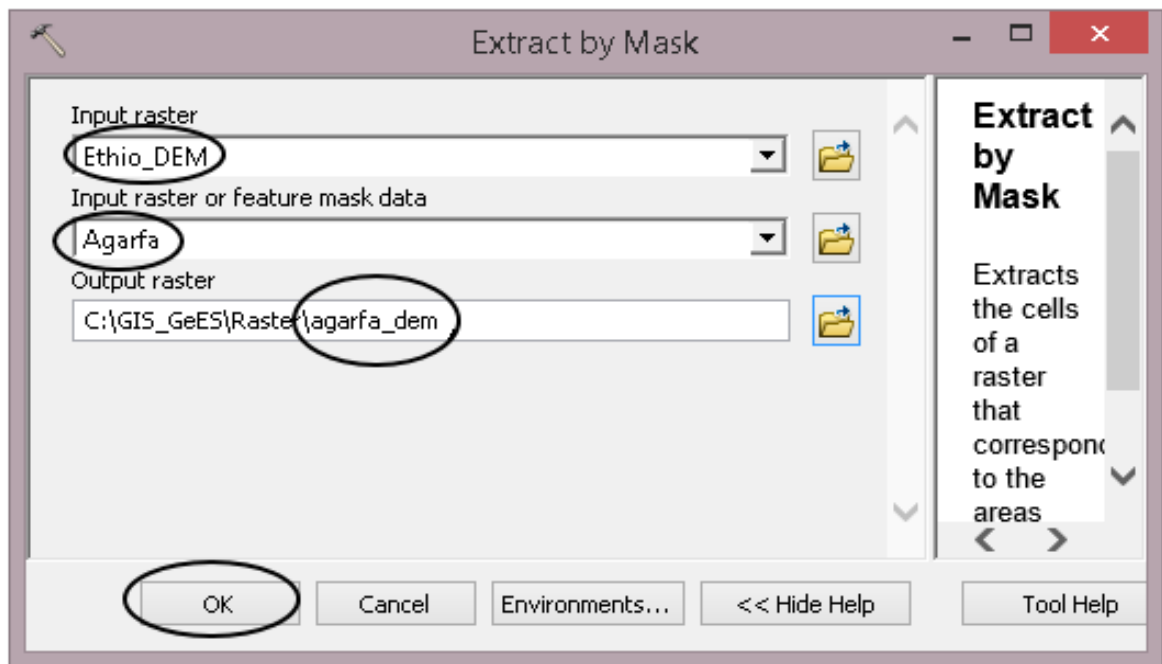
Dear trainee, before you start 3D analyst, you are advised to extract a portion of Ethio_DEM covering your area of interest (Agarfa Woreda).



Steps.

- a. Click on Geoprocessing ArcToolbox
- b. Click Spatial Analyst Tool → Extraction → Extract by Mask





- c. Click “**Ok**” to extract
- d. Agarfa_DEM is automatically is extracted and listed in the table of content
To see the Added DEM, un check all except “**Agarfa_DEM**”

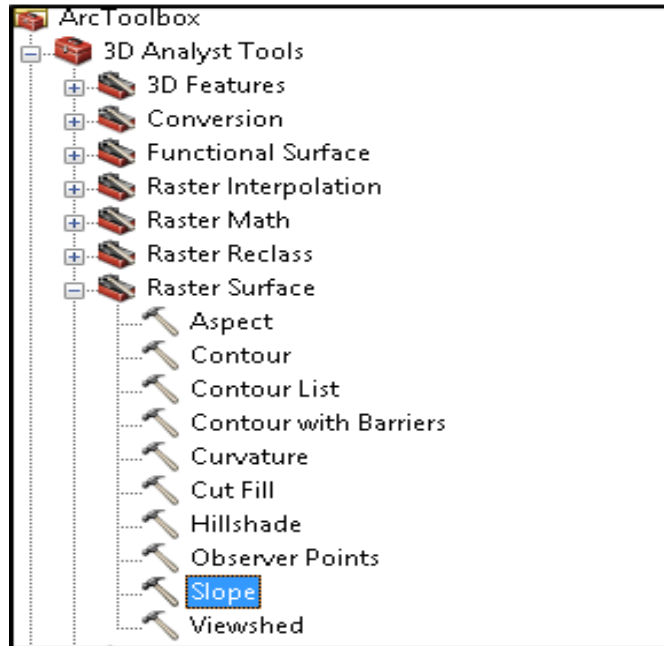
Question: *What is the difference between Extract and Clip function you have practiced?*

9.1.2. Generating Slope

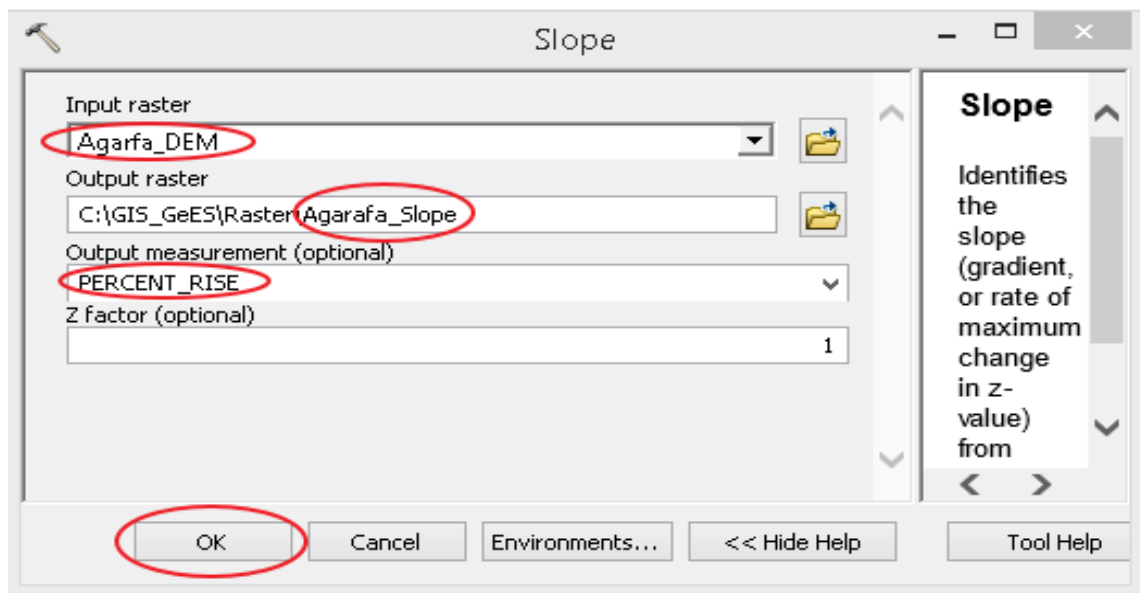
Slope is the rate of maximum change in z-value from each cell. The slope of the landscape is a key variable in a wide variety of landscape analysis. It affects where structures or trails can be built, crops can be planted or harvested, the speed of flowing water and consequent erosion, landslide potential, and the list just goes on and on. Slope data is calculated on a cell by cell basis. It is done by averaging the change in elevation measurements between one cell and those surrounding it. Slope may be presented as a degree of change or a percent change (Theobald pp.273-274).

Steps

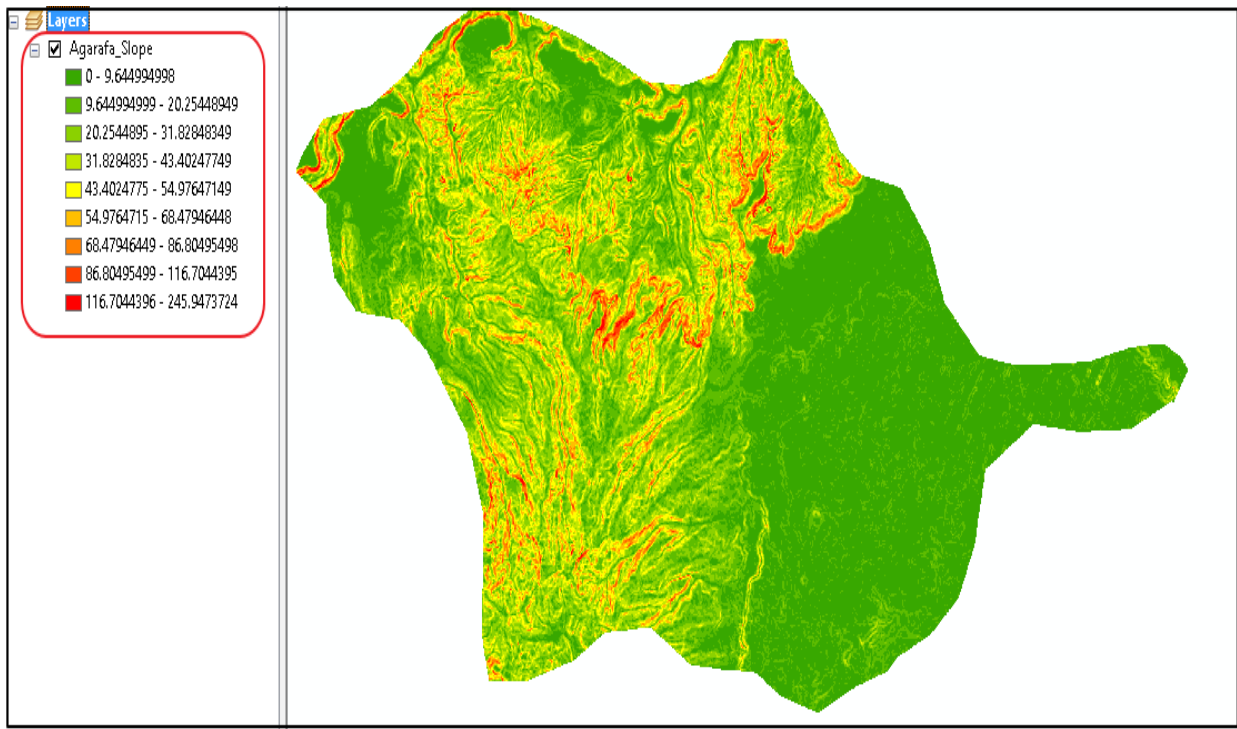
- a. **Got ArcToolbox → 3D Analyst → Raster Surface→ Slope**



- b. Make input raster Agarfa_DEM
- c. Save the Output Raster to your working directory C:\GIS\GeES\Raster and name it Agarfa_Slope. If you do not, you have merely generated a temporary raster that will be stored in Default Geodatabase.
- d. Make your output measurement **Percent_Rise**.



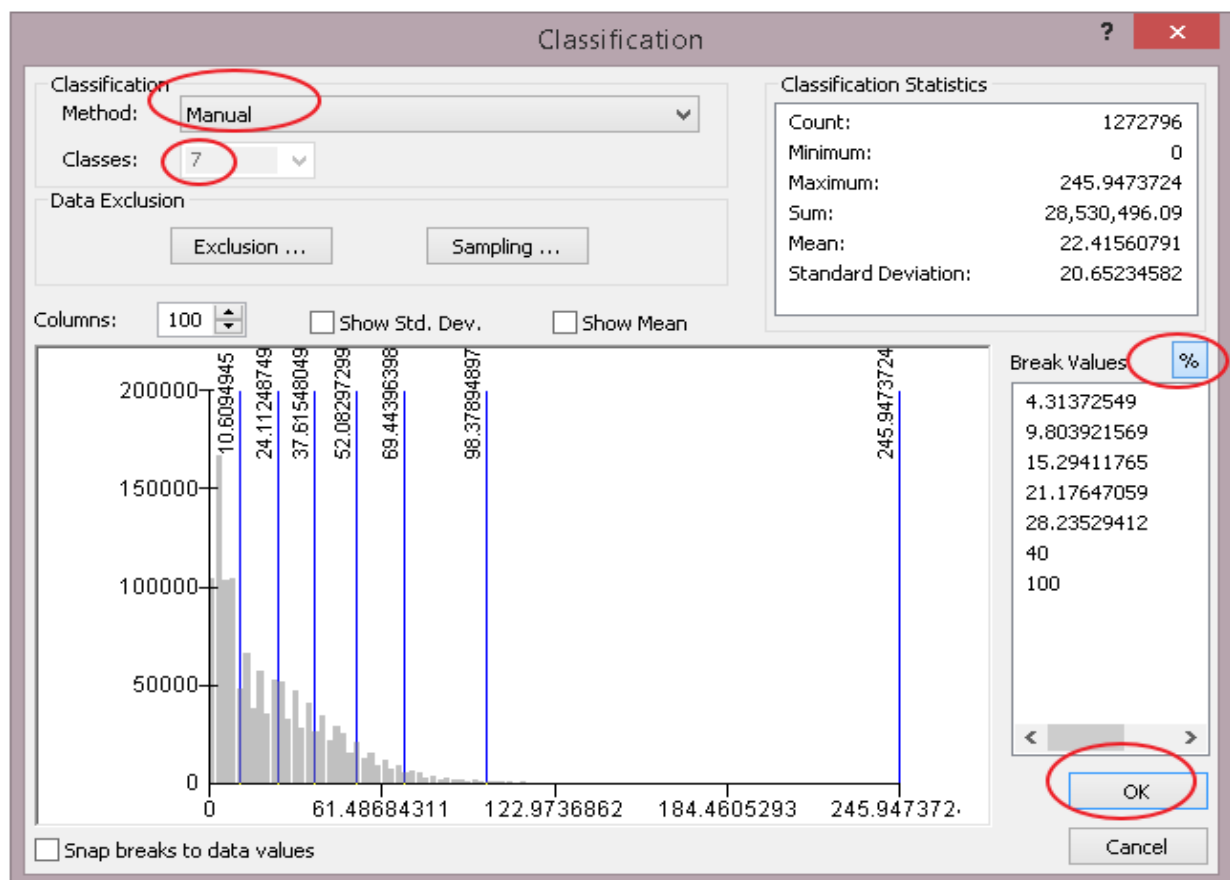
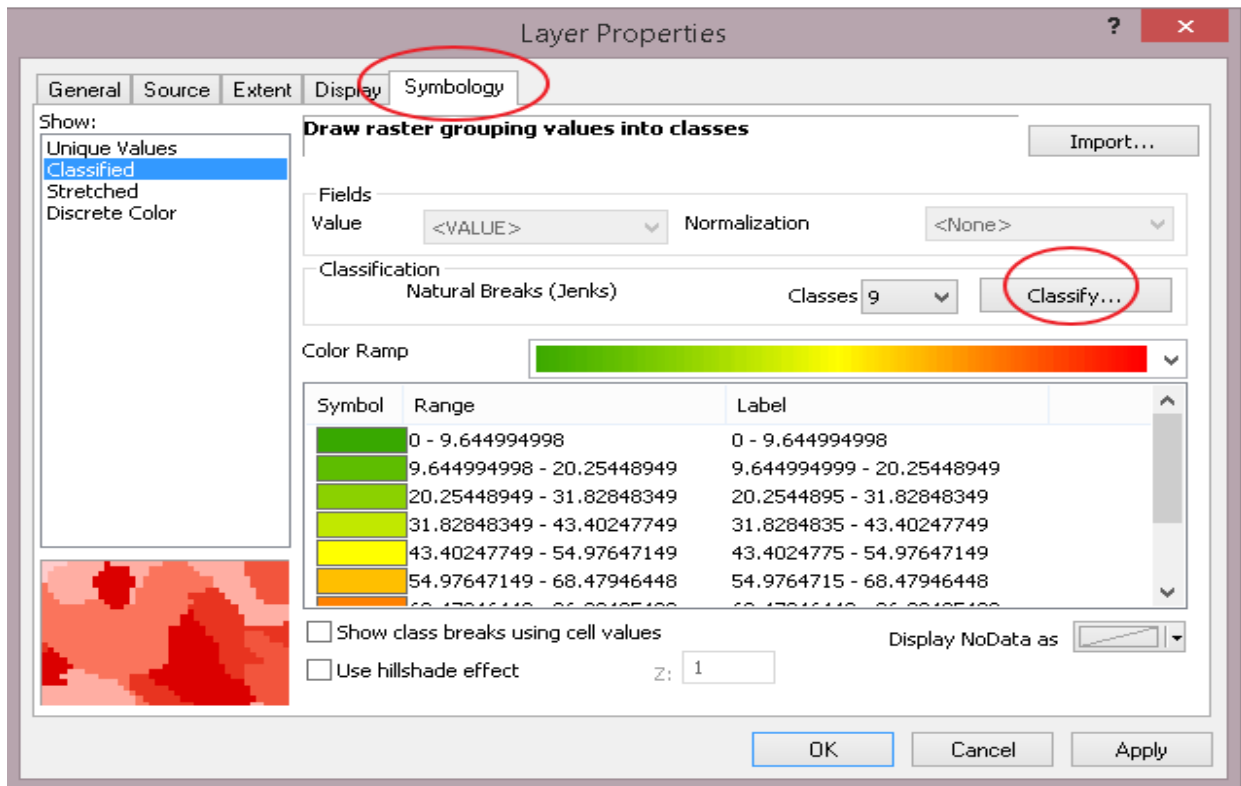
e. Click OK. You should now see your **slope GRID** in your TOC.



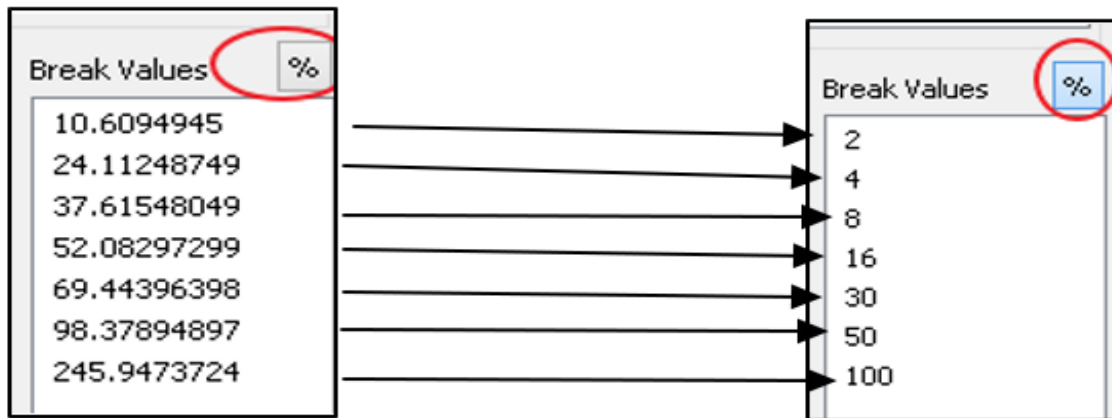
Now let us classify the slope you derived into **seven Classes**

Slope in %	Inclination degree	Description	Code
0-2	0 - 1.15 ⁰	Level	LE
2-4	1.15 - 2.29 ⁰	Very gently Sloping	VG
4-8	2.29- 4.57 ⁰	Gently Sloping	GS
8-16	4.57- 9 ⁰	Sloping	SL
16-30	9 - 16.7 ⁰	Strongly Sloping	SS
30-50	16.7- 26.56 ⁰	Moderate Steep	MS
>50	>26.56 ⁰	Very Steep	VS

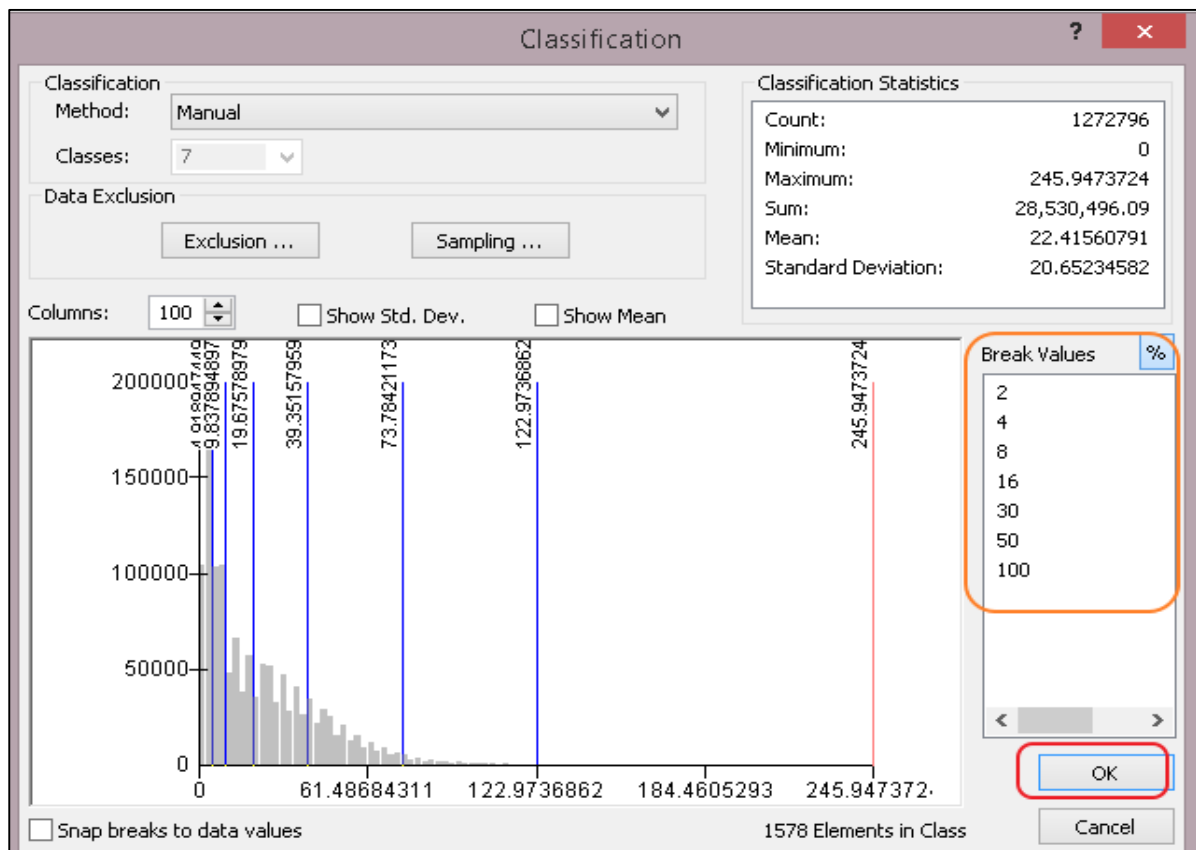
f. To do so, double click on **Agarfa_Slope** layer from Table of Content → Click **Symbology** tab → Click **Classify**



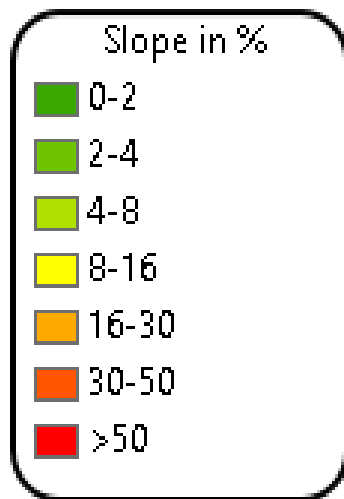
- g. Under the classification window, you have to select Manual Method with 7 Class
Accordingly, you have to type the upper limits over the default break values percent classified



In the layer property window, change the label class ranging between 0-2, 2-4, 4-16, 16-30, 30-50, >50.



Finally, when you click OK, the slope class are displayed in the Table of Content (TOC)

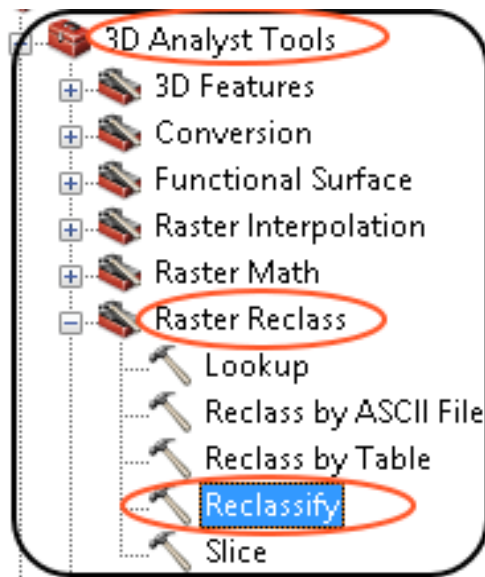


Question: Which Slope Class Covers Large percentage area and which slope Class cover small percentage?

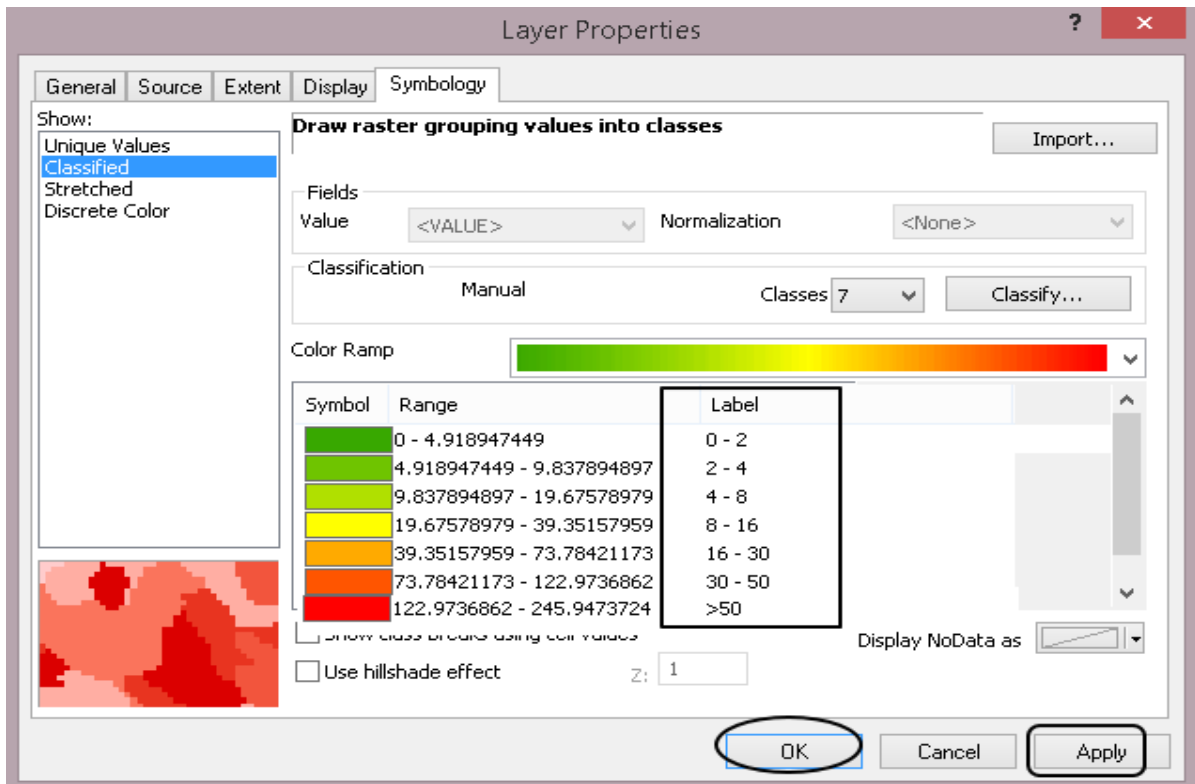
In order to answer the above question we need to raster data and give cells a different value obviously into seven class of **L, VGS, GS, S, SS, MS and VS**.

Steps

- Go to **ArcToolbox** → Click on **3D Analyst** → **Raster Reclass** → **Reclassify**

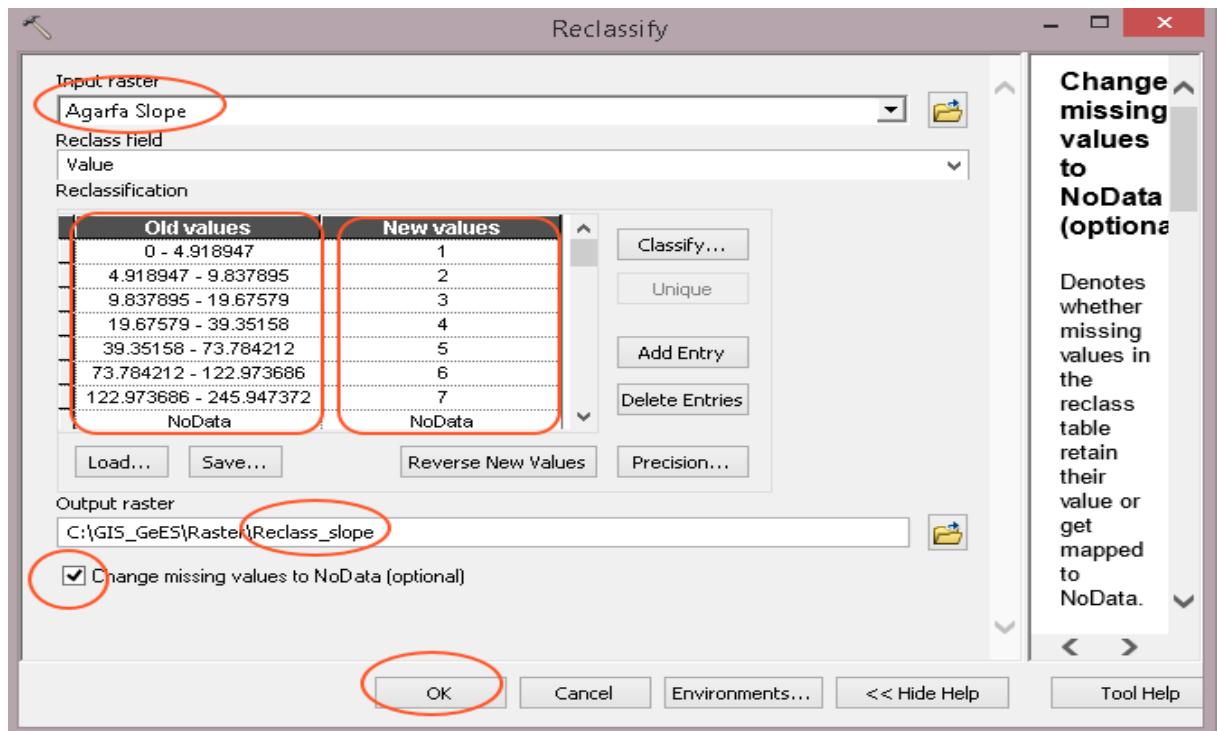


- b. In the reclassify window, the input raster is Agarfa_Slope

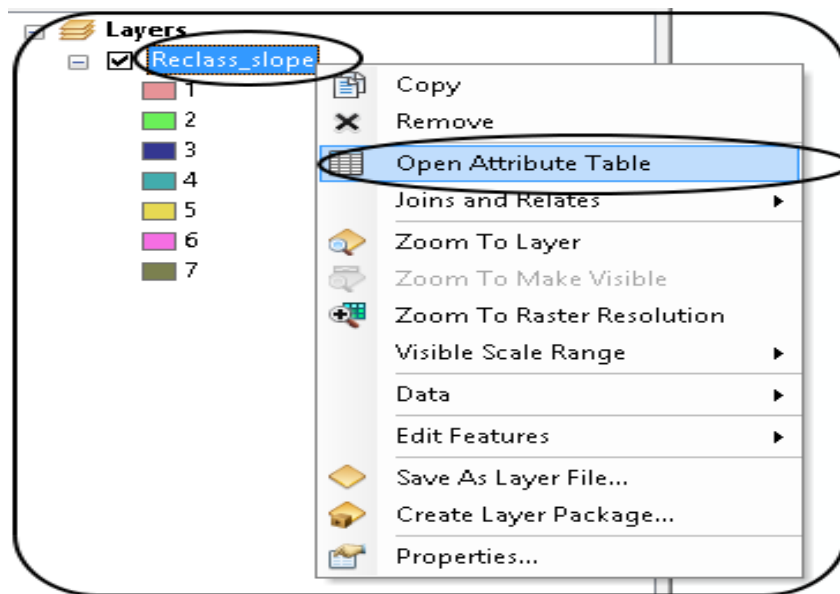


- c. Check the existence of “**Old Value**” and “**New value**” from 1-7.
- d. Check the box change missing values to **No Data**.
- e. Make sure to save the out raster to your working directory (**C:\GIS_GeES\Raster**) and name it something meaningful(**Reclass_Slope**)
- f. Click **OK** when finished

- g. In the TOC, rename your numerical values to something the map reader can understand

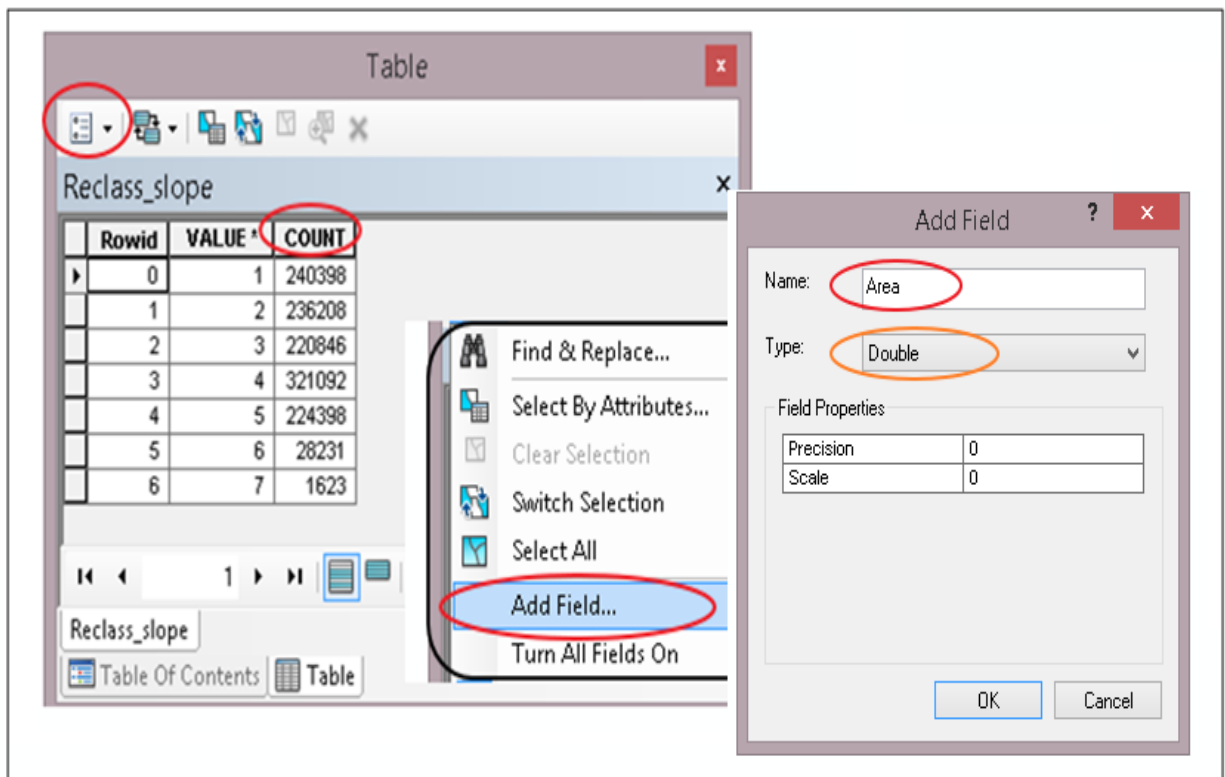


- h. To do so right click on the **Reclass_Slope** from TOC → **Open attribute Table**

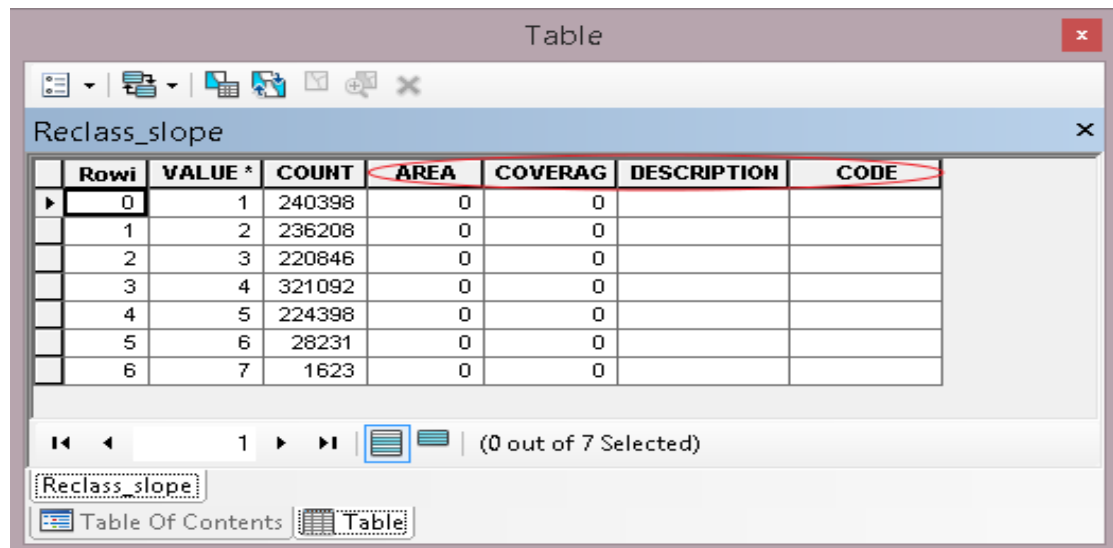


In the attribute of Reclass slope window, you can see the number of pixel values counted in each of the seven class. Therefore, you can add field names called “Area”, “Coverage”, “Description” and “Code”

- i. Click Table options → add field
- j. In add field Window, type “Area” and select “Double” in the Type and Click **OK**.

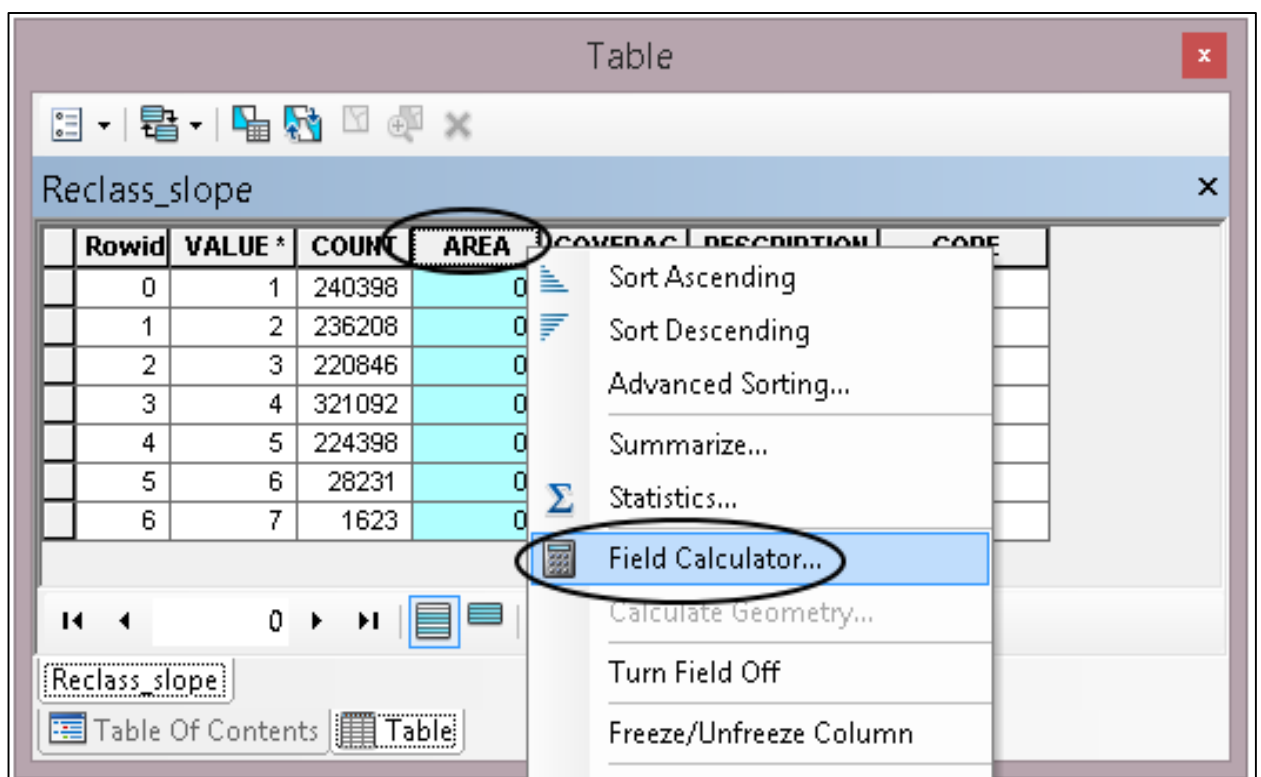


- k. Do step “i” and type “Coverage” in the Name and select **Float** and Click **OK**.
- l. Do step “j” and type “Description” in the name and Select “Text” in the text type and length should be changed from **50** to **20** then click **OK**.
- m. Finally, you will have the attribute table with the following added field names that you are supposed to calculate and field.



Rowid	VALUE *	COUNT	AREA	COVERAG	DESCRIPTION	CODE
0	1	240398	0	0		
1	2	236208	0	0		
2	3	220846	0	0		
3	4	321092	0	0		
4	5	224398	0	0		
5	6	28231	0	0		
6	7	1623	0	0		

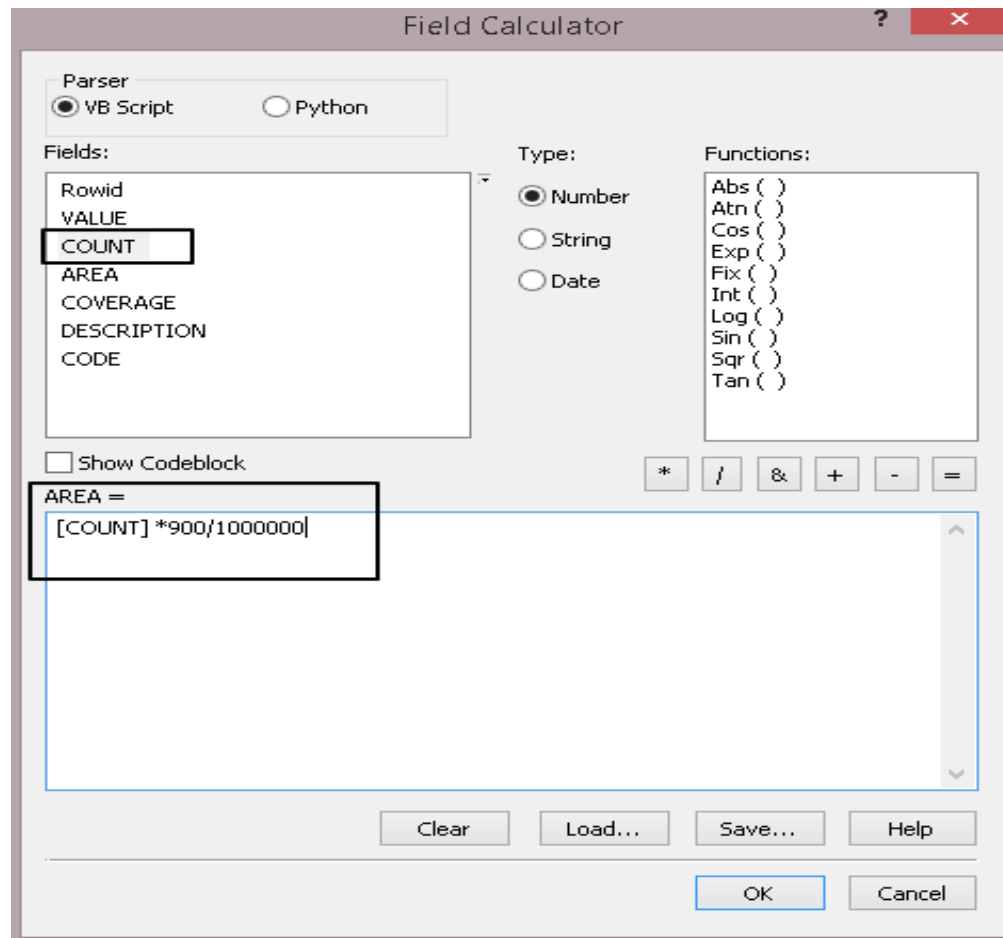
n. Right click on the **Area** field name → Click field Calculator



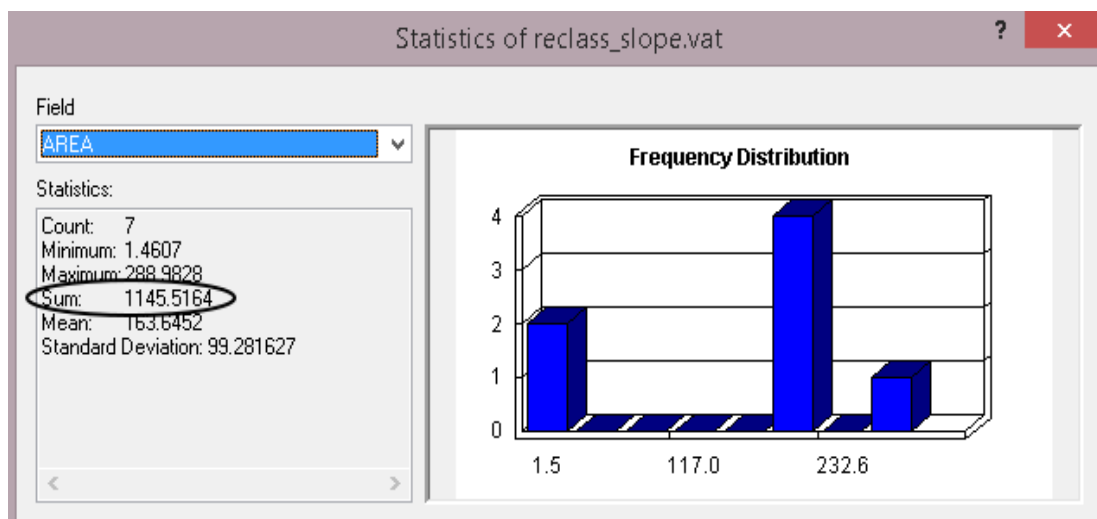
o. In the field **calculator window** appeared, write the formula to compute the **area** by double clicking **Count** and then type ***900/1000000**

Area= [Count]*900/1000000

This is because the size of one pixel in the raster source data is 30m by 30m and if we multiply it, it will become 900 m². Therefore, the number of pixel counted in each class category multiply by 900m² and then divide it by 1000000 to state area in KM²



- m. Click ok and your area will be calculated automatically.
- n. To calculate the area coverage of each class **Right Click on area** field name → Select **Statistics** and copy and save the summation of all class area i.e. **1145.5164**



Field Calculator

Parser: ☒ VB Script ☐ Python

Fields:

- Rowid
- VALUE
- COUNT
- AREA
- COVERAGE
- DESCRIPTION
- CODE

Type: ☒ Number ☐ String ☐ Date

Functions:

- Abs ()
- Atn ()
- Cos ()
- Exp ()
- Fix ()
- Int ()
- Log ()
- Sin ()
- Sqr ()
- Tan ()

☐ Show Codeblock

COVERAGE =

[AREA] / 1145.5164 * 100

Clear Load... Save... Help

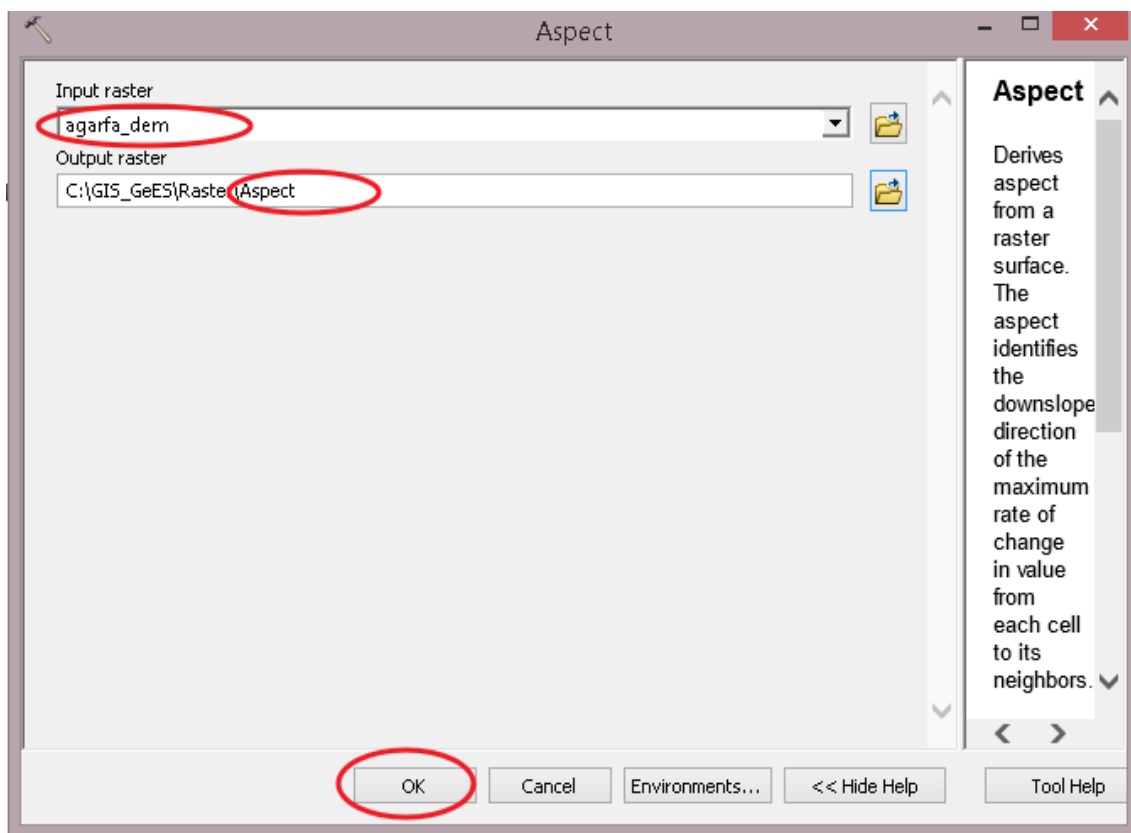
OK Cancel

Rowid	VALUE	COUNT	AREA	COVERAGE	DESCRIPTION	CODE
0	1	240398	216.3582	18.887394	Level	LE
1	2	236208	212.5872	18.558197	Very Gently Sloping	VG
2	3	220846	198.7614	17.35125	Gently Sloping	GS
3	4	321092	288.9828	25.227295	Sloping	SL
4	5	224398	201.9582	17.63032	Strongly Sloping	SS
5	6	28231	25.4079	2.21803	Moderate Steep	MS
6	7	1623	1.4607	0.127515	Very Steep	VS

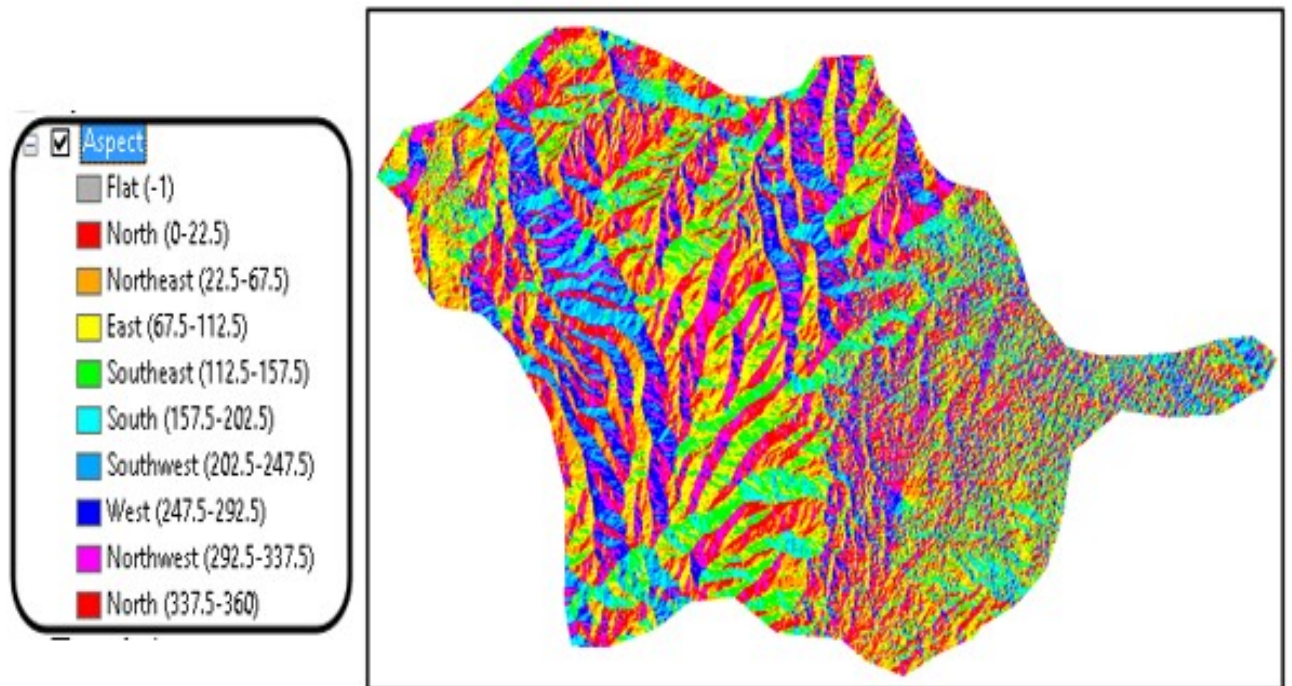
9.1.3. Generating Aspect

Aspect is an important contributor to vegetation and habitat type, as north-facing slopes often have very different conditions and temperatures than south-facing slopes, and is also important in determining whether significant features are visible from a particular point on the landscape. Aspect is calculated for each cell as the direction of maximum slope, expressed in degrees (0-360, increasing clockwise) and by -1 for a flat surface.

- Open the Aspect (**ArcToolBox box**) → **3D analyst Tool** → **Raster Surface→Aspect**).
- Specify your reprojected **Agarfa Elevation** data for the input.
- Save your output to your workspace and name it something useful (Aga_Aspect). This is because you are not allowed to give file name more than 13 characters.
- **Click OK.**



- Your aspect raster should now be visible in the TOC. The data is categorized by direction.



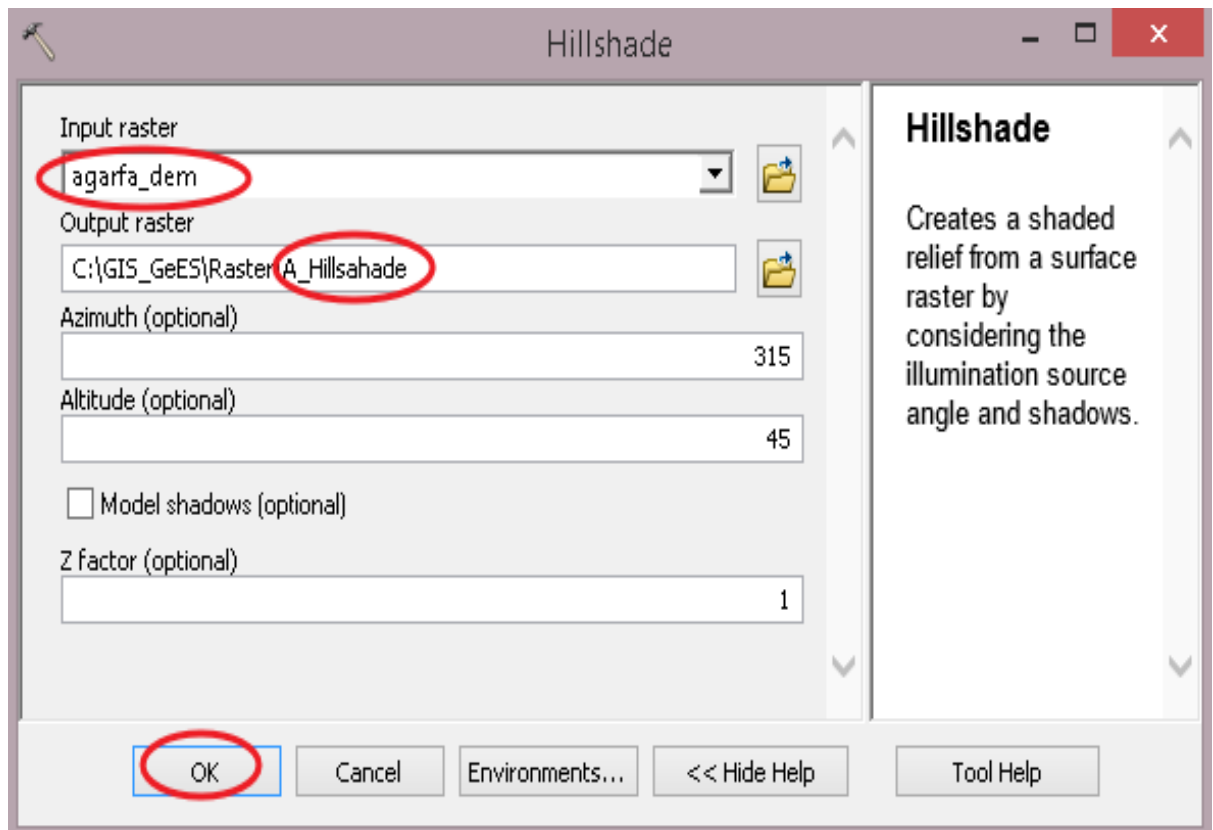
Questions:

1. How many Aspect have the terrain?
2. Which Aspect class Covers large percentage of area?
3. Which Aspect class covers large percentage of area?

9.1.4. Generating Hillshade Data

A hill shade map is used to help visualize a topographic surface. Each pixel value represents the shadow/light that would fall upon that surface area given a constant azimuth of sunlight (location of the sun given seasons) and zenith/altitude (position of the sun given time of day). Hill shade can be computed by using digital terrain model or digital elevation model (DEM) which is proficient to capture the reflectance of the sun ray at different angles. For graphical display, Hillshade can greatly enhance the relief of the surface with three-dimensional effect that provides a sense of visual relief.

- Open the Hillshade Surface Analyzer (**ArcToolbox box → 3D analyst Tool → Raster Surface → Hill Shade**).
- Specify your reprojected Elevation (Agarfa_DEM) for the input.
- Save your output to your workspace and name it something useful.
- Leave the azimuth and altitude default values. Z-Factor as 1.
- Click OK.
- Your hillshade raster should now be visible in the TOC.

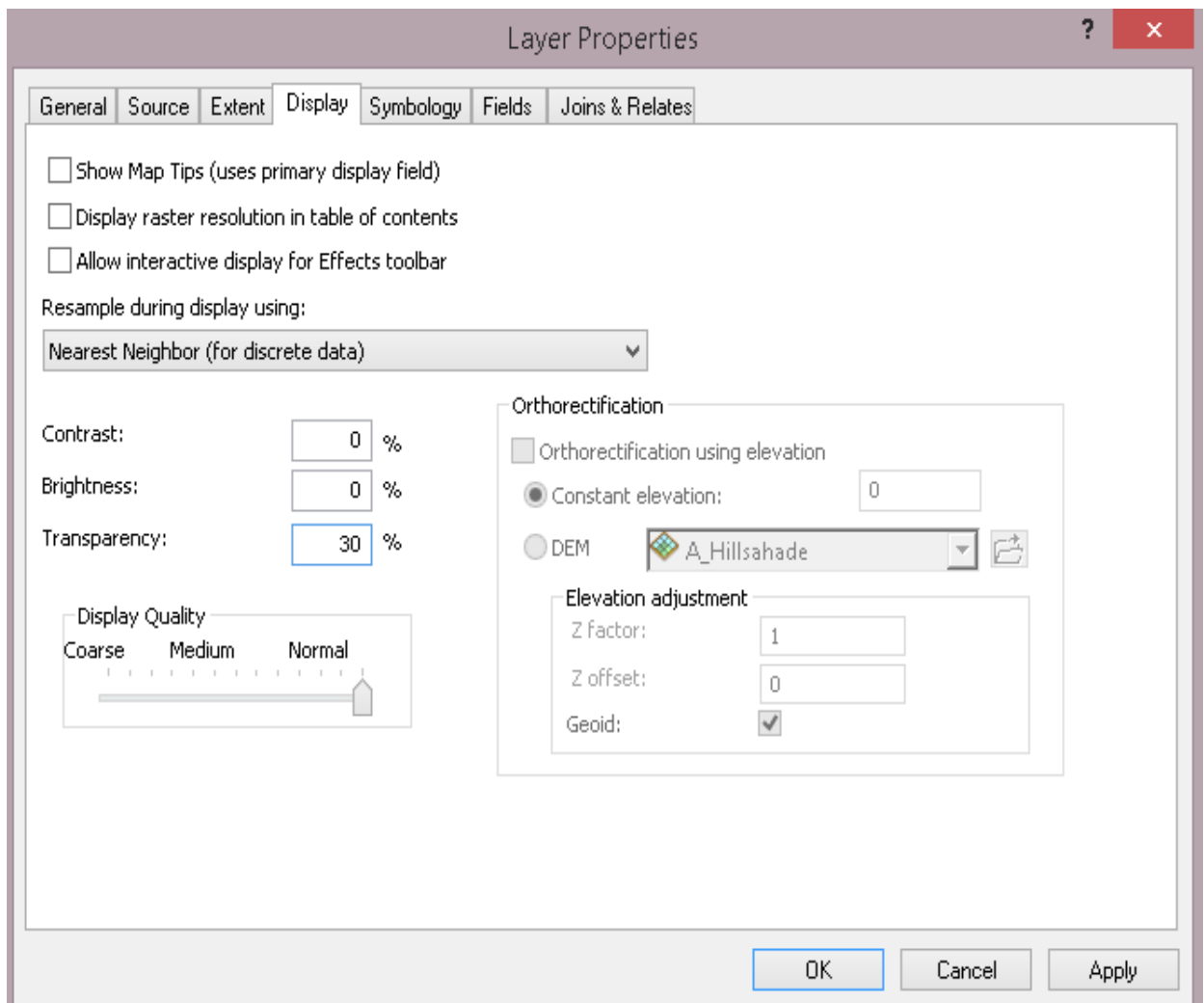


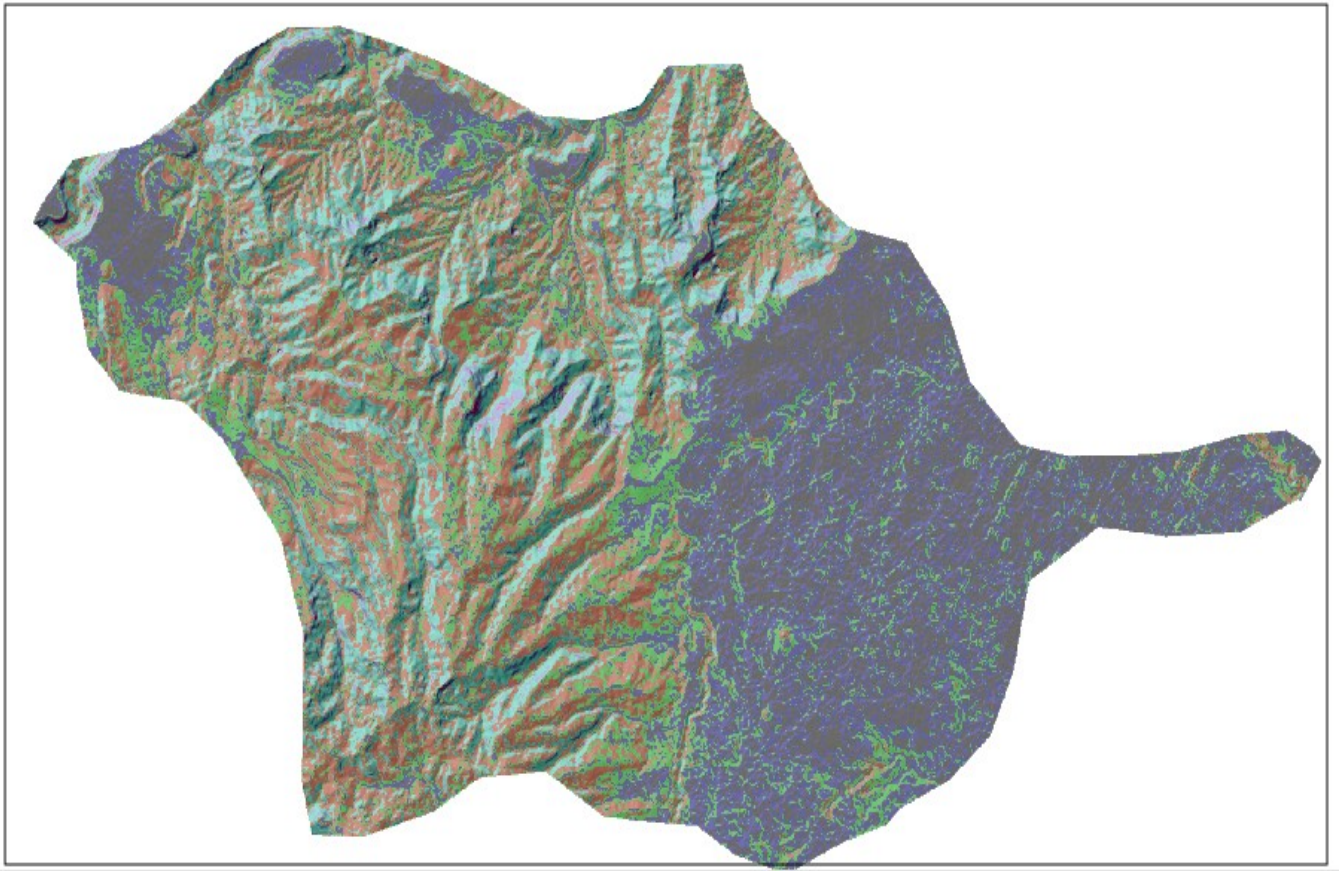
- Once we have opened the hillshade in Arc Map, place it above all other raster images and below any vector layers.

9.1.5. Visualize Reclassified slope and the Hillshade

- Make Slope transparent (Right Click on the slope Layer → properties Display) inorder to view both layers.
- Make the transparency 30% . To see the difference change the transparency to 70% adjust this value to create the best looking map). *To see what transparency level works best. The greater the percentage transparency used the more the transparent the layer).*

- h. *NO_SHADOWS* —the output raster only considers local illumination angles; the effects of shadows are not considered. The output values can range from 0 to 180, with 0 representing the darkest areas, and 180 the brightest.
- i. *SHADOWS* — the output shaded raster considers both local illumination angles and shadows. The output values range from 0 to 180, with 0 representing the shadow areas, and 180 the brightest.





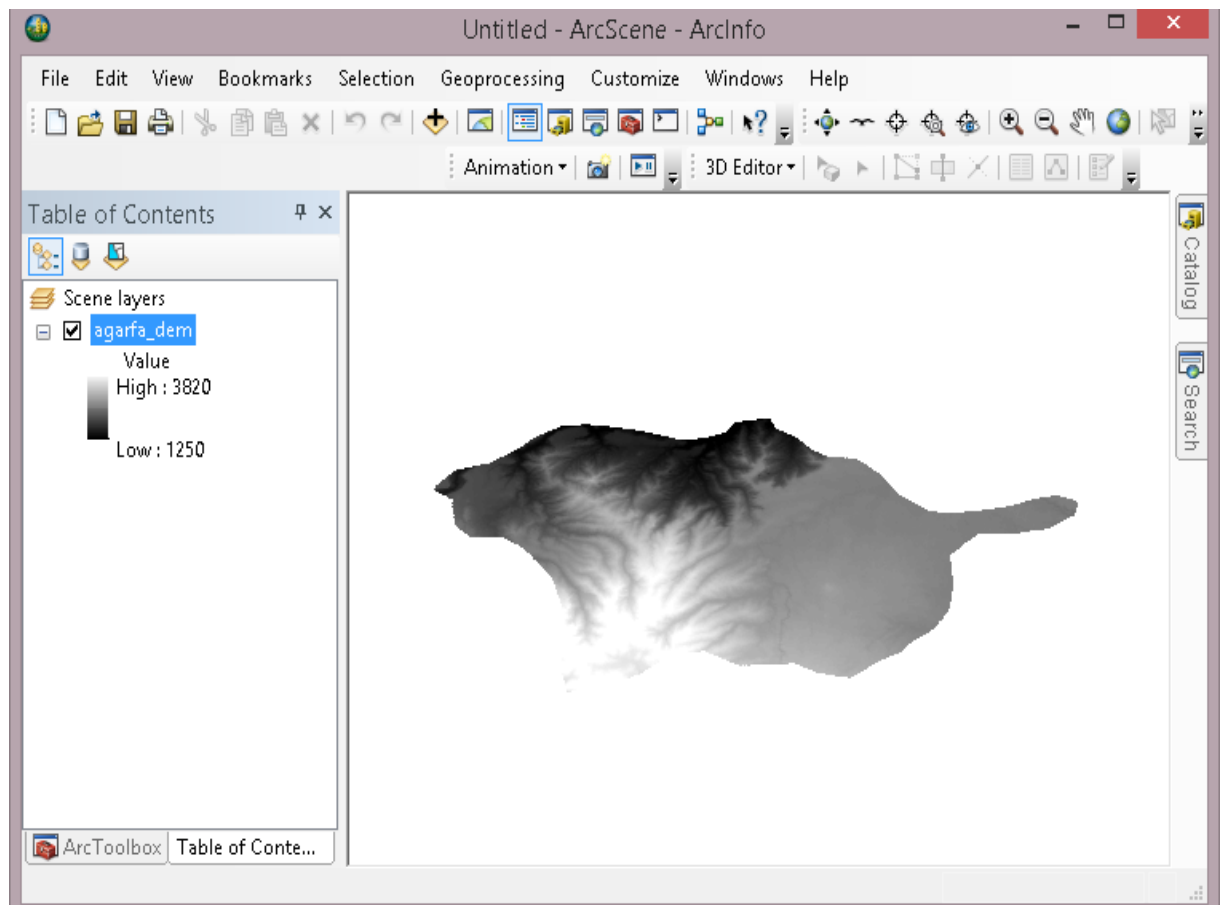
9.2. Display Landscape data in 3D

- A. Open ArcScene (From the same place where ArcMap is started). Double click the **ArcScene** icon.

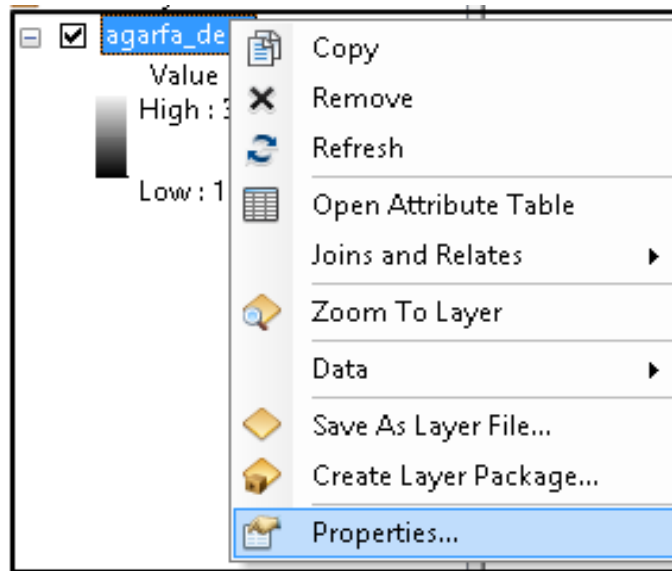


- B. In the ArcScene window appeared, add the grid layer **Agarfa_DEM**.

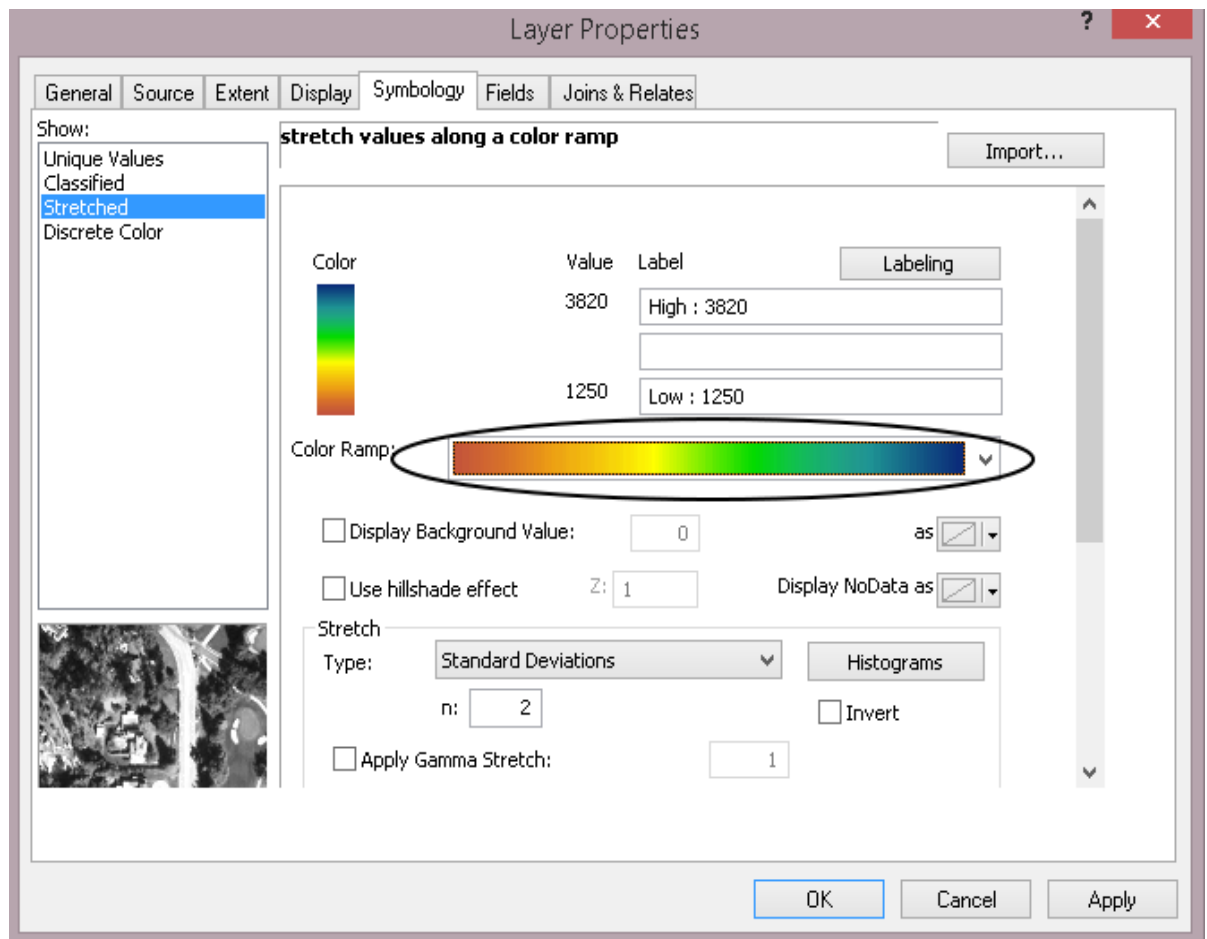
Question: What is the highest and the lowest Elivation of **Agarfa_DEM**?



- C. Right Click and open the properties of Agarfa_DEM



D. Alter the **Symbology** to use the color ramp

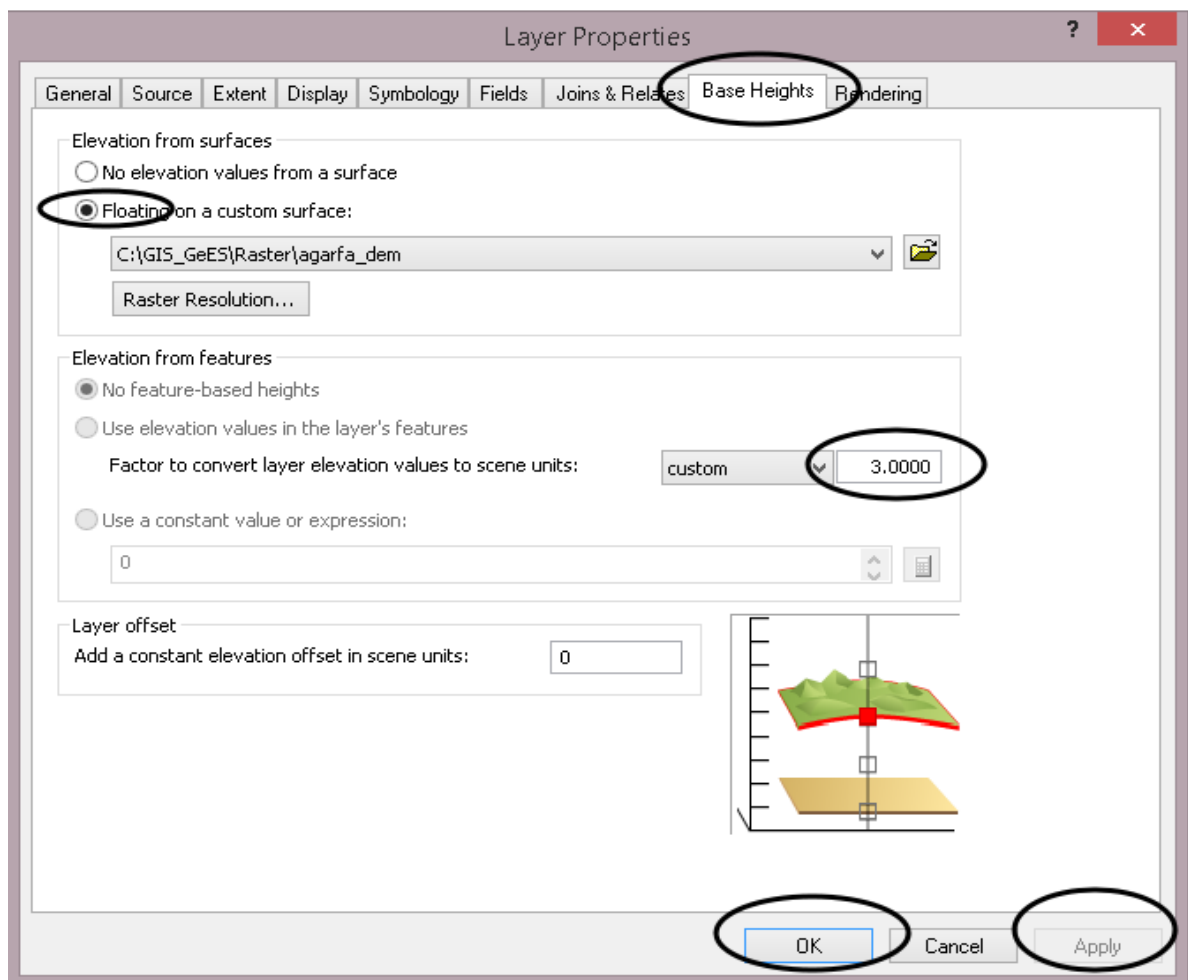


E. Click Base heights tab.

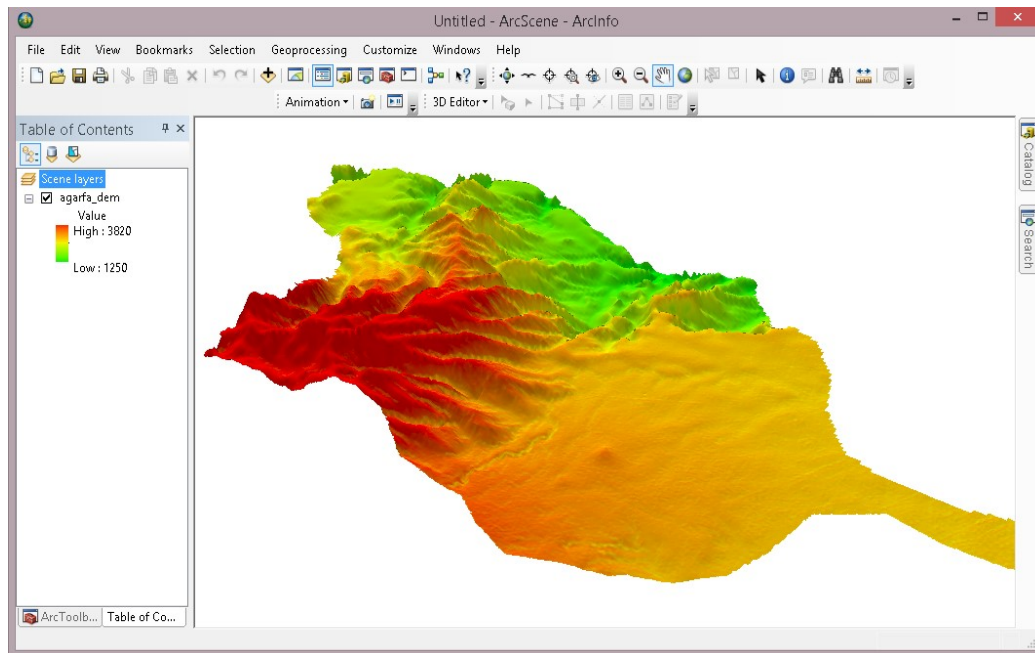
- Click **obtain heights for...** and make sure it is set to **Agarfa_DEM**
- Set a Z unit conversion of 3. This sets a **vertical exaggeration** factor of 3.
- Click **ok** and then **Apply**.

N.B. If vertically exaggerated map does not appear on the ArcScene window, there might be due to the following problem.

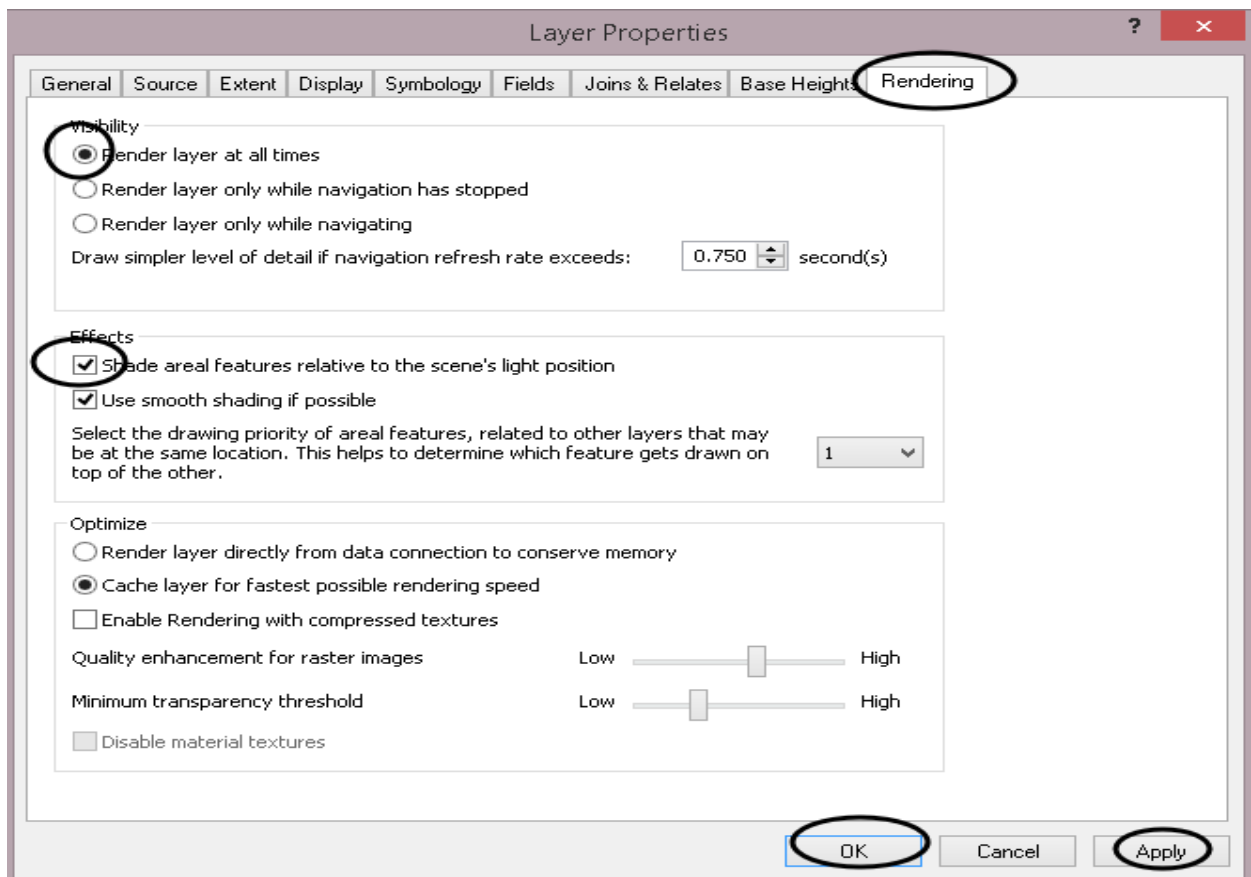
- *The coordinate system of Agarfa_DEM is Geographic coordinate System (GCS)*
- *The vertical Exaggeration value may be too large to be displayed in the ArcScene.*



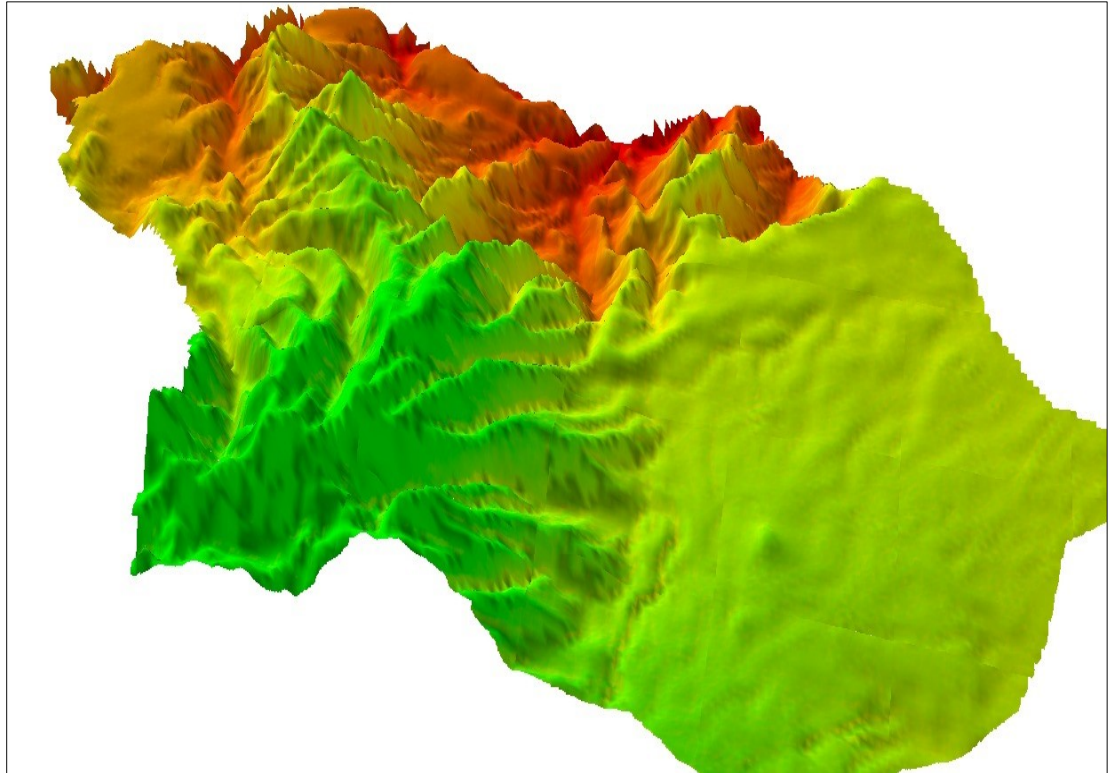
F. Your 3D landscape is displayed properly as follows.



G. Check **Rendering → Shade areal features...** to apply analytical Hillshade to the model.



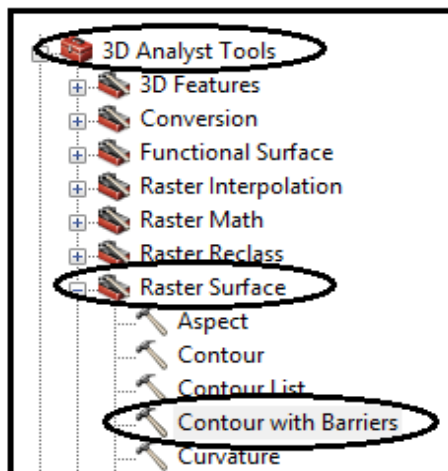
H. Now you see the landforms looks more realistic.



9.2.1. Create Contour and Triangulated Irregular Network

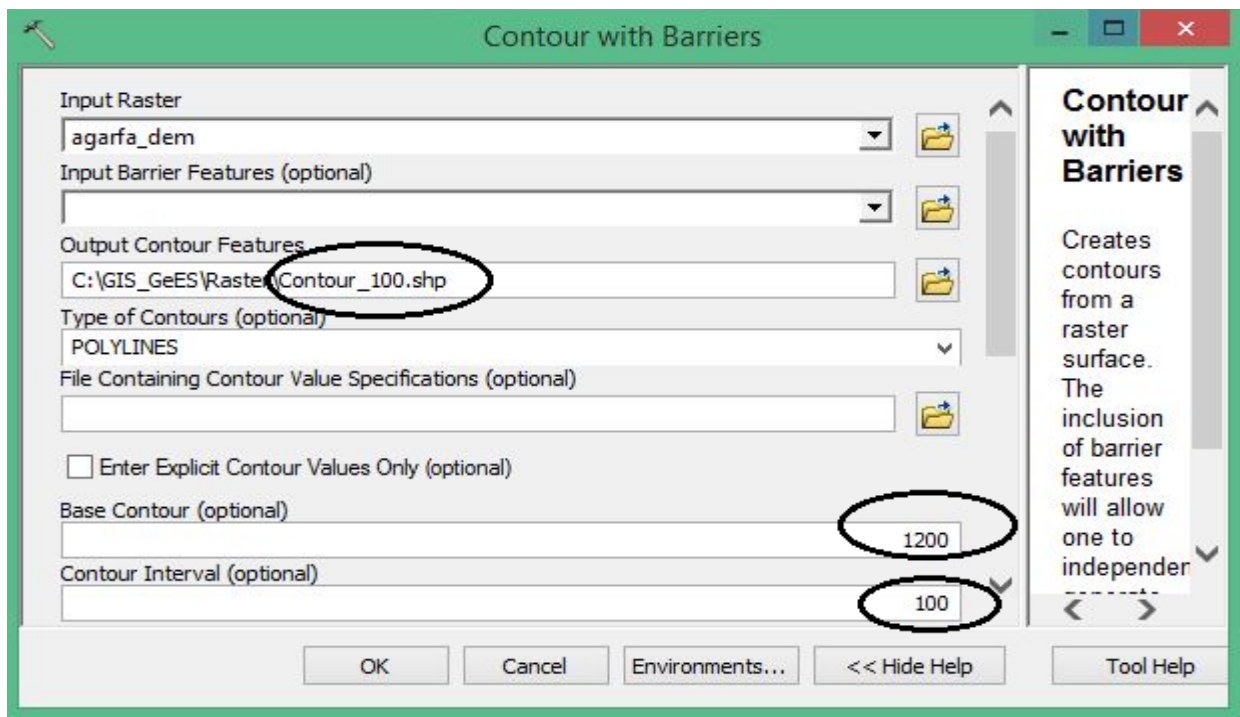
9.2.1.1. Create Contour

- a. Open the Contour with Barrier (ArcToolbox box) → 3D analyst Tool → Raster Surface → Contour with Barrier).



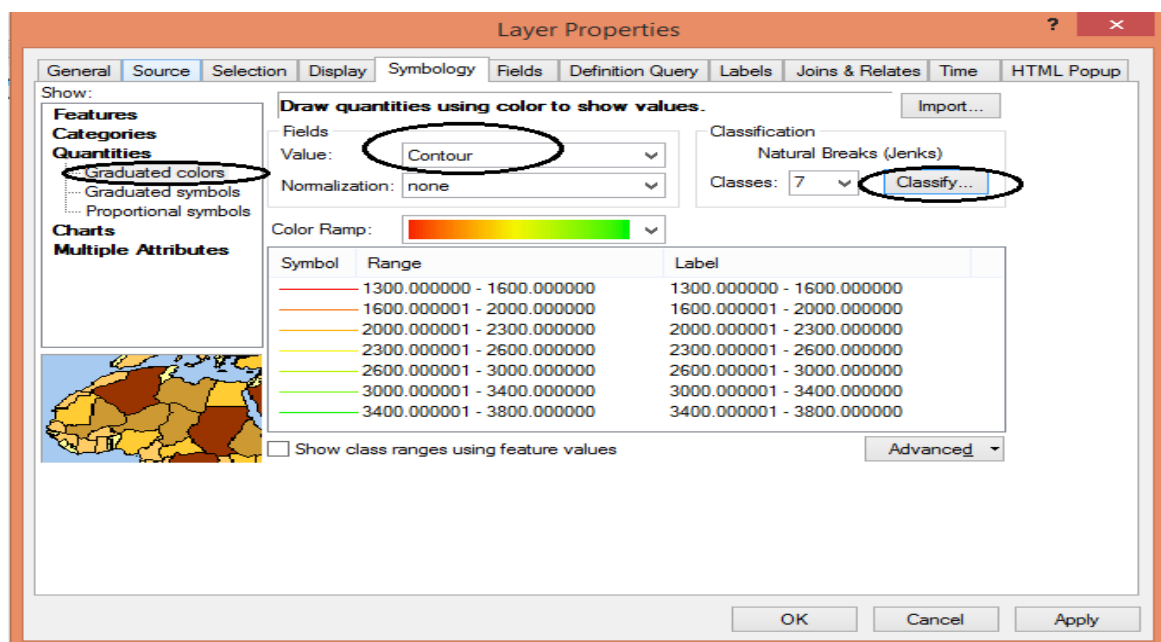
- e. Specify your reprojected **Agarfa DEM** data for the input.
- f. Specify base contour to 1200 and contour interval to 100m
Save your output to your workspace and name it something useful (Contour100.

- j. Click apply | ok.

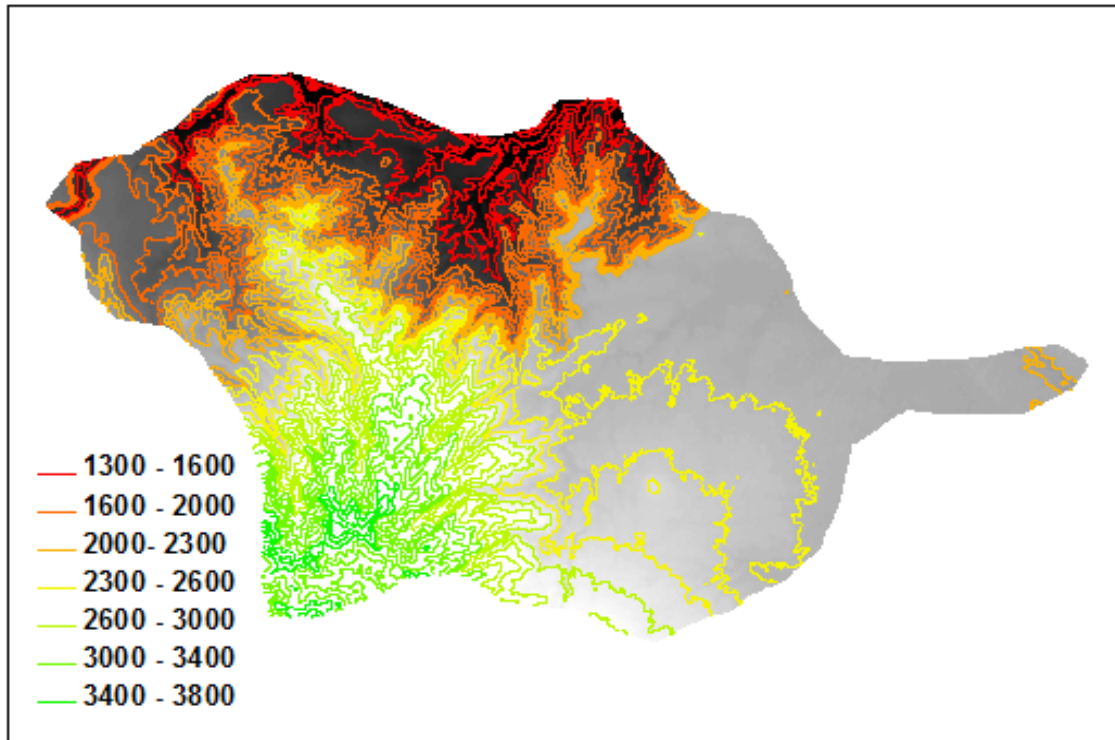


Step 2 Classify contour with graduate color

- k. To do so, double click on Contour_100 layer from Table of Content → Click **Symbology** tab → Click **Quantities**



- l. Select **Quantities – Graduated color** from the left hand side of the window. **For the value field, select “contour” and normalization field as “none”.**
- m. **Classify into 7 using Natural break**



3.9.1.2. Create the TIN

- a. Open the “3D Analyst” tool from **ArcToolbox box → TIN Management→ Create TIN**.
- b. In the “**Input feature class**” dialogue box →choose “**Contour_100**”.
- c. Choose “**Contour_100**” as **height field** from the column under input feature class.
- d. **Set the spatial reference s UTM_WGS1984_ Zone 37N**
- e. **Save your TIN under your Working directory**

Questions

What do you observe the relationship between the contour lines map and the TIN?

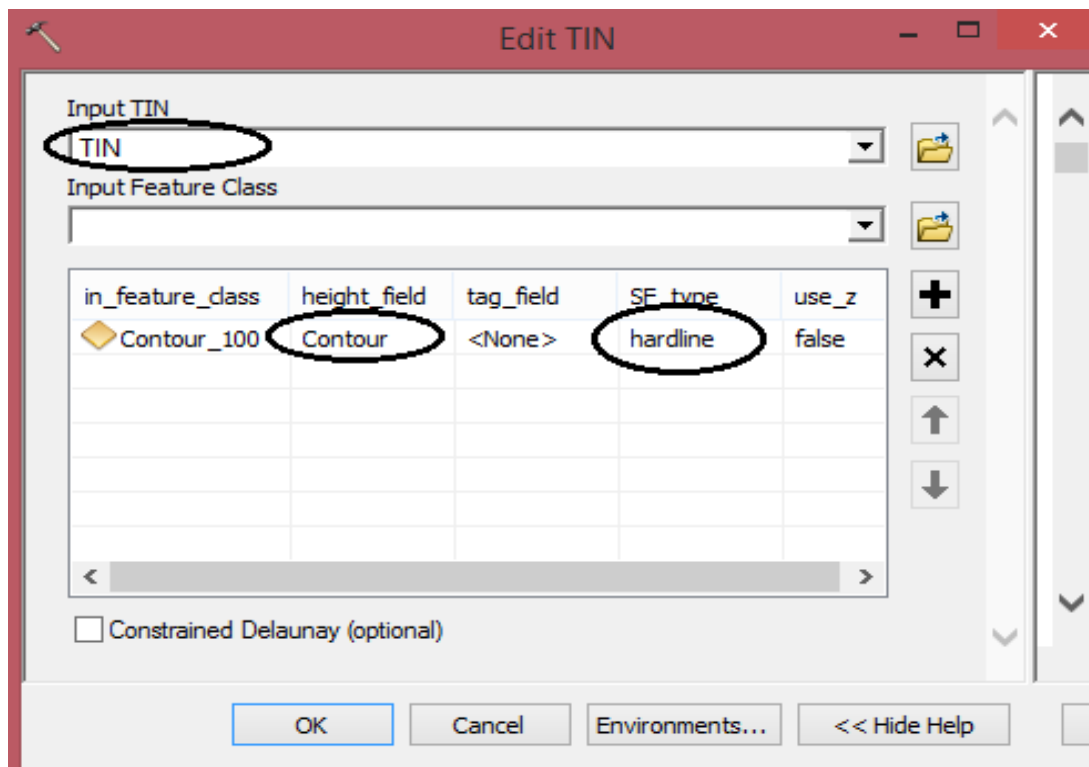
Where is the highest and the lowest elevation on the TIN Map?

Where do you think the river flow direction in the tin Map?

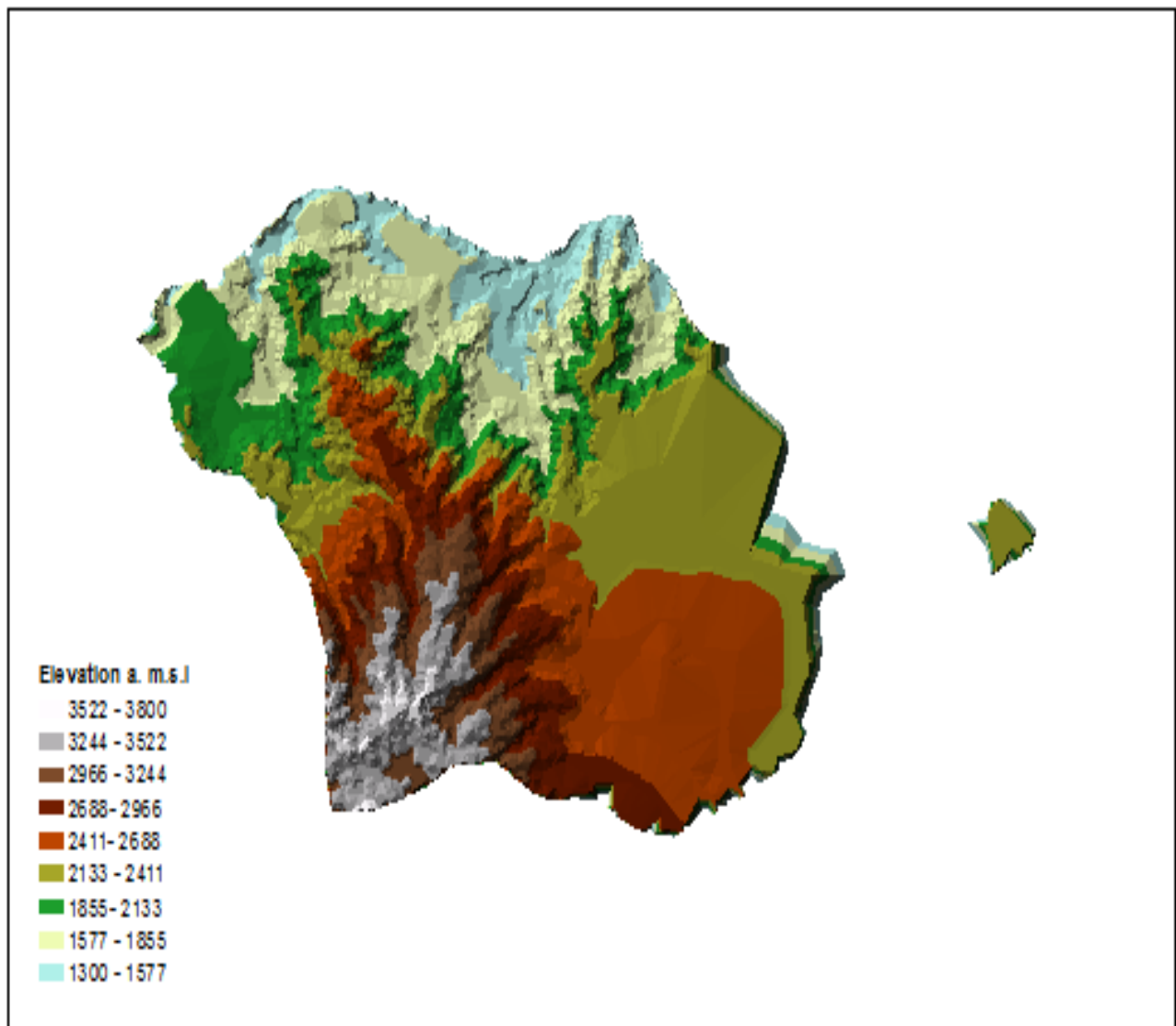
Edit TIN using contour Line

The derived TIN map is generalized into grid lines which cannot reflect the topography of the area comprehensively. Hence we need to refine the TIN by using Contour lines.

- Open the “**3D Analyst**” tool from **ArcToolbox box** → **TIN Management**→ **Edit TIN**.
- In the “**Edit TIN**” dialogue box →choose “**Tin**” in the Input Tin dialogue box.
- Choose “**Contour_100**” in the input feature class dialogue box which we will use to refine the TIN.
- Choose “**Contour**” as height source and triangulate as “**hardline**” form the column under input feature class dialogue box.



The result is shown below



3.10. Spatial Interpolation

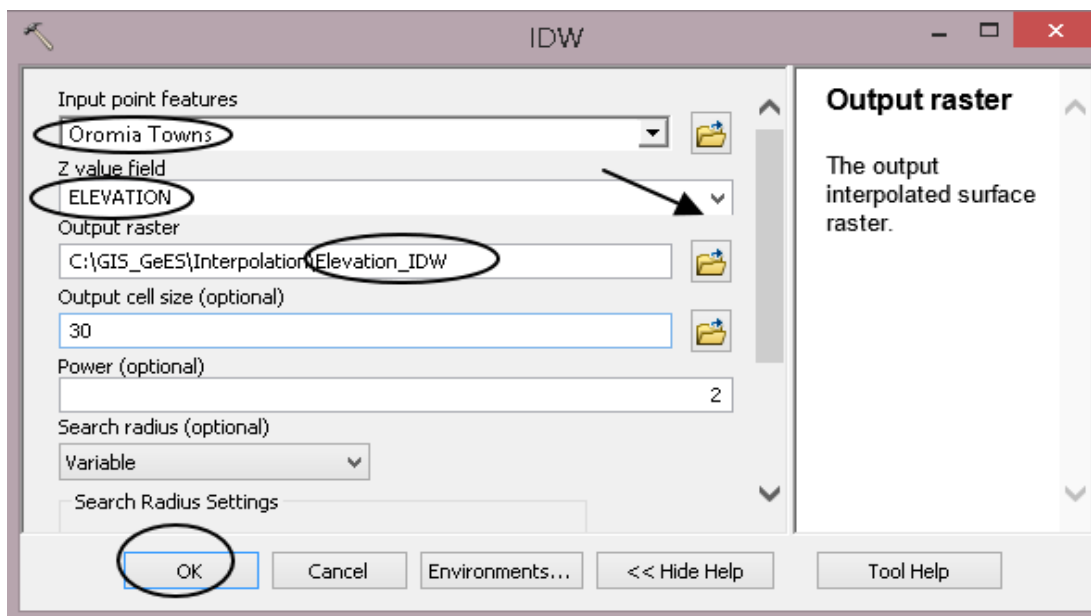
Spatial continuous data (spatial continuous surfaces) play a significant role in planning, risk assessment and decision making in environmental management. They are, however, usually not readily available for all places. Therefore, spatial interpolation techniques are essential for estimating variables for the unsampled locations needed accurate spatial continuous data across a region to make justified interpretations i.e. Use points with known values to estimate values at other points. Thus, spatial interpolation is the procedure of estimating the value of properties at unsampled sites within the area covered by existing observations. In almost all cases the property must be interval or ratio scaled. Rationale behind spatial interpolation is the observation that points close together in space are more likely to have similar values than points far apart (Tobler's Law of Geography). Spatial interpolation may be used in GISs:

- To provide contours for displaying data graphically
- To calculate some property of the surface at a given point
- To change the unit of comparison when using different data structures in different layers
- Frequently is used as an aid in the spatial decision making process both in physical & human geography as well as interrelated disciplines such as mineral prospecting &
- Weather station readings, spot heights, oil well readings, porosity measurement.
- precipitation, snow accumulation, water table, population density

Steps

- a. With ArcMap open, and **Oromia Town** data added from your working directory **C:\GIS_GeES\Interpolation**,
- b. Open up the ArcToolbox → **Spatial Analyst Tools** → **Interpolation** → **IDW**).
- c. Confirm that the input point features **Oromia Town**.
- d. Make your **Z-Value** field **Elevation_IDW**.
- e. Save your result file to your workspace and name it something useful  (Altitude). If you do not, you will have merely generated a temporary raster that will be deleted once ArcMap closes.

- f. Click **OK**. *This process takes a few minutes* and finally you should see your **Elevation GRID** in your **TOC**.



CHAPTER TEN

IMAGE CLASSIFICATION USING ArcGIS

Image classification refers to the task of extracting information classes from a multiband raster image. The resulting raster from image classification can be used to create thematic maps. Depending on the interaction between the analyst and the computer during classification, there are two types of classification: supervised and unsupervised.

In ArcGIS Spatial Analyst, there is a full suite of tools in the Multivariate toolset to perform supervised and unsupervised classification. The classification process is a multistep workflow, therefore, the Image Classification toolbar has been developed to provide an integrated environment to perform classifications with the tools. Not only does the toolbar help with the workflow for performing unsupervised and supervised classification, it also contains additional functionality for analyzing input data, creating training samples and signature files, and determining the quality of the training samples and signature files. The recommended way to perform classification and multivariate analysis is through the Image Classification toolbar.

10.1. Supervised classification

Supervised classification uses the spectral signatures obtained from training samples to classify an image. With the assistance of the Image Classification toolbar, you can easily create training samples to represent the classes you want to extract. You can also easily create a signature file from the training samples, which is then used by the Spatial Analyst multivariate classification tools to classify the image.

10.1.1. Adding the Image Classification toolbar

After enabling the Spatial Analyst extension, you can add the Image Classification toolbar and begin to use its tools to work with your data.

Steps:

- a. Click the **Customize** menu in ArcMap.
- b. Click **Toolbars > Image Classification**. The toolbar is added in ArcMap.

10.1.2. Adding a multiband image for the classification

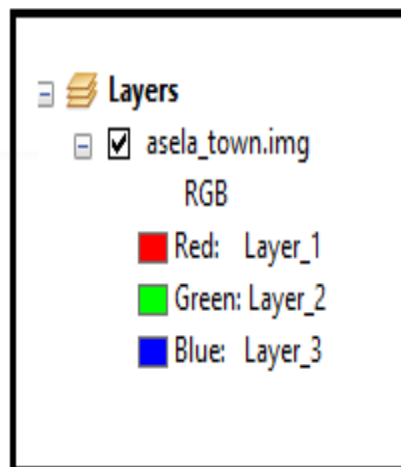
The Image Classification toolbar allows you to classify a multiband raster. A multiband satellite image is a typical example of a multiband raster. To specify a source image for the classification analysis, add the image to ArcMap. Then choose the image layer in the Layer drop-down list on the Image Classification toolbar.

Steps:

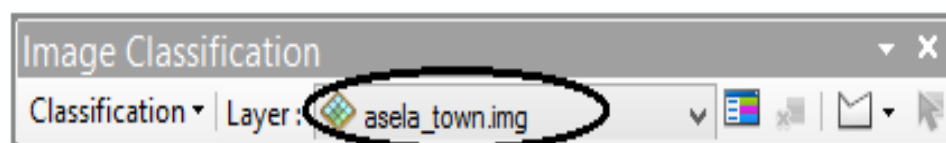
- Start ArcMap.
- Add the Image Classification toolbar to ArcMap. Refer to the topic Adding the Image Classification toolbar for how to add it.
- On the Standard toolbar, click the Add Data button.



- On the Add Data dialog box, browse to the image on the disk and click Add.
- The image is added as a layer in the table of contents. The layer points to all the bands on the disk.



- On the Image Classification toolbar, click the Layer arrow and click the image layer you just added.



- h. This specifies the source image for all the subsequent image classification tasks. All the bands associated with the image layer are used in the classification analysis.

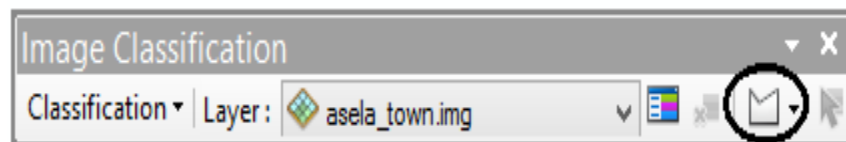
10.2. Creating Training Samples

To create training samples, use the training sample drawing tools on the Image Classification toolbar.

Steps:

The steps below show how to create training samples with the Image Classification toolbar:

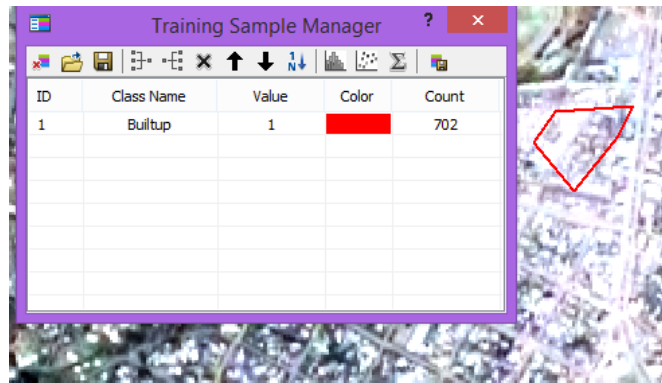
- a. On the Image Classification toolbar, choose an appropriate image layer in the Layer list.
- b. Click one of the drawing tools on the Image Classification toolbar. Notice that there are three drawing tools available, for drawing *polygons*, *circles*, and *rectangles*.



- c. In ArcMap, identify an area that belongs to a known class. Draw a training sample to enclose it. The following image shows a polygon training sample in ArcMap
- d. Once you finish drawing the training sample, a new class is created in Training Sample Manager with a default *name, value, and color*. Open Training Sample Manager and change the class name, value, and color if desired.
- g. Repeat steps 2 to 6 to create a few more training samples to represent the rest of the classes on the image. The following is what Training Sample Manager looks like after six classes are created:

10.2.1. Managing training samples

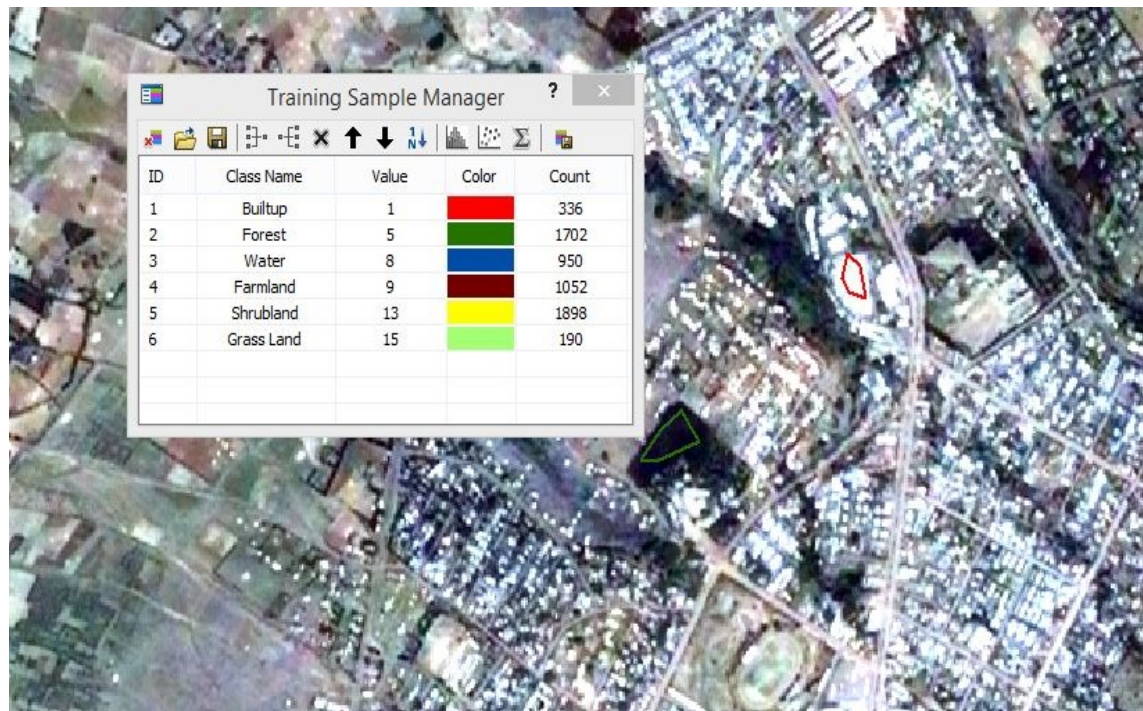
During the process of creating and evaluating training samples, you will need to use the tools on the Training Sample Manager to maintain the list of classes. For example, you may need to change the class name, value, and color. Also, classes may need to be *merged, deleted, and reorganized, and so on*. See the following on how to perform some common training sample management tasks.



10.2.2. Managing training samples

The first three tools on the toolbar are used for managing the training samples. Clearing all training samples.

- In the Training Sample Manager, click the Clear Training Samples button. Loading training samples
- In the Training Sample Manager, click the Load button. A file browser dialog box appears.
- In the file browser dialog box, select a training sample feature class and click Add.
- In the Training Sample Manager, click the Save button. A file browser dialog box appears.



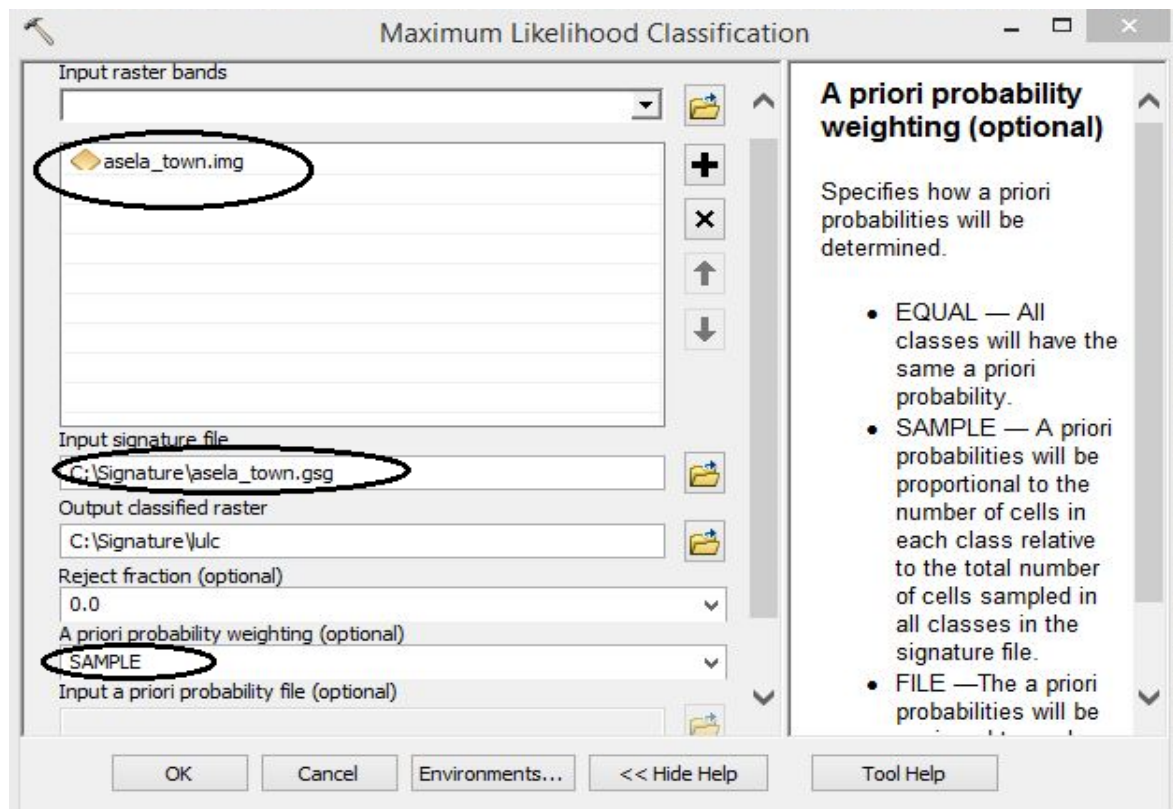
- e. In the file browser dialog box, navigate to the desired location and specify a name for the feature class or shapefile, then click Save.

10.3. Executing the Maximum Likelihood Classification tool

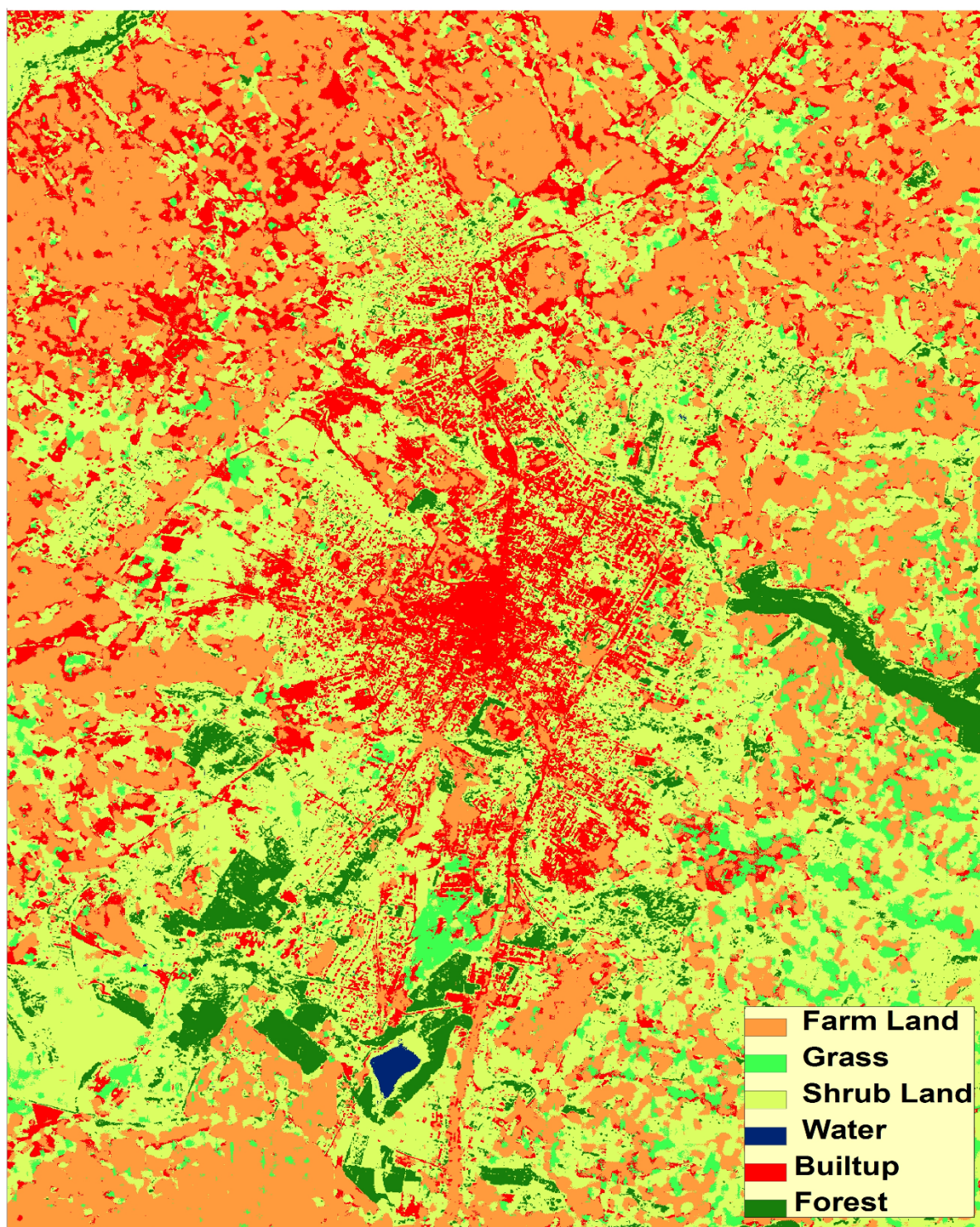
When you have a signature file ready, you can use the Maximum Likelihood Classification tool to classify the input image.

Steps:

- a. Create training samples if no training samples have been created. Refer to the topic creating training samples on how to create training samples.
- b. Create a signature file for the training samples if it has not been created already.
- c. On the Image Classification toolbar, click Classification → Maximum Likelihood Classification to open the Maximum Likelihood Classification tool.
- d. In the tool dialog box, specify values for the three required parameters —Input raster bands, Input signature file, and Output classified raster. Accept the default values for other parameters.
- e. Click **OK** to run the tool.



The output classified raster will be automatically added to ArcMap when the tool finishes



CHAPTER ELEVEN

SPATIAL ANALYSIS AND MODELING

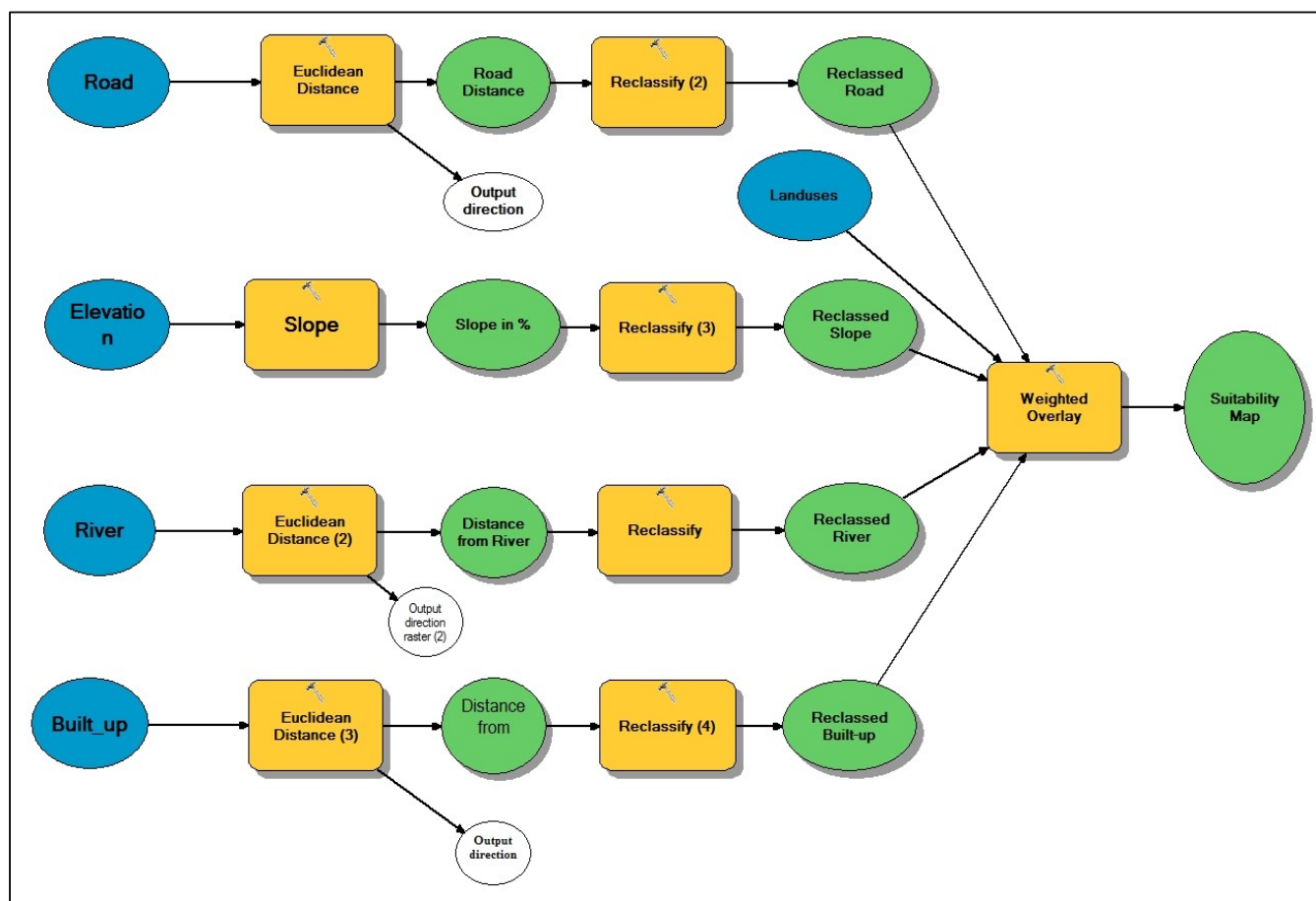
Introduction

Using the Spatial Analyst tools available with ArcGIS, you can easily perform spatial analysis on your data. You can provide answers to simple spatial questions, such as how steep is it at this location? or what direction is this location facing? You can also find answers to more complex spatial questions, such as where is the best location for a new facility? or what is the least costly path from 'A' to 'B'? The comprehensive set of Spatial Analyst tools within ArcGIS allows you to explore and analyze your spatial data and enables you to find solutions to your spatial problems. You can run tools from the Spatial Analyst toolbox or the command line, accessible via any ArcGIS Desktop application, or you can run tools from the Spatial Analyst toolbar in Arc Map. You can also create your own custom tools (models or scripts) to run a sequence of tools at one time.

11.1. Tutorial scenario

Dire Dawa city, has exhibited a considerable increase in population in recent years. The latest demographic data suggests this increase has occurred because natural increase and unrestricted migration from surrounding woreda's strive to find better job, better schooling and quality of life. In due course, this problem attracts the municipality and decided to identify new housing sites. Suppose, as a town planner, you have been assigned the task of finding the potential sites for new housing construction. This tutorial will show you how to use many of the available tools and will give you a solid basis from which you can start to think about how to solve your own specific spatial problems. It is assumed that you have installed ArcGIS Desktop (ArcView, ArcEditor, or ArcInfo) and activate ArcGIS Spatial Analyst extension before you begin this tutorial. You'll access tools in the toolbox from the ArcToolbox window to execute the tasks. The ArcToolbox window can be docked in any ArcGIS Desktop application. The idea is pretty simple but there are lots of steps in the process. This is how the logic of the final model looks for you to have a general idea of what we are about to do:

This is the power of GIS in general, and the ModelBuilder in particular. The model builder helps to do all these processes in one “model run” and you can return and run it again, or change the logic, in the future. **The full model we’ll produce looks like this:**



The data required is included On the ArcGIS Tutorial Data Setup wizard, check to install the Spatial Analyst data (the default installation path is C:\GIS_GeES\Spatial Analyst.

FACTORS	DESCRIPTION
LAND USE	All urban land use class which includes Business centers, Green areas, Manufacturing, Reserved areas, Residence and administration, Municipal service and Terminal areas.
ROAD	All transportation facilities such as Principal Roads and minor local Streets
SLOPE	The inclines, or steepness of the land surface in the town.
RIVER	Feature class representing river channels and flood sheets
BUILTUP	Feature class representing developed area

N.B. The datasets were not comprehensive and partially selected from the local parameters/factors in Dire Dawa City for use in this tutorial only.

This tutorial is divided into three section

- **In part 1**, you'll prepare for analysis. You'll copy the tutorial data locally and create a Geodatabase to hold your results.
- **In part 2**, you'll create a suitability map to help you find the best location for a new school. You'll derive datasets of distance and slope, reclassify datasets to a common scale, then weight those that are more important to consider and combine them to find the most suitable locations. You'll then locate the optimal site using the selection tools within ArcMap.
- **In part 3**, you'll find the least costly route for an alternate access road to the new school site.

♣ It is recommended to execute each tasks under all sections in sequence and don't forget to save your results along the way.

11.2. Preparing for analysis

In this exercise, you will prepare for analysis by first copying the tutorial data locally, then creating a Geodatabase for your results.

✓ Arranging Datasets

Before working with Spatial Analyst tools, you will expected to arrange your input datasets.

✓ Locating Datasets

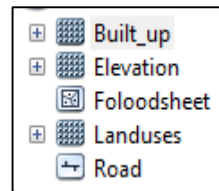
Steps:

- a. Navigate to the location in your file browser where you installed the tutorial data, C:\GIS_GeES\Spatial Analyst
- b. Right-click the Spatial Analyst folder and select **Copy**.
- c. Browse to your working directory, for example, your C:\ drive.
- d. Right-click C:\ (or an alternative drive) \ Create New folder\ Rename as *Exercise* and select Paste

11.3. Starting New ArcMap

Steps:

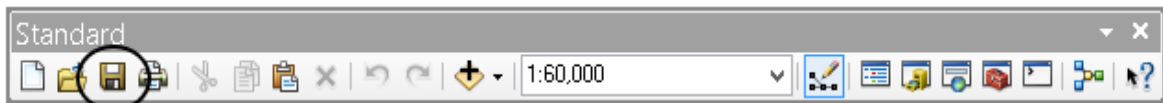
- a. Start ArcMap by either double-clicking a shortcut installed on your desktop or clicking **Start→ All Programs → ArcGIS →ArcMap**.
- b. Click **New Maps** in the *ArcMap - Getting Started* window if it is not already highlighted.
- c. Click the **Open** button
- d. Click the **Connect to folder** button and Navigate to C:\GIS_GeES\Spatial Analyst.
- e. Click **Datasets.gdb**
- f. Select all the datasets and Click ok



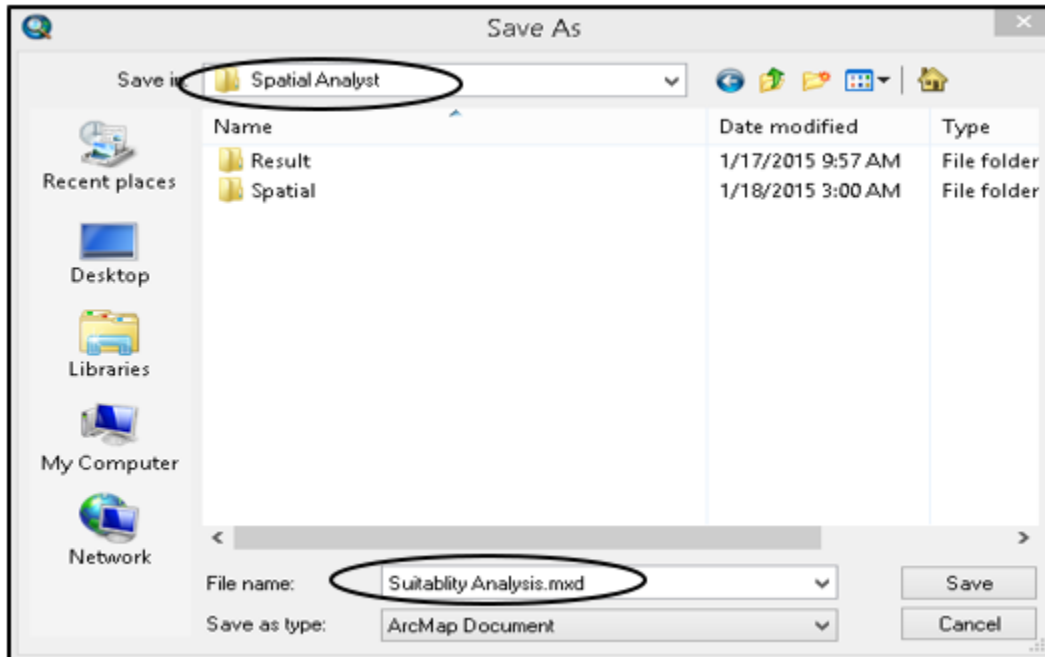
You should see two feature classes and three raster's in the table of content

11.4. Saving your map document

- a. On the Standard toolbar, click the **Save** button. 



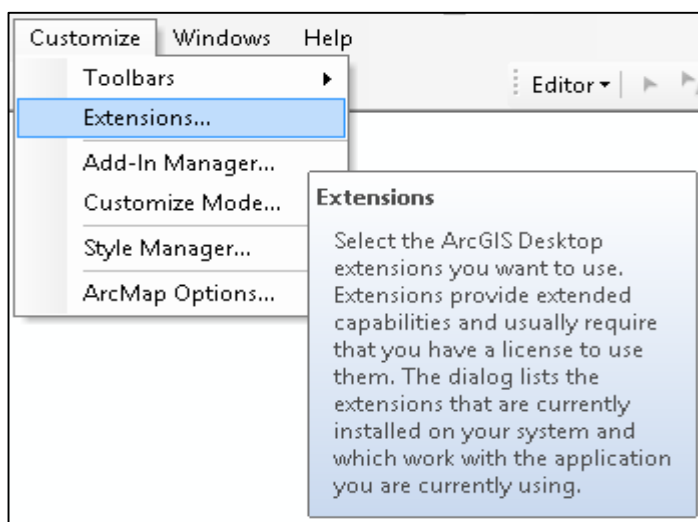
- b. Browse to the working copy of the Spatial Analyst folder.
- c. For **File name**, enter *Suitability Analysis. mxd*.
- d. Click save



You have prepared a workspace in which the datasets created by following the tutorial workflow will be created. You can now proceed to Activity 2.

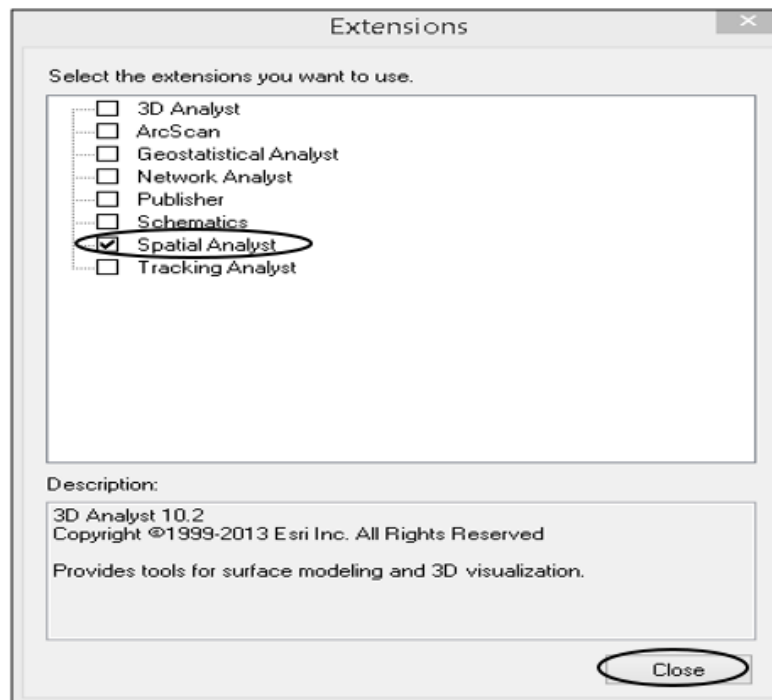
11.5. Accessing Spatial Analyst and Data Exploration

You will learn how to turn on the Spatial Analyst extension, access the Spatial Analyst toolbar, and search for Geoprocessing tools. **Checking out a Spatial Analyst license**



- Check the **Spatial Analyst** check box.

- Click **Close**.



11.1. Finding a site for a new Housing Sites

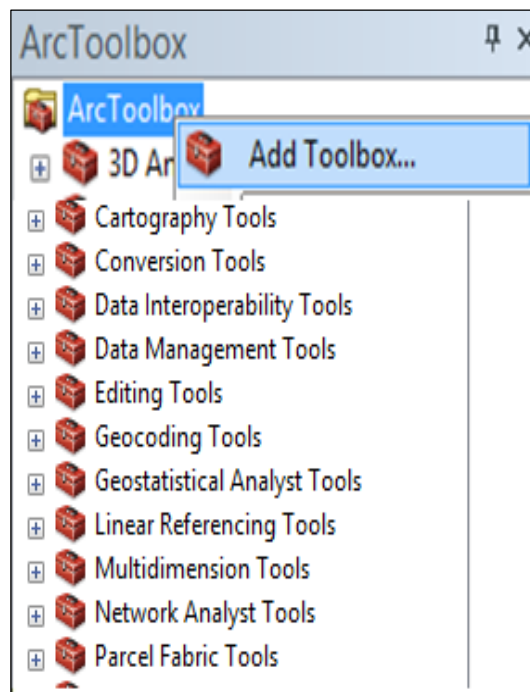
In this exercise, you will build a suitability model that finds suitable locations for a new housing development. The steps to produce such a suitability model are outlined below. Your input datasets in this exercise are landuse, elevation, River, built-up and Roads. You will derive slope, distance to road, distance from River and distance to existing built-up, then reclassify these derived datasets to a common scale from 1 to 5. You'll weight them and the land-uses dataset according to a percentage of influence and combine them to produce a map displaying suitable locations for housing development. You'll then select the optimal site for housing.

11.1.1. Creating a new toolbox

You'll first create a new toolbox to hold the models you will create in this exercise and the next exercise.

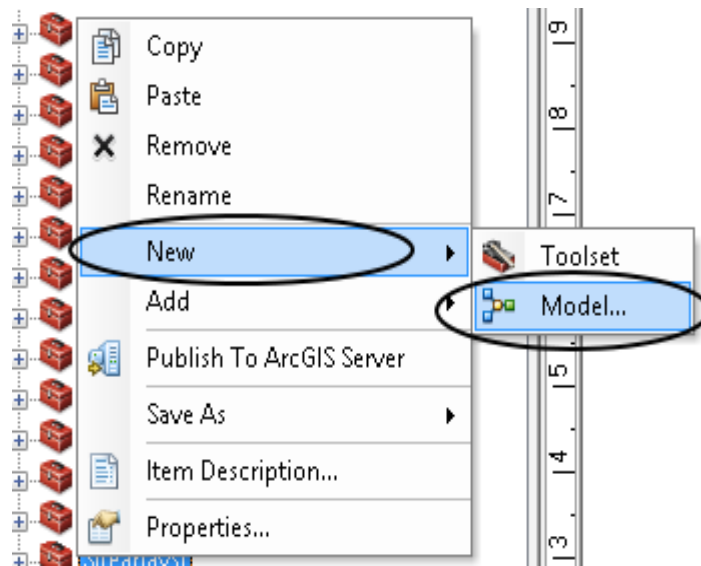
- Create a new toolbox in your Spatial Analyst folder. Name the toolbox Site Analysis.

- b. Right click on ArcToolbox → Add Toolbox



11.1.2. Creating a new model

You will create a model to perform Spatial Analyst tasks. A model is built by stringing tools together in ModelBuilder. Once your model is created, you can easily experiment with parameter values, use different input data, run the model over and over again, and share it with others. In this exercise, you will create a model to find a suitable location for a new school site.

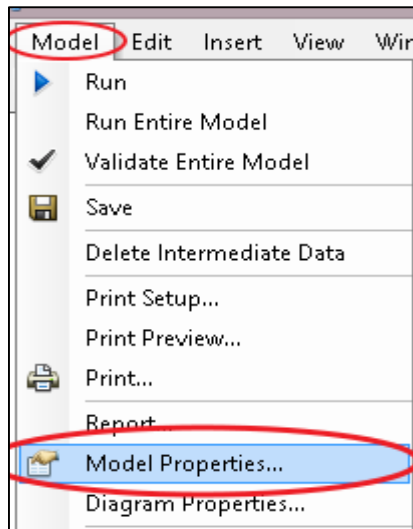


An empty ModelBuilder session will open.

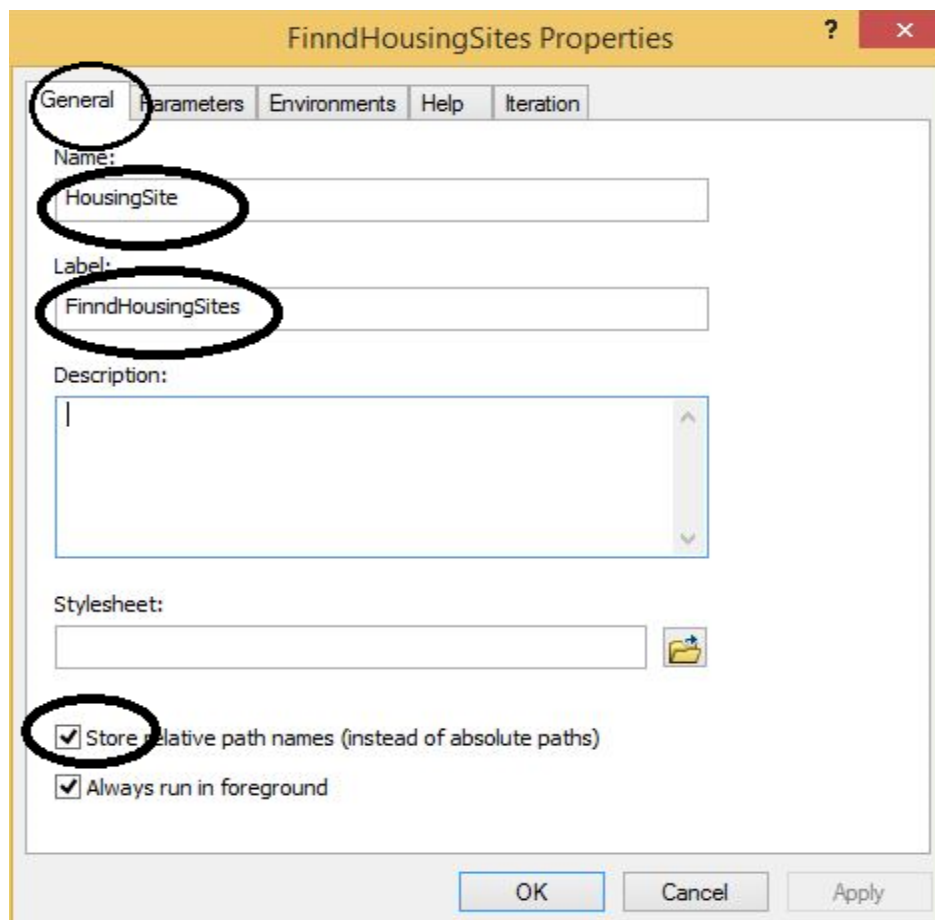
11.1.3. Renaming the model

Steps

- a) On the model's main menu, click **Model** → **Model Properties**



- b) Click the **General** tab.



- c) Type *housing Site* in the **Name** text box and *Find Housing Site* in the **Label** text box.

- d) Check the **Store relative path names (instead of absolute paths)** check box

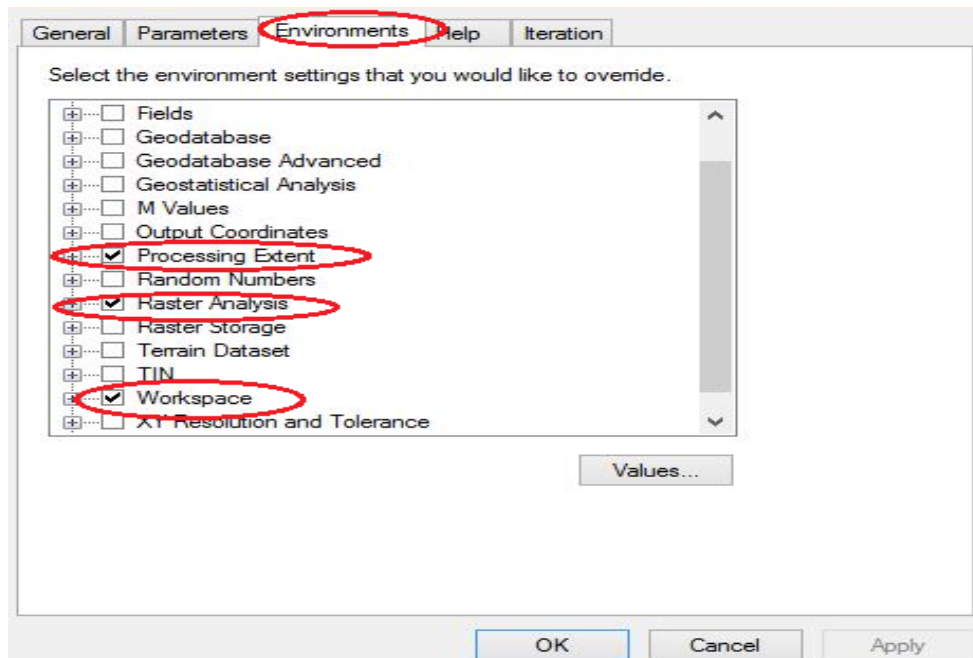
11.1.4. Specifying environment settings

Before you start to perform analysis on your data, you should set any relevant environment settings.

Steps:

- a. Click the **Environments** tab.

- b. check **Processing Extent** and **Raster analysis and workspace**

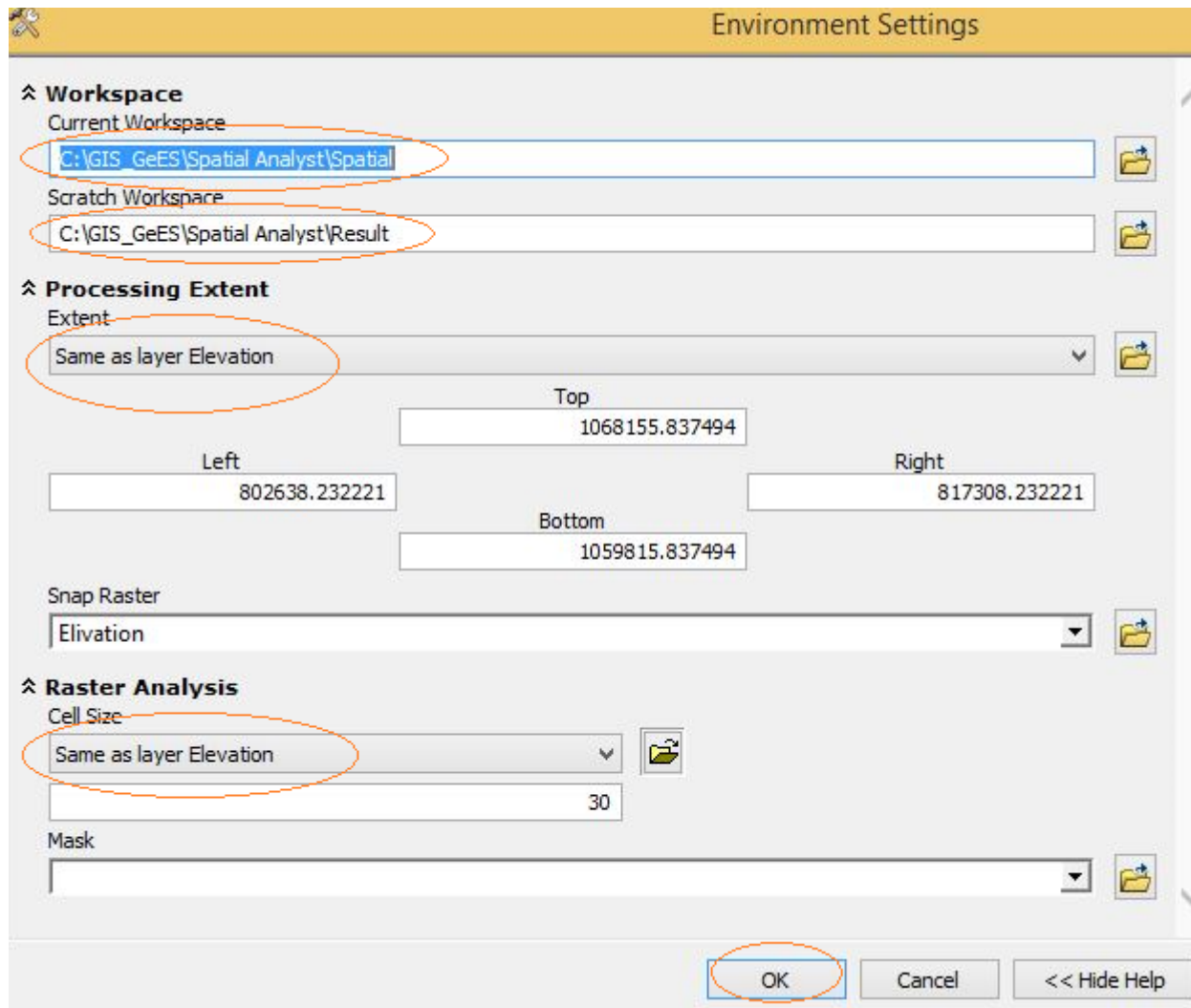


- c. Click **Values**.

- d. Expand **Processing Extent**. Set the **Extent** by clicking the drop-down arrow and selecting **same as Layer elevation**.
- e. Expand **Raster Analysis**. Set the **Cell Size** by clicking the drop-down arrow and selecting **same as Layer elevation**.

The cell size of your elevation layer will be applied to all subsequent raster outputs. Your elevation dataset has the largest cell size (30 meters).

- f. Click **OK** on the **Environment Settings** window.



8. Click **OK** on the *Model Properties* window.

9. On the toolbar, click the **Save**  button. .

***N.B.** The model's properties are updated. If at any point you want to close the model and carry on later, right-click the model in the **ArcToolbox** window and click **Edit** to continue with the tutorial.*

11.2. Deriving datasets

You are ready to start to process your project data to locate suitable areas for the new school.

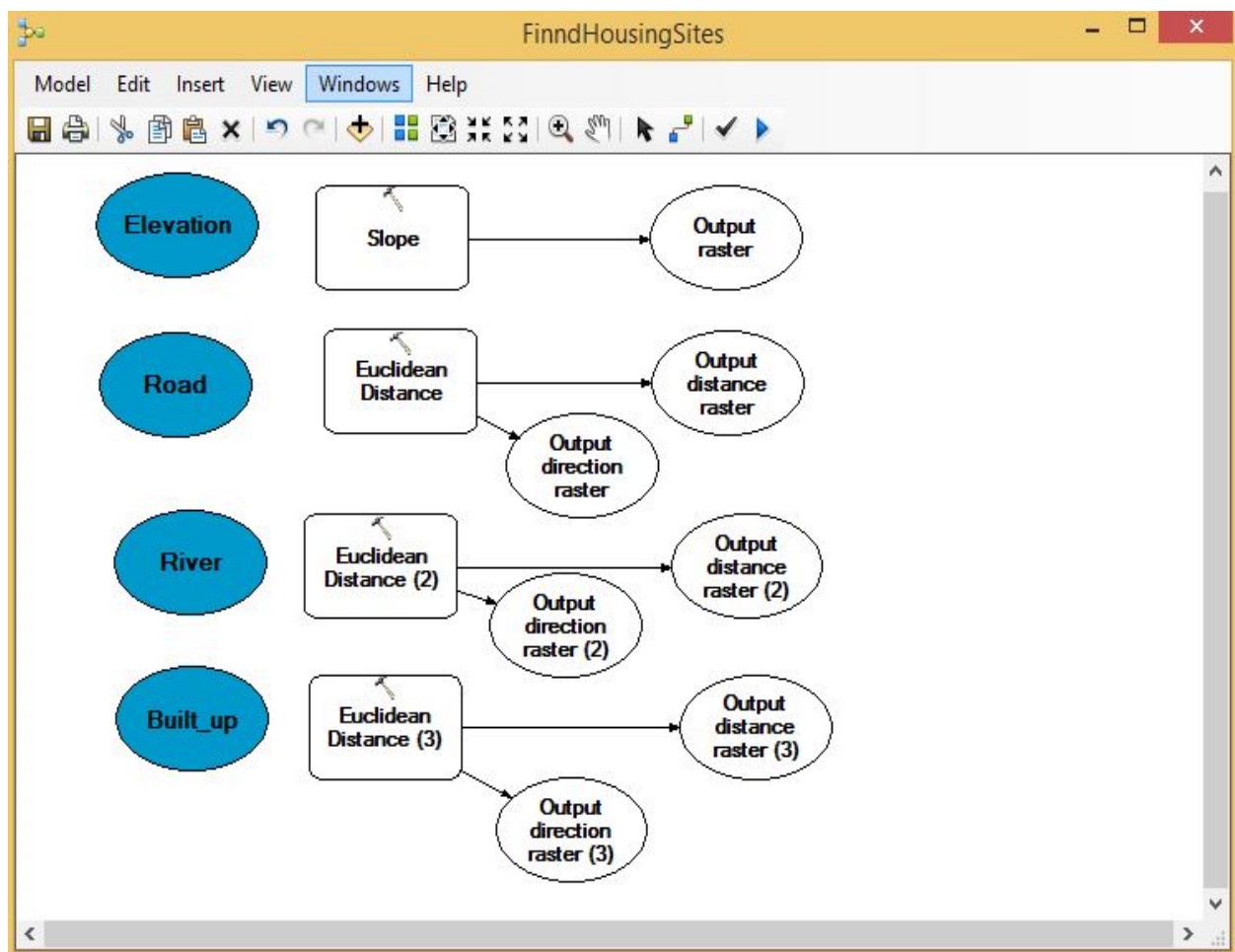
You'll derive the following from your project data:

- ✓ ***Derive Slope from the elevation dataset.***

- ✓ *Derive Distance from River.*
- ✓ *Derive Distance from Road dataset.*
- ✓ *Derive Distance from Built-up dataset*

Steps:

- a. From your **table of contents**, drag the layers elevation, River, Road, and built-up onto your model.
- b. Click and drag the Slope tool from the Spatial Analyst Surface toolset onto your model and place it in line with your elevation data.
- c. Locate the Euclidean Distance tool in the Spatial Analyst Tools toolbox Distance toolset. Click and drag the Euclidean Distance tool onto your model and place it in line with Roads, River and Built-up layers



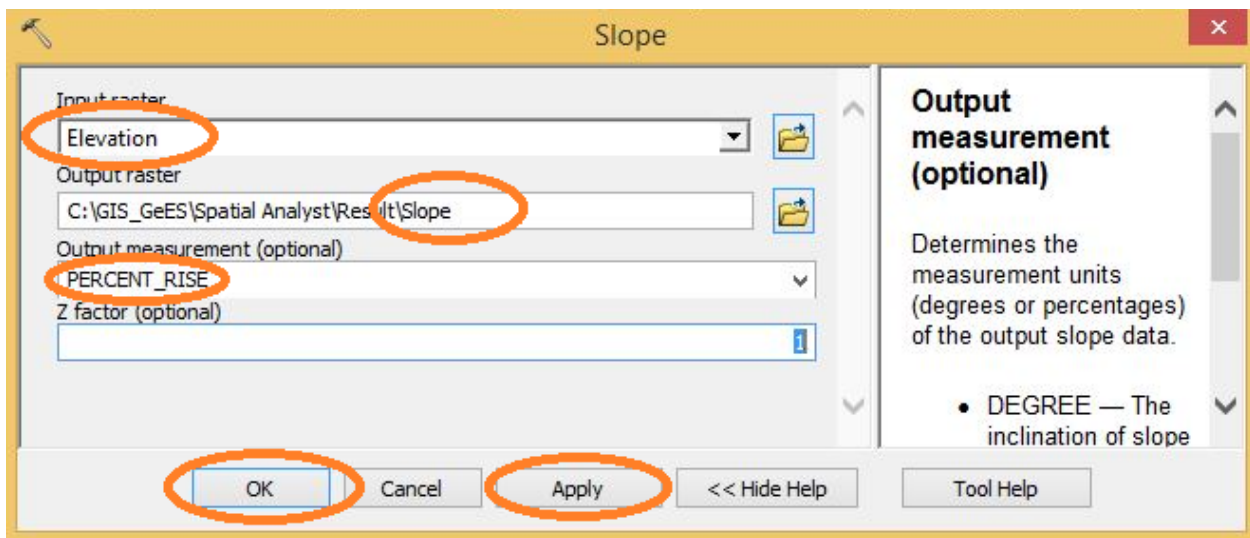
- d. Click the **Add Connection** tool.
- e. Use the **Add Connection** tool to connect to the *Elevation* dataset to the Slope tool. To do this, click elevation, then click the Slope tool.
- f. Repeat the previous step, this time connecting *Road*, *River* and *Built-up* to the Euclidean Distance tool.

11.3. Deriving Slope from Elevation

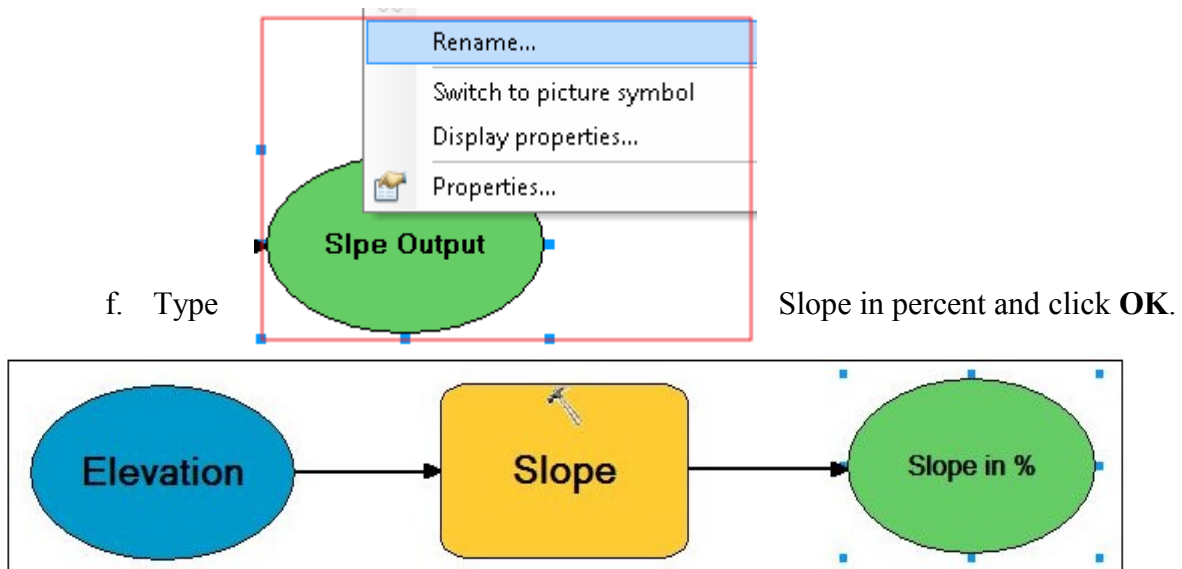
Since the area is mountainous, you need to find areas of *relatively flat land* .

Steps:

- a. Right-click the Slope tool and click **Open**, or double-click the Slope tool.
- b. Leave the **Input raster** and the **Output measurement** as the default values.
- c. Accept the default location for the value of the **Output raster** parameter, *Slope output*



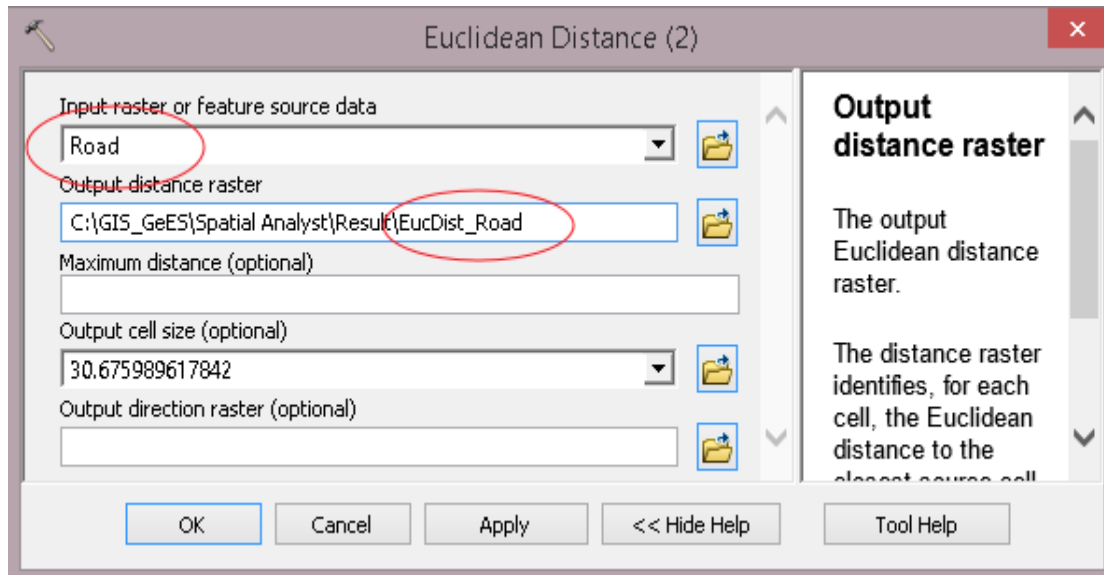
- d. Click **OK**.
- e. Right-click the output variable from the Slope tool and click **Rename**.



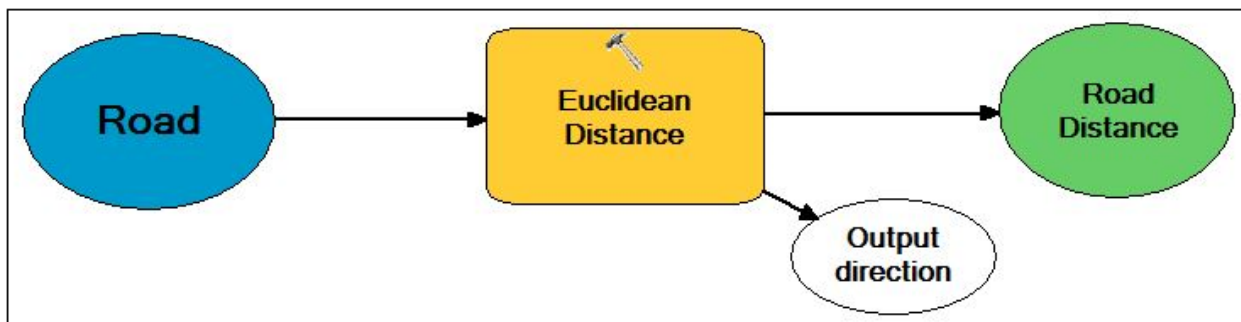
11.3.1. Deriving distance from Road

To find locations close to Road, you must first calculate the Euclidean (straight-line) distance from Road. Hence, the identification of suitable land for future development must consider the accessibility to road. The roads for suitable site may not affect strongly but the more the site is closer to the street the more likely suitable for residential house because of transportation access with short distances without extra costs. Therefore, the road accessibility is the most important factor for residential area since it facilitate the linkage between the settlements.

- a. Right-click the Road tool and click Open, or double-click the Road tool.
- b. You can easily see all the default parameters set for this tool. There is no need to adjust any of these parameters.
- c. Click OK.
- d. Right-



Rename the output variable from the Euclidean Distance tool to **Road Distance**



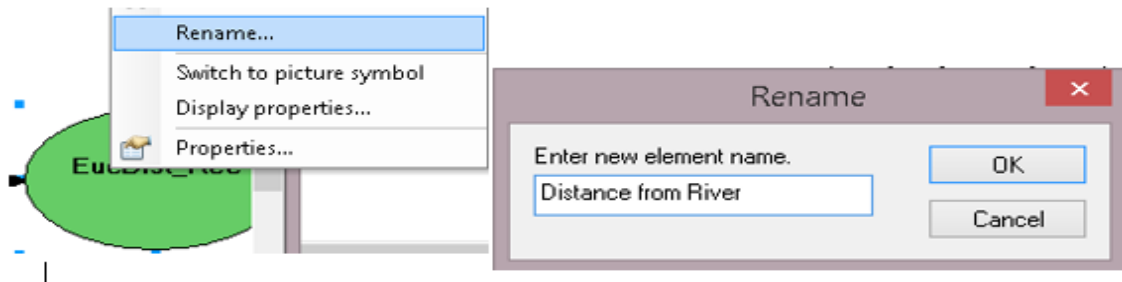
11.3.2. Deriving distance from River

From the goal of safety the candidate suitable site for residential housing area should be free from risk aroused by river flooding during rainy seasons.

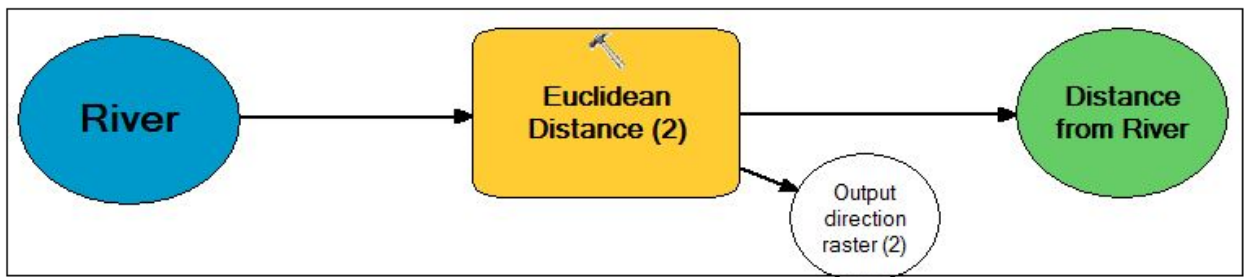
Steps:

- b. The Euclidean Distance (2) tool connected to River Channels You can easily see all the default parameters set for this tool. There is no need to adjust any of these parameters.

- c. Rename the output variable from the EucDist_River (2) tool to Distance from River.



You accepted the default for the **Maximum distance**, thus leaving this parameter empty. Therefore, the edge of the output raster is used as the maximum distance. The **Output cell size** is taken from the environment setting previously set to that of your elevation data. In this exercise, the **Output direction raster** is not required.

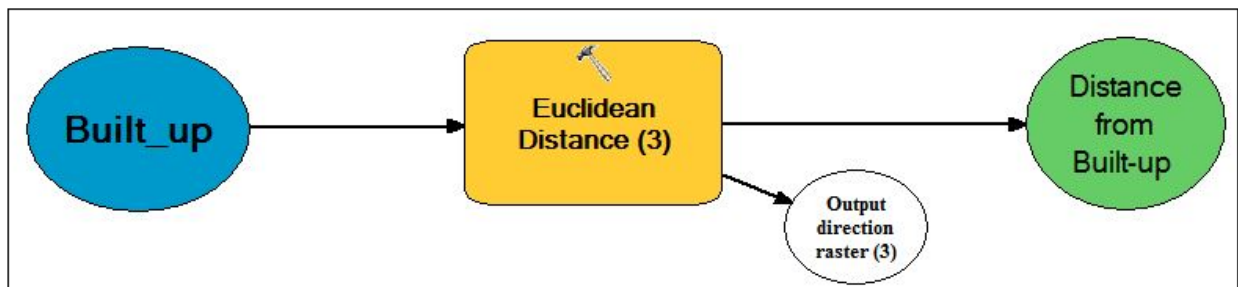


11.3.3. Deriving Distance from Built-up (Developed area)

To find locations away from existing **Built-up** (Developed area, you must first calculate the Euclidean (straight-line) distance from schools.

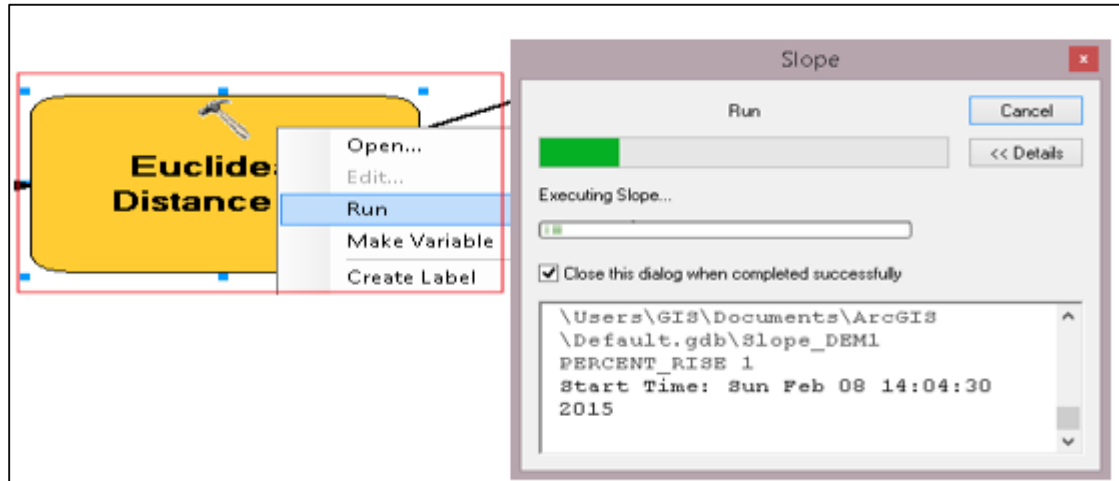
Steps:

- However the Euclidean Distance tool connected to Built-up. You can easily see all the default parameters set for this tool. There is no need to adjust any of these parameters.
- Rename the output variable from the Euclidean Distance tool to Distance from Built-up.

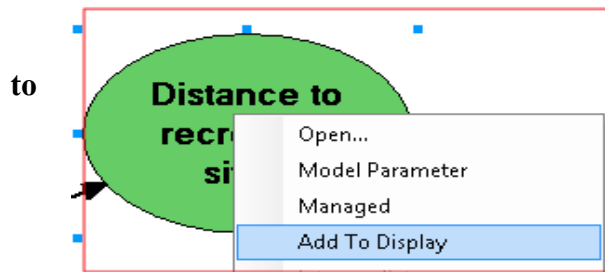


11.4. Run Model to Derive Datasets

- Right-click each of the output variables (Slope in percent, Road Distance, Distance from River and Distance from Built-up) and right click on each layer to **Add onto display**



Notice that as the tool runs, its progress is documented on the progress dialog box, and the tool that references the tool is highlighted in red. When the tools have finished running, the tool and its output become shaded, indicating that the output has been created on disk. If the progress dialog box is present, check the **Close this dialog when completed successfully**



remaining three layers.

check box, then click **Close**. With the **Add Display** property on, the data referenced by the variable will be added to the display each time the model is run. Click the **Run** button on the model toolbar to execute the

11.5. Reclassifying Datasets

Deriving datasets, such as slope, is the first step when building a suitability model. Each cell in your study area now has a value for each input criteria (slope, distance to road and distance from river and built-up). You need to combine the derived datasets so you can create your suitability map that will identify the potential locations for housing development. To combine the datasets, they first need to be set to a common measurement scale, such as 1 to 5. That common measurement scale is what determines how suitable a particular location for housing. Higher values indicate more suitable locations for housing development. Using the Weighted Overlay tool, you can weigh the values of each dataset, then combine them. However, the inputs for the Weighted Overlay tool must contain discrete, integer values. However, for each Land use class, we are expected to assign values sets in measurement scale of 1 to 5. The values in the datasets you derived in previous steps are all floating-point, continuous datasets, categorized into ranges, and they must first be reclassified so that each range of values is assigned one discrete integer value.

However, it is easier to weight the cell values for derived datasets while reclassifying. In the Weighted Overlay tool, you can accept the default and leave the scale values the same as the input values. You will reclassify each derived dataset to a common measurement scale, giving each range a discrete integer value between 1 and 5. Higher values will be given to attributes within each dataset that are more suitable for housing.

Steps:

- a. Locate the **Reclassify tool** in the **Spatial Analyst Tools** toolbox **Reclass toolset**. **Click and drag the Reclassify tool onto Model Builder** in line with **Slope in percent**. Add another Reclassify tool in line with **Distance to Road** and another in line with **Distance from River** and do the same for **Distance from built-up**.
- b. Click the **Add Connection** tool. Use the connect tool to connect:
 - ✓ **Slope in percent** to the Reclassify tool
 - ✓ **Distance to Road** to the Reclassify tool
 - ✓ **Distance from River** to the Reclassify tool
 - ✓ **Distance from built-up** to the Reclassify tool

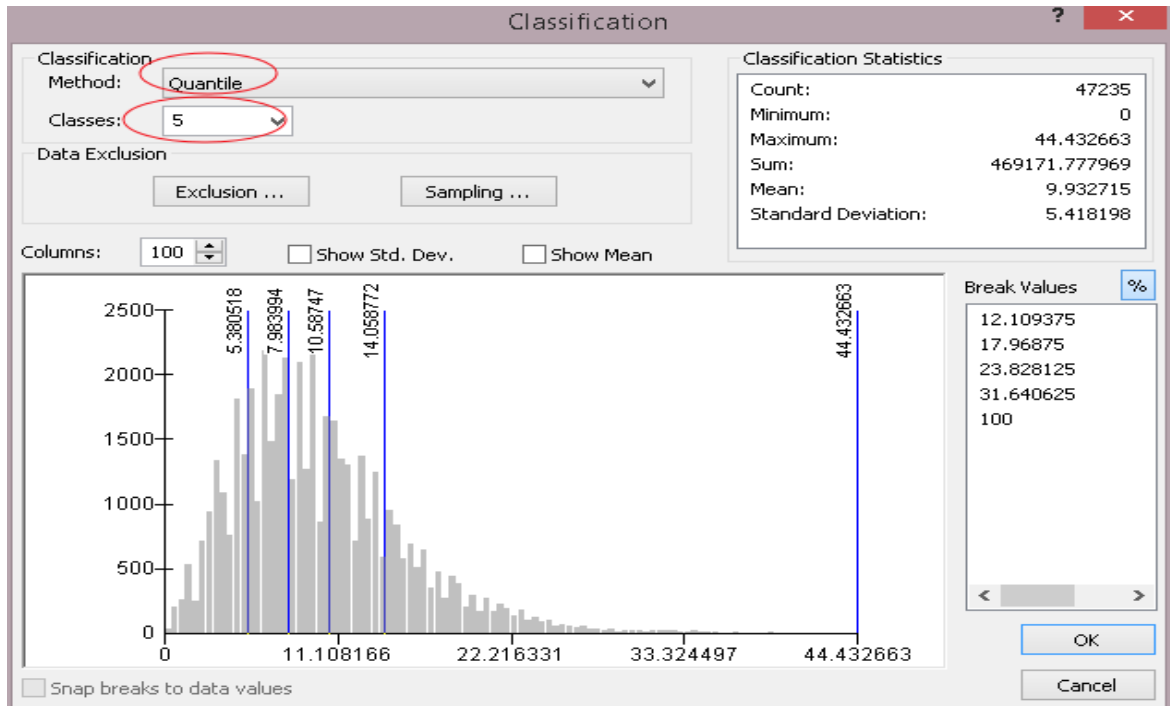
- c. On the model toolbar, click the **Select** tool.
- d. Click the **Auto Layout** button, then click the **Full View** button.

11.5.1. Reclassifying slope

It is preferable that the new school site be located on relatively flat ground. You'll reclassify the slope in %, dividing the values into five intervals using Quantile Classification Method. You'll assign a value of 5 to the most suitable range of slopes (those with the lowest angle of slope) and 1 to the least suitable range of slopes (those with the steepest angle of slope) and rank the values in between linearly.

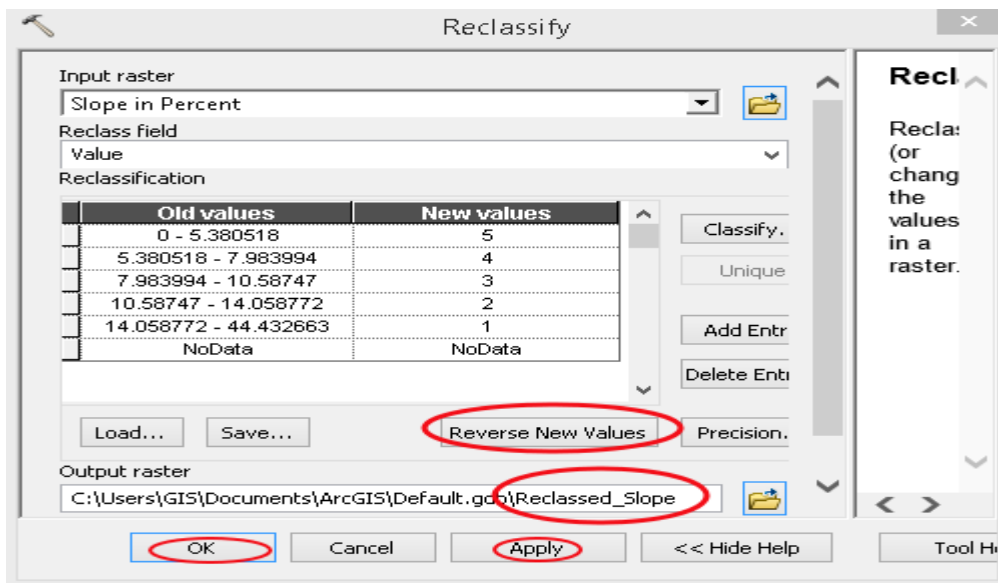
Steps:

- a. Open the Reclassify tool connected to the Slope Output variable.
- b. Accept the default for the **Reclass field** parameter so the Value field will be used.
- c. Click **Classify**.
- d. Click the **Method** drop-down arrow and click **Quantile**.
- e. Click the **Classes** drop-down arrow and click **5**.



- f. Click Ok

- g. Click **Reverse New Values**. Reversing the values applies higher new values to the values representing less steep slope, since these areas are more suitable for building.



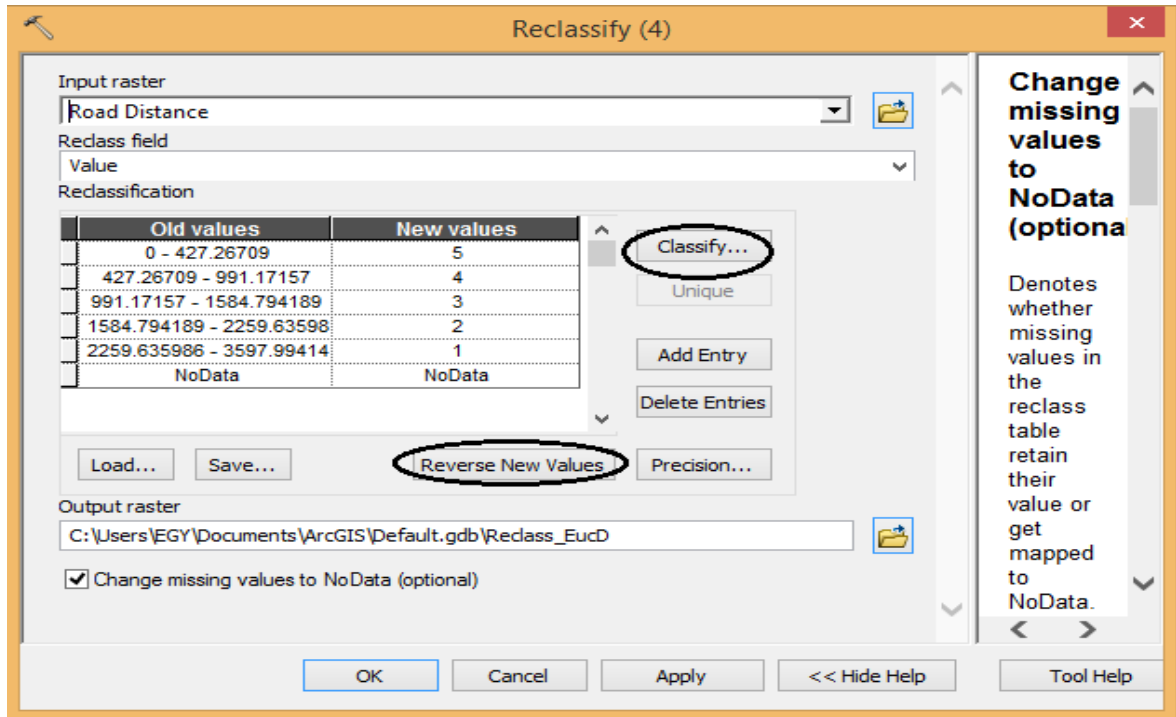
- h. Accept the default for the **Output raster** parameter
 i. Click **OK**.
 j. Rename the output variable from the Reclassify tool to Reclassed slope.

11.5.2. Reclassifying distance from Road

Reclassify the distance from Roads, assigning the number 5 to ranges of values that represent areas closest to road (the most suitable locations), assigning the number 1 to ranges of values that represent areas far from Road (the least suitable locations).

Steps:

- Open the Reclassify tool connected to the Road Distance sites variable.
- Accept the default for the **Reclass field** parameter so the Value field will be used.
- Click **Classify**.
- Set the **Method** to **Quantile** and the number of **Classes** to **5**.
- Click **OK**.
- Click **Reverse New Values**. Clicking **Reverse New Values** makes it so that distances close to Road network receive a higher new value, since these areas are more desirable.
- Accept the default path and name for the **Output raster** parameter.
- Click **OK**.



- i. Rename the output variable from the Reclassify tool to ***Reclassified Road distance***.

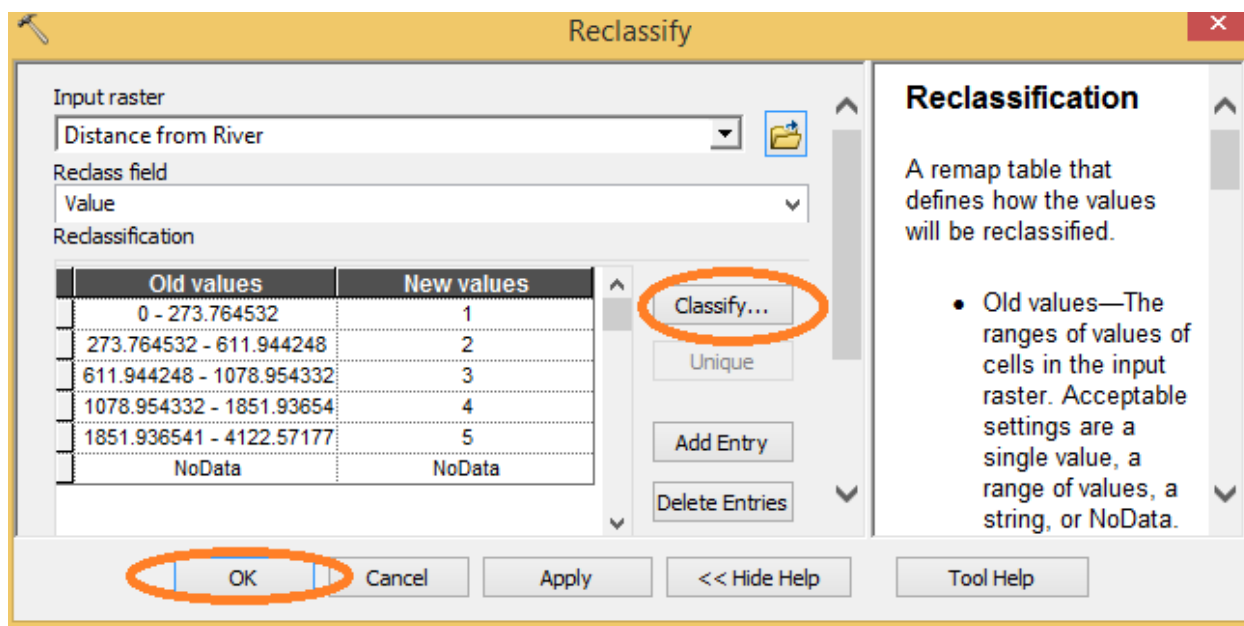
11.5.3. Reclassifying Distance from River

Reclassify the Distance from river layer, assigning a value of 5 to areas farthest from existing river (the most suitable locations), assigning a value of 1 to areas near river (the least suitable locations). By doing this, you will determine which areas are near and which areas are far from river

Steps:

- a. Open Reclassify tool.
- b. Accept the default for the **Reclassified field** parameter so the Value field will be used.
- c. Click **Classify**.
- d. Set the **Method** to **Quantile** and the number of **Classes** to **5**.
- e. Click **OK**.

You want to position the sites away from river, so you will assign larger numbers to ranges of values that represent locations farther away, because these locations are most desirable. Since the default assigns high **new values** (more suitable locations) to higher range of **Old values** (locations farther away from river), you do not need to change any values at this time.



- f. Accept the default for the **Output raster** parameter.
- g. Click **OK**.
- h. Rename the output variable from the Reclassify tool to **Reclassified River**.

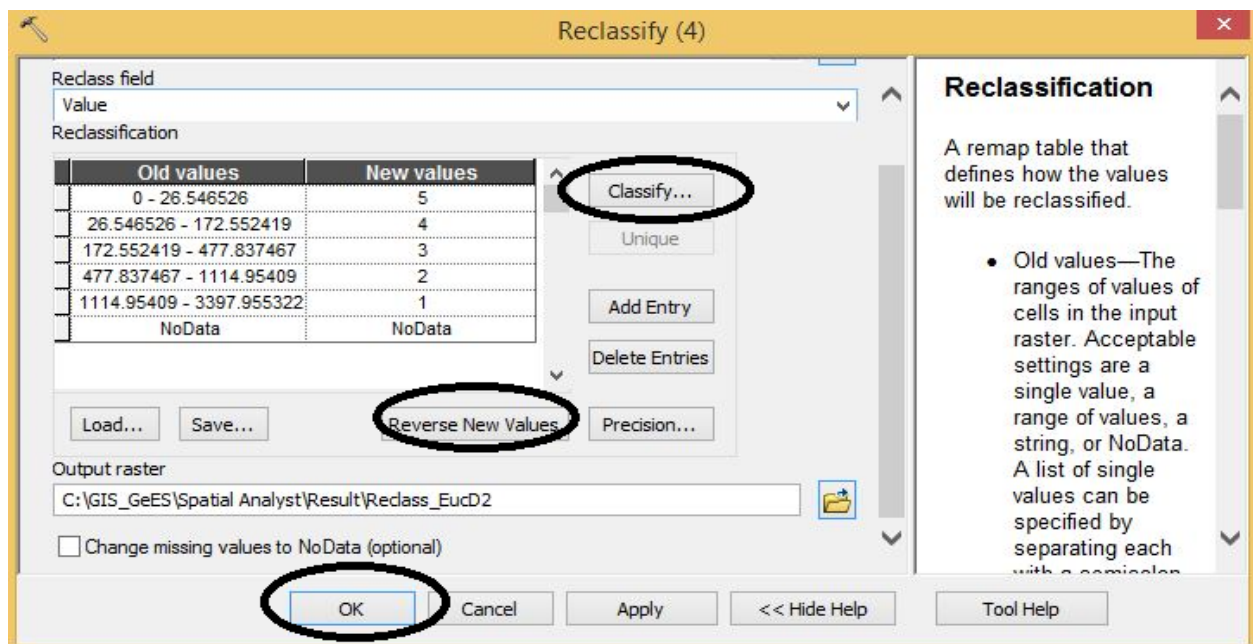
11.5.4. Reclassifying Distance to Built-up area

The new housing project sites should be located as close as possible to built-up (developed area). Reclassify the distance to recreation sites output, assigning the number 5 to ranges of values that represent areas closest to built-up sites (the most suitable locations), assigning the number 1 to ranges of values that represent areas far from built-up sites (the least suitable locations).

Steps:

- a. Open the Reclassify tool connected to the Distance to recreation sites variable.
- b. Accept the default for the **Reclassified field** parameter so the Value field will be used.
- c. Click **Classify**.
- d. Set the **Method** to **Quantile** and the number of **Classes** to **5**.
- e. Click **OK**.
- f. Click **Reverse New Values**.
- g. Clicking **Reverse New Values** makes it so that distances close to built-up a higher new value, since these areas are more desirable.
- h. Accept the default path and name for the **Output raster** parameter.

- i. Click **OK**.
- j. Rename the output variable from the Reclassify tool to **Reclassified - Built-up**

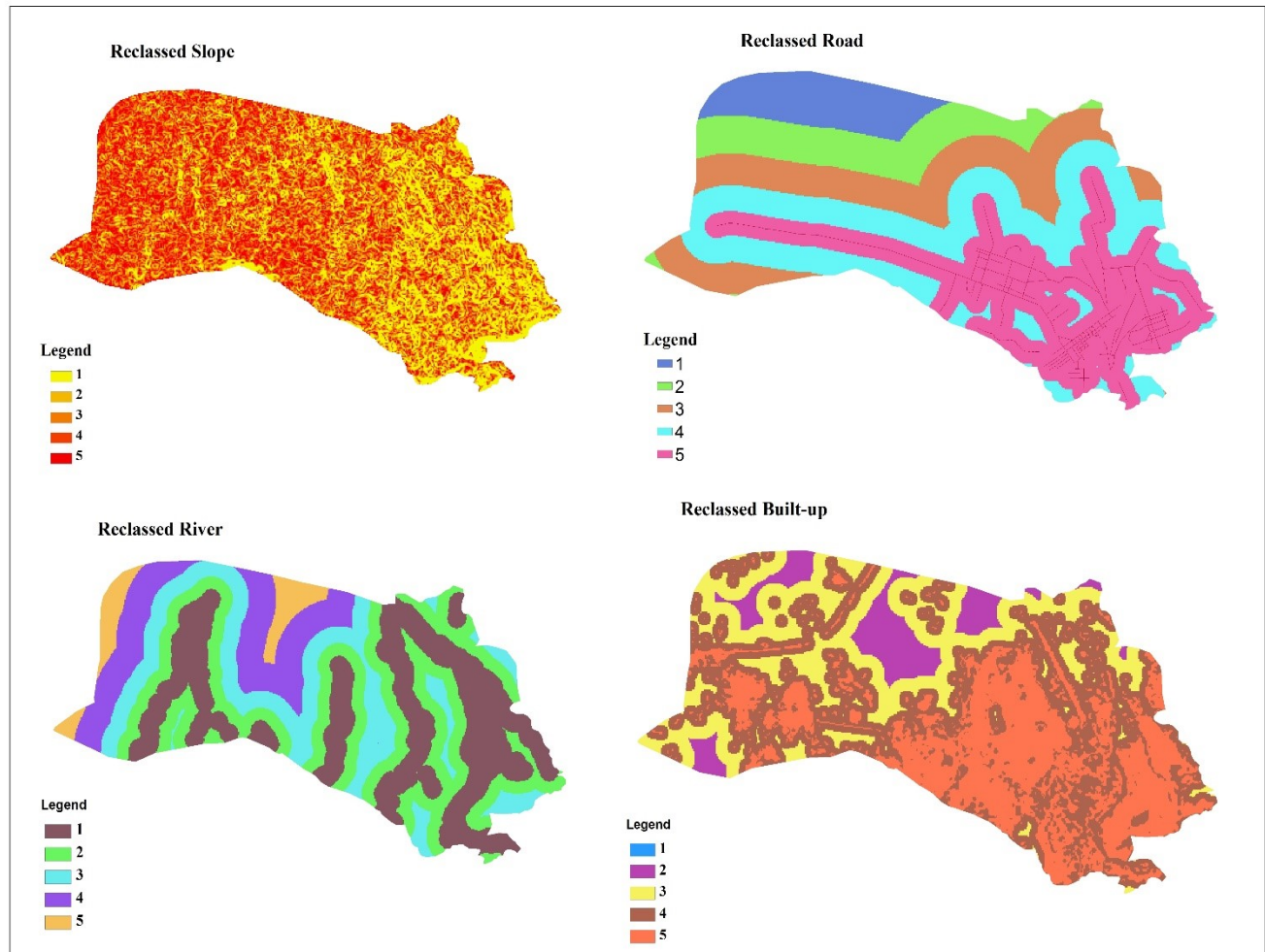


11.6. Executing Reclassify

Steps:

- a. Right-click each of the variable outputs—Reclassified slope, Reclassed Road, Reclassed river and Reclassed built-up .
- b. Click the Run button to execute four Reclassify tools in your model.
- c. click Add To Display.
- d. On the toolbar, click the Save  button.

e. Examine the layers added to your ArcMap display.

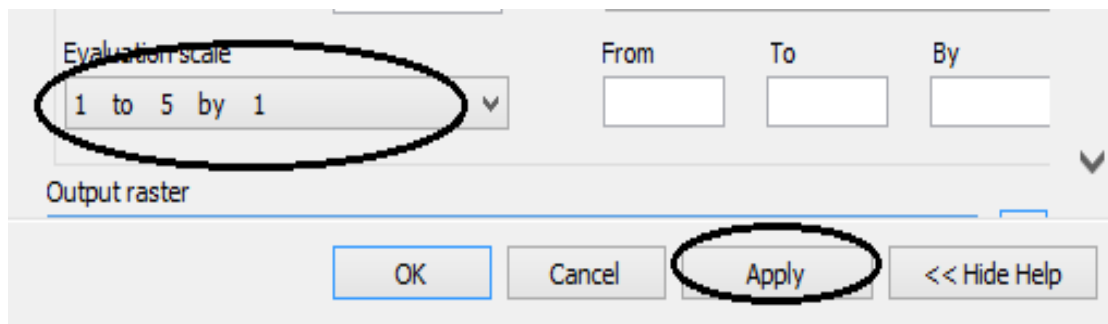


11.7. Weighting and Combining Datasets

You are now ready to combine the reclassified datasets and land use to find the most suitable locations. The values of the reclassified datasets representing **slope**, **distance to road**, **Distance from river** and **built-up distance**- all been reclassified to a common measurement scale (more suitable cells have higher values). The **Landuses** dataset is still in its original form because you can weigh the cell values for this dataset as part of the weighted overlay process. You will weight all the inputs, assigning each a percentage of influence. The higher the percentage, the more influence a particular input will have in the suitability model. You will assign the inputs the following percentages of influence:

Steps:

- a. Click and drag the Weighted Overlay tool, located in the Spatial Analyst toolbox Overlay toolset, into ModelBuilder.
- b. Open the Weighted Overlay tool.
- c. Type 1, 5, and 1 in the **From**, **To**, and **By** text boxes. The default **evaluation scale** is from 1 to 5 by 1.
- d. Set the evaluation scale from 1 to 5 by 1. This means you will avoid having to update the scale values after adding your input datasets.
- e. Click **Apply**.



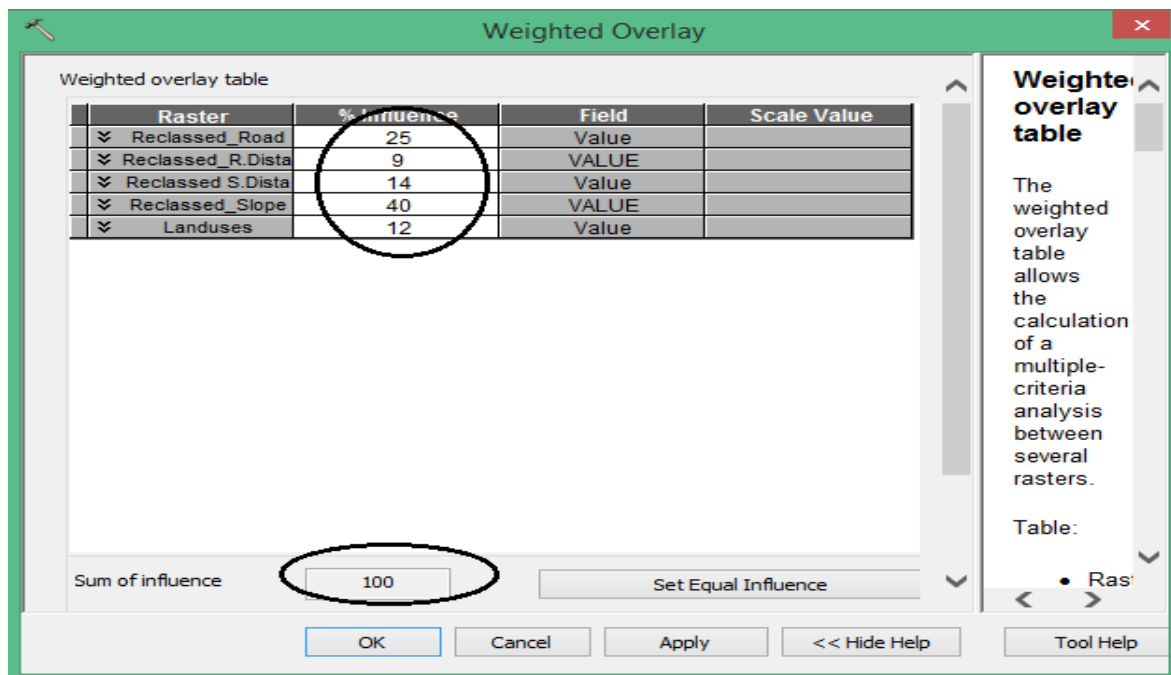
Steps.

- Add the Reclassed slope to the Weighted Overlay tool.
- Click the **Add raster row** button.
- For the input raster, select Reclassed slope from the drop-down list and leave the input field as Value.
- Click **OK**.
- Repeat the previous step to add each of the reclassified datasets including **Reclassified distance to road**, **Reclassified distance from river**, and **Reclassified built-up**.
- Add the **land use layer**, this time setting the **Input field** to **Land use**. Click **OK**. You'll now weight the scale values of the land use layer so they are comparable with the other inputs. A lower value indicates that a particular land-use type is less suitable for housing. Change the default **Scale Values** for the land use layer to the following values:

- ♣ Forest—1
- ♣ Shrub—3
- ♣ Agriculture—2
- ♣ Bare land—5
- ♣ Builtup -Restricted

You'll now assign a percentage of influence to each raster, based on how much importance (Or weight) each should have in the final suitability map. In the **% Influence** column, type the percentages for each of the input raster's:

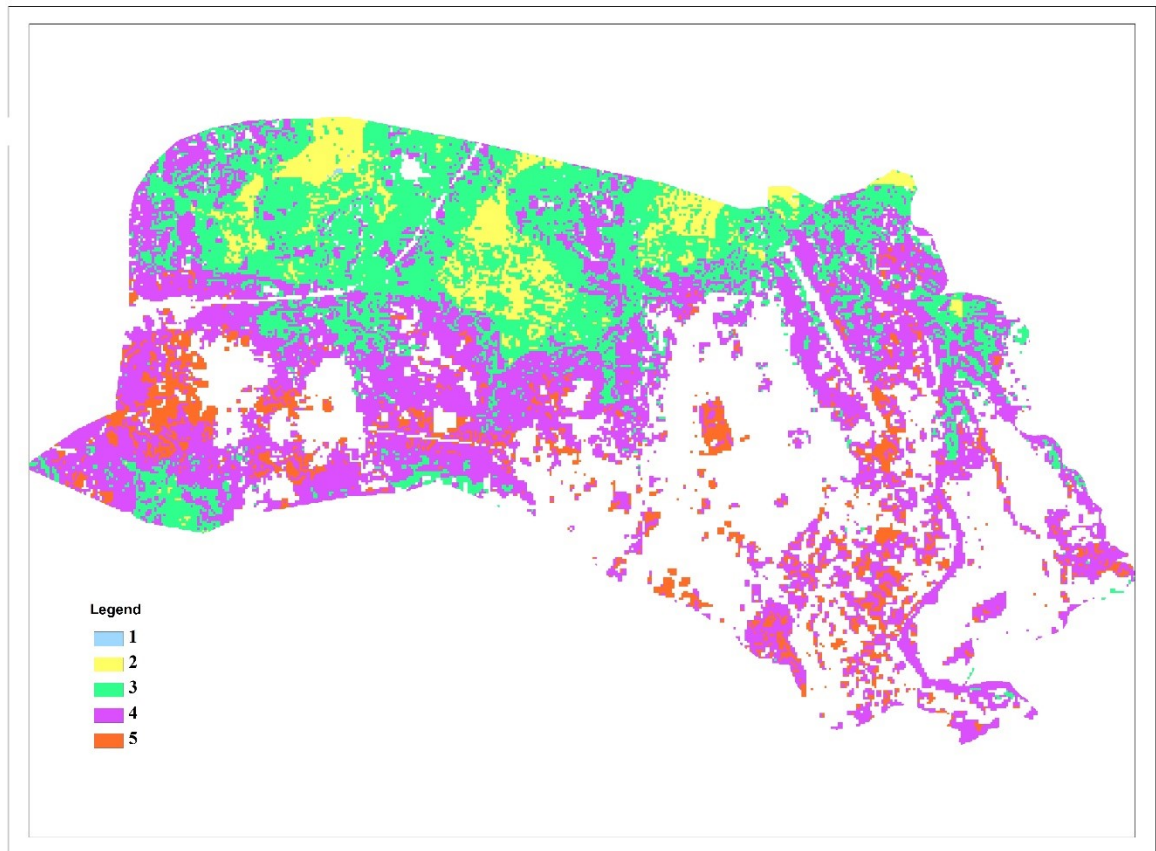
- ✚ Reclassed slope 14
- ✚ Reclassed road to 25
- ✚ Reclassed river to 5
- ✚ Reclassed built-up to 45
- ✚ Landuse to 12
- ✚ Accept the default for the **Output raster** parameter.
- ✚ Click **OK**.



11.8. Executing Weighted Overlay

Steps:

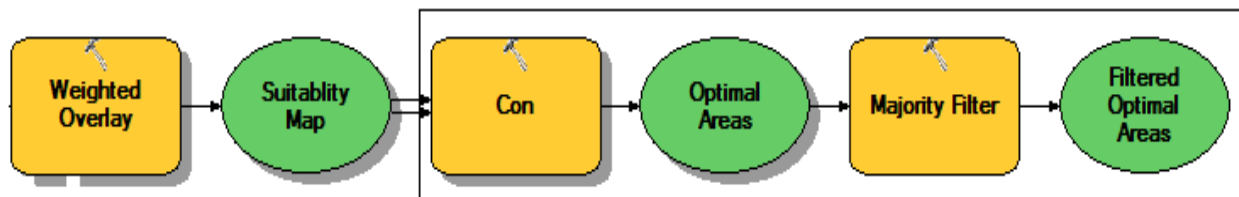
- Click the **Auto Layout** button, then click the **Full View** button.
- Rename the output variable from the Weighted Overlay tool to Suitable Areas and click **OK**.
- Right-click the Suitable Areas variable and click **Add To Display**.
- Run the Weighted Overlay tool.
- On the toolbar, click the **Save**  button. Examine the layer added to your ArcMap display. Locations with higher values indicate more suitable sites—areas that are on less steep slopes of suitable land-use types, closer to road, away from river channels and away from developed area.



11.9. Selecting optimal sites

On your layer, each pixel has a value that indicates how suitable that location is for a new housing. Pixels with the value of **5** are most suitable, and pixels with the value of **0** are not suitable. Therefore, the optimal site location for a housing has the value of **5**. Another criteria for an optimal location is the size of the suitable area. A suitable location would include several pixels with value of **5** being connected. This section of your model will look like the following:

11.9.1. Extract optimal sites using the Con tool

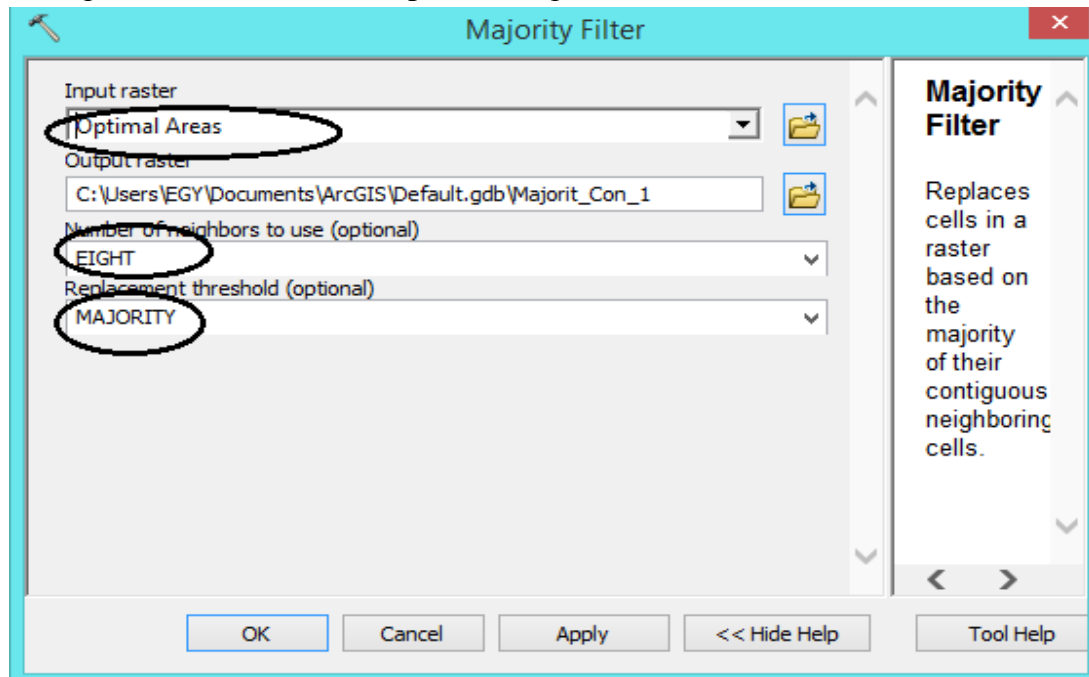


You will use a conditional expression in the Con tool to extract only the optimal sites. It has been decided that those sites that are considered optimal must have a suitability value of 5 (the highest value in the suitability map). In the conditional expression, all areas with a value of 5 will retain their original value (5). Areas with a value of less than 5 will be changed to **NoData**.

Steps:

- Click and drag the **Con** tool, located in the Conditional toolset, into ModelBuilder.
- Open the Con tool.
- Click the **Input Conditional Raster** value drop-down arrow and click the Suitable Areas variable.
- Type the condition **Value = 5** for the value of the **Expression** parameter.
- Click the **Input true raster or constant** value drop-down arrow and click the **Suitable Areas variable**. If the condition you enter is true, then the value of the cells of the Input true raster or constant value will be applied to the cells of the output raster.
- Leave the value for the **Input false raster or constant value** parameter blank.
- Leaving the **Input false raster or constant value** parameter blank will apply the default.
- The default is that if any value in the **Input conditional raster** that doesn't meet the condition you enter will be assigned **NoData** in the output raster.

- i. Accept the default for the **Output raster** parameter.



- j. Click **OK**.
- k. Click the **Auto Layout** button, then click the **Full View** button.
- l. Rename the output variable from the Con tool to **optimal areas** and click **OK**.
- m. Right-click **optimal areas** and click **Add To Display**.
- n. Run the Con tool.

Examine the layer added to your ArcMap display. These are the optimal site locations for the new school. There are many single cells representing optimal locations. These 30-meter cells are too small for the school site. You'll clean up the result, removing these small areas, using the Majority Filter tool.

11.9.2. Refine the optimal areas using Majority Filter tool

Steps:

- Click the Majority Filter tool, located in the Spatial Analyst Tools toolbox Generalization toolset, and add it to ModelBuilder.
- Open the Majority Filter tool.
- Click the **Input raster** drop-down arrow and click the optimal areas raster variable.
- Accept the default **Output raster** parameter.

- e. Click the **Number of neighbors to use** drop-down arrow and click **EIGHT**. This option specifies the number of neighboring cells to use in the kernel of the filter. The kernel of the filter will be the **eight nearest neighbors** (a **3-by-3** cell window) to the present cell.
- f. Accept the default to use the **MAJORITY** as the **Replacement threshold**. Using **MAJORITY** as the **Replacement threshold** means five out of eight connected cells must have the same value for the present cell to retain its value.
- g. Click **OK**.
- h. Rename the output variable from the Majority Filter tool to Filtered optimal areas.
- i. Right-click the Filtered optimal areas and click **Add To Display**.
- j. Run the Majority Filter tool

Examine the layer added to your ArcMap display. Compare Filtered optimal areas and optimal areas. Many optimal areas that were considered too small in area have been removed

11.10. Selecting the Best site

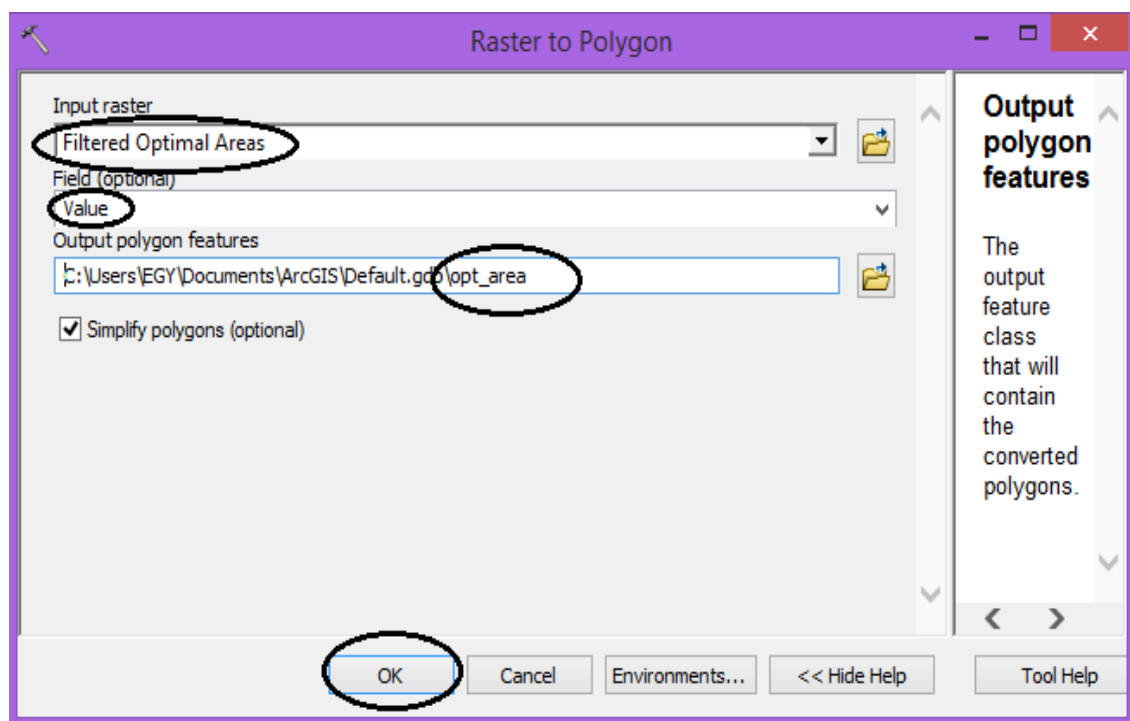
All the locations in the Filtered optimal areas layer are suitable. The last step in this exercise is to locate the best site out of the alternatives. The roads layer displays the roads within the town of Stowe. By examining the Filtered optimal areas layer with the roads layer, you'll see that there are some suitable areas that are not close to roads within the city. You'll first exclude these areas by locating suitable sites that are intersected by roads. Then you'll locate the best site based on area. An optimal housing site is larger than 50,000 square meters. You'll first convert the Filtered optimal areas raster to a feature class inside a Geodatabase so you can use the area field that is generated. You'll use the Select Layer By Location tool to select features that are intersected by roads. Then you'll use the Select Layer By Attribute tool to identify the optimal site from the alternatives, based on area. Finally, you'll create a new feature class from the selection that you'll use in the next exercise.

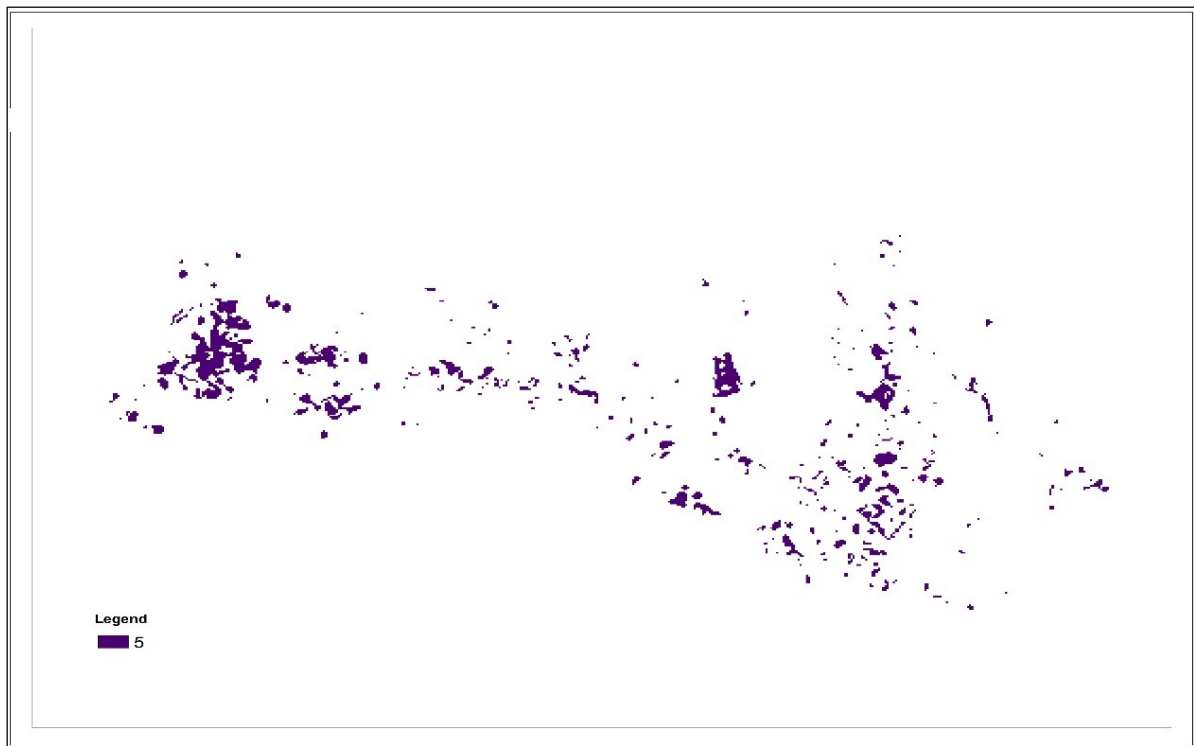
11.10.1. Raster to Polygon

Steps:

- a. Open the Raster To Polygon tool in the Conversion Tools toolbox, inside the From Raster toolset.

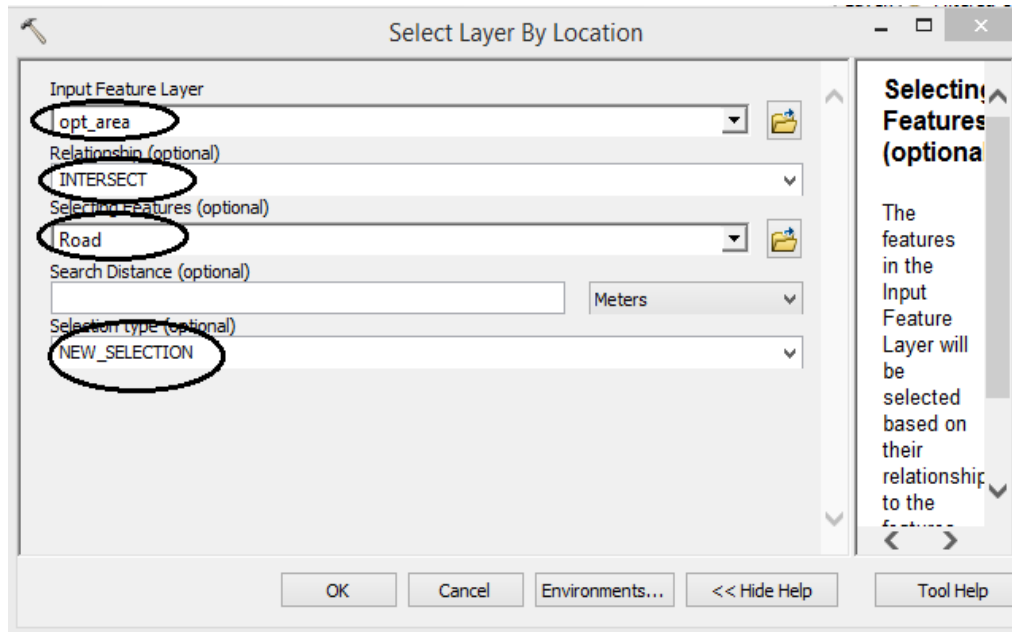
- b. Click the **Input raster** drop-down arrow and click the Filtered optimal areas raster layer.
- c. Leave the default for the **Field** parameter as VALUE.
- d. Accept the default **Output polygon features** parameter path, but change the name to opt_area.
- e. Leave the default checked to **Simplify polygons**. The polygons will be simplified to reduce the stair-step effect when a raster is converted to a polygon.
- f. Click **OK**.





11.10.2. Select By Location

- Open the Select Layer By Location tool in the Data Management toolbox, Layers and Table
- Views toolset.
- For the **Input Feature Layer**, select **opt_area** from the drop-down list.
- For the **Relationship** parameter, accept the default, which is **INTERSECT**.
- Click the **Selecting Features** drop-down arrow and click the Roads layer.
- Accept the default for the **Selection type** parameter, which is **NEW_SELECTION**.
- Click **OK**. All features that intersect roads are selected in the display.

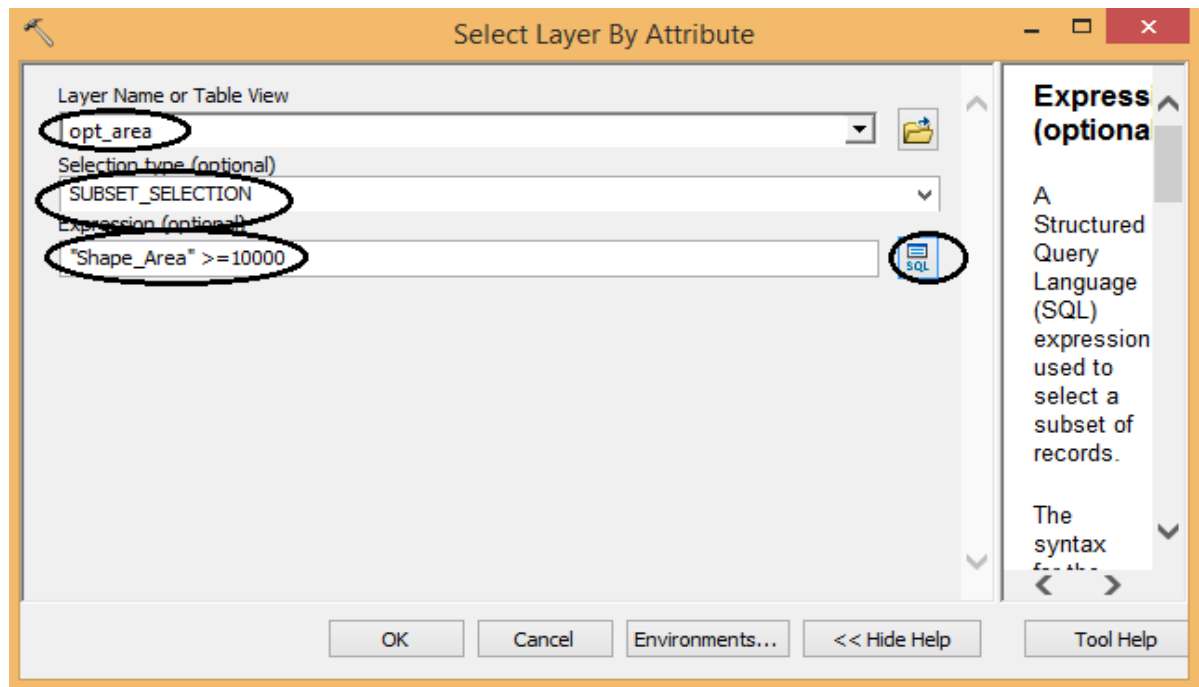


11.10.3. Select Layer By Attribute

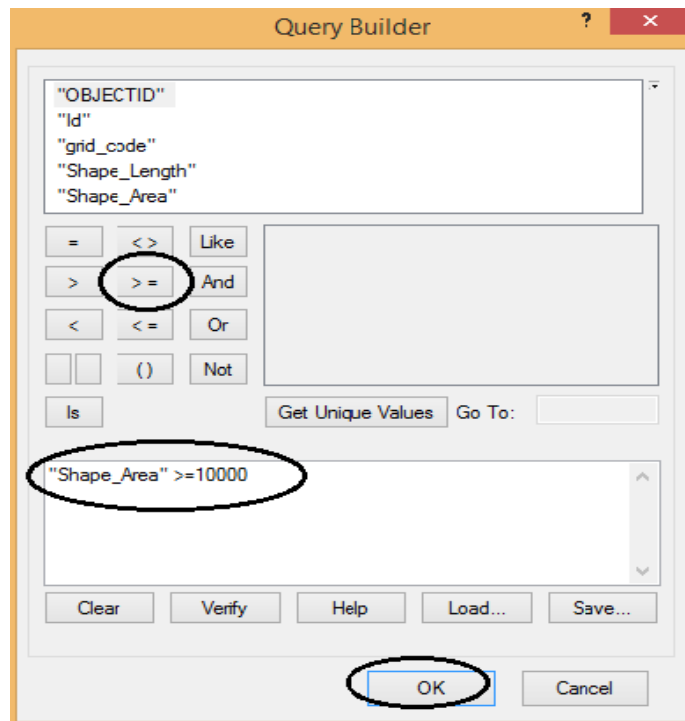
Steps:

- Open the **Select Layer By Attribute** tool in the Data Management toolbox, Layers and Table
- Views toolset.
- For the **Layer Name of Table View**, select **opt_area** from the drop-down list.
- For the **Selection type** parameter, choose **SUBSET_SELECTION**.
- Click the query builder button.
- Double-click Shape Area in the list of fields to enter the field into the expression text box.
- Click the greater than or equal to button.

- h. Type **50,000**



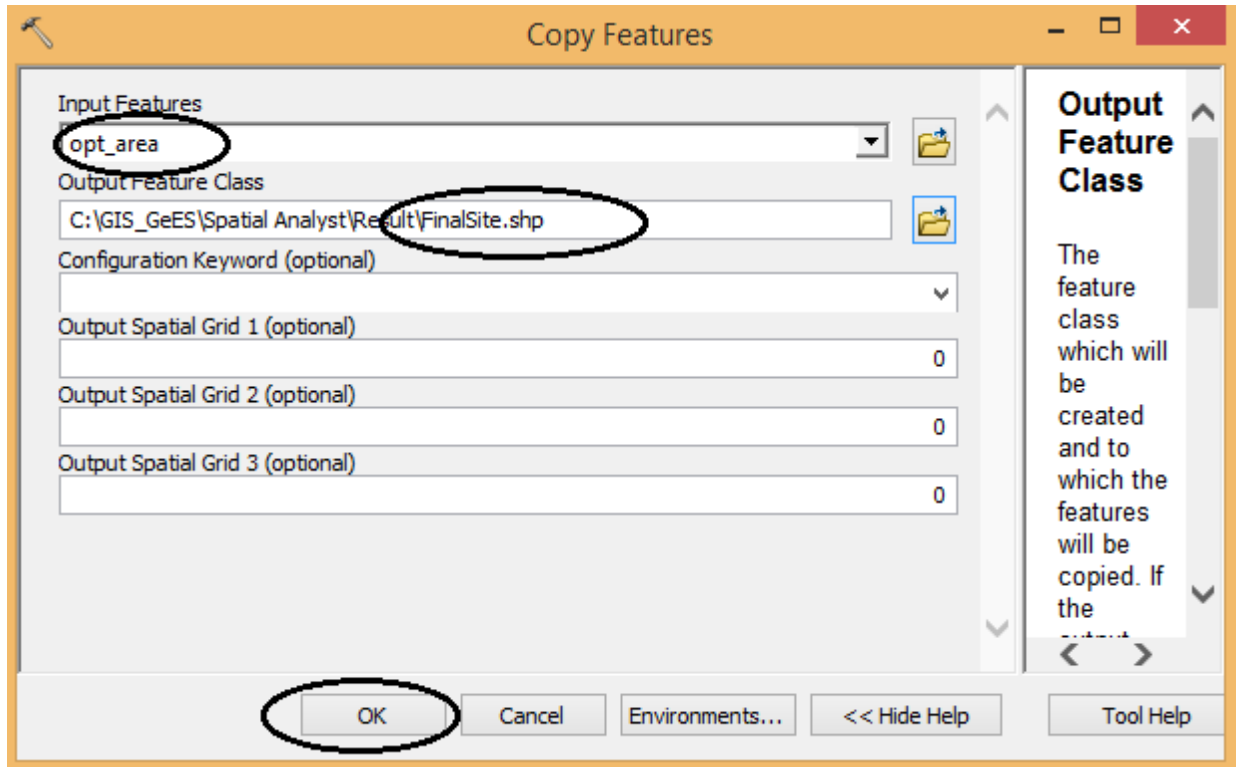
- i. Click **OK**, then click **Close**.



11.10.4. Copy Features Tool

Steps:

- Open the Copy Features tool in the Data Management toolbox, **Features toolset**.
- Click the **Input Features** drop-down arrow and click the **opt_area** layer.
- Click the **Browse** button next to the **Output Feature Class** parameter and navigate to
- C:\Spatial Analyst\Result. This stores your final result.
- Type **final site** for the name of the output feature class and click **Save**.
- Click **OK** to run the tool



- On the Standard toolbar, click the **Save** button 

CHAPTER TWELVE

WATERSHED DELINEATION USING ARC HYDRO TOOLS

Different hydrologic modeling involves delineating streams and watersheds, and getting some basic watershed properties such as area, slope, flow length, stream network density, etc. Traditionally this was (and still is!) being done manually by using topographic/contour maps. With the availability of digital elevation models (DEM) and GIS tools, watershed properties can be extracted by using automated procedures. The processing of DEM to delineate watersheds is referred to as *terrain pre-processing*. We will use Arc Hydro tools to process a DEM to delineate watershed, sub-watersheds, stream network and some other watershed characteristics that collectively describe the drainage patterns of a basin. Thus, the purpose of this training is in order to give a high light about how to delineate watersheds using ArcGIS software.

The prime Objective of this Chapter is to create watershed boundaries from raster digital elevation model datasets and analyze stream networks characteristics within those boundaries

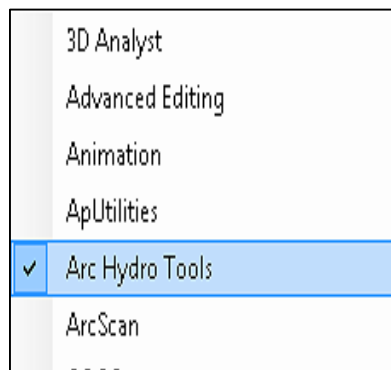
Computer Requirements

You must have a computer with windows operating system, and the following programs installed:

1. ArcGIS 10
2. Arc Hydro tools2.0

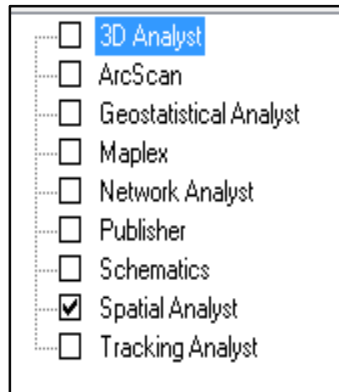
12.1. Getting Started

Open ArcMap. Create a new empty map, and save it as *Watershade.mxd* in C:\GIS_GeES\Watershade folder. **Right click** on the menu bar to pop up the context menu showing available tools as shown below.



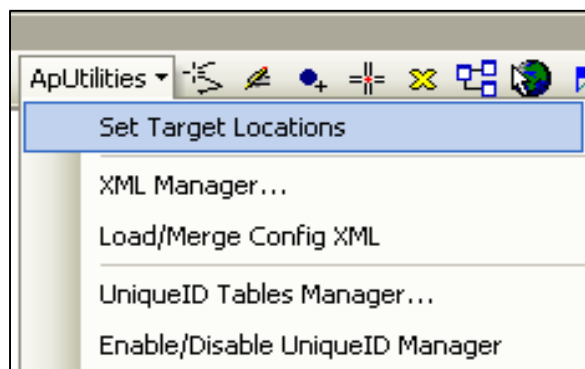


Also activate the Spatial Analyst Extension needs to be activated, by clicking Customize → Extensions..., and checking the box next to Spatial Analyst.



12.2. Dataset Setup

All vector data created with the Arc Hydro tools will be stored in a new Geodatabase that has the same name as the stored project or ArcMap document (unless pointed to an existing Geodatabase) and in the same directory where the project has been saved (C:\GIS_GeES\Watershade folder). By default, the new raster data are stored in a subdirectory with the same name as the dataset or Data Frame in the ArcMap document (called Layers by default and under the directory where the project is stored). The location of the vector, raster, and time series data can be explicitly specified using the function *ApUtilities* → *Set Target Locations*.



You can leave the default settings if they are pointing to the same directory where the ArcMap document is saved.

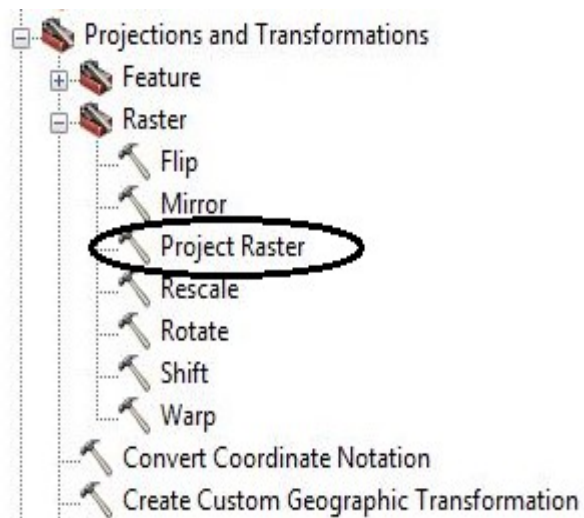
12.3. Adding Elevation data to ArcMap

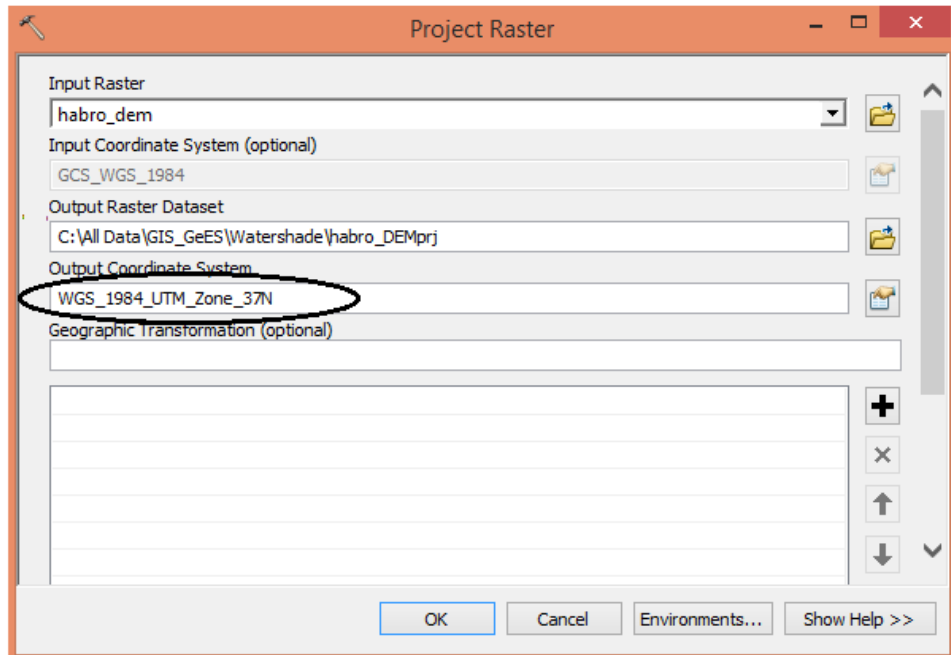
Elevation data (DEM) is a raster image that can be downloaded from USGS Website/www.usgs.gov or from DEM explorer website with different resolution/pixel size. The spectral resolution of the elevation data for this training is with, 90*90m resolution.

Click on the Add icon  to add the raster data. In the dialog box, **navigate** to the location of the data *C:\GIS_GeES\Watershade\DEM*; **select** the raster file *elevation* containing the DEM of Gobeles river watershed and **click** on the Add button. The added file will then be listed in the Arc Map Table of contents.

Set Projection to *UTM _ WGS1984 _ Zone 38N*

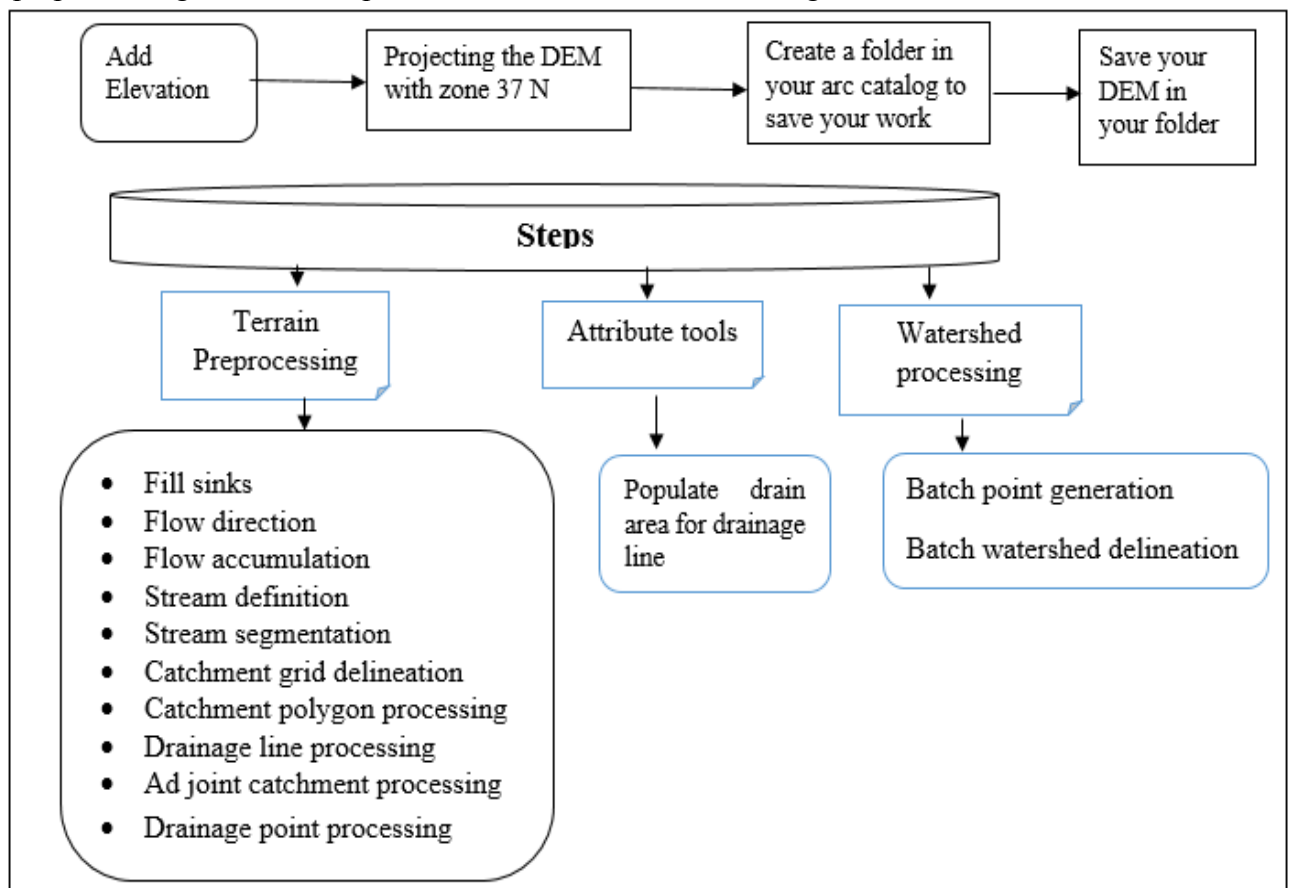
- i. Project the image using the specified zone and hemisphere. If the image is new and not projected you should do it as follows.
- ii. *Go to Arc tool box → data management tools → projection and transformation → Raster → Project raster.*
- iii. *When 'define projection' window popup add an image by locating in your folder that you same it in.*
- iv. *Locate your data under 'Input data set' → 'specify the 'coordinate system' in to projected coordinate system → northern hemisphere → UTM → UTM _ WGS1984 _ Zone 38N proj. → Add → OK.*





12.4. The process of Watershade Delineation

Arc Hydro Terrain Preprocessing should be performed in sequential order. All of the preprocessing must be completed before Watershed Processing functions can be used.



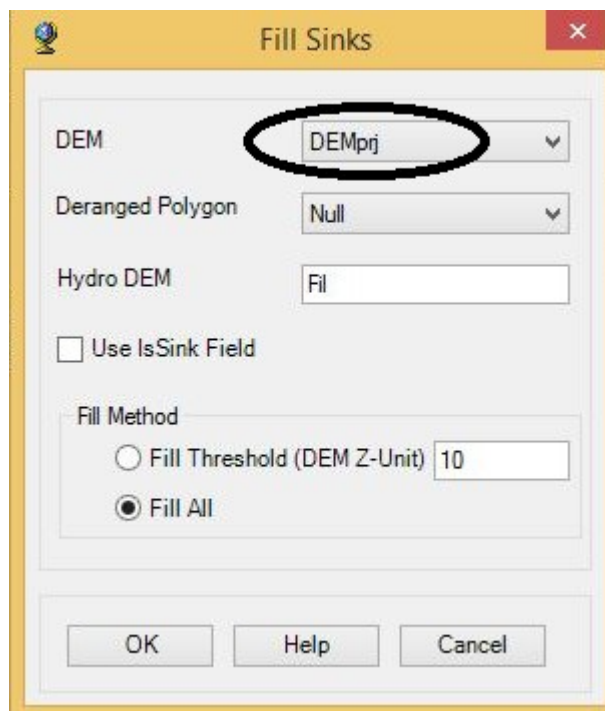
Watershed delineation in ARCGIS environment with an extension of Arc hydro software that which can be downloaded freely from ESRI website and it can be easily installed in to your ARCGIS software ware involves the following procedures.

12.4.1. Fill Sinks

This tool fills in the sinks in the input DEM and generates the output Fill grid. This function fills the sinks in a grid. If cells with higher elevation surround a cell, the water is trapped in that cell and cannot flow. The Fill Sinks function modifies the elevation value to eliminate these problems.

On the ArcHydro Toolbar, select Terrain Preprocessing →Data Manipulation →Fill Sinks.

Confirm that the input for DEMprj is Elevation and the output is the Hydro DEM layer, named by default *Fil*. Leave Fill Threshold and Input Deranged Polygon blank and click OK. This default name can be overwritten. **Leave** the other options **unchanged**.

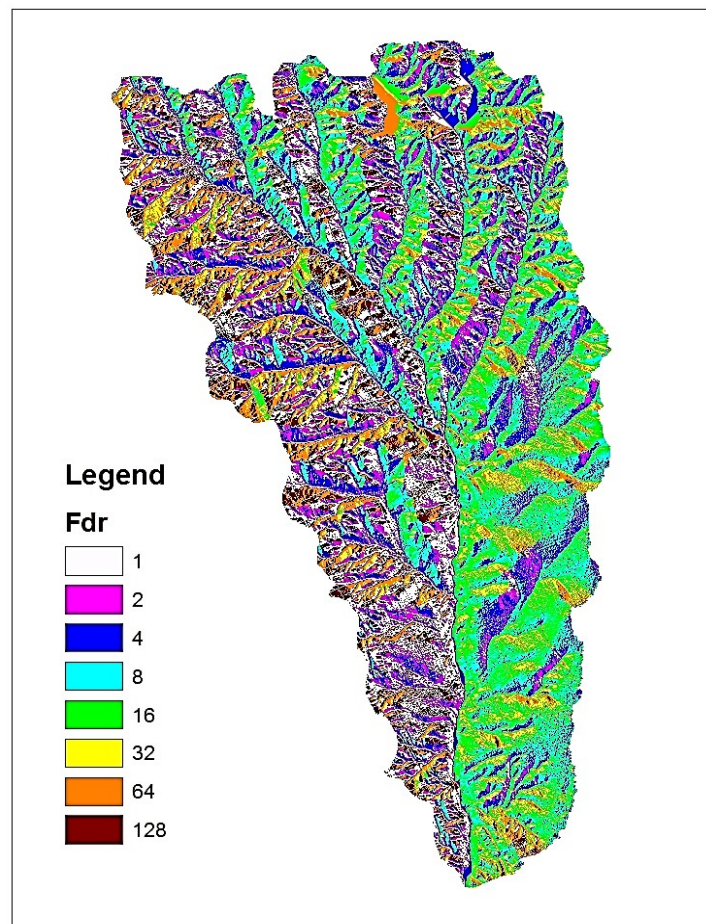
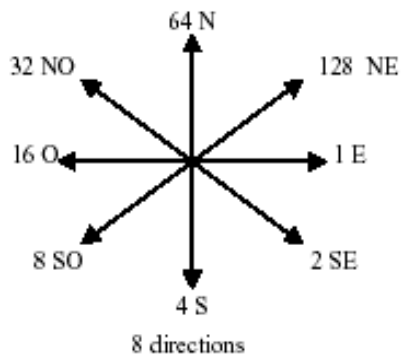
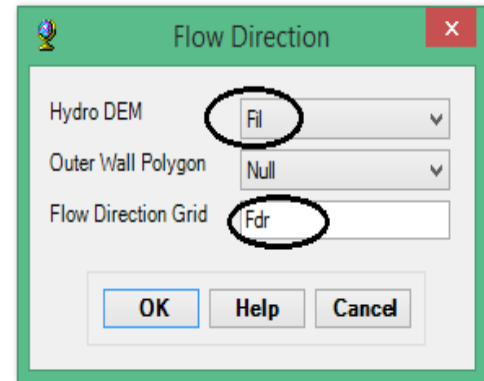


Press OK. Upon successful completion of the process, the *Fil* layer is added to the map. This process takes a few minutes.

12.4.2. Flow Direction

This function computes the flow direction for a given grid. The values in the cells of the flow direction grid indicate the direction of the steepest descent from that cell.

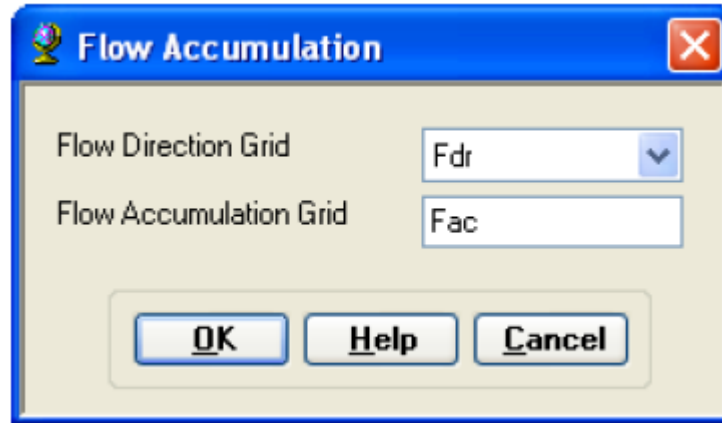
- Click Terrain Preprocessing → Flow Direction.
- Specify Fill All as input Hydro DEM and rename the output flow direction grid FdrFilled.
- Leave the optional input Outer Wall Polygon blank and click OK.



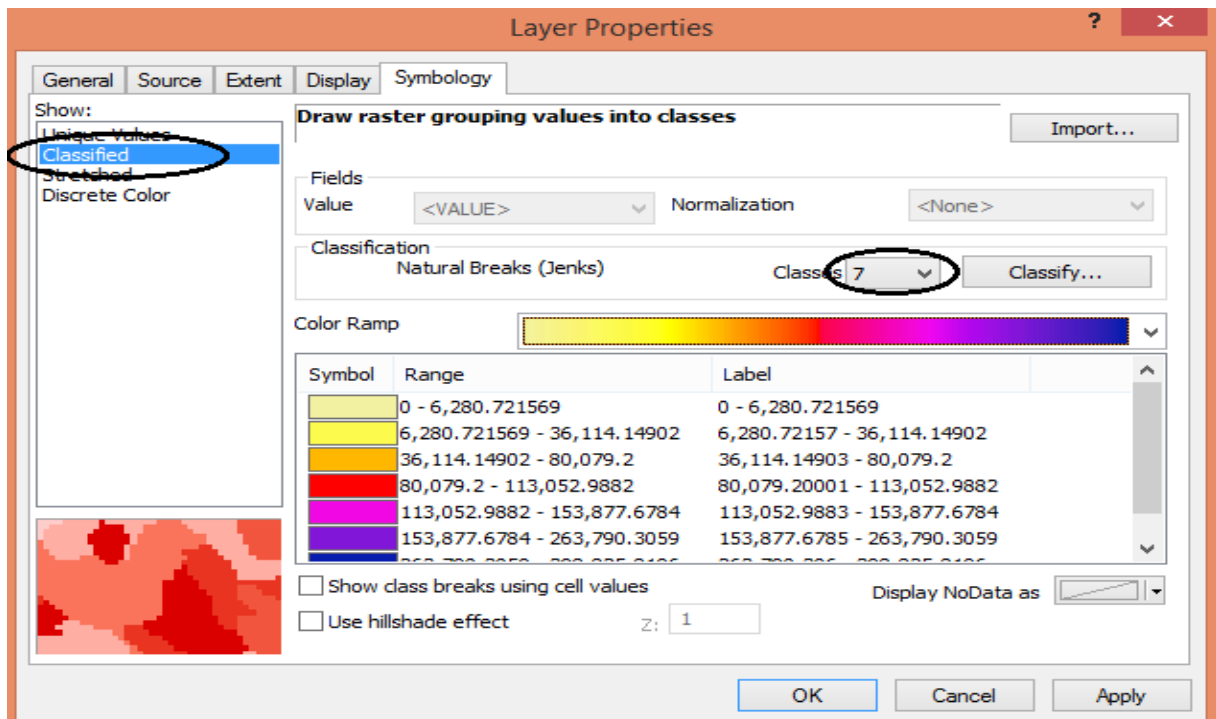
12.4.3. Flow Accumulation

This tool generates the Flow Accumulation grid associated to the input flow direction grid: each cell in the flow accumulation grid stores the number of cells located upstream of that cell.

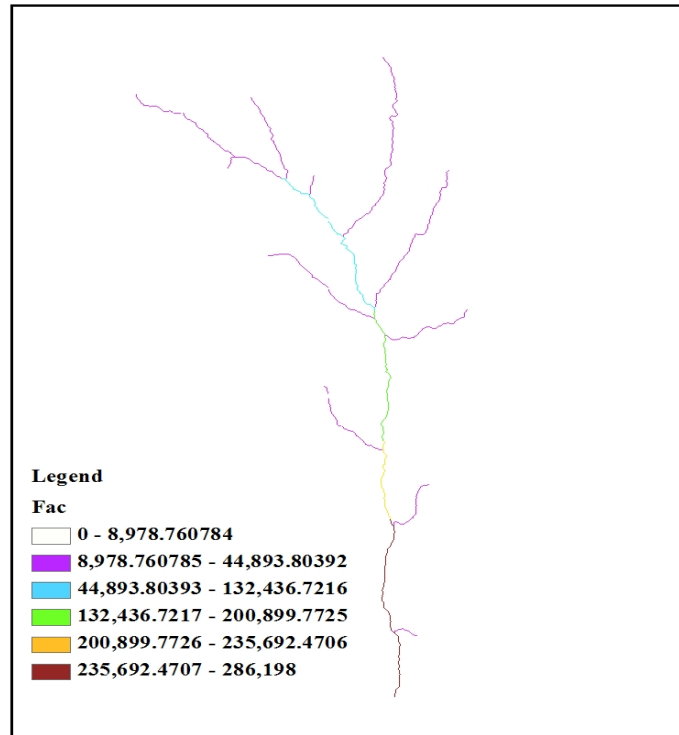
- Click Terrain Preprocessing → Flow Accumulation.
- Confirm** the FdrFilled as input Flow Direction Grid. Click OK.
- The tool generates the output flow accumulation grid and adds it to the map



Press OK. Upon successful completion of the process, the flow accumulation grid *Fac* is added to the map. This process may take several minutes for a large grid! **Adjust** the symbology of the Flow Accumulation layer *Fac* to a multiplicatively increasing scale to illustrate the increase of flow accumulation as one descends into the grid flow network.



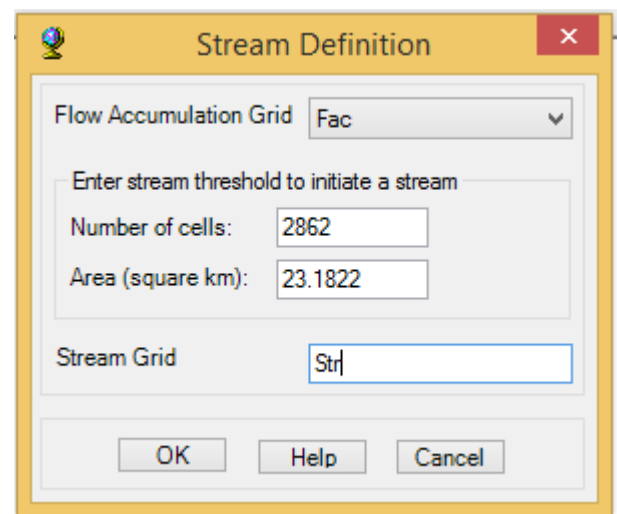
Zoom-in to a stream network junction to see how the symbology changes from light to dark color as the number of upstream cells draining to a stream increase from upstream to downstream as shown below.



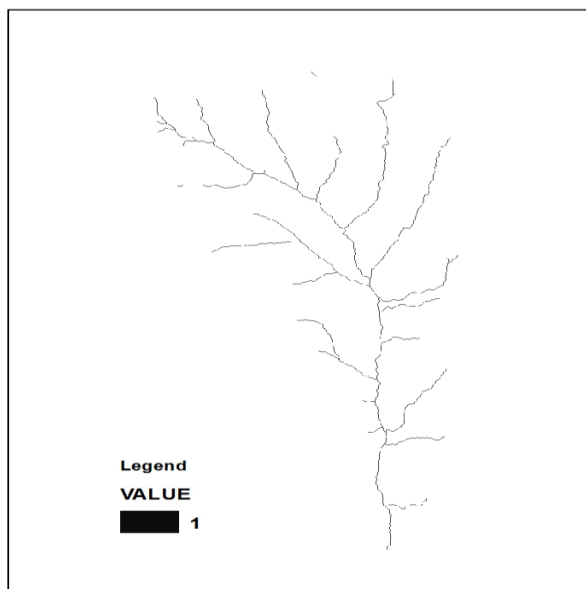
12.4.4. Stream Definition

This tool generates the stream grid for an input flow accumulation grid and a threshold. Unlike the related function in the standard Arc Hydro tools, you need to specify the threshold in number of cells defining where a stream should start without any feedback from the function. The recommended value is usually around 1% of the maximum flow accumulation value.

- Click **Terrain Preprocessing** → **Stream Definition**.
- Select Fac as input Flow Accumulation Grid and specify a threshold that is approximately 1% of the maximum value of the Flow Accumulation grid/Cell a default value is displayed for the river threshold (i.e. 2862 for the tutorial data).
- The output is the Stream Grid named *Str*, default name that can be overwritten. Click OK.



The USGS Elevation Derivatives for National Applications (EDNA <http://edna.usgs.gov/>) approach uses a threshold of 5000 30 x 30 m cells (an area of 4.5 km²) for catchment definition. A smaller threshold will result in a denser stream network and usually in a greater number of delineated catchments. Upon successful completion of the process, the stream grid *Str* is added to the map. This *Str* grid contains a value of "1" for all the cells in the input flow accumulation grid (Fac) that have a value greater than the given threshold (3299 as shown in above figure). All other cells in the Stream Grid contain no data. The cells in *Str* grid with a value of 1 are symbolized with black color to get a stream network as shown below:

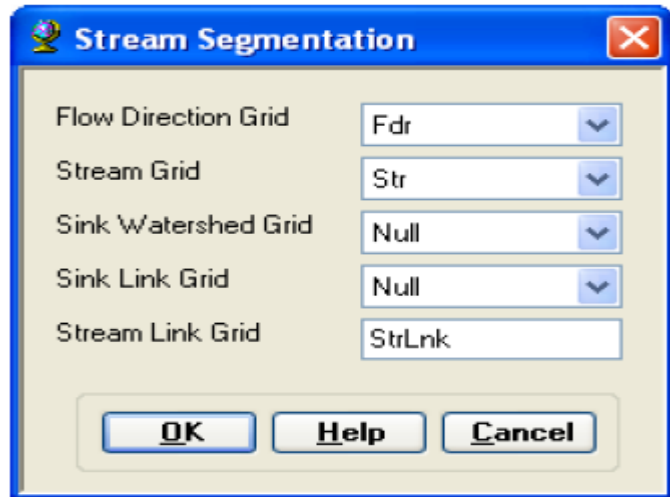


12.4.5. Stream Segmentation

This function creates a grid of stream segments that have a unique identification. Either a segment may be a head segment, or it may be defined as a segment between two segment junctions. All the cells in a particular segment have the same grid code that is specific to that segment. This tool generates the stream link grid for an input stream grid and flow direction grid. All cells in a given link segment have the same value that uniquely identify the segment.

- a. Double-click **Terrain Preprocessing** → **Stream Segmentation**

- b. **Confirm** that the input to the Flow Direction Grid and Link Grid are *Fdr* and *Lnk* respectively. The output is the Catchment Grid layer. *Cat* is its default name that can be overwritten by the user.

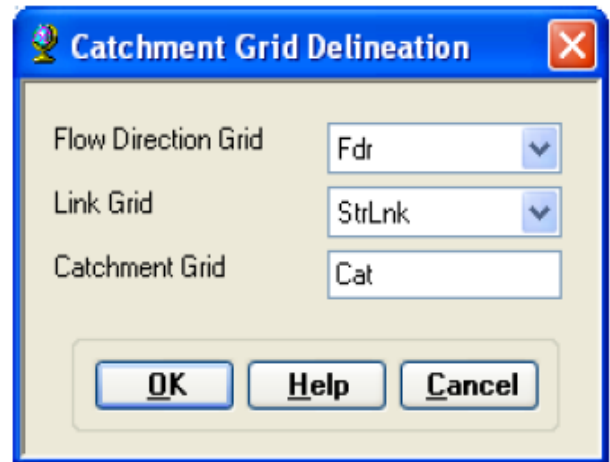


- c. **Press OK.** Upon successful completion of the process, the Catchment grid *Cat* is added to the map. If you want, you can recolor the grid with unique values to get a nice display (use *properties* → *symbolology*).

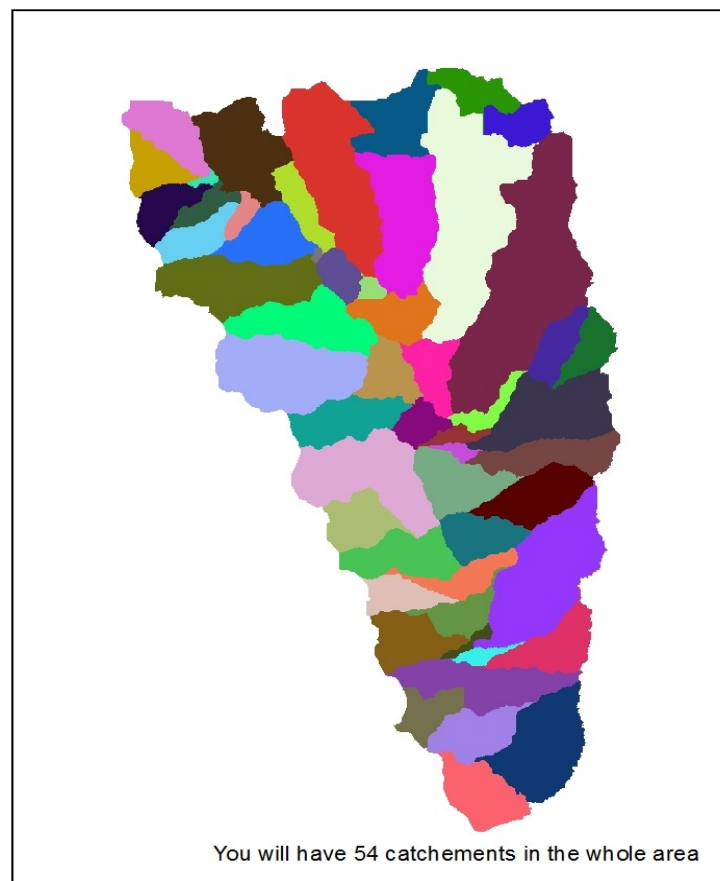
12.4.6. Catchment Grid Delineation

This tool generates the catchment grid associated to an input flow direction grid and link grid.

- a. Click Terrain Preprocessing → Catchment Grid Delineation.
- b. Select FdrFilled as input Flow Direction grid and StrLink as input Link Grid. Click OK.



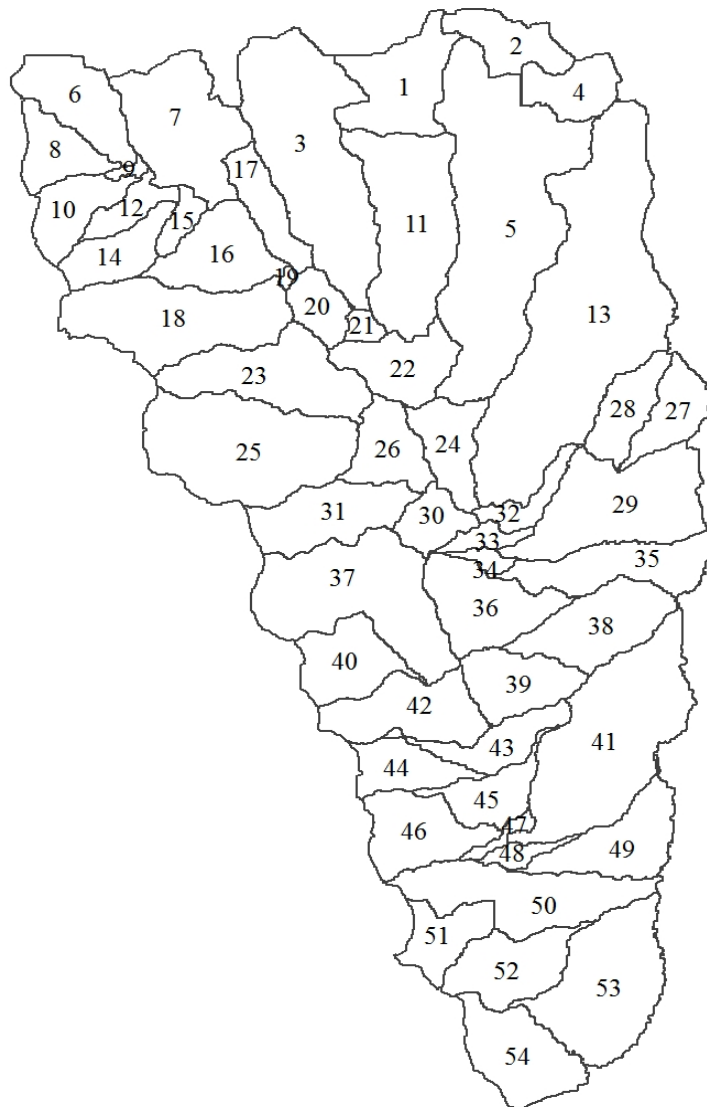
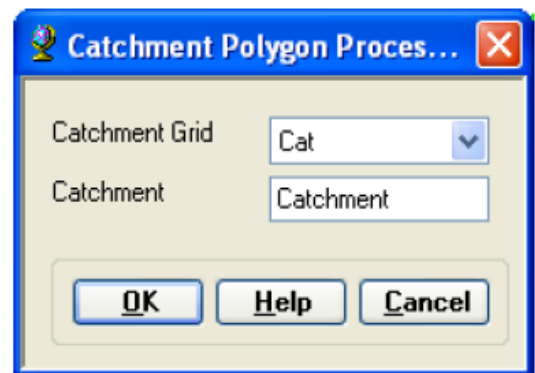
- c. **Press OK.** Upon successful completion of the process, the Catchment grid *Cat* is added to the map. If you want, you can recolor the grid with unique values to get a nice display (**use properties → symbology**).



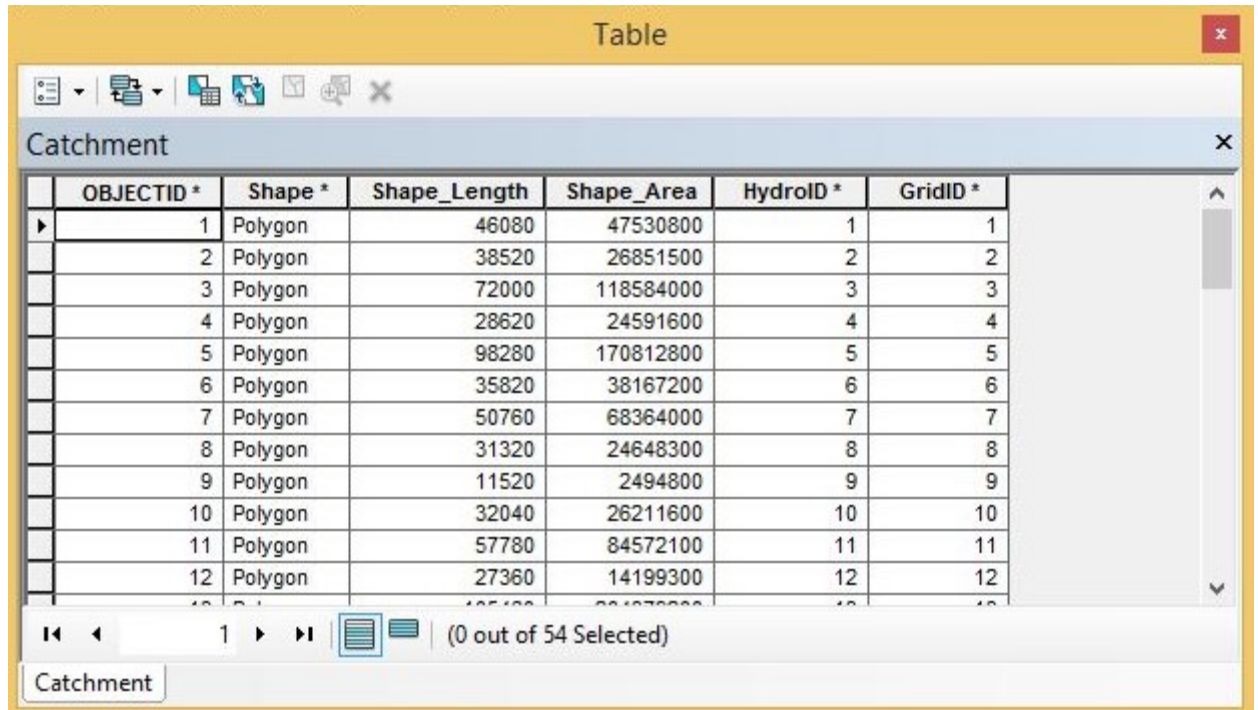
12.4.7. Catchment Polygon Processing

This tool generates the catchment polygon feature classes corresponding to the input catchment grid.

- Click Terrain Preprocessing → **Catchment Polygon Processing**.
- Specify the input catchment grid and output catchment polygon feature class.
- Press OK**. Upon successful completion of the process, the polygon feature class *Catchment* is added to the map.



Open the attribute table of *Catchment*. Notice that each catchment has a *HydroID* assigned that is the unique identifier of each catchment within *Arc Hydro*. Each catchment also has its Length and Area attributes. These quantities are automatically computed when a feature class becomes part of a Geodatabase.

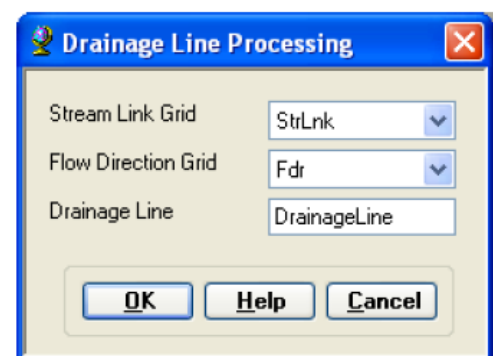


OBJECTID *	Shape *	Shape_Length	Shape_Area	HydroID *	GridID *
1	Polygon	46080	47530800	1	1
2	Polygon	38520	26851500	2	2
3	Polygon	72000	118584000	3	3
4	Polygon	28620	24591600	4	4
5	Polygon	98280	170812800	5	5
6	Polygon	35820	38167200	6	6
7	Polygon	50760	68364000	7	7
8	Polygon	31320	24648300	8	8
9	Polygon	11520	2494800	9	9
10	Polygon	32040	26211600	10	10
11	Polygon	57780	84572100	11	11
12	Polygon	27360	14199300	12	12

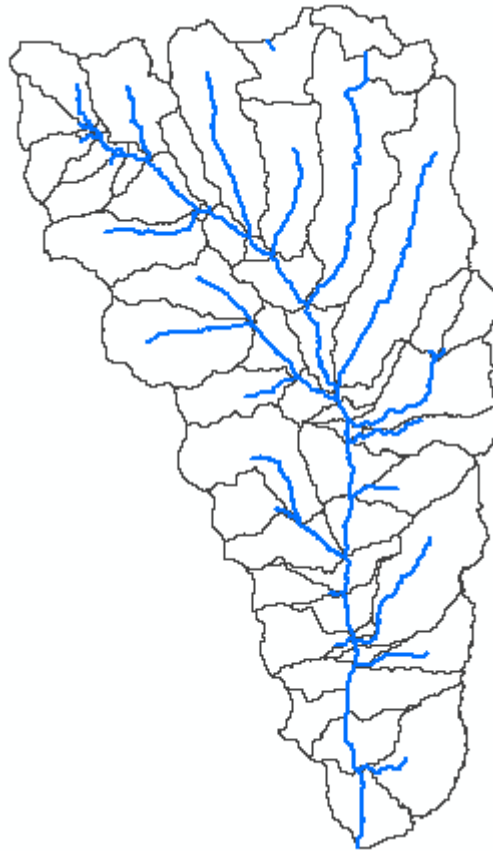
12.4.8. Drainage Line Processing

This tool generates the drainage line feature class associated to an input stream link grid and flow direction grid.

- Click Terrain Preprocessing → Drainage Line Processing.
- Select StrLnk as input Stream Link Grid and FdrFilled as input Flow Direction Grid.
- Rename the output Drainage Line and click OK.



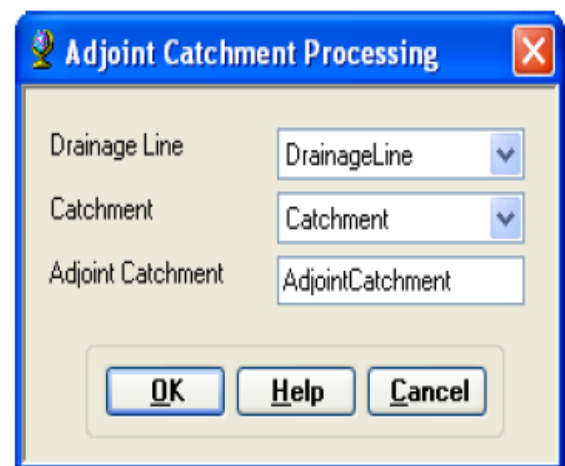
Press OK. Upon successful completion of the process, the linear feature class *Drainage Line* is added to the map.



12.4.9. Adjoint Catchment Processing

This function generates the aggregated upstream catchments from the *Catchment* feature class. For each catchment that is not a head catchment, a polygon representing the whole upstream area draining to its inlet point is constructed and stored in a feature class that has an *Adjoint Catchment* tag. This feature class is used to speed up the point delineation process.

- On the ArcHydro toolbar, **select** *Terrain Preprocessing* → *Adjoint Catchment Processing*.
- **Confirm** that the inputs to Drainage Line and Catchment are respectively *Drainage Line* and *Catchment*. The output is Adjoint Catchment, with a default name *AdjointCatchment* that can be overwritten



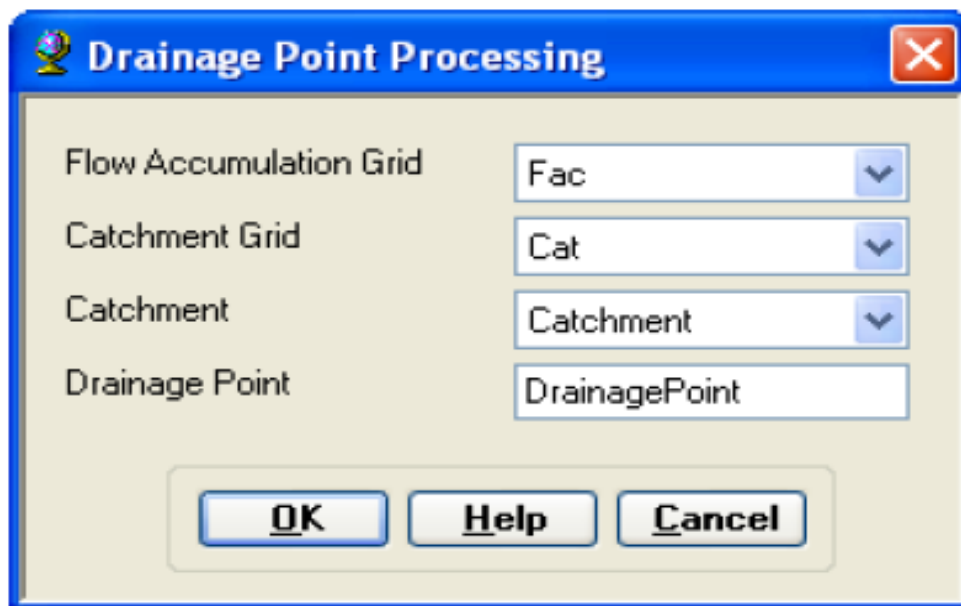
Press OK. Upon successful completion of the process, you will see a message box similar to the one below that will give you a summary of the number of catchments that were aggregated to create the Adjoint Catchments.

11.4.10. Drainage Point Processing

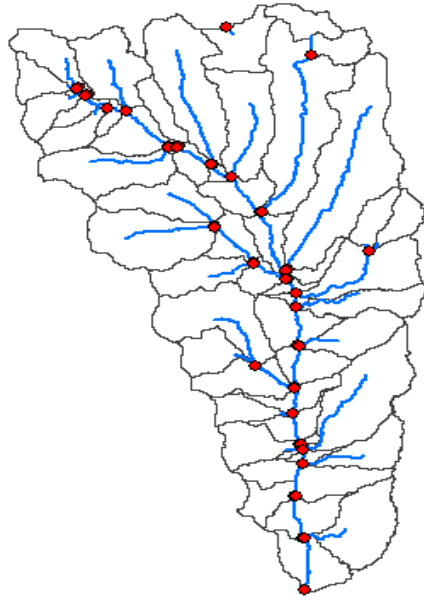
This tool generates the drainage point feature class associated to an input catchment feature class and flow accumulation grid. Drainage Point features represent the location of the cell with the maximum flow accumulation value within each catchment.

✚ Click Terrain Preprocessing → Drainage Point Processing and click OK.

Confirm that the inputs are as below. The output is Drainage Point with the default name *DrainagePoint* that can be overwritten.



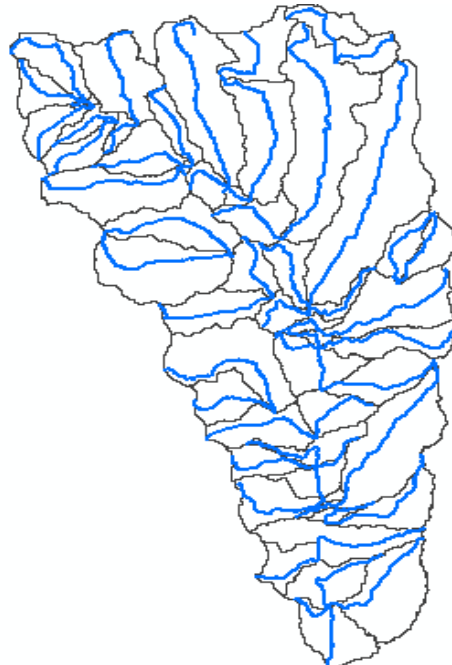
Press OK. Upon successful completion of the process, the point feature class “DrainagePoint” is added to the map.



12.5. Longest flow path for catchments.

This tool generates the Longest Flow Path associated to the input Drainage Area features. It works on the selected set or on all features if there is no selection

- Add the flow direction grid (fdr) into the map.
- Click **terrain preprocessing** → **Longest Flow Path for catchment**.
- Specify Catchment as Input Drainage Area, fdr as Input Flow Direction Grid and keep the default name for the output. Click OK.



12.6. Watershed Processing

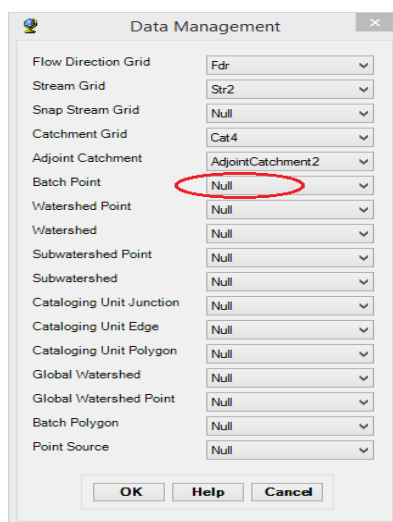
Arc Hydro toolbar also provides an extensive set of tools for delineating watersheds. These tools rely on the datasets derived during terrain processing. This part of the exercise will expose you to some of the Watershed Processing functionality in Arc Hydro.

12.6.1. Batch watershed delineation

Arc Hydro manages the input/output to the tools by using tags that are automatically assigned by the functions to the selected inputs and outputs. A tag may be used as input by one function and as output by another one. For example, the "Longest Flow Path" is an output from Longest Flow Path, and an input to Flow Path Parameters. The Data Management function in the [Watershed Processing](#) menu provides a global view of the tags assignments for that menu in the active Map/Data Frame. The function also allows assigning, reassigning or resetting the tags. A tag may be reset by selecting "Null" as the corresponding layer. When a reset tag is used as output, the function presents the user with default layer name associated to the tag. This default is defined in the XML file and may be modified (see [XML Manager](#)).

This tool allows creating the Batch Point feature class that is used as input by the batch delineation functions in the Watershed Processing menu. You will not be prompted for the name of the output Batch Point feature class if it is already set. You can reset it by using the Data Management functions.

- Select Watershed Processing → Data Management Watershed Delineation and reset Batch Point to Null

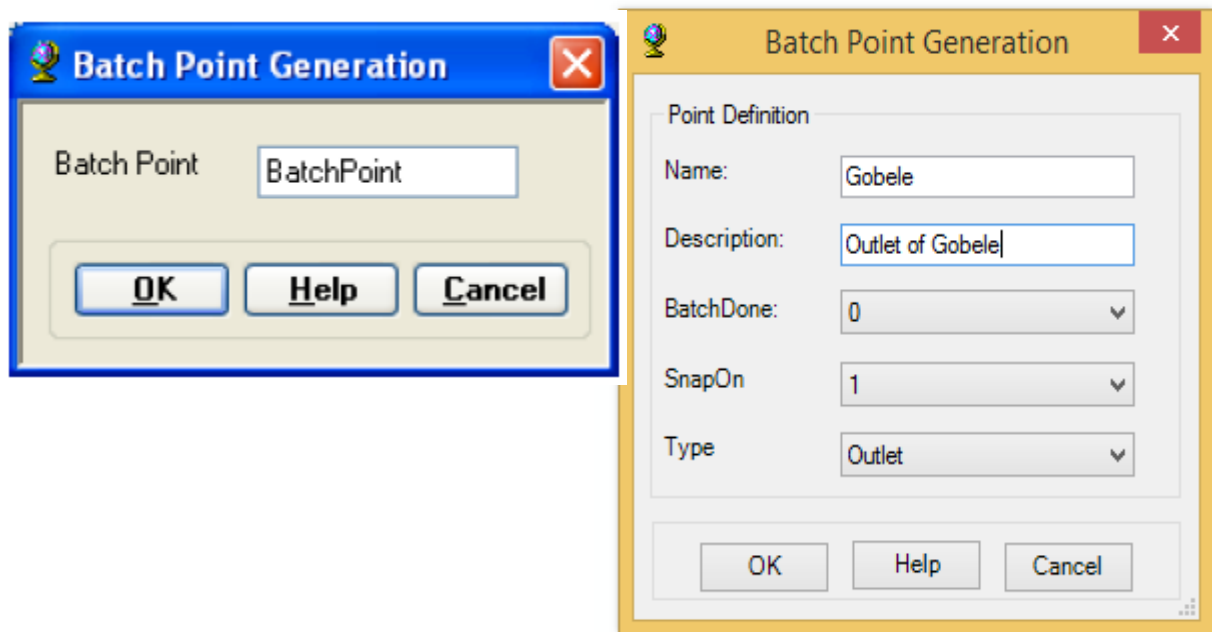


Note

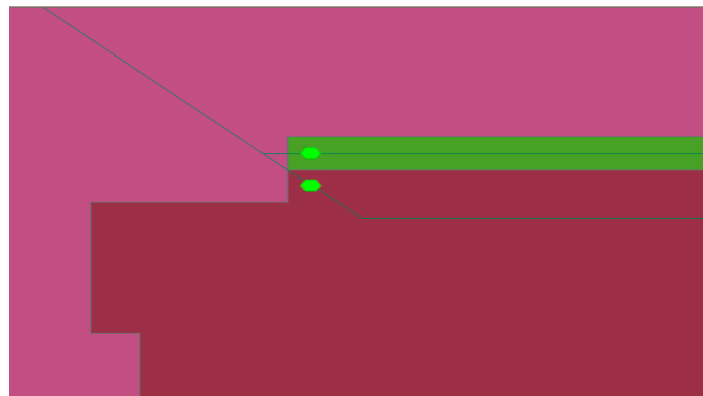
If you previously used Batch Point, which is the default name for the Batch Point feature class, the function will automatically use Batch Point and not prompt you for a name. In this case, you need to change the default name in the XML to be able to reset Batch Point:

After resetting Batch Point, follow the steps below:

- Click on the icon  in the Arc Hydro Tools toolbar.
- Click with the mouse to create a point on the map. Confirm, if prompted, that the name of the batch point feature class is “Batch Point Tool” and click OK.



While we do on batch point generation we zoom in until we separate each drainage point in one point which can more one drainage point as shown below.

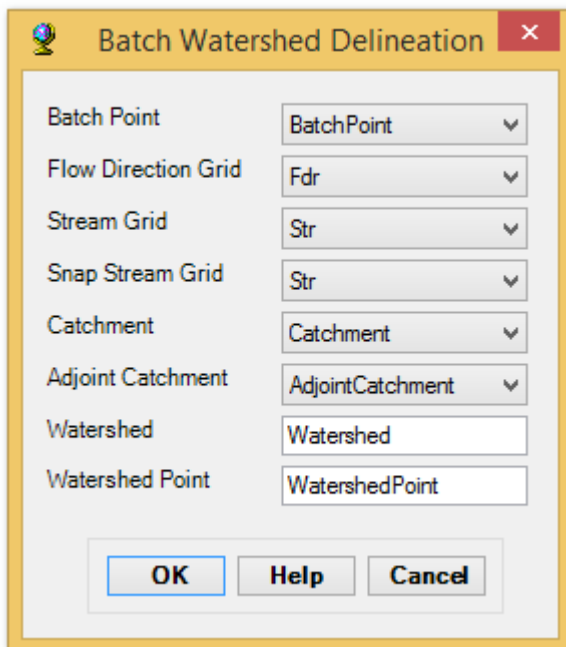


Confirm that the name of the batch point feature class is *BatchPoint*.

The following form is displayed. Fill in the fields Name and Description. Both are text fields. The Batch done and Snap On options can be used to turn on (select 1) or off (select 0) the batch processing and stream snapping for that point. Select the options shown in the form. The point type may be Inlet (1) or Outlet (0). The inlet type is used only by the function Delineated from Multiple Inlets and Outlets in the Watershed Processing menu.

12.6.2. Delineating the watershed using batch point.

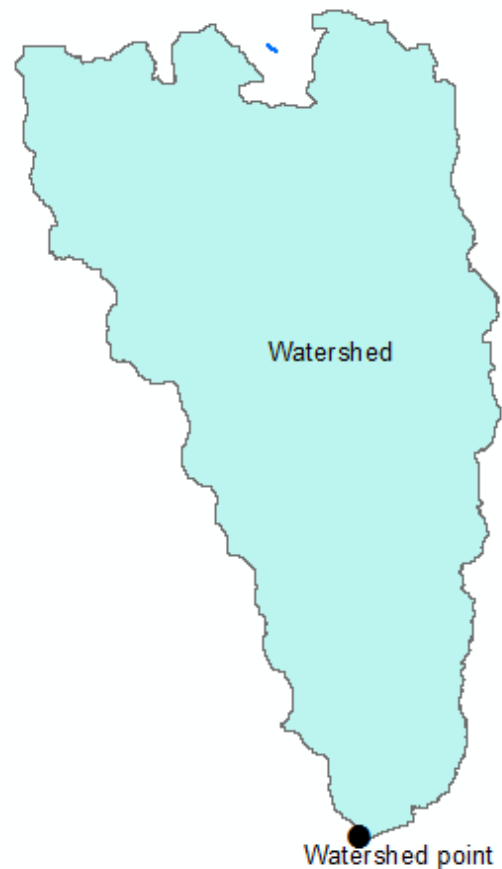
Finally, move to watershed shed processing tool and click on batch watershed generation. The following windows will pop up immediately. Make sure that all the field is filled with appropriate variables.



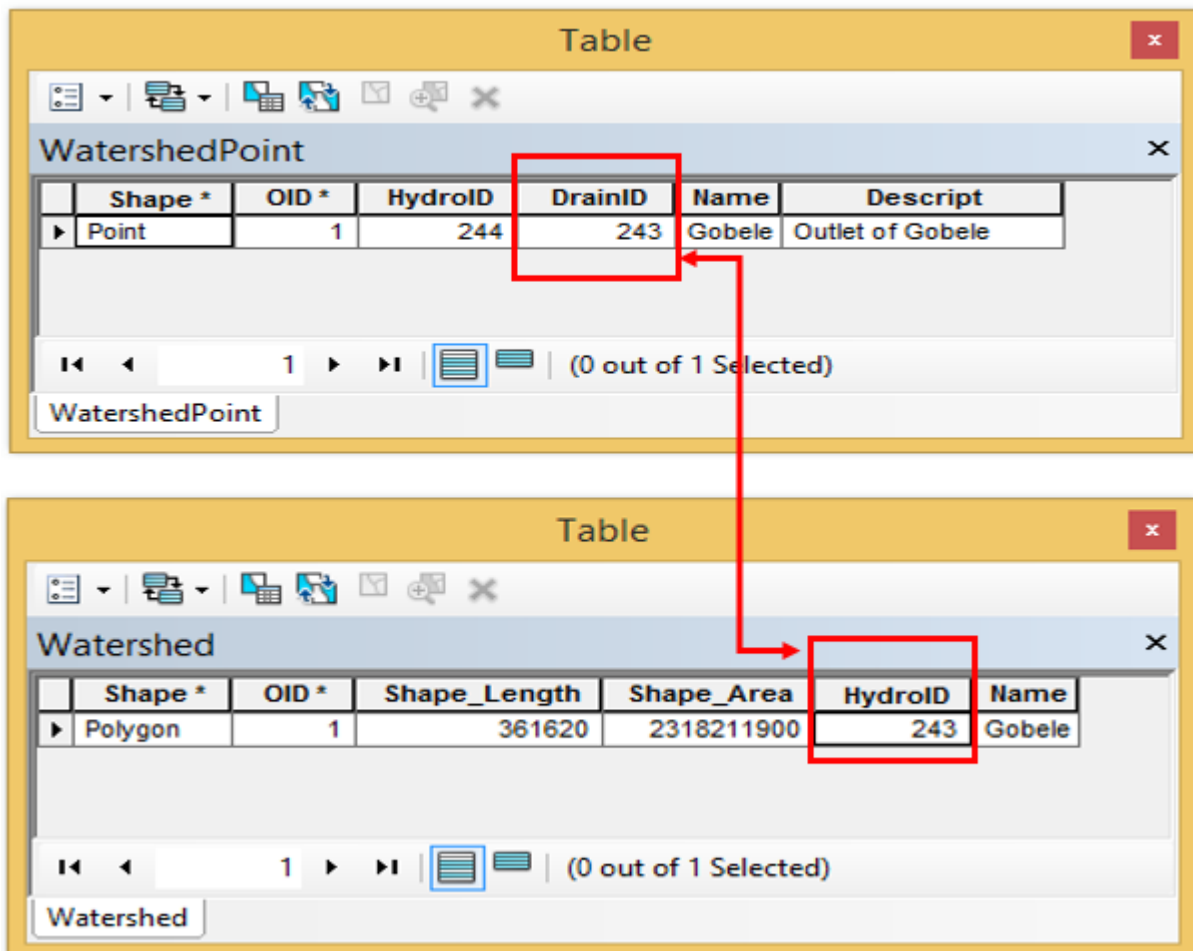
The dialog box titled "Batch Watershed Delineation" contains the following fields and options:

Field	Value
Batch Point	BatchPoint
Flow Direction Grid	Fdr
Stream Grid	Str
Snap Stream Grid	Str
Catchment	Catchment
Adjoint Catchment	AdjointCatchment
Watershed	Watershed
Watershed Point	WatershedPoint

Buttons: OK, Help, Cancel



The new watershed is stored in the *Watershed* feature class and another feature class named *WatershedPoint* is also added to the map. **Open** the attribute table of *Watershed* and *WatershedPoint*, and you will see that these two are related through *HydroID* – the *DrainID* of *WatershedPoint* is equal to the *HydroID* of the *watershed*.



The image displays two screenshots of the ArcGIS attribute table interface, illustrating the relationship between the *WatershedPoint* and *Watershed* feature classes.

Top Screenshot: WatershedPoint Attribute Table

Shape *	OID *	HydroID	DrainID	Name	Descript
Point	1	244	243	Gobebe	Outlet of Gobebe

The **DrainID** value (243) is highlighted with a red box.

Bottom Screenshot: Watershed Attribute Table

Shape *	OID *	Shape_Length	Shape_Area	HydroID	Name
Polygon	1	361620	2318211900	243	Gobebe

The **HydroID** value (243) is highlighted with a red box. A red arrow points from the **DrainID** box in the top screenshot to the **HydroID** box in the bottom screenshot, demonstrating that the *DrainID* of a *WatershedPoint* is equal to the *HydroID* of the corresponding *Watershed*.

References

- Abdelkader M. 2010. The contribution of the digital elevation models and geographic information systems in a watershed hydrologic research. *Appl Geomat* 2, 33–42.
- Amy Hillier. 2011. *Manual for working with ArcGIS 10*. University of Pennsylvanian.
- ESRI. 2010. *3D Analyst Tutorial*.
- Kennedy, M., 1996. *The Global Positioning System and GIS*. Ann Arbor Press, Chelsea, MI.
- Lyon, J., Ed., 2003. *GIS for Water Resources and Watershed Management*. Taylor & Francis, London.
- Maidment, D. and Djokic, D. 2000. *Hydrologic and Hydraulic Modeling Support with Geographic Information Systems*. ESRI Press, Redlands, CA.
- Phyllis T. 2003. 3D Models Using ArcGIS.
- Timothy James. 2010. *Gefting to know ArcGIS desktop ESRI Press*. ISBN 97&1-58948-260-9 (pbk.: alk. paper).
- Van Sickle, J., 2001. *GPS for Land Surveyors*. Ann Arbor Press, Chelsea, MI.
- Venkatesh M. 2012. *Watershed and Stream Network Delineation using Arc Hydro Tools*. School of Civil Engineering, Purdue University.
- Ward, A. and W. Elliot, 1995. *Environmental Hydrology*. Lewis Publishers, Boca Raton, FL.